

Equation-Free function toolbox for Matlab/Octave:

Summary User Manual

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Abstract

This ‘equation-free toolbox’ empowers the computer-assisted analysis of complex, multiscale systems. Its aim is to enable you to use microscopic simulators to perform system level tasks and analysis, because microscale simulations are often the best available description of a system. The methodology bypasses the derivation of macroscopic evolution equations by computing only short bursts of the microscale simulator (Kevrekidis & Samaey 2009, Kevrekidis et al. 2004, 2003, e.g.), and often only computing on small patches of the spatial domain (Roberts et al. 2014, e.g.). This suite of functions empowers users to start implementing such methods in their own applications. Download via <https://github.com/uoa1184615/EquationFreeGit>

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1 Introduction

Users Download via <https://github.com/uoa1184615/EquationFreeGit>. Ensure the folders `Patch`, `ProjInt` and `AutoDiff` of this toolbox are in a path searched by MATLAB/Octave. Then read the section(s) that documents the function of interest.

Quick start Maybe start by adapting one of the included examples. Many of the main functions include, at their beginning, example code of their use—code which is executed when the function is invoked without any arguments.

- To projectively integrate over time a multiscale, slow-fast, system of ODEs you could use `PIRK2()`, or `PIRK4()` for higher-order accuracy: adapt the Michaelis–Menten example at the beginning of `PIRK2.m` ([Section 2.1.2](#)).
- You may use forward bursts of simulation in order to simulate the slow dynamics backward in time, as in `egPIMM.m` ([Section 2.2](#)).
- To only resolve the slow dynamics in the projective integration, use lifting and restriction functions by adapting the singular perturbation ODE example at the beginning of `PIG.m` ([Section 2.3.2](#)).

Space-time systems Consider an evolving system over a large spatial domain when all you have is a microscale code. To efficiently simulate over the large domain, one can simulate in just small patches of the domain, appropriately coupled.

- In 1D space adapt the code at the beginning of `configPatches1.m` for Burgers' PDE ([Section 3.1.1](#)).
- In 2D space adapt the code at the beginning of `configPatches2.m` for nonlinear diffusion ([Section 4.1.1](#)).
- In 3D space adapt the code at the beginning of `configPatches3.m` for wave propagation through a heterogeneous medium ([Section 5.1.1](#)), or the patches of the 3D heterogeneous diffusion of `homoDiffEdgy3.m` ([Section 5.4](#)).
- Other provided examples include cases of macroscale *computational homogenisation* of microscale heterogeneity.
- In MATLAB, the axis labelling works best if one executes `set(groot,'defaultTextInterpreter','latex')`

Verification Most of these schemes have analytically proven ‘accuracy’ when compared to the underlying specified microscale system. In the spatial patch schemes, we measure ‘accuracy’ by the order of consistency between macroscale dynamics and the given specified microscale.

- [Roberts & Kevrekidis \(2007\)](#) and [Roberts et al. \(2014\)](#) proved reasonably general high-order consistency for the 1D and 2D patch schemes, respectively.
- In wave-like systems, [Cao & Roberts \(2016\)](#) established high-order consistency for the 1D staggered patch scheme, and [Divahar et al. \(2022\)](#), [Bunder et al. \(2021\)](#) established excellent 2D staggered patch schemes for waves.
- A heterogeneous microscale is more difficult, but [Bunder et al. \(2020\)](#), [Bunder & Roberts \(2022\)](#) developed a new ‘edgy’ inter-patch interpolation that is analytically proven to be excellent for simulating the macroscale homogenised dynamics of microscale heterogeneous systems in multiple space dimensions—now coded in the toolbox.

Blackbox scenarios Suppose that you have a *detailed and trustworthy* computational simulation of some problem of interest. Let’s say the simulation is coded in terms of detailed (microscale) variable values $\vec{u}(t)$, in $\mathbb{R}^p$ for some number p of field variables, and evolving in time t . The details $\vec{u}$ could represent particles, agents, or states of a system. When the computation is too time consuming to simulate all the times of interest, then Projective Integration may be able to predict long-time dynamics, both forward and backward in time. In this case, provide your detailed computational simulation as a ‘black box’ to the Projective Integration functions of [Chapter 2](#).

In many scenarios, the problem of interest involves space or a ‘spatial’ lattice. Let’s say that indices i correspond to ‘spatial’ coordinates $\vec{x}_i(t)$, which are often fixed: in lattice problems the positions $\vec{x}_i$ would be fixed in time (unless employing a moving mesh on the microscale); however, in particle problems the positions would evolve. And suppose your detailed and trustworthy simulation is coded also in terms of micro-field variable values $\vec{u}_i(t) \in \mathbb{R}^p$ at time t . Often the detailed computational simulation is too expensive over all the desired spatial domain $\vec{x} \in \mathbb{X} \subset \mathbb{R}^d$. In this case, the toolbox functions of [Chapter 3](#) empower you to simulate on only small, well-separated, patches of space by appropriately coupling between patches your simulation code, as a ‘black box’, executing on each small patch. The computational savings may be enormous, especially if combined with projective integration.

[Chapter 6](#) provides small examples of how to parallelise the patch computations over multiple processors. But such parallelisation may be only useful for scenarios where the microscale code has many millions of operations per time-step.

Contributors The aim of this project is to collectively develop a powerful and flexible MATLAB/Octave toolbox of equation-free algorithms. Initially

the algorithms are basic, and the ongoing program is developing more and more capability.

MATLAB appears a good choice for a first version since it is widespread, efficient, supports various parallel modes, and development costs are reasonably low. Further it is built on BLAS and LAPACK so the cache and superscalar CPU are well utilised. We aim to develop functions that work for MATLAB/Octave.

2 Projective integration of deterministic ODEs

This chapter provides some good projective integration functions ([Gear & Kevrekidis 2003b,c](#), [Givon et al. 2006](#), [Marschler et al. 2014](#), [Maclean & Gottwald 2015](#), [Sieber et al. 2018](#), e.g.). The goal is to enable computationally expensive multiscale dynamic simulations/integrations to efficiently compute over very long time scales.

Quick start [Section 2.1.2](#) shows the most basic use of a projective integration function. [Section 2.2](#) shows how to code more variations of the introductory example of a long time simulation of the Michaelis–Menton multiscale system of differential equations. Then see [Figures 2.1](#) and [2.2](#)

Scenario When you are interested in a complex system with many interacting parts or agents, you usually are primarily interested in the self-organised emergent macroscale characteristics. Projective integration empowers us to efficiently simulate such long-time emergent dynamics. We suppose you have coded some accurate, fine-scale, microscale simulation of the complex system, and call such code a microsolver.

The Projective Integration section of this toolbox consists of several functions. Each function implements over a long-time scale a variant of a standard numerical method to simulate/integrate the emergent dynamics of the complex system. Each function has standardised inputs and outputs.

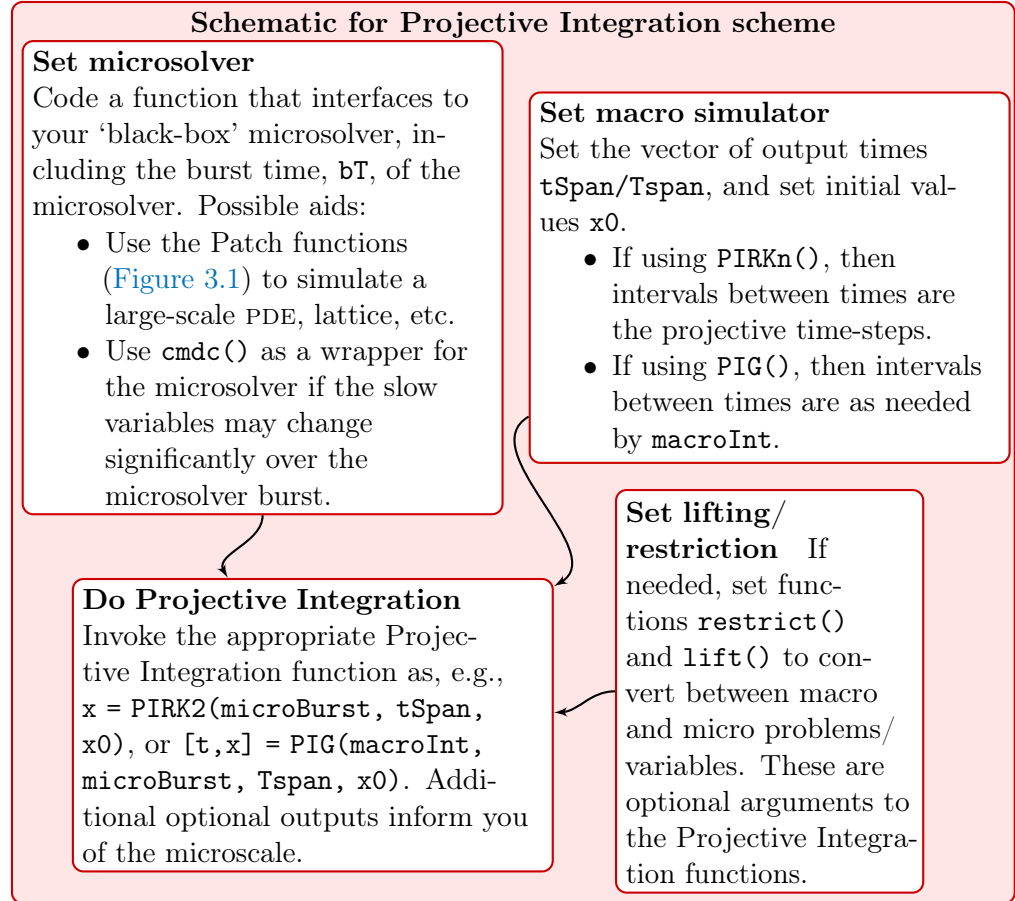
[Petersik \(2019–\)](#) is also developing, in python, some projective integration functions.

Time homogenisation [Section 2.5](#) introduces projective integration to efficiently time-homogenise rapid microscale fluctuations in a system. Inspired by [Bunder & Roberts \(2015\)](#), [von Oertzen & Kiefer \(2026\)](#).

Main functions

- Projective Integration by second or fourth-order Runge–Kutta is implemented by `PIRK2()` or `PIRK4()` respectively. These schemes are suitable for precise simulation of the slow dynamics, provided the time period spanned by an application of the microsolver is not too large.
- Projective Integration with a General method, `PIG()`. This function enables a Projective Integration implementation of any integration method over macroscale time-steps. It does not matter whether the method is a standard MATLAB/Octave algorithm, or one supplied by the user. `PIG()` should only be used directly in very stiff systems, less stiff systems additionally require `cdmc()`.

Figure 2.1: The Projective Integration method greatly accelerates simulation/integration of a system exhibiting multiple time scales. The Projective Integration [Chapter 2](#) presents several separate functions, as well as several optional wrapper functions that may be invoked. This chart overviews constructing a Projective Integration simulation, whereas [Figure 2.2](#) roughly guides which top-level Projective Integration functions should be used. [Chapter 2](#) fully details each function.

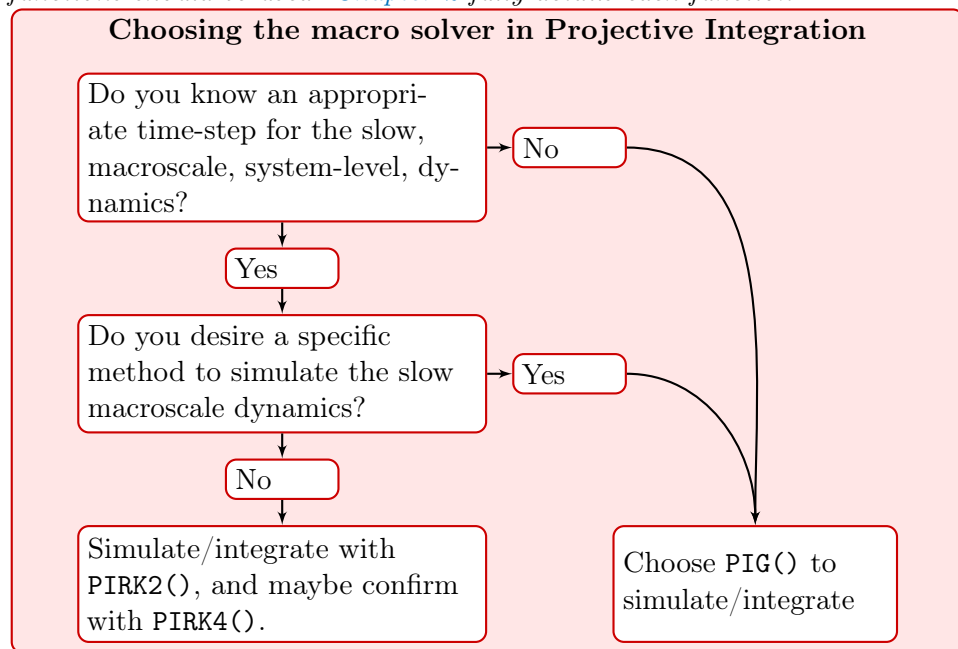


- *Constraint-defined manifold computing*, `cdmc()`, is a helper function, based on the method introduced in [Gear et al. \(2005a\)](#), that iteratively applies the microsolver and backward projection in time. The result is to project the fast variables close to the slow manifold, without advancing the current time by the burst time of the microsolver. This function reduces errors related to the simulation length of the microsolver in the `PIG` function. In particular, it enables `PIG()` to be used on problems that are not particularly stiff.

The above functions share dependence on a user-specified *microsolver* that accurately simulates some problem of interest.

The following sections describe the `PIRK2()` and `PIG()` functions in detail, providing an example for each. The function `PIRK4()` is very similar to `PIRK2()`. Descriptions for the minor functions follow, and an example using `cdmc()`.

Figure 2.2: The Projective Integration method greatly accelerates simulation/integration of a system exhibiting multiple time scales. In conjunction with Figure 2.1, this chart roughly guides which top-level Projective Integration functions should be used. Chapter 2 fully details each function.



2.1 PIRK2(): projective integration of second-order accuracy

2.1.1 Introduction

This Projective Integration scheme implements a macroscale scheme that is analogous to the second-order Runge–Kutta Improved Euler integration.

```
21 function [x, tms, xms, rm, svf] = PIRK2(microBurst, tSpan, x0, bT)
```

Input If there are no input arguments, then this function applies itself to the Michaelis–Menton example: see the code in [Section 2.1.2](#) as a basic template of how to use.

- `microBurst()`, a user-coded function that computes a short-time burst of the microscale simulation.

```
[tOut, xOut] = microBurst(tStart, xStart, bT)
```

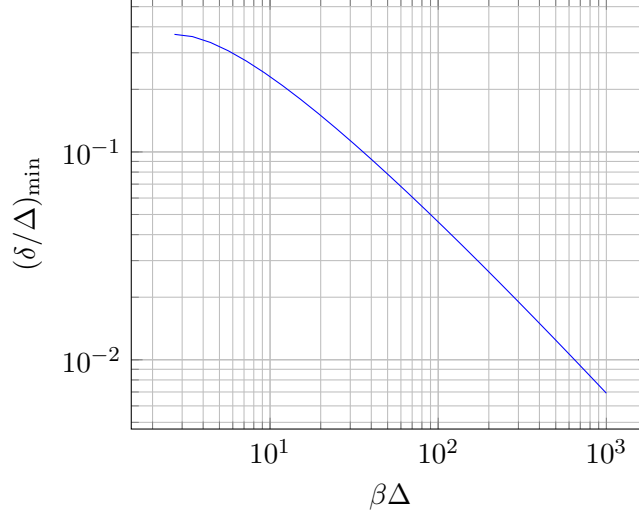
- Inputs: `tStart`, the start time of a burst of simulation; `xStart`, the row n -vector of the starting state; `bT`, *optional*, the total time to simulate in the burst—if your `microBurst()` determines the burst time, then replace `bT` in the argument list by `varargin`.
- Outputs: `tOut`, the column vector of solution times; and `xOut`, an array in which each *row* contains the system state at corresponding times.

Be wary that for very large scale separations (such as `MMepsilon < 1e-5` in the Michaelis–Menten example), microscale integration by error-controlled variable-step routines (such as `ode23/45`) often generate microscale variations that ruin the projective extrapolation of `PIRK2()`. In such cases, a fixed time-step microscale integrator is much better (such as `rk2Int()`).

- `tSpan` is an ℓ -vector of times at which the user requests output, of which the first element is always the initial time. `PIRK2()` does not use adaptive time-stepping; the macroscale time-steps are (nearly) the steps between elements of `tSpan`.
- `x0` is an n -vector of initial values at the initial time `tSpan(1)`. Elements of `x0` may be `NaN`: such `Nans` are carried in the simulation through to the output, and often represent boundaries/edges in spatial fields.
- `bT`, *optional*, either missing, or empty (`[]`), or a scalar: if a given scalar, then it is the length of the micro-burst simulations—the minimum amount of time needed for the microscale simulation to relax to the slow manifold; else if missing or `[]`, then `microBurst()` must itself determine the length of a burst.

```
77 if nargin < 4, bT = []; end
```

Figure 2.3: Need macroscale step Δ such that $|\alpha\Delta| \lesssim \sqrt{6\varepsilon}$ for given relative error ε and slow rate α , and then $\delta/\Delta \gtrsim \frac{1}{\beta\Delta} \log |\beta\Delta|$ determines the minimum required burst length δ for every given fast rate β .



Choose a long enough burst length Suppose: firstly, you have some desired relative accuracy ε that you wish to achieve (e.g., $\varepsilon \approx 0.01$ for two digit accuracy); secondly, the slow dynamics of your system occurs at rate/frequency of magnitude about α ; and thirdly, the rate of *decay* of your fast modes are faster than the lower bound β (e.g., if three fast modes decay roughly like $e^{-12t}, e^{-34t}, e^{-56t}$ then $\beta \approx 12$). Then set

1. a macroscale time-step, $\Delta = \text{diff}(\text{tSpan})$, such that $\alpha\Delta \approx \sqrt{6\varepsilon}$, and
2. a microscale burst length, $\delta = \text{bT} \gtrsim \frac{1}{\beta} \log |\beta\Delta|$, see [Figure 2.3](#).

Output If there are no output arguments specified, then a plot is drawn of the computed solution $\mathbf{x}$ versus tSpan .

- $\mathbf{x}$, an $\ell \times n$ array of the approximate solution vector. Each row is an estimated state at the corresponding time in tSpan . The simplest usage is then $\mathbf{x} = \text{PIRK2}(\text{microBurst}, \text{tSpan}, \mathbf{x0}, \text{bT})$.

However, microscale details of the underlying Projective Integration computations may be helpful. `PIRK2()` provides up to four optional outputs of the microscale bursts.

- $\mathbf{tms}$, optional, is an L dimensional column vector containing the microscale times within the burst simulations, each burst separated by `NaN`;
- $\mathbf{xms}$, optional, is an $L \times n$ array of the corresponding microscale states—each rows is an accurate estimate of the state at the corresponding time $\mathbf{tms}$ and helps visualise details of the solution.
- $\mathbf{rm}$, optional, a struct containing the ‘remaining’ applications of the `microBurst` required by the Projective Integration method during the calculation of the macrostep:

- `rm.t` is a column vector of microscale times; and
- `rm.x` is the array of corresponding burst states.

The states `rm.x` do not have the same physical interpretation as those in `xms`; the `rm.x` are required in order to estimate the slow vector field during the calculation of the Runge–Kutta increments, and do *not* accurately approximate the macroscale dynamics.

- `svf`, optional, a struct containing the Projective Integration estimates of the slow vector field.
 - `svf.t` is a 2ℓ dimensional column vector containing all times at which the Projective Integration scheme is extrapolated along microBurst data to form a macrostep.
 - `svf.dx` is a $2\ell \times n$ array containing the estimated slow vector field.

2.1.2 If no arguments, then execute an example

```
182 if nargin==0
```

Example code for Michaelis–Menton dynamics The Michaelis–Menten enzyme kinetics is expressed as a singularly perturbed system of differential equations for $x(t)$ and $y(t)$:

$$\frac{dx}{dt} = -x + (x + \frac{1}{2})y \quad \text{and} \quad \frac{dy}{dt} = \frac{1}{\epsilon} [x - (x + 1)y]$$

(encoded in function `MMburst()` in the next paragraph). With initial conditions $x(0) = 1$ and $y(0) = 0$, the following code computes and plots a solution over time $0 \leq t \leq 6$ for parameter $\epsilon = 0.05$. Since the rate of decay is $\beta \approx 1/\epsilon$ we choose a burst length $\epsilon \log(\Delta/\epsilon)$ as here the macroscale time-step $\Delta = 1$.

```
203 global MMepsilon
204 MMepsilon = 0.05
205 ts = 0:6
206 bT = MMepsilon*log( (ts(2)-ts(1))/MMepsilon )
207 [x,tms,xms] = PIRK2(@MMburst, ts, [1;0], bT);
208 figure, plot(ts,x,'o:',tms,xms)
209 title('Projective integration of Michaelis--Menten enzyme kinetics')
210 xlabel('time t'), legend('x(t)','y(t)')
```

Upon finishing execution of the example, exit this function.

```
216 return
217 end%if no arguments
```

Code a burst of Michaelis–Menten enzyme kinetics Integrate a burst of length `bT` of the ODEs for the Michaelis–Menten enzyme kinetics at parameter ϵ inherited from above. Code ODEs in function `dMMdt` with variables $x = x(1)$ and $y = x(2)$. Starting at time `ti`, and state `xi` (row), we here simply use MATLAB/Octave’s `ode23/lsode` to integrate a burst in time.

```
15 function [ts, xs] = MMburst(ti, xi, bT)
16     global MMepsilon
17     dMMdt = @(t,x) [ -x(1)+(x(1)+0.5)*x(2)
18                     1/MMepsilon*( x(1)-(x(1)+1)*x(2) ) ];
19     if ~exist('OCTAVE_VERSION','builtin')
20         [ts, xs] = ode23(dMMdt, [ti ti+bT], xi);
21     else % octave version
22         [ts, xs] = odeOdt(dMMdt, [ti ti+bT], xi);
23     end
24 end

8 function [ts,xs] = odeOdt(dxdt,tSpan,x0)
9     if length(tSpan)>2, ts = tSpan;
10    else ts = linspace(tSpan(1),tSpan(end),21);
11    end
12    % mimic ode45 and ode23, but much slower for non-PI
13    lsode_options('integration method','non-stiff');
14    xs = lsode(@(x,t) dxdt(t,x),x0,ts);
15 end
```

2.2 egPIMM: Example projective integration of Michaelis–Menton kinetics

The Michaelis–Menten enzyme kinetics is expressed as a singularly perturbed system of differential equations for $x(t)$ and $y(t)$:

$$\frac{dx}{dt} = -x + (x + \tfrac{1}{2})y \quad \text{and} \quad \frac{dy}{dt} = \frac{1}{\epsilon} [x - (x + 1)y]$$

(encoded in function `MMburst()` below). As illustrated by [Figure 2.5](#), the slow variable $x(t)$ evolves on a time scale of one, whereas the fast variable $y(t)$ evolves on a time scale of the small parameter ϵ .

Invoke projective integration Clear, and set the scale separation parameter ϵ to something small like 0.01. Here use $\epsilon = 0.1$ for clearer graphs.

```
31 clear all, close all
32 global MMepsilon
33 MMepsilon = 0.1
```

First, the end of this section encodes the computation of bursts of the Michaelis–Menten system in a function `MMburst()`. Second, here set macroscale times of computation and interest into vector `ts`. Then, invoke Projective Integration with `PIRK2()` applied to the burst function, say using bursts of simulations of length 2ϵ , and starting from the initial condition for the Michaelis–Menten system, at time $t = 0$, of $(x, y) = (1, 0)$ (off the slow manifold).

```
48 ts = 0:6
49 xs = PIRK2(@MMburst, ts, [1;0], 2*MMepsilon)
50 plot(ts,xs,'o:')
51 xlabel('time t'), legend('x(t)', 'y(t)')
52 title('macroscale points only')
53 ifOurCf2eps([mfilename '1'])
54 pause(1)
```

[Figure 2.4](#) plots the macroscale results showing the long time decay of the Michaelis–Menten system on the slow manifold. [Sieber et al. \(2018\)](#) [§4] used this system as an example of their analysis of the convergence of Projective Integration.

Request and plot the microscale bursts Because the initial conditions of the simulation are off the slow manifold, the initial macroscale step appears to ‘jump’ ([Figure 2.4](#)). In order to see the initial transient attraction to the slow manifold we plot some microscale data in [Figure 2.5](#). Two further output variables provide this microscale burst information.

```
80 [xs,tMicro,xMicro] = PIRK2(@MMburst, ts, [1;0], 2*MMepsilon);
81 figure, plot(ts,xs,'o:',tMicro,xMicro)
82 xlabel('time t'), legend('x(t)', 'y(t)')
83 title('macroscale points with microscale bursts')
84 ifOurCf2eps([mfilename '2'])
85 pause(1)
```

Figure 2.4: Michaelis–Menten enzyme kinetics simulated with the projective integration of `PIRK2()`: macroscale samples.

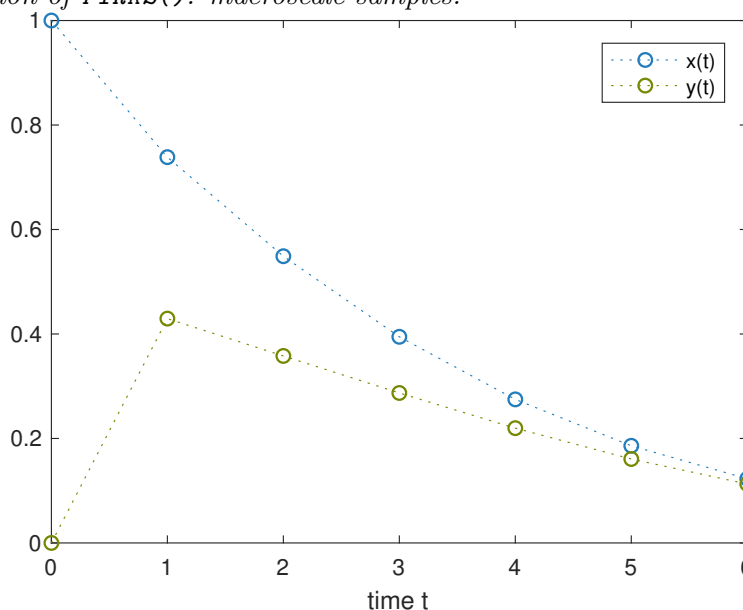


Figure 2.5 plots the macroscale and microscale results—also showing that the initial burst is by default twice as long. Observe the slow variable $x(t)$ is also affected by the initial transient (hence other schemes which ‘freeze’ slow variables are less accurate).

Simulate backward in time Figure 2.6 shows that projective integration even simulates backward in time along the slow manifold using short forward bursts (Gear & Kevrekidis 2003a, Frewen et al. 2009). Such backward macroscale simulations succeed despite the fast variable $y(t)$, when backward in time, being viciously unstable. However, backward integration appears to need longer bursts, here 3ϵ .

```

115 ts = 0:-1:-5
116 [xs,tMicro,xMicro] = PIRK2(@MMburst, ts, 0.2*[1;1], 3*MMepsilon);
117 figure, plot(ts,xs,'o:',tMicro,xMicro)
118 xlabel('time t'), legend('x(t)','y(t)')
119 title('backward integration showing points with bursts')
120 ifOurCf2eps([mfilename '3'])

```

Code a burst of Michaelis–Menten enzyme kinetics Integrate a burst of length bT of the ODEs for the Michaelis–Menten enzyme kinetics at parameter ϵ inherited from above. Code ODEs in function `dMMdt` with variables $x = x(1)$ and $y = x(2)$. Starting at time t_i , and state x_i (row), we here simply use MATLAB/Octave’s `ode23/lode` to integrate a burst in time.

```

15 function [ts, xs] = MMburst(ti, xi, bT)
16     global MMeppsiion
17     dMMdt = @(t,x) [ -x(1)+(x(1)+0.5)*x(2)
18                     1/MMeppsiion*( x(1)-(x(1)+1)*x(2) ) ];

```

Figure 2.5: Michaelis–Menten enzyme kinetics simulated with the projective integration of `PIRK2()`: the microscale bursts show the initial transients on a time scale of $\epsilon = 0.1$, and then the alignment along the slow manifold.

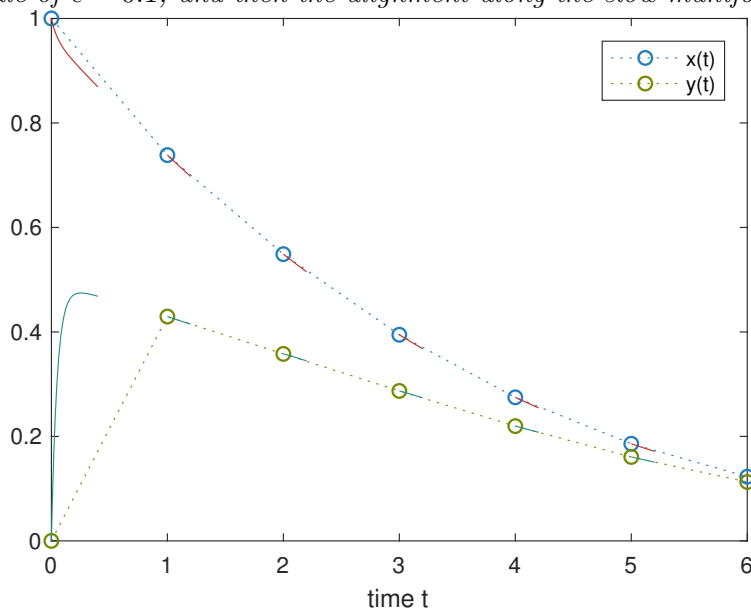
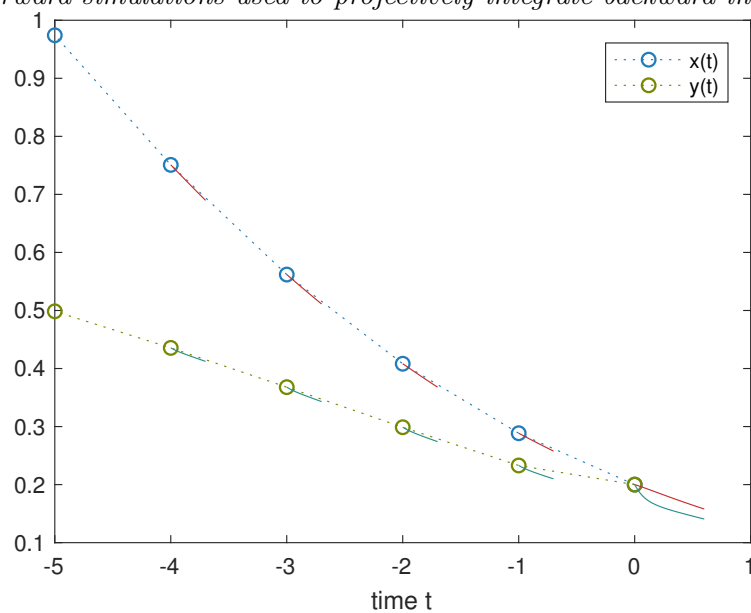


Figure 2.6: Michaelis–Menten enzyme kinetics at $\epsilon = 0.1$ simulated backward with the projective integration of `PIRK2()`: the microscale bursts show the short forward simulations used to projectively integrate backward in time.




```
19     if ~exist('OCTAVE_VERSION','builtin')
20         [ts, xs] = ode23(dMMdt, [ti ti+bT], xi);
21     else % octave version
22         [ts, xs] = odeOct(dMMdt, [ti ti+bT], xi);
23     end
24 end

8  function [ts,xs] = odeOct(dxdt,tSpan,x0)
9      if length(tSpan)>2, ts = tSpan;
10     else ts = linspace(tSpan(1),tSpan(end),21);
11     end
12     % mimic ode45 and ode23, but much slower for non-PI
13     lsode_options('integration method','non-stiff');
14     xs = lsode(@(x,t) dxdt(t,x),x0,ts);
15 end
```

2.3 PIG(): Projective Integration via a General macroscale integrator

2.3.1 Introduction

This is a Projective Integration scheme when the macroscale integrator is any specified coded method. The advantage is that one may use MATLAB/Octave's inbuilt integration functions, with all their sophisticated error control and adaptive time-stepping, to do the macroscale integration/simulation.

By default, for the microscale simulations PIG() uses 'constraint-defined manifold computing', `cdmc()` (Section 2.6). This algorithm, initiated by Gear et al. (2005b), uses a backward projection so that the simulation time is unchanged after running the microscale simulator.

```

30 function [T,X,tms,xms,svf] = PIG(macroInt,microBurst,Tspan,x0 ...
31                                ,restrict,lift,cdmcFlag)

```

Inputs:

- `macroInt()`, the numerical method that the user wants to apply on a slow-time macroscale. Either specify a standard MATLAB/Octave integration function (such as 'ode23' or 'ode45'), or code your own integration function using standard arguments. That is, if you code your own, then it must be

$$[Ts,Xs] = \text{macroInt}(F,Tspan,X0)$$

where

- function $F(T,X)$ notionally evaluates the time derivatives $d\vec{X}/dt$ at any time;
- `Tspan` is either the macro-time interval, or the vector of macroscale times at which macroscale values are to be returned; and
- `X0` are the initial values of $\vec{X}$ at time `Tspan(1)`.

Then the i th row of `Xs`, `Xs(i,:)`, is to be the vector $\vec{X}(t)$ at time $t = Ts(i)$. Remember that in PIG() the function $F(T,X)$ is to be estimated by Projective Integration.

- `microBurst()` is a function that produces output from the user-specified code for a burst of microscale simulation. The function must internally specify/decide how long a burst it is to use. Usage

$$[tbs,xbs] = \text{microBurst}(tb0,xb0)$$

Inputs: `tb0` is the start time of a burst; `xb0` is the n -vector microscale state at the start of a burst.

Outputs: `tbs`, the vector of solution times; and `xbs`, the corresponding microscale states.

- **Tspan**, a vector of macroscale times at which the user requests output. The first element is always the initial time. If **macroInt** reports adaptively selected time steps (e.g., **ode45**), then **Tspan** consists of an initial and final time only.
- **x0**, the n -vector of initial microscale values at the initial time **Tspan**(1).

Optional Inputs: **PIG()** allows for none, two or three additional inputs after **x0**. If you distinguish distinct microscale and macroscale states and your aim is to do Projective Integration on the macroscale only, then lifting and restriction functions must be provided to convert between them. Usage **PIG(...,restrict,lift)**:

- **restrict(x)**, a function that takes an input high-dimensional, n -D, microscale state $\vec{x}$ and computes the corresponding low-dimensional, N -D, macroscale state $\vec{X}$;
- **lift(X,xApprox)**, a function that converts an input low-dimensional, N -D, macroscale state $\vec{X}$ to a corresponding high-dimensional, n -D, microscale state $\vec{x}$, given that **xApprox** is a recently computed microscale state on the slow manifold.

Either both **restrict()** and **lift()** are to be defined, or neither. If neither are defined, then they are assumed to be identity functions, so that $N=n$ in the following.

If desired, the default constraint-defined manifold computing microsolver may be disabled, via **PIG(...,restrict,lift,cdmcFlag)**

- **cdmcFlag**, *any* seventh input to **PIG()**, will disable **cdmc()**, e.g., the string '**cdmc off**'.

If the **cdmcFlag** is to be set without using a **restrict()** or **lift()** function, then use empty matrices **[]** for the restrict and lift functions.

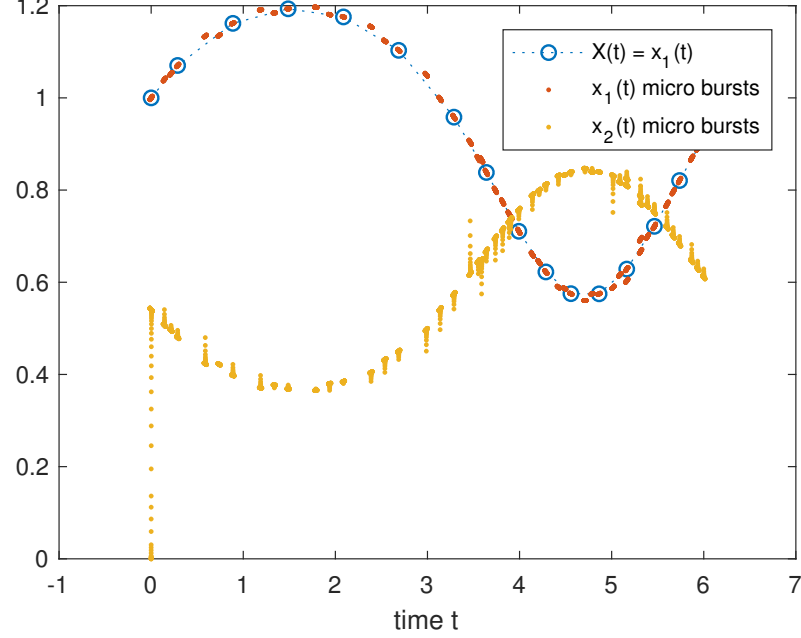
Output Between zero and five outputs may be requested. If there are no output arguments specified, then a plot is drawn of the computed solution **X** versus **T**. Most often you would store the first two output results of **PIG()**, via say **[T,X] = PIG(...)**.

- **T**, an L -vector of times at which **macroInt** produced results.
- **X**, an $L \times N$ array of the computed solution: the i th row of **X**, **X(i,:)**, is to be the macro-state vector $\vec{X}(t)$ at time $t = T(i)$.

However, microscale details of the underlying Projective Integration computations may be helpful, and so **PIG()** provides some optional outputs of the microscale bursts, via **[T,X,tms,xms] = PIG(...)**

- **tms**, optional, is an ℓ -dimensional column vector containing microscale times with bursts, each burst separated by **NaN**;
- **xms**, optional, is an $\ell \times n$ array of the corresponding microscale states.

Figure 2.7: Projective Integration by *PIG* of the example system (2.1) with $\epsilon = 10^{-3}$ (Section 2.3.2). The macroscale solution $X(t)$ is represented by just the blue circles. The microscale bursts are the microscale states $(x_1(t), x_2(t)) = (\text{red, yellow})$ dots.



In some contexts it may be helpful to see directly how Projective Integration approximates a reduced slow vector field, via `[T,X,tms,xms,svf] = PIG(...)` in which

- **svf**, optional, a struct containing the Projective Integration estimates of the slow vector field.
 - **svf.T** is a $\hat{L}$ -dimensional column vector containing all times at which the microscale simulation data is extrapolated to form an estimate of $d\vec{x}/dt$ in `macroInt()`.
 - **svf.dX** is a $\hat{L} \times N$ array containing the estimated slow vector field.

If `macroInt()` is, for example, the forward Euler method (or the Runge–Kutta method), then $\hat{L} = L$ (or $\hat{L} = 4L$).

2.3.2 If no arguments, then execute an example

180 `if nargin==0`

As a basic example, consider a microscale system of the singularly perturbed system of differential equations

$$\frac{dx_1}{dt} = \cos(x_1) \sin(x_2) \cos(t) \quad \text{and} \quad \frac{dx_2}{dt} = \frac{1}{\epsilon} [\cos(x_1) - x_2]. \quad (2.1)$$

The macroscale variable is $X(t) = x_1(t)$, and the evolution dX/dt is unclear. With initial condition $X(0) = 1$, the following code computes and plots a solution of the system (2.1) over time $0 \leq t \leq 6$ for parameter

$\epsilon = 10^{-3}$ (Figure 2.7). Whenever needed by `microBurst()`, the microscale system (2.1) is initialised ('lifted') using $x_2(t) = x_2^{\text{approx}}$ (yellow dots in Figure 2.7).

First we code the right-hand side function of the microscale system (2.1) of ODEs.

```
214 epsilon = 1e-3;
215 dxdt=@(t,x) [ cos(x(1))*sin(x(2))*cos(t)
216               ( cos(x(1))-x(2) )/epsilon ];
```

Second, we code microscale bursts, here using the standard `ode45()`. We choose a burst length $2\epsilon \log(1/\epsilon)$ as the rate of decay is $\beta \approx 1/\epsilon$ but we do not know the macroscale time-step invoked by `macroInt()`, so blithely assume $\Delta \leq 1$ and then double the usual formula for safety.

```
227 bT = 2*epsilon*log(1/epsilon)
228 if ~exist('OCTAVE_VERSION','builtin')
229     micB='ode45'; else micB='rk2Int'; end
230 microBurst = @(tb0, xb0) feval(micB,dxdt,[tb0 tb0+bT],xb0);
```

Third, code functions to convert between macroscale and microscale states.

```
237 restrict = @(x) x(1);
238 lift = @(X,xApprox) [X; xApprox(2)];
```

Fourth, invoke PIG to use MATLAB/Octave's `ode23/lode`, say, on the macroscale slow evolution. Integrate the micro-bursts over $0 \leq t \leq 6$ from initial condition $\vec{x}(0) = (1, 0)$. You could set `Tspan=[0 -6]` to integrate backward in macroscale time with forward microscale bursts (Gear & Kevrekidis 2003a, Frewen et al. 2009).

```
250 Tspan = [0 6];
251 x0 = [1;0];
252 if ~exist('OCTAVE_VERSION','builtin')
253     macInt='ode23'; else macInt='odeOct'; end
254 [Ts,Xs,tms,xms] = PIG(macInt,microBurst,Tspan,x0,restrict,lift);
```

Plot output of this projective integration.

```
260 figure, plot(Ts,Xs,'o:',tms,xms,'.')
261 title('Projective integration of singularly perturbed ODE')
262 xlabel('time t')
263 legend('X(t) = x_1(t)', 'x_1(t) micro bursts', 'x_2(t) micro bursts')
```

Upon finishing execution of the example, exit this function.

```
269 return
270 end%if no arguments
```

2.4 PIRK4(): projective integration of fourth-order accuracy

2.4.1 Introduction

This Projective Integration scheme implements a macrosolver analogous to the fourth-order Runge–Kutta method.

```
19 function [x, tms, xms, rm, svf] = PIRK4(microBurst, tSpan, x0, bT)
```

See [Section 2.1](#) as the inputs and outputs are the same as PIRK2().

If no arguments, then execute an example

```
29 if nargin==0
```

Example of Michaelis–Menton backwards in time The Michaelis–Menten enzyme kinetics is expressed as a singularly perturbed system of differential equations for $x(t)$ and $y(t)$ (encoded in function `MMburst`):

$$\frac{dx}{dt} = -x + (x + \frac{1}{2})y \quad \text{and} \quad \frac{dy}{dt} = \frac{1}{\epsilon} [x - (x + 1)y].$$

With initial conditions $x(0) = y(0) = 0.2$, the following code uses forward time bursts in order to integrate backwards in time to $t = -5$ ([Frewen et al. 2009](#), e.g.). It plots the computed solution over time $-5 \leq t \leq 0$ for parameter $\epsilon = 0.1$. Since the rate of decay is $\beta \approx 1/\epsilon$ we choose a burst length $\epsilon \log(|\Delta|/\epsilon)$ as here the macroscale time-step $\Delta = -1$.

```
50 global MMepsilon
51 MMepsilon = 0.05
52 ts = 0:-1:-5
53 bT = MMepsilon*log(abs(ts(2)-ts(1))/MMepsilon)
54 [x,tms,xms,rm,svf] = PIRK4(@MMburst, ts, 0.2*[1;1], bT);
55 figure, plot(ts,x,'o:',tms,xms)
56 xlabel('time t'), legend('x(t)','y(t)')
57 title('Backwards-time projective integration of Michaelis--Menten')
```

Upon finishing execution of the example, exit this function.

```
63 return
64 end%if no arguments
```

Code a burst of Michaelis–Menten enzyme kinetics Integrate a burst of length `bT` of the ODEs for the Michaelis–Menten enzyme kinetics at parameter ϵ inherited from above. Code ODEs in function `dMMdt` with variables $x = x(1)$ and $y = x(2)$. Starting at time `ti`, and state `xi` (row), we here simply use MATLAB/Octave’s `ode23/lsode` to integrate a burst in time.

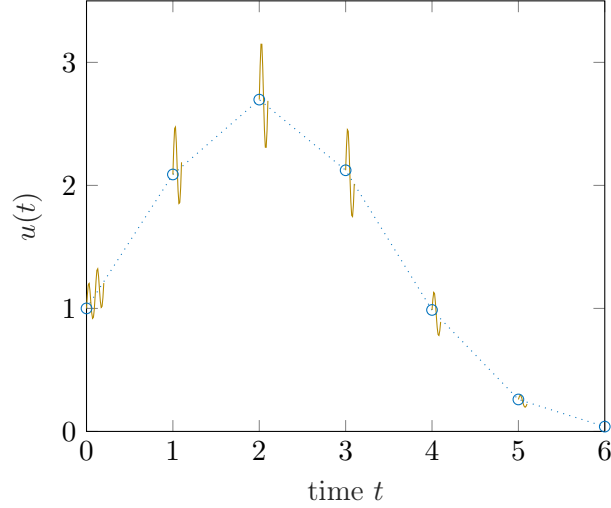
```
15 function [ts, xs] = MMburst(ti, xi, bT)
16     global MMepsilon
17     dMMdt = @(t,x) [ -x(1)+(x(1)+0.5)*x(2)
18                     1/MMepsilon*( x(1)-(x(1)+1)*x(2) ) ];
19     if ~exist('OCTAVE_VERSION','builtin')
```

```
20     [ts, xs] = ode23(dMMdt, [ti ti+bT], xi);
21     else % octave version
22     [ts, xs] = odeOct(dMMdt, [ti ti+bT], xi);
23     end
24 end

8  function [ts,xs] = odeOct(dxdt,tSpan,x0)
9      if length(tSpan)>2, ts = tSpan;
10     else ts = linspace(tSpan(1),tSpan(end),21);
11     end
12     % mimic ode45 and ode23, but much slower for non-PI
13     lsode_options('integration method','non-stiff');
14     xs = lsode(@(x,t) dxdt(t,x),x0,ts);
15 end
```

Figure 2.8: Time homogenisation of rapidly fluctuating ODE (2.2) by projective integration of short burst: blue, the macroscale homogenisation solution; yellow, the micro-scale bursts of rapid fluctuations.

Projective integration of rapidly fluctuating ODE



2.5 timeHomogenise: example time-homogenisation by projective integration of a rapidly fluctuating ODE

Any one of three Projective Integration schemes are invoked to integrate an ordinary differential equation which has a rapidly fluctuating component (Figure 2.8). Here

$$\frac{du}{dt} = (1 - t/2)u + \frac{1}{\tau}u \cos(2\pi t/\tau), \quad u(0) = 1, \quad \text{over } 0 \leq t \leq 6. \quad (2.2)$$

The fluctuation, of microscale period τ , is of large strength $1/\tau$ in order for the fluctuations to have a strong effect on the solutions (Figure 2.8). The trick to this application is to sub-sample the micro-burst to one data-point each microscale period (aka the micro-period return map): then the projective integration functions just operates on the ‘smooth’ return-map data.

Set the period τ as global variable `mPeriod`. Also (optional) define and reset global cells in order to store the details of the micro-bursts *before* sub-sampling.

```
43 global ums tms mPeriod
44 ums={}; tms={};
```

Select the projective integration method with `pirk`, set macro-time-steps to be one (low accuracy), and set micro-period τ here to be relatively big so we can see the microscale fluctuations in the plots

```
53 pirk = 4    % select 1=PIG or 2nd or 4th order method
54 ts = 0:6    % set macroscale time steps
55 mPeriod = 0.1 % microscale period of fluctuations
```


Execute selected projective integration scheme using `timeHomogeniseBurst` function. Use a burst of length just *one period* long because here there are no exp-decaying transients to overcome, and because one period is sufficient for the projections.

```

65 switch pirk
66 case 2, u = PIRK2(@timeHomogeniseBurst, ts, 1, mPeriod);
67 case 4, u = PIRK4(@timeHomogeniseBurst, ts, 1, mPeriod);
68 case 1, mPeriod = 0.01 % shorter microscale
69     [ts,u] = PIG(@ode23 ...
70         ,@(t0, u0) timeHomogeniseBurst(t0,u0,mPeriod) ...
71         ,ts([1 end]),1);
72 end

```

Plot the PI macroscale solution (blue). Add plots of the stored microscale bursts (dark yellow).

```

79 figure(1), plot(ts,u,'o:')
80 title('Projective integration of rapidly fluctuating ODE')
81 xlabel('time $t$'), ylabel('$u(t)$')
82 hold on
83 arrayfun(@(x,y) plot(x{:},y{:},'Color',[0.71 0.54 0]) ...
84     ,tms([1 2:pirk:end]),ums([1 2:pirk:end]))
85 hold off

```

Code a burst of a rapidly fluctuating ODE Integrate a burst of length `bT` of the ODE (2.2) adapted from [Bunder & Roberts \(2015\)](#).

```

107 function [ts, us] = timeHomogeniseBurst(ti, ui, bT)
108     global ums tms mPeriod

```

Code ODE in internal function `dudt` with variable $u = u(1)$. For fluctuating ODE need the phase of micro-fluctuation to be synchronised with each burst.

```

116     dudt = @(t,u) (1-t/2)*u+u*cos(2*pi/mPeriod*(t-ti))/mPeriod;

```

Use `rk2Int` for the burst because it empowers us to force that a burst is an integral number of micro-periods. Ensure there are at least 10 time-steps per micro-period (`rk2Int` enforces at least 10).

```

125     nPeriods = ceil(bT/mPeriod);
126     ts = ti+linspace(0,nPeriods*mPeriod,nPeriods*10+1);
127     [ts, us] = rk2Int(dudt, ts, ui);

```

We then sub-sample the micro-burst at one phase in each period. To avoid discarding sub-period details, here store sub-period details in global cell-arrays. Since `rkInt` may increase the number of (equal-spaced) micro-steps until accurate, we have to recompute the number of micro-steps per micro-period.

```

138     tms{end+1}=ts; ums{end+1}=us;
139     mSteps = (length(ts)-1)/nPeriods;
140     assert(round(mSteps)==mSteps,"not integer micro-steps")
141     ts = ts(1:mSteps:end);

```

```
142         us = us(1:mSteps:end);  
143     end
```

2.6 `cdmc()`: constraint defined manifold computing

The function `cdmc()` iteratively applies the given micro-burst and then projects backward to the initial time. The cumulative effect is to relax the variables to the attracting slow manifold, while keeping the ‘final’ time for the output the same as the input time.

```
17 function [ts, xs] = cdmc(microBurst, t0, x0)
```

Input

- `microBurst()`, a black-box micro-burst function suitable for Projective Integration. See any of `PIRK2()`, `PIRK4()`, or `PIG()` for a description of `microBurst()`.
- `t0`, an initial time.
- `x0`, an initial state vector.

Output

- `ts`, a vector of times.
- `xs`, an array of state estimates produced by `microBurst()`.

This function is a wrapper for the micro-burst. For instance if the problem of interest is a dynamical system that is not too stiff, and which is simulated by the micro-burst function `sol(t,x)`, one would invoke `cdmc()` by defining

```
cdmcSol = @(t,x) cdmc(sol,t,x) |
```

and thereafter use `cdmcSol()` in place of `sol()` as the `microBurst` in any Projective Integration scheme. The original `microBurst sol()` could create large errors if used in the `PIG()` scheme, but the output via `cdmc()` should not.

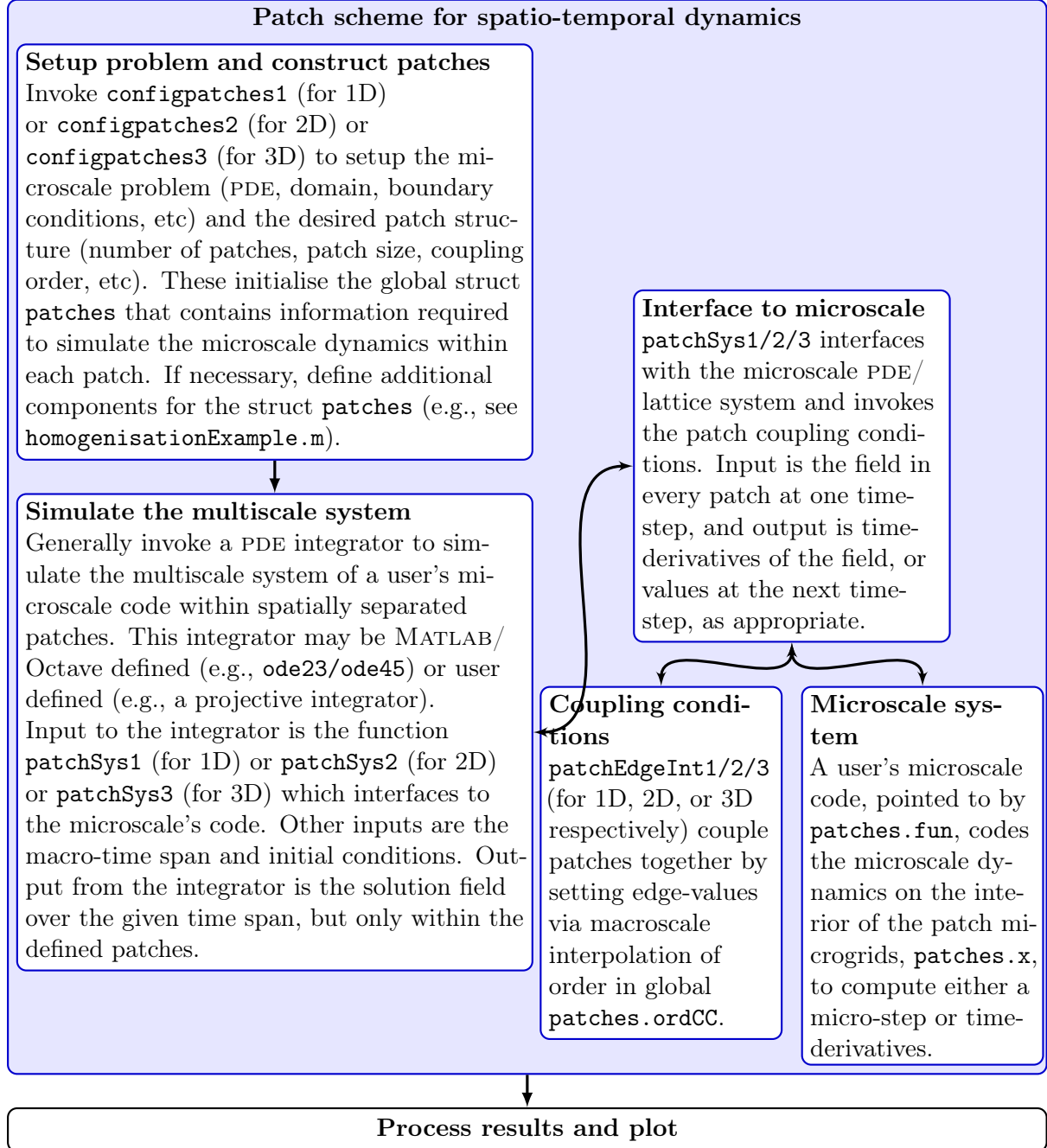
3 Patch scheme for given microscale discrete space system

Consider spatio-temporal multiscale systems where the spatial domain is so large that a given microscale code cannot be computed in a reasonable time. The *patch scheme* computes the microscale details only on small patches of the space-time domain, and produce correct macroscale predictions by craftily coupling the patches across unsimulated space (Hyman 2005, Samaey et al. 2005, 2006, Roberts & Kevrekidis 2007, Liu et al. 2015, e.g.). The resulting macroscale predictions were generally proved to be consistent with the microscale dynamics, to some specified order of accuracy, in a series of papers: 1D-space dissipative systems (Roberts & Kevrekidis 2007, Bunder et al. 2017); 2D-space dissipative systems (Roberts et al. 2014, Bunder et al. 2020); and 1D-space wave-like systems (Cao & Roberts 2016).

The microscale spatial structure is to be on a lattice such as obtained from finite difference/element/volume approximation of a PDE. The microscale is either continuous or discrete in time.

Quick start See Sections 3.1.1 and 4.1.1 which respectively list example basic code that uses the provided functions to simulate the 1D Burgers' PDE, and a 2D nonlinear 'diffusion' PDE. Then see Figure 3.1.

Figure 3.1: The Patch methods, [Chapter 3](#), accelerate simulation/integration of multiscale systems with interesting spatial/network structure/patterns. The methods use your given microsimulators whether coded from PDEs, lattice systems, or agent/particle microscale simulators. The patch functions require that a user configure the patches, and interface the coupled patches with a time integrator/simulator. This chart overviews the main functional recursion involved.



3.1 configPatches1(): configure spatial patches in 1D

Makes the struct `patches` for use by the patch/gap-tooth time derivative/step function `patchSys1()`. [Section 3.1.1](#) lists an example of its use.

```

19 function patches = configPatches1(fun,Xlim,Dom ...
20     ,nPatch,ordCC,dx,nSubP,varargin)
21 version = '2023-10-12';

```

Input If invoked with no input arguments, then executes an example of simulating Burgers' PDE—see [Section 3.1.1](#) for the example code.

- `fun` is the name of the user function, `fun(t,u,patches)` or `fun(t,u)` or `fun(t,u,patches,...)`, that computes time derivatives (or time-steps) of quantities on the 1D micro-grid within all the 1D patches.
- `Xlim` give the macro-space spatial domain of the computation, namely the interval `[Xlim(1),Xlim(2)]`.

But, if `nPatch = 1`, then the domain is the mean of `Xlim` plus/minus half the patch-width.

- `Dom` sets the type of macroscale conditions for the patches, and reflects the type of microscale boundary conditions of the problem. If `Dom` is `NaN` or `[]`, then the field `u` is macro-periodic in the 1D spatial domain, and resolved on equi-spaced patches. If `Dom` is a character array, then that specifies the `.type` of the following structure, with `.bcOffset` set to the default zero. Otherwise `Dom` is a structure with the following components.
 - `.type`, char-array, of either `'periodic'` (the default), `'equispace'`, `'chebyshev'`, `'usergiven'`. For all cases except `'periodic'`, users *must* code into `fun` the micro-grid boundary conditions that apply at the left(right) edge of the leftmost(rightmost) patches.
 - `.bcOffset`, optional one or two element array, in the cases of `'equispace'` or `'chebyshev'` the patches are placed so the left/right macroscale boundaries are aligned to the left/right edges of the corresponding extreme patches, but offset by `bcOffset` of the sub-patch micro-grid spacing. For example, use `bcOffset=0` when applying Dirichlet boundary values on the extreme edge micro-grid points, whereas use `bcOffset=0.5` when applying Neumann boundary conditions halfway between the extreme edge micro-grid points.
 - `.X`, optional array, in the case `'usergiven'` it specifies the locations of the centres of the `nPatch` patches—the user is responsible it makes sense.
- `nPatch` is the number of spatial patches: an integer ≥ 1 .
- `ordCC`, must be ≥ -1 , is the 'order' of interpolation across empty space of the macroscale patch values to the edge of the patches for inter-patch

coupling: where `ordCC` of 0 or -1 gives spectral interpolation; and `ordCC` being odd specifies staggered spatial grids.

- `dx` (real) is usually the sub-patch micro-grid spacing in x .

However, if `Dom` is NaN (as for pre-2023), then `dx` actually is `ratio`, namely the ratio of (depending upon `EdgyInt`) either the half-width or full-width of a patch to the equi-spacing of the patch mid-points—adjusted a little when `nEdge` > 1 . So either `ratio` $= \frac{1}{2}$ means the patches abut and `ratio` $= 1$ is overlapping patches as in holistic discretisation, or `ratio` $= 1$ means the patches abut. Small `ratio` should greatly reduce computational time.

- `nSubP` is the number of equi-spaced microscale lattice points in each patch. If not using `EdgyInt`, then `nSubP/nEdge` must be odd integer so that there is/are centre-patch lattice point(s). So for the defaults of `nEdge` $= 1$ and not `EdgyInt`, then `nSubP` must be odd.
- `'nEdge'`, *optional*, default=1, the number of edge values set by interpolation at the edge regions of each patch. The default is one (suitable for microscale lattices with only nearest neighbour interactions).
- `EdgyInt`, true/false, *optional*, default=false. If true, then interpolate to left/right edge-values from right/left next-to-edge values. If false or omitted, then interpolate from centre-patch values.
- `nEnsem`, *optional-experimental*, default one, but if more, then an ensemble over this number of realisations.
- `hetCoeffs`, *optional*, default empty. Supply a 1D or 2D array of microscale heterogeneous coefficients to be used by the given microscale `fun` in each patch. Say the given array `cs` is of size $m_x \times n_c$, where n_c is the number of different sets of coefficients. The coefficients are to be the same for each and every patch; however, macroscale variations are catered for by the n_c coefficients being n_c parameters in some macroscale formula.
 - If `nEnsem` $= 1$, then the array of coefficients is just tiled across the patch size to fill up each patch, starting from the first point in each patch. Best accuracy usually obtained when the periodicity of the coefficients is a factor of `nSubP-2*nEdge` for `EdgyInt`, or a factor of $(\text{nSubP}-\text{nEdge})/2$ for not `EdgyInt`.
 - If `nEnsem` > 1 (value immaterial), then reset `nEnsem` $:= m_x$ and construct an ensemble of all m_x phase-shifts of the coefficients. In this scenario, the inter-patch coupling couples different members in the ensemble. When `EdgyInt` is true, and when the coefficients are diffusivities/elasticities, then this coupling cunningly preserves symmetry.
- `nCore`, *optional-experimental*, default one, but if more, and only for non-`EdgyInt`, then interpolates from an average over the core of a patch, a core of size `??`. Then edge values are set according to interpolation of the averages`??` or so that average at edges is the interpolant`??`

- `'parallel'`, true/false, *optional*, default=false. If false, then all patch computations are on the user's main CPU—although a user may well separately invoke, say, a GPU to accelerate sub-patch computations.

If true, and it requires that you have MATLAB's Parallel Computing Toolbox, then it will distribute the patches over multiple CPUs/cores. In MATLAB, only one array dimension can be split in the distribution, so it chooses the one space dimension x . A user may correspondingly distribute arrays with property `patches.codist`, or simply use formulas invoking the preset distributed arrays `patches.x`. If a user has not yet established a parallel pool, then a 'local' pool is started.

Output The struct `patches` is created and set with the following components. If no output variable is provided for `patches`, then make the struct available as a global variable.¹

```
190 if nargout==0, global patches, end
191 patches.version = version;
```

- `.fun` is the name of the user's function `fun(t,u,patches)` or `fun(t,u)` or `fun(t,u,patches,...)`, that computes the time derivatives (or steps) on the patchy lattice.
- `.ordCC` is the specified order of inter-patch coupling.
- `.periodic`: either true, for interpolation on the macro-periodic domain; or false, for general interpolation by divided differences over non-periodic domain or unevenly distributed patches.
- `.stag` is true for interpolation using only odd neighbouring patches as for staggered grids, and false for the usual case of all neighbour coupling.
- `.Cwtsr` and `.Cwtsl`, only for macro-periodic conditions, are the `ordCC`-vector of weights for the inter-patch interpolation onto the right and left edges (respectively) with patch:macroscale ratio as specified or as derived from `dx`.
- `.x` (4D) is `nSubP × 1 × 1 × nPatch` array of the regular spatial locations x_{iI} of the i th microscale grid point in the I th patch.
- `.ratio`, only for macro-periodic conditions, is the size ratio of every patch.
- `.nEdge` is, for each patch, the number of edge values set by interpolation at the edge regions of each patch.
- `.le`, `.ri` determine inter-patch coupling of members in an ensemble. Each a column vector of length `nEnsem`.
- `.cs` either

¹When using `spmd` parallel computing, it is generally best to avoid global variables, and so instead prefer using an explicit output variable.

- [] 0D, or
- if `nEnsem = 1`, $(\text{nSubP}(1) - 1) \times n_c$ 2D array of microscale heterogeneous coefficients, or
- if `nEnsem > 1`, $(\text{nSubP}(1) - 1) \times n_c \times m_x$ 3D array of m_x ensemble of phase-shifts of the microscale heterogeneous coefficients.
- `.parallel`, logical: true if patches are distributed over multiple CPUs/cores for the Parallel Computing Toolbox, otherwise false (the default is to activate the *local* pool).
- `.codist`, *optional*, describes the particular parallel distribution of arrays over the active parallel pool.

3.1.1 If no arguments, then execute an example

```

264 if nargin==0
265 disp('With no arguments, simulate example of Burgers PDE')

```

The code here shows one way to get started: a user's script may have the following three steps (“ $\mapsto$ ” denotes function recursion).

1. `configPatches1`
2. `ode15s` integrator $\mapsto$ `patchSys1` $\mapsto$ user's PDE
3. process results

Establish global patch data struct to point to and interface with a function coding Burgers' PDE: to be solved on 2π -periodic domain, with eight patches, spectral interpolation couples the patches, with micro-grid spacing 0.06, and with seven microscale points forming each patch.

```

285 global patches
286 patches = configPatches1(@BurgersPDE, [0 2*pi], ...
287     'periodic', 8, 0, 0.06, 7);

```

Set some initial condition, with some microscale randomness.

```

293 u0=0.3*(1+sin(patches.x))+0.1*randn(size(patches.x));

```

Simulate in time using a standard stiff integrator and the interface function `patchSys1()` (Section 3.2).

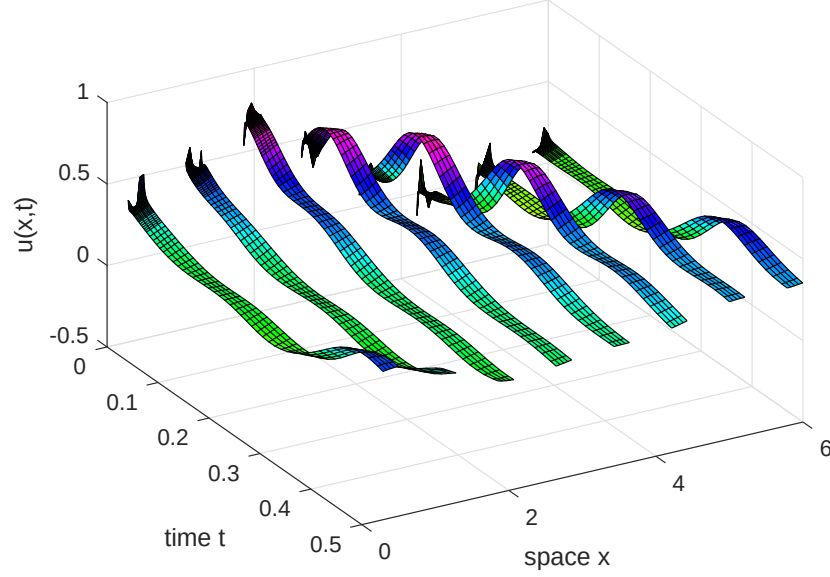
```

301 if ~exist('OCTAVE_VERSION','builtin')
302 [ts,us] = ode15s( @patchSys1,[0 0.5],u0(:));
303 else % octave version
304 [ts,us] = odeOcts(@patchSys1,[0 0.5],u0(:));
305 end

```

Plot the simulation using only the microscale values interior to the patches: either set x -edges to `nan` to leave the gaps; or use `patchEdgyInt1` to re-interpolate correct patch edge values and thereby join the patches. Figure 3.2 illustrates an example simulation in time generated by the patch scheme applied to Burgers' PDE.

Figure 3.2: field $u(x,t)$ of the patch scheme applied to Burgers' PDE.
Burgers PDE: patches in space, continuous time



```

317 figure(1),clf
318 if 1, patches.x([1 end],:,:)=nan; us=us.';
319 else us=reshape(patchEdgyInt1(us.'),[],length(ts));
320 end
321 mesh(ts,patches.x(:),us)
322 view(60,40), colormap(0.7*hsv)
323 title('Burgers PDE: patches in space, continuous time')
324 xlabel('time $t$'), ylabel('space $x$'), zlabel('$u(x,t)$')

```

Upon finishing execution of the example, optionally save the graph to be shown in [Figure 3.2](#), then exit this function.

```

338 if0urCf2eps(mfilename)
339 return
340 end%if nargin==0

```

Example of Burgers PDE inside patches As a microscale discretisation of Burgers' PDE $u_t = u_{xx} - 30uu_x$, here code $\dot{u}_{ij} = \frac{1}{\delta x^2}(u_{i+1,j} - 2u_{i,j} + u_{i-1,j}) - 30u_{ij}\frac{1}{2\delta x}(u_{i+1,j} - u_{i-1,j})$. Here there is only one field variable, and one in the ensemble, so for simpler coding of the PDE we squeeze them out (with no need to reshape when via `patchSys1()`).

```

15 function ut=BurgersPDE(t,u,patches)
16     u=squeeze(u); % omit singleton dimensions
17     dx=diff(patches.x(1:2)); % microscale spacing
18     i=2:size(u,1)-1; % interior points in patches
19     ut=nan+u; % preallocate output array
20     ut(i,:)=diff(u,2)/dx^2 ...
21         -30*u(i,:).*(u(i+1,:)-u(i-1,:))/(2*dx);
22 end

```

```
10 function [ts,xs] = ode0cts(dxdt,tSpan,x0)
11     if length(tSpan)>2, ts = tSpan;
12     else ts = linspace(tSpan(1),tSpan(end),21)';
13     end
14     lsode_options('integration method','non-stiff');
15     xs = lsode(@(x,t) dxdt(t,x),x0,ts);
16 end
```

3.2 patchSys1(): interface 1D space to time integrators

To simulate in time with 1D spatial patches we often need to interface a user's time derivative function with time integration routines such as `ode23` or `PIRK2`. This function provides an interface. Communicate patch-design variables (Section 3.1) either via the global struct `patches` or via an optional third argument. `patches` is required for the parallel computing of `spmd`, or if parameters are to be passed though to the user microscale function.

```

24 function dudt=patchSys1(t,u,patches,varargin)
25 if nargin<3, global patches, end

```

Input

- `u` is a vector/array of length $n_{\text{SubP}} \cdot n_{\text{Vars}} \cdot n_{\text{Ensem}} \cdot n_{\text{Patch}}$ where there are $n_{\text{Vars}} \cdot n_{\text{Ensem}}$ field values at each of the points in the $n_{\text{SubP}} \times n_{\text{Patch}}$ grid.
- `t` is the current time to be passed to the user's time derivative function.
- `patches` a struct set by `configPatches1()` with the following information used here.
 - `.fun` is the name of the user's function `fun(t,u,patches,...)` that computes the time derivatives on the patchy lattice. The array `u` has size $n_{\text{SubP}} \times n_{\text{Vars}} \times n_{\text{Ensem}} \times n_{\text{Patch}}$. Time derivatives should be computed into the same sized array, then herein the patch edge values are overwritten by zeros.
 - `.x` is $n_{\text{SubP}} \times 1 \times 1 \times n_{\text{Patch}}$ array of the spatial locations x_i of the microscale grid points in every patch. Currently it *must* be an equi-spaced lattice on the microscale.
- `varargin`, optional, is arbitrary number of parameters to be passed onto the users time-derivative function as specified in `configPatches1`.

Output

- `dudt` is a vector/array of of time derivatives, but with patch edge-values set to zero. It is of total length $n_{\text{SubP}} \cdot n_{\text{Vars}} \cdot n_{\text{Ensem}} \cdot n_{\text{Patch}}$ and the same dimensions as `u`.

3.3 patchEdgeInt1(): sets patch-edge values from interpolation over the 1D macroscale

Couples 1D patches across 1D space by computing their edge values from macroscale interpolation of either the mid-patch value (Roberts 2003, Roberts & Kevrekidis 2007), or the patch-core average (Bunder et al. 2017), or the opposite next-to-edge values (Bunder et al. 2020)—this last alternative often maintains symmetry. This function is primarily used by patchSys1() but is also useful for user graphics. When using core averages (not fully implemented), assumes the averages are sensible macroscale variables: then patch edge values are determined by macroscale interpolation of the core averages (Bunder et al. 2017).²

Communicate patch-design variables via a second argument (optional, except required for parallel computing of spmd), or otherwise via the global struct patches.

```
32 function u=patchEdgeInt1(u,patches)
33 if nargin<2, global patches, end
```

Input

- **u** is a vector/array of length $nSubP \cdot nVars \cdot nEnsem \cdot nPatch$ where there are $nVars \cdot nEnsem$ field values at each of the points in the $nSubP \times nPatch$ multiscale spatial grid.
- **patches** a struct largely set by configPatches1(), and which includes the following.
 - **.x** is $nSubP \times 1 \times 1 \times nPatch$ array of the spatial locations x_{iI} of the microscale grid points in every patch. Currently it *must* be an equi-spaced lattice on the microscale index i , but may be variable spaced in macroscale index I .
 - **.ordCC** is order of interpolation, integer ≥ -1 .
 - **.periodic** indicates whether macroscale is periodic domain, or alternatively that the macroscale has left and right boundaries so interpolation is via divided differences.
 - **.stag** in $\{0, 1\}$ is one for staggered grid (alternating) interpolation, and zero for ordinary grid.
 - **.Cwtsr** and **.Cwtsl** are the coupling coefficients for finite width interpolation—when invoking a periodic domain.
 - **.EdgyInt**, true/false, for determining patch-edge values by interpolation: true, from opposite-edge next-to-edge values (often preserves symmetry); false, from centre-patch values (original scheme).
 - **.nEdge**, for each patch, the number of edge values set by interpolation at the edge regions of each patch (default is one).

² Script patchEdgeInt1test.m verifies this code.

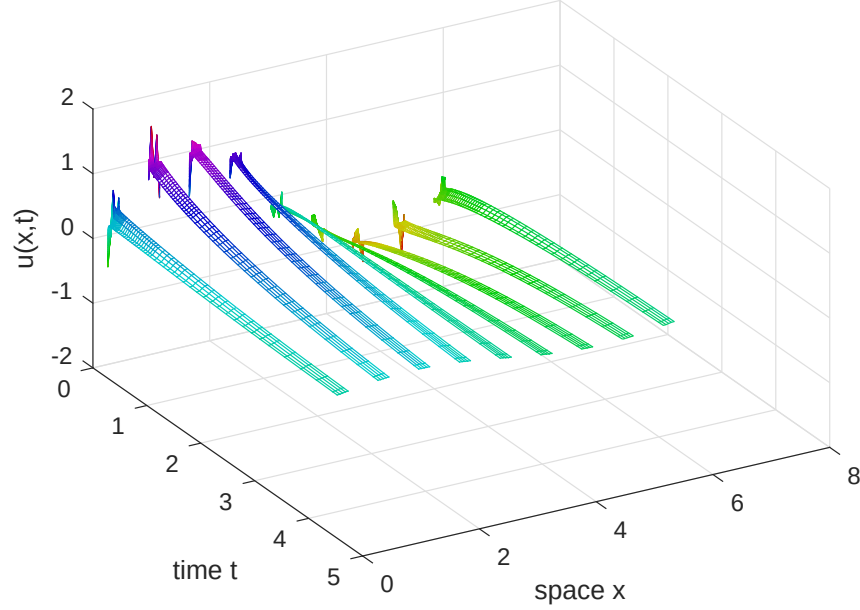
- `.nEnsem` the number of realisations in the ensemble.
- `.parallel` whether serial or parallel.
- `.nCore` ³

Output

- `u` is 4D array, $\text{nSubP} \times \text{nVars} \times \text{nEnsem} \times \text{nPatch}$, of the fields with edge values set by interpolation.

³**ToDo:** introduced sometime but not fully implemented yet, because prefer ensemble, now disabled

Figure 3.3: the diffusing field $u(x,t)$ in the patch (gap-tooth) scheme applied to microscale heterogeneous diffusion (Section 3.4).



3.4 homogenisationExample: simulate heterogeneous diffusion in 1D on patches

Figure 3.3 shows an example simulation in time generated by the patch scheme applied to heterogeneous diffusion. That such simulations of heterogeneous diffusion makes valid predictions was established by Bunder et al. (2017) who proved that the scheme is accurate when the number of points in a patch is one more than a multiple of the periodic of the microscale heterogeneity.

The first part of the script implements the following gap-tooth scheme (left-right arrows denote function recursion).

1. configPatches1
2. ode15s $\mapsto$ patchSys1 $\mapsto$ heteroDiff
3. process results

Consider a lattice of values $u_i(t)$, with lattice spacing dx , and governed by the heterogeneous diffusion

$$\dot{u}_i = [c_{i-1/2}(u_{i-1} - u_i) + c_{i+1/2}(u_{i+1} - u_i)]/dx^2. \quad (3.1)$$

In this 1D space, the macroscale, homogenised, effective diffusion should be the harmonic mean of these coefficients.

3.4.1 Script to simulate via stiff or projective integration

Set the desired microscale periodicity, and correspondingly choose random microscale diffusion coefficients (with subscripts shifted by a half).

```

53 mPeriod = 3
54 cDiff = exp(randn(mPeriod,1))
55 cHomo = 1/mean(1./cDiff)

```

Establish global data struct `patches` for heterogeneous diffusion on 2π -periodic domain. Use nine patches, each patch of half-size ratio 0.2. Quartic (fourth-order) interpolation `ordCC = 4` provides values for the inter-patch coupling conditions. Here include the diffusivity coefficients, repeated to fill up a patch.

```

67 global patches
68 nPatch = 9
69 ratio = 0.2
70 nSubP = 2*mPeriod+1
71 Len = 2*pi;
72 ordCC = 4;
73 configPatches1(@heteroDiff,[0 Len],nan,nPatch ...
74               ,ordCC,ratio,nSubP,'hetCoeffs',cDiff);

```

For comparison: conventional integration in time Set an initial condition, and here integrate forward in time using a standard method for stiff systems—because of the simplicity of linear problems this method works quite efficiently here. Integrate the interface `patchSys1` (Section 3.2) to the microscale differential equations.

```

88 u0 = sin(patches.x)+0.3*randn(nSubP,1,1,nPatch);
89 if ~exist('OCTAVE_VERSION','builtin')
90 [ts,ucts] = ode15s(@patchSys1, [0 2/cHomo], u0(:));
91 else % octave version
92 [ts,ucts] = ode0cts(@patchSys1, [0 2/cHomo], u0(:));
93 end
94 ucts = reshape(ucts,length(ts),length(patches.x(:)),[]);

```

Plot the simulation in Figure 3.3.

```

101 figure(1),clf
102 xs = patches.x; xs([1 end],:) = nan;
103 mesh(ts,xs(:),ucts'), view(60,40)
104 xlabel('time $t$'), ylabel('space $x$'), zlabel('$u(x,t)$')
105 ifOurCf2eps([mfilename 'CtsU'])
106 title('space-time solution $u(x,t)$ on patches in space')

```

The code may invoke this integration interface.

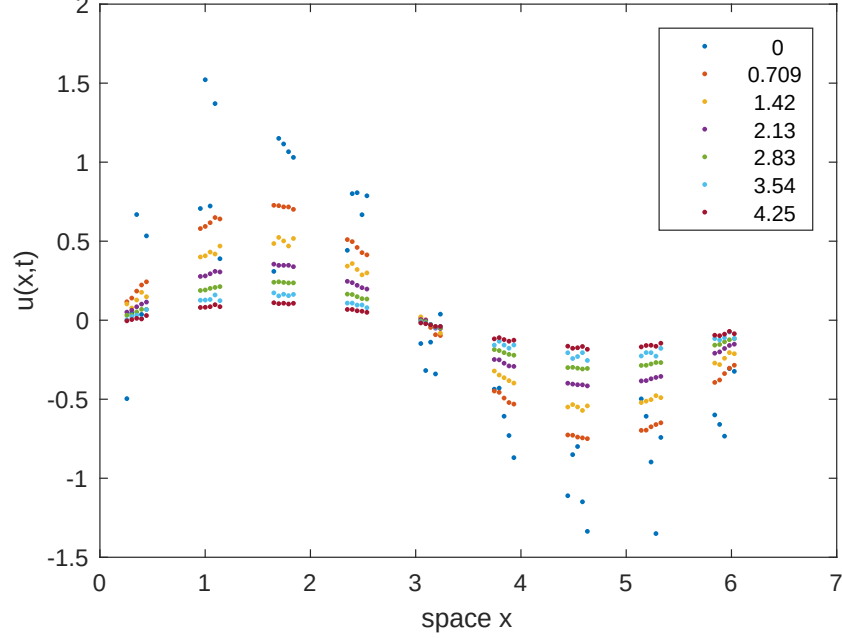
```

10 function [ts,xs] = ode0cts(dxdt,tSpan,x0)
11     if length(tSpan)>2, ts = tSpan;
12     else ts = linspace(tSpan(1),tSpan(end),21)';
13     end
14     lsode_options('integration method','non-stiff');
15     xs = lsode(@(x,t) dxdt(t,x),x0,ts);
16 end

```

Use projective integration in time Now take `patchSys1`, the interface to the time derivatives, and wrap around it the projective integration `PIRK2`

Figure 3.4: field $u(x, t)$ shows basic projective integration of patches of heterogeneous diffusion: different colours correspond to the times in the legend. This field solution displays some fine scale heterogeneity due to the heterogeneous diffusion.



(Section 2.1), of bursts of simulation from `heteroBurst` (Section 3.4.3), as illustrated by Figure 3.4.

This second part of the script implements the following design, where the micro-integrator could be, for example, `ode45` or `rk2int`.

1. `configPatches1` (done in first part)
2. `PIRK2` $\mapsto$ `heteroBurst` $\mapsto$ micro-integrator $\mapsto$ `patchSys1` $\mapsto$ `heteroDiff`
3. process results

Mark that edge of patches are not to be used in the projective extrapolation by setting initial values to NaN.

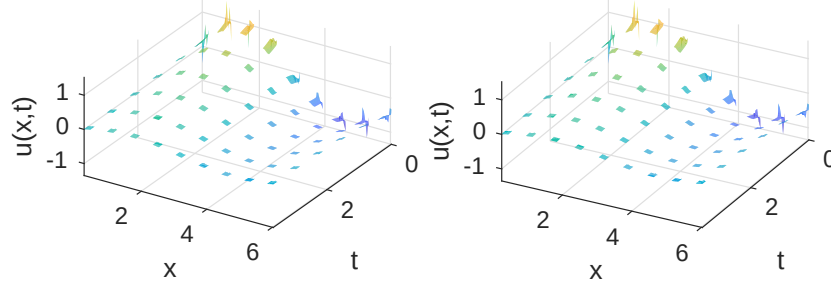
```
142 u0([1 end], :) = nan;
```

Set the desired macro- and microscale time-steps over the time domain: the macroscale step is in proportion to the effective mean diffusion time on the macroscale; the burst time is proportional to the intra-patch effective diffusion time; and lastly, the microscale time-step is proportional to the diffusion time between adjacent points in the microscale lattice.

```
154 ts = linspace(0, 2/cHomo, 7)
155 bT = 3*( ratio*Len/nPatch )^2/cHomo
156 %addpath(' ../ProjInt')
157 [us, tss, uss] = PIRK2(@heteroBurst, ts, u0(:), bT);
```

Plot the macroscale predictions to draw Figure 3.4.

Figure 3.5: cross-eyed stereo pair of the field $u(x,t)$ during each of the microscale bursts used in the projective integration of heterogeneous diffusion.



```

164 figure(2),clf
165 plot(xs(:),us','.')
166 ylabel('$u(x,t)$'), xlabel('space $x$')
167 legend(num2str(ts',3))
168 ifOurCf2eps([mfilename 'U'])
169 title('spatial structure of $u(x,t)$ in patches at selected times')

```

Also plot a surface detailing the microscale bursts as shown in the stereo [Figure 3.5](#).

```

184 figure(3),clf
185 for k = 1:2, subplot(2,2,k)
186     surf(tss,xs(:),uss', 'EdgeColor','none')
187     ylabel('$x$'), xlabel('$t$'), zlabel('$u(x,t)$')
188     axis tight, view(126-4*k,45)
189     title('stereo of $u(x,t)$ on space-time patches')
190 end
191 ifOurCf2eps([mfilename 'Micro'])

```

End of this example script.

3.4.2 heteroDiff(): heterogeneous diffusion

This function codes the lattice heterogeneous diffusion inside the patches. For 2D input arrays u and x (via edge-value interpolation of `patchSys1`, [Section 3.2](#)), computes the time derivative (3.1) at each point in the interior of a patch, output in ut . The column vector of diffusivities c_i , and possibly Burgers' advection coefficients b_i , have previously been stored in struct `patches.cs`.

```

22 function ut = heteroDiff(t,u,patches)
23     if ~patches.periodic % apply simple Dirichlet BCs
24         u(1, :, :, 1)=0; % left-edge of leftmost is zero
25         u(end, :, :, end)=0; % right-edge of rightmost is zero
26     end%if not periodic
27
28     dx = diff(patches.x(2:3)); % space step
29     i = 2:size(u,1)-1; % interior points in a patch
30     ut = nan(size(u),'like',u); % preallocate output array, nan+u poss

```

```

31     ut(i, :, :, :) = diff(patchcs(:, 1, :)).*diff(u))/dx^2;
32     % possibly include heterogeneous Burgers' advection
33     if size(patchcs, 2) > 1 % check for advection coeffs
34         buu = patchcs(:, 2, :). * u.^2;
35         ut(i, :) = ut(i, :) - (buu(i+1, :) - buu(i-1, :))/(dx*2);
36     end
37 end% function

```

3.4.3 heteroBurst(): a burst of heterogeneous diffusion

This code integrates in time the derivatives computed by `heteroDiff` from within the patch coupling of `patchSys1`. Try `ode23` or `rk2Int`, although `ode45` may give smoother results.

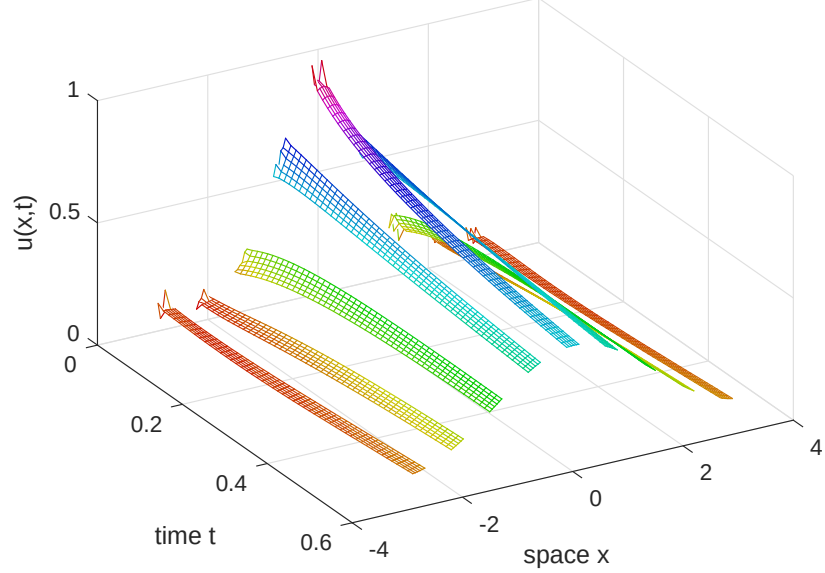
```

15 function [ts, ucts] = heteroBurst(ti, ui, bT)
16     if ~exist('OCTAVE_VERSION', 'builtin')
17         [ts, ucts] = ode23( @patchSys1, [ti ti+bT], ui(:));
18     else % octave version
19         [ts, ucts] = rk2Int(@patchSys1, [ti ti+bT], ui(:));
20     end
21 end

```

Fin.

Figure 3.6: diffusion field $u(x,t)$ of the gap-tooth scheme applied to the diffusion (3.2). The microscale random component to the initial condition, the sub-patch fluctuations, decays, leaving the emergent macroscale diffusion. This simulation uses nine patches of ‘large’ size ratio 0.25 for visibility.



3.5 homoDiffEdgy1: computational homogenisation of a 1D heterogeneous diffusion by simulation on small patches

Figure 3.6 shows an example simulation in time generated by the patch scheme applied to macroscale diffusion propagation through a medium with microscale heterogeneity. The inter-patch coupling is realised by quartic interpolation of the patch’s next-to-edge values to the patch opposite edges. Such coupling preserves symmetry in many systems, and quartic appears to be the lowest order that generally gives good accuracy.

Suppose the spatial microscale lattice is at points x_i , with constant spacing dx . With dependent variables $u_i(t)$, simulate the microscale lattice diffusion system

$$\frac{\partial u_i}{\partial t} = \frac{1}{dx^2} \delta[c_{i-1/2} \delta u_i], \quad (3.2)$$

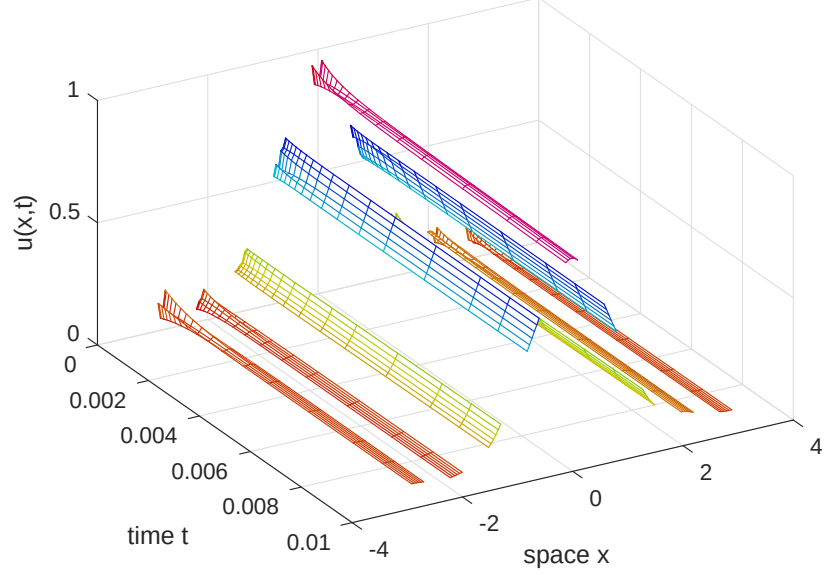
in terms of the centred difference operator δ . The system has a microscale heterogeneity via the coefficients $c_{i+1/2}$ which we assume to have some given known periodicity. Figure 3.6 shows one patch simulation of this system: observe the effects of the heterogeneity within each patch.

3.5.1 Script code to simulate heterogeneous diffusion systems

This example script implements the following patch/gap-tooth scheme (left-right arrows denote function recursion).

1. configPatches1
2. ode15s $\mapsto$ patchSys1 $\mapsto$ heteroDiff
3. plot the simulation

Figure 3.7: diffusion field $u(x,t)$ of the gap-tooth scheme applied to the diffusive (3.2). Over this short meso-time we see the macroscale diffusion emerging from the damped sub-patch fast quasi-equilibration.



4. use patchSys1 to explore the Jacobian

First establish the microscale heterogeneity has micro-period `mPeriod` on the lattice, and random log-normal values, albeit normalised to have harmonic mean one. This normalisation then means that macroscale diffusion on a domain of length 2π should have near integer decay rates, the squares of $0, 1, 2, \dots$. Then the heterogeneity is repeated to fill each patch, and phase-shifted for an ensemble.

```

90 clear all
91 mPeriod = randi([2 5])
92 % set random diffusion coefficients
93 cHetr=exp(0.3*randn(mPeriod,1));
94 %cHetr = [3.966;2.531;0.838;0.331;7.276];
95 cHetr = cHetr*mean(1./cHetr) % normalise

```

Establish the global data struct `patches` for the microscale heterogeneous lattice diffusion system (3.2) solved on 2π -periodic domain, with seven patches, here each patch of size ratio 0.25 from one side to the other, with five micro-grid points in each patch, and quartic interpolation (4) to provide the edge-values of the inter-patch coupling conditions. Setting `patches.EdgeyInt` to one means the edge-values come from interpolating the opposite next-to-edge values of the patches (not the mid-patch values). In this case we appear to need at least fourth order (quartic) interpolation to get reasonable decay rate for heterogeneous diffusion. When simulating an ensemble of configurations, `nSubP` (the number of points in a patch) need not be dependent on the period of the heterogeneous diffusion.

```

117 global patches
118 nPatch = 9

```

```

119 ratio = 0.25;
120 nSubP = randi([mPeriod+1 2*mPeriod+2])
121 if mod(nSubP,mPeriod)==2, nEnsem=1,
122 else nEnsem = mPeriod % number realisations in ensemble
123 end
124 configPatches1(@heteroDiff,[-pi pi],nan,nPatch ...
125               ,4,ratio,nSubP,'EdgyInt',true,'nEnsem',nEnsem ...
126               ,'hetCoeffs',cHetr);

```

Simulate Set the initial conditions of a simulation to be that of a lump perturbed by significant random microscale noise, via `randn`.

```

137 u0 = 0.8*exp(-patches.x.^2)+0.2*rand(nSubP,1,nEnsem,nPatch);
138 du0dt = patchSys1(0,u0(:));

```

Integrate using standard integrators.

```

144 if ~exist('OCTAVE_VERSION','builtin')
145     [ts,us] = ode23(@patchSys1, [0 0.6], u0(:));
146 else % octave version
147     [ts,us] = ode0cts(@patchSys1, 0.6*linspace(0,1).^2, u0(:));
148 end

```

Plot space-time surface of the simulation We want to see the edge values of the patches, so we adjoin a row of `nans` in between patches. For the field values (which are rows in `us`) we need to reshape, permute, interpolate to get edge values, pad with `nans`, and reshape again. In the case of an ensemble of phase-shifts, we plot the mean over the ensemble.

```

161 xs = squeeze(patches.x);
162 us = patchEdgeInt1( permute( reshape(us ...
163     ,length(ts),nSubP,nEnsem,nPatch) ,[2 1 3 4]) );
164 ustd = squeeze(std(us,0,3));
165 us = squeeze(mean(us,3));
166 if 0, % omit interpolated edges
167     us([1 end],:,:) = nan;
168     ustd([1 end],:,:) = nan;
169 else % insert nans between patches
170     xs(end+1,:) = nan;
171     us(end+1,:,:) = nan;
172     ustd(end+1,:,:) = nan;
173 end
174 us=reshape(permute(us,[1 3 2]),[],length(ts));
175 ustd=reshape(permute(ustd,[1 3 2]),[],length(ts));

```

Now plot two space-time graphs. The first is every time step over a meso-time to see the oscillation and decay of the fast sub-patch diffusions. The second is subsampled surface over the macroscale duration of the simulation to show the propagation of the macroscale diffusion over the heterogeneous lattice.

```

187 for p=1:2
188     switch p
189         case 1, j=find(ts<0.01);
190         case 2, [~,j]=min(abs(ts(:)-linspace(ts(1),ts(end),50)));
191         end
192         figure(p),clf
193         mesh(ts(j),xs(:),us(:,j))
194         view(60,40), colormap(0.8*hsv)
195         xlabel('time $t$'), ylabel('space $x$'), zlabel('$u(x,t)$')
196         ifOurCf2eps([mfilename 'U' num2str(p)])
197         switch p
198             case 1, title('short time simulation shows any transients')
199             case 2, title('simulation of $u(x,t)$ on patches in space')
200         end
201     end
202     pause(3)

```

Compute Jacobian and its spectrum Let's explore the Jacobian dynamics for a range of orders of interpolation, all for the same patch design and heterogeneity. Here use a smaller ratio, and more patches, as we do not plot.

```

215 nPatch = 13
216 ratio = 0.01;
217
218 leadingEvals=[];
219 for ord=0:2:8
220     ordInterp=ord
221     configPatches1(@heteroDiff,[-pi pi],nan,nPatch ...
222         ,ord,ratio,nSubP,'EdgyInt',true,'nEnsem',nEnsem ...
223         , 'hetCoeffs',cHetr);

```

Form the Jacobian matrix, linear operator, by numerical construction about a zero field. Use `i` to store the indices of the micro-grid points that are interior to the patches and hence are the system variables.

```

233 u0 = zeros(nSubP,1,nEnsem,nPatch);
234 u0([1 end],:,:,:) = nan; u0=u0(:);
235 i=find(~isnan(u0));
236 tic
237 nJ=length(i);
238 Jac=nan(nJ);
239 for j=1:nJ
240     u0(i)=((1:nJ)==j);
241     dudt=patchSys1(0,u0);
242     Jac(:,j)=dudt(i);
243 end
244 finiteDiffTime=toc
245 nonSymmetric=norm(Jac-Jac')
246 assert(nonSymmetric<5e-9,'failed symmetry')

```

Table 3.1: example parameters and list of eigenvalues (every fourth one listed is sufficient due to symmetry): `nPatch = 19`, `ratio = 0.1`, `nSubP = 5`. The columns are for various `ordCC`, in order: 0, spectral interpolation; 2, quadratic; 4, quartic; and 6, sixth order.

```

cHetr =
    6.9617
    0.4217
    2.0624
leadingEvals =
    2e-11    -2e-12    4e-12    -2e-11
   -0.9999   -1.5195   -1.0127   -1.0003
   -3.9992   -11.861    -4.7785   -4.0738
   -8.9960   -45.239    -17.164   -10.703
  -15.987    -116.27    -56.220   -30.402
  -24.969    -230.63    -151.74   -92.830
  -35.936    -378.80    -327.36   -247.37
  -48.882    -535.89    -570.87   -521.89
  -63.799    -668.21    -818.33   -855.72
  -80.678    -743.96    -976.57  -1093.4
  -29129     -29233     -29227    -29222
  -29151     -29234     -29229    -29223

```

```

247     Jac(abs(Jac)<1e-12)=0;

```

If AutoDiff is available, then compare the Jacobian computed by AutoDiff. This confirms the two Jacobians are the same—generally AutoDiff should be more accurate. In this case it is usually slower to compute than the finite-difference Jacobian, but in large problems is quicker.

```

257     if exist('AutoDiff','class')
258         disp('**** starting AutoDiff Jac')
259         tic
260         Jad=AutoDiffJacobianAutoDiff(@(u) patchSys1(0,u),u0,i);
261         autoDiffTime=toc
262         Jad=Jad(i,:);
263         disp('**** finished AutoDiff Jac')
264         figure(3),spy(Jac,'b+'),hold on,spy(Jad,'rx'),hold off
265         JacDiff=norm(Jac-Jad)
266         assert(JacDiff<1e-7,'jacobians differ significantly')
267     end%if exist

```

Find the eigenvalues of the Jacobian, and list for inspection in [Table 3.1](#): the spectral interpolation is effectively exact for the macroscale; quadratic interpolation is usually quantitatively in error; quartic interpolation appears to be the lowest order for reliable quantitative accuracy.

The number of zero eigenvalues, `nZeroEv`, indicates the number of decoupled systems in this patch configuration.


```

308     [evecs,evals]=eig(Jac);
309     eval=-sort(-diag(real(evals)));
310     nZeroEv=sum(eval(:)>-1e-5)
311     leadingEvals=[leadingEvals eval(1:nPatch)];

```

End of the for-loop over orders of interpolation, and output the tables of eigenvalues.

```

318 end
319 disp(' spectral quadratic    quartic    sixth    ...')
320 leadingEvals=leadingEvals

```

End of the main script.

3.5.2 heteroDiff(): heterogeneous diffusion

This function codes the lattice heterogeneous diffusion inside the patches. For 2D input arrays u and x (via edge-value interpolation of `patchSys1`, [Section 3.2](#)), computes the time derivative (3.1) at each point in the interior of a patch, output in ut . The column vector of diffusivities c_i , and possibly Burgers' advection coefficients b_i , have previously been stored in struct `patches.cs`.

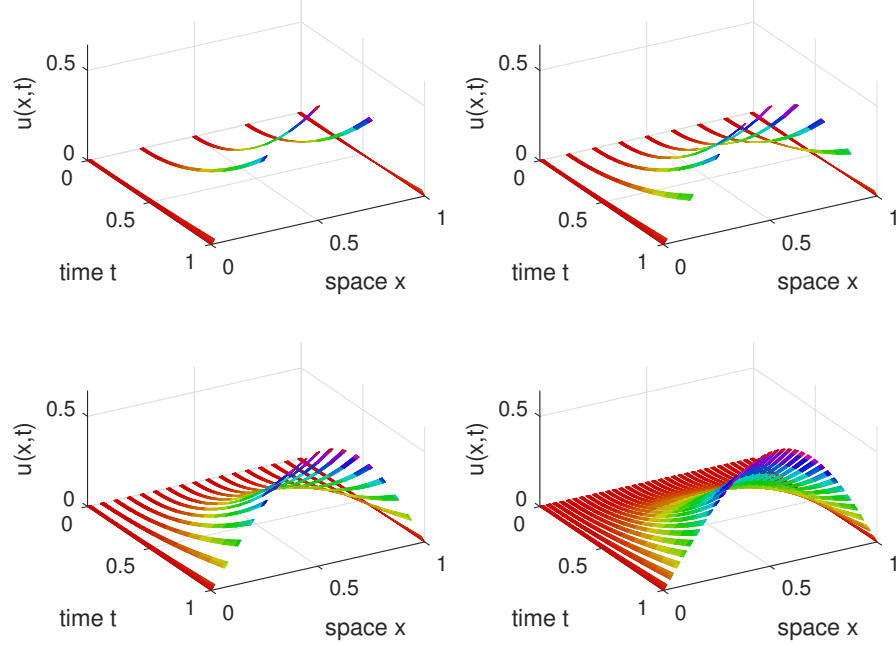
```

22 function ut = heteroDiff(t,u,patches)
23     if ~patches.periodic % apply simple Dirichlet BCs
24         u( 1 ,:,:, 1 )=0; % left-edge of leftmost is zero
25         u(end,:,:,end)=0; % right-edge of rightmost is zero
26     end%if not periodic
27
28     dx = diff(patches.x(2:3)); % space step
29     i = 2:size(u,1)-1; % interior points in a patch
30     ut = nan(size(u),'like',u); % preallocate output array, nan+u poss
31     ut(i,:,:,) = diff(patches.cs(:,1,:).*diff(u))/dx^2;
32     % possibly include heterogeneous Burgers' advection
33     if size(patches.cs,2)>1 % check for advection coeffs
34         buu = patches.cs(:,2,:).*u.^2;
35         ut(i,:) = ut(i,)-(buu(i+1,:)-buu(i-1,:))/(dx*2);
36     end
37 end% function

```

Fin.

Figure 3.8: diffusion field $u(x,t)$ of the patch scheme applied to the forced heterogeneous diffusive (3.3). Simulate for 5, 9, 17, 33 patches and compare to the full-domain simulation (65 patches, not shown).



3.6 Eckhardt2210eg2: example of a 1D heterogeneous diffusion by simulation on small patches

Plot an example simulation in time generated by the patch scheme applied to macroscale forced diffusion through a medium with microscale heterogeneity in space. This is more-or-less the second example of Eckhardt & Verfürth (2022) [§6.2.1].

Suppose the spatial microscale lattice is at points x_i , with constant spacing dx . With dependent variables $u_i(t)$, simulate the microscale lattice forced diffusion system

$$\frac{\partial u_i}{\partial t} = \frac{1}{dx^2} \delta[a_{i-1/2} \delta u_i] + f_i(t), \quad (3.3)$$

in terms of the centred difference operator δ . The system has a microscale heterogeneity via the coefficients $a_{i+1/2}$ which has some given known periodicity ϵ .

Here use period $\epsilon = 1/130$ (so that computation completes in seconds). The patch scheme computes only on a fraction of the spatial domain, see Figure 3.8. Compute *errors* as the maximum difference (at time $t = 1$) between the patch scheme prediction and a full-domain simulation of the same underlying spatial discretisation (which here has space step 0.00128).

patch spacing H	0.25	0.12	0.06	0.03
exp-sine-forcing error	8E-3	2E-3	3E-4	2E-5
parabolic-forcing error	9E-9	4E-9	1E-9	0.06E-9

The smooth sine-forcing leads to errors that appear due to patch scheme

and its interpolation. The parabolic-forcing errors appear to be due to the integration errors of `ode15s` and not at all due to the patch scheme. In comparison, [Eckhardt & Verfürth \(2022\)](#) reported much larger errors in the range 0.001–0.1 (Figure 3).

3.6.1 Simulate heterogeneous diffusion systems

First establish the microscale heterogeneity has micro-period `mPeriod` on the lattice, and coefficients to match [Eckhardt20210.04536 §6.2.1](#). Set the phase of the heterogeneity so that each patch centre is a point of symmetry of the diffusivity. Then the heterogeneity is repeated to fill each patch.

```

78 clear all
79 %global OurCf2eps, OurCf2eps=true %option to save plots
80 mPeriod = 6
81 y = linspace(0,1,mPeriod+1)';
82 a = 1./(2-cos(2*pi*y(1:mPeriod)))
83 global microTimePeriod; microTimePeriod=0;

Set the spatial period  $\epsilon$ , via integer  $1/\epsilon$ , and other parameters.

91 maxLog2Nx = 6
92 nPeriodsPatch = 2 % any integer
93 rEpsilon = nPeriodsPatch*(2^maxLog2Nx+1) % up to 200 say
94 dx = 1/(mPeriod*rEpsilon+1)
95 nSubP = nPeriodsPatch*mPeriod+2
96 tol=1e-9;
```

Loop to explore errors on various sized patches.

```

102 Us=[]; DXs=[]; % for storing results to compare
103 iPP=0; I=nan;
104 for log2Nx = 2:maxLog2Nx
105     nP = 2^log2Nx+1
```

Determine indices of patches that are common in various resolutions

```

112     if isnan(I), I=1:nP; else I=2*I-1; end
```

Establish the global data struct `patches` for the microscale heterogeneous lattice diffusion system [\(3.3\)](#) solved on domain $[0, 1]$, with `nP` patches, and say fourth order interpolation to provide the edge-values. Setting `patches.EdgyInt` true means the edge-values come from interpolating the opposite next-to-edge values of the patches (not the mid-patch values).

```

127     global patches
128     ordCC = 4
129     configPatches1(@heteroDiffF,[0 1],'equispace',nP ...
130         ,ordCC,dx,nSubP,'EdgyInt',true,'hetCoeffs',a);
131     DX = mean(diff(squeeze(patches.x(1,1,1,:))))
132     DXs=[DXs;DX];
```

Set the forcing coefficients, either the original parabolic, or exp-sinusoidal.

```

140     if 0 % given forcing is exact
141         patches.f1=2*( patches.x-patches.x.^2 );
142         patches.f2=2*0.5+0*patches.x;
143     else% simple exp.sine forcing
144         patches.f1=sin(pi*patches.x).*exp(patches.x);
145         patches.f2=pi/2*sin(pi*patches.x).*exp(patches.x);
146     end%if

```

Simulate Set the initial conditions of a simulation to be zero. Integrate to time 1 using standard integrators.

```

157     u0 = 0*patches.x;
158     tic
159     [ts,us] = ode15s(@patchSys1, [0 1], u0(:));
160     cpuTime=toc

```

Plot space-time surface of the simulation We want to see the edge values of the patches, so adjoin a row of nans in between patches. For the field values (which are rows in us) we need to reshape, permute, interpolate to get edge values, pad with nans, and reshape again.

```

173     xs = squeeze(patches.x);
174     us = patchEdgeInt1( permute( reshape(us ...
175         ,length(ts),nSubP,1,nP) ,[2 1 3 4]) );
176     us = squeeze(us);
177     xs(end+1,:) = nan; us(end+1,:,:) = nan;
178     uss=reshape(permute(us,[1 3 2]),[],length(ts));

```

Plot a space-time surface of field values over the macroscale duration of the simulation.

```

186     iPP=iPP+1;
187     if iPP<=4 % only draw four subplots
188         figure(1), if iPP==1, clf(), end
189         subplot(2,2,iPP)
190         mesh(ts,xs(:),uss)
191         if iPP==1, uMax=ceil(max(uss(:))*100)/100, end
192         view(60,40), colormap(0.8*hsv), zlim([0 uMax])
193         xlabel('time $t$'), ylabel('space $x$'), zlabel('$u(x,t)$')
194         drawnow
195     end%if

```

At the end of the log2Nx-loop, store field at the end-time from centre region of each patch for comparison.

```

203     i=nPeriodsPatch/2*mPeriod+1+(-mPeriod/2+1:mPeriod/2);
204     Us(:, :, iPP)=squeeze(us(i,end,I));
205     Xs=squeeze(patches.x(i,1,1,I));
206     if iPP>1
207         assert(norm(Xs-Xsp)<tol,'sampling error in space')
208     end

```

```

209     Xsp=Xs;
210 end%for log2Nx
211 ifOurCf2eps(mfilename) %optionally save plot

    Assess errors by comparing to the full-domain solution

217 DXs=DXs
218 Uerr=squeeze(max(max(abs(Us-Us(:, :, end))))))
219 figure(2),clf,
220 loglog(DXs,Uerr,'o:')
221 xlabel('$H$'),ylabel('error')
222 ifOurCf2eps([mfilename 'Errs']) %optionally save plot

```

3.6.2 heteroDiffF(): forced heterogeneous diffusion

This function codes the lattice heterogeneous diffusion inside the patches with forcing and with microscale boundary conditions on the macroscale boundaries. Computes the time derivative at each point in the interior of a patch, output in `ut`. The column vector of diffusivities a_i has been stored in struct `patches.cs`, as has the array of forcing coefficients.

```

17 function ut = heteroDiffF(t,u,patches)

```

Cater for the two cases: one of a non-autonomous forcing oscillating in time when `microTimePeriod` > 0, or otherwise the case of an autonomous diffusion constant in time.

```

26     global microTimePeriod
27     if microTimePeriod>0 % optional time fluctuations
28         at = cos(2*pi*t/microTimePeriod)/30;
29     else at=0; end

```

Two basic parameters, and initialise result array to NaNs.

```

35     dx = diff(patches.x(2:3)); % space step
36     i = 2:size(u,1)-1; % interior points in a patch
37     ut = nan(size(u),'like',u); % preallocate output array

```

The macroscale Dirichlet boundary conditions are zero at the extreme edges of the two extreme patches.

```

44     u( 1 ,:,:, 1 )=0; % left-edge of leftmost is zero
45     u(end,:,:,end)=0; % right-edge of rightmost is zero

```

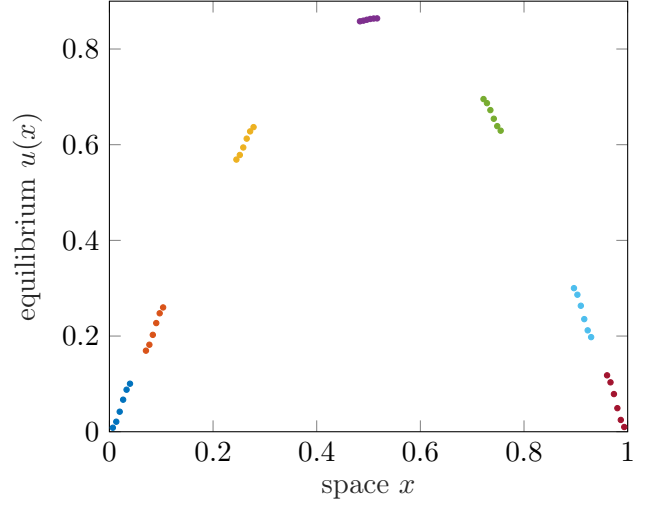
Code the microscale forced diffusion.

```

51     ut(i,:,:,) = diff((patches.cs(:,1,:)+at).*diff(u))/dx^2 ...
52         +patches.f2(i,:,:,)*t^2+patches.f1(i,:,:,)*t;
53 end% function

```

Figure 3.9: Equilibrium of the heterogeneous diffusion problem with forcing the same as that applied at time $t = 1$, and for relatively large $\epsilon = 0.04$ so we can see the patches. By default this code sets $\epsilon = 0.004$ whence the microscale heterogeneity and patches are tiny.



3.7 EckhardtEquilib: find an equilibrium of a 1D heterogeneous diffusion via small patches

Sections 3.6 and 3.6.2 describe details of the problem and more details of the following configuration. The aim is to find the equilibrium, Figure 3.9, of the forced heterogeneous system with a forcing corresponding to that applied at time $t = 1$. Computational efficiency comes from only computing the microscale heterogeneity on small spatially sparse patches, potentially much smaller than those shown in Figure 3.9.

First configure the patch system Establish the microscale heterogeneity has micro-period `mPeriod` on the lattice, and coefficients to match Eckhardt & Verfürth (2022) [§6.2.1].

```

46 clear all
47 global patches
48 %global OurCf2eps, OurCf2eps=true %option to save plots
49 mPeriod = 6
50 y = linspace(0,1,mPeriod+1)';
51 a = 1./(2-cos(2*pi*y(1:mPeriod)))
52 global microTimePeriod; microTimePeriod=0;

```

Set the number of patches, the number of periods per patch, and the spatial period ϵ , via integer $1/\epsilon$.

```

61 nPatch = 7
62 nPeriodsPatch = 1 % any integer
63 rEpsilon = 25 % 25 for graphic, up to 2000 say
64 dx = 1/(mPeriod*rEpsilon+1)
65 nSubP = nPeriodsPatch*mPeriod+2

```

Establish the global data struct `patches` for the microscale heterogeneous lattice diffusion system (3.3) solved on domain $[0, 1]$, with Chebyshev-like distribution of patches, and say fourth order interpolation to provide the

edge-values. Use 'edgy' interpolation.

```

77 ordCC = 4
78 configPatches1(@heteroDiffF,[0 1],'chebyshev',nPatch ...
79     ,ordCC,dx,nSubP,'EdgyInt',true,'hetCoeffs',a);

```

Set the forcing coefficients, either the original parabolic, or exp-sinusoidal. At time $t = 1$ the resultant forcing we actually apply here is simply the sum of the two components.

```

88 if 0 % given forcing
89     patches.f1 = 2*( patches.x-patches.x.^2 );
90     patches.f2 = 2*0.5+0*patches.x;
91 else% simple exp-sine forcing
92     patches.f1 = sin(pi*patches.x).*exp(patches.x);
93     patches.f2 = pi/2*sin(pi*patches.x).*exp(patches.x);
94 end%if

```

Find equilibrium with fsolve We seek the equilibrium for the forcing that applies at time $t = 1$ (as if that specific forcing were applying for all time). For this linear problem, it is computationally quicker using a linear solver, but `fsolve` is quicker in human time, Start the search from a zero field.

```

107 u = 0*patches.x;

```

But set patch-edge values to `Nan` in order to use `patches.i` to index the interior sub-patch points as they are the variables.

```

115 u([1 end],:,:) = nan;
116 patches.i = find(~isnan(u));

```

Seek the equilibrium, and report the norm of the residual, via the generic patch system wrapper `theRes` ([Section 3.15](#)).

```

124 [u(patches.i),res] = fsolve(@theRes,u(patches.i));
125 normRes = norm(res)

```

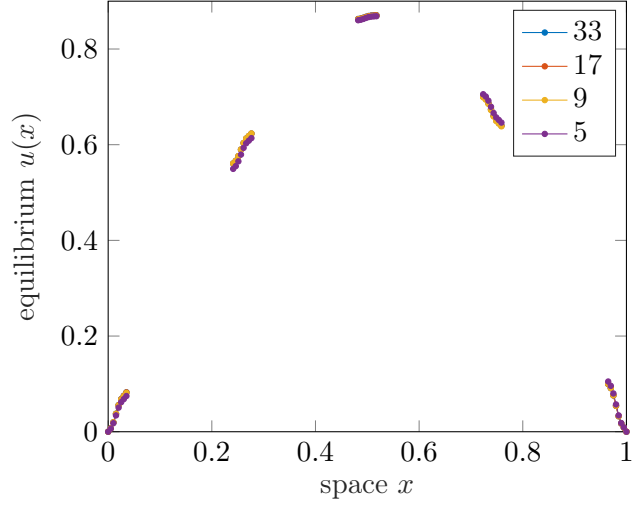
Plot the equilibrium see [Figure 3.9](#).

```

132 clf, plot(squeeze(patches.x),squeeze(u),'.')
133 xlabel('space $x$'),ylabel('equilibrium $u(x)$')
134 ifOurCf2tex(mfilename)%optionally save

```

Figure 3.10:
Equilibrium of the
heterogeneous diffusion
problem for relatively
large $\epsilon = 0.03$ so we
can see the patches.
The solution is obtained
with various numbers
of patches, but we only
compare solutions in
these five common
patches.



3.8 EckhardtEquilibErrs: explore errors in equilibria of a 1D heterogeneous diffusion on small patches

Section 3.7 finds the equilibrium, of the forced heterogeneous system with a forcing corresponding to that applied at time $t = 1$. Computational efficiency comes from only computing the microscale heterogeneity on small spatially sparse patches. Here we explore the errors as the number N of patches increases, see Figures 3.10 and 3.11. Find mean-abs errors to be the following:

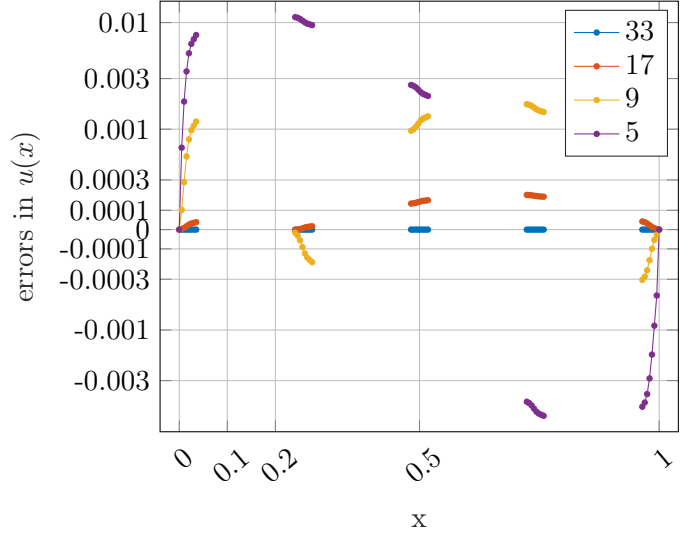
	$N =$	5	9	17	33	65
equispace	second-order	8E-3	1E-2	1E-2	4E-3	9E-4
equispace	fourth-order	2E-3	7E-4	1E-4	9E-6	5E-7
equispace	sixth-order	2E-3	2E-5	4E-7	1E-8	2E-10
chebyshev	second-order	4E-2	6E-2	3E-2	2E-2	2E-2
chebyshev	fourth-order	9E-4	3E-3	6E-4	3E-4	2E-4
chebyshev	sixth-order	9E-4	3E-5	1E-5	4E-6	1E-6
usergiven	second-order	4E-2	6E-2	3E-2	9E-3	2E-3
usergiven	fourth-order	8E-4	3E-3	6E-4	4E-5	2E-6
usergiven	sixth-order	8E-4	3E-5	1E-5	2E-7	3E-9

For ‘chebyshev’ this assessment of errors is a bit dodgy as it is based only on the centre and boundary patches. The ‘usergiven’ distribution is for overlapping patches with Chebyshev distribution of centres—a spatial ‘christmas tree’⁴. Curiously, and with above caveats, here my ‘smart’ chebyshev is the worst, the overlapping Chebyshev is good, but *equispace appears usually the best*.

The above errors are for simple sin forcing. What if we make not so simple with exp modification of the forcing? The errors shown below are very little

⁴ But the error assessment is with respect to finest patch-grid, no longer with a full domain solution

Figure 3.11: Errors in the equilibrium of the heterogeneous diffusion problem for relatively large $\epsilon = 0.03$. The solution is obtained with various numbers of patches, but we only plot the errors within these five common patches.



different (despite the magnitude of the solution being a little larger).

	$N =$	5	9	17	33	65
equispace	fourth-order	4E-3	7E-4	1E-4	8E-6	5E-7
chebyshev	fourth-order	7E-4	2E-3	5E-4	3E-4	1E-4
usergiven	fourth-order	2E-3	3E-3	5E-4	4E-5	2E-6

Clear, and initiate global patches. Choose the type of patch distribution to be either 'equispace', 'chebyshev', or 'usergiven'. Also set order of interpolation (fourth-order is good start).

```

136 clear all
137 global patches
138 %global OurCf2eps, OurCf2eps=true %option to save plots
139 switch 1
140     case 1, Dom.type = 'equispace'
141     case 2, Dom.type = 'chebyshev'
142     case 3, Dom.type = 'usergiven'
143 end% switch
144 ordInt = 4

```

First configure the patch system Establish the microscale heterogeneity has micro-period `mPeriod` on the lattice, and coefficients to match Eckhardt2210.04536 §6.2.1.

```

155 mPeriod = 6
156 z = (0.5:mPeriod)'/mPeriod;
157 a = 1./(2-cos(2*pi*z))
158 global microTimePeriod; microTimePeriod=0;

```

To use a hierarchy of patches with `nPatch` of 5, 9, 17, ..., we need up to N patches plus one `dx` to fit into the domain interval. Cater for up to some full-domain simulation—can compute $\log_2 N_{\max} = 129$ ($\epsilon = 0.008$) in a few seconds:

```

169 log2Nmax = 8 % 5 for plots, 7 for choice
170 nPatchMax=2^log2Nmax+1

Set the periodicity  $\epsilon$ , and other microscale parameters.

177 nPeriodsPatch = 1 % any integer
178 nSubP = nPeriodsPatch*mPeriod+2 % for edgy int
179 epsilon = 1/(nPatchMax*nPeriodsPatch+1/mPeriod)
180 dx = epsilon/mPeriod

```

For various numbers of patches Choose five to be the coarsest number of patches. Want place to store common results for the solutions. Assign Ps to be the indices of the common patches: for equispace set to the five common patches, but for chebyshev the only common ones are the three centre and boundary-adjacent patches.

```

193 us=[]; xs=[]; nPs=[];
194 for log2N=log2Nmax:-1:2
195     if log2N==log2Nmax
196         Ps=linspace(1,nPatchMax ...
197             ,5-2*all(Dom.type=='chebyshev'))
198     else Ps=(Ps+1)/2
199     end

```

Set the number of patches in (0,1):

```

205 nPatch = 2^log2N+1

```

In the case of 'usergiven', we choose standard Chebyshev distribution of the centre of the patches, which involves overlapping of patches near the boundaries! (instead of the coded chebyshev which has a boundary layer of non-overlapping patches and a Chebyshev within the interior).

```

216 if all(Dom.type=='usergiven')
217     halfWidth=dx*(nSubP-1)/2;
218     X1 = 0+halfWidth; X2 = 1-halfWidth;
219     Dom.X = (X1+X2)/2-(X2-X1)/2*cos(linspace(0,pi,nPatch));
220 end

```

Configure the patches:

```

226 configPatches1(@heteroDiffF,[0 1],Dom,nPatch ...
227     ,ordInt,dx,nSubP,'EdgyInt',true,'hetCoeffs',a);

```

Set the forcing coefficients, either the original parabolic, or sinusoidal. At time $t = 1$ the resultant forcing we actually apply here is simply the sum of the two components.

```

236 if 0 %given forcing gives exact answers for ordInt=4 !
237     patches.f1 = 2*( patches.x-patches.x.^2 );
238     patches.f2 = 2*0.5+0*patches.x;
239 else% simple exp-sine forcing
240     patches.f1 = sin(pi*patches.x).*exp(patches.x);

```

```

241     patches.f2 = pi/2*sin(pi*patches.x).*exp(patches.x);
242     end%if

```

Solve for steady state Set initial guess of either zero or a subsample of the next finer solution, with NaN to indicate patch-edge values. Index *i* are the indices of patch-interior points, and the number of unknowns is then its length.

```

254     if log2N==log2Nmax
255         u0 = zeros(nSubP,1,1,nPatch);
256     else u0 = u0(:,:,:,1:2:end);
257     end
258     u0([1 end],:) = nan;
259     i = find(~isnan(u0));
260     nVariables = numel(i)

```

Could solve via `fsolve` for simplicity and robustness, via the generic patch wrapper `theRes` (Section 3.15). However, for such linear problems it is much faster to use the `AutoDiff` package to get the Jacobian, and then solve directly.

```

270     U0=AutoDiff(0*u0); % set an AD array of zeros
271     tic
272     F0 = patchSys1(1,U0); % evaluate patch fn and its derivatives
273     Jac = getderivs(F0); % the Jacobian is the derivatives
274     uSoln = -Jac(i,i)\getvalue(F0(i)); % solve linear system
275     ADsolveTime = toc

```

Store the solution into the patches, and give magnitudes—Inf norm is `max(abs())`.

```

282     normSoln = norm(uSoln,Inf)
283     u0(i) = uSoln;
284     f0 = patchSys1(1,u0);
285     normResidual = norm(f0(i),Inf)
286     u0 = patchEdgeInt1(u0);
287     u0( 1 ,:,:, 1 ) = 0;
288     u0(end,:,:,end) = 0;

```

Concatenate the solution on common patches into stores.

```

294     us=cat(3,us,squeeze(u0(:,:,:,Ps)));
295     xs=cat(3,xs,squeeze(patches.x(:,:,:,Ps)));
296     nPs = [nPs;nPatch];

```

End loop. Check grids were aligned, then compute errors compared to the full-domain solution.

```

304     end%for log2N
305     assert(max(abs(reshape(diff(xs,1,3),[],1)))<1e-12,'x-coord failure')
306     errs = us-us(:,:,1);
307     meanAbsErrs = mean(abs(reshape(errs,[],size(us,3))))
308     ratioErrs = meanAbsErrs(2:end)./meanAbsErrs(1:end-1)

```

Plot solution in common patches First adjoin NaNs to separate patches, and reshape.

```

318 x=xs(:,:,1); u=us;
319 x(end+1,:)=nan; u(end+1,:)=nan;
320 u=reshape(u,numel(x),[]);

```

Reshape solution field.

```

326 figure(1),clf
327 plot(x(:),u,'.-'), legend(num2str(nPs))
328 xlabel('space $x$'),ylabel('equilibrium $u(x)$')
329 ifOurCf2tex([mfilename 'us'])%optionally save

```

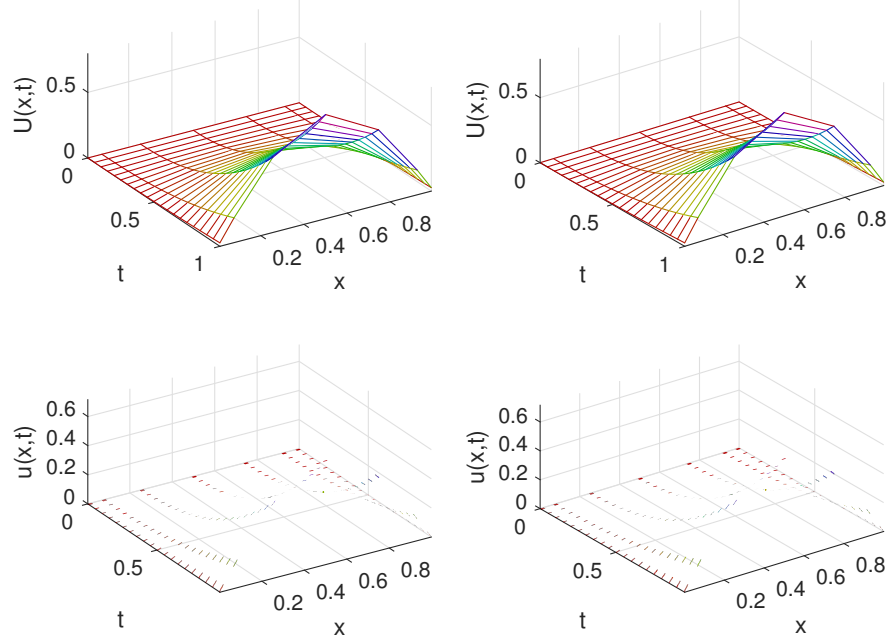
Plot errors Use quasi-log axis to separate the errors.

```

337 err = u(:,1)-u;
338 figure(2), clf
339 plot(x(:),err,'.-'); legend(num2str(nPs))
340 quasiLogAxes(10,sqrt(prod(meanAbsErrs(2:3))))
341 xlabel('space $x$'), ylabel('errors in $u(x)$')
342 ifOurCf2tex(mfilename)%optionally save

```

Figure 3.12: diffusion field $u(x, t)$ of the patch scheme applied to the forced space-time heterogeneous diffusive (3.4). Simulate for seven patches (with a ‘Chebyshev’ distribution): the top stereo pair is a mesh plot of a macroscale value at the centre of each spatial patch at each projective integration time-step; the bottom stereo pair shows the corresponding tiny space-time patches in which microscale computations were carried out.



3.9 Eckhardt2210eg1: example of 1D space-time heterogeneous diffusion via computational homogenisation with projective integration and small patches

An example simulation in time generated by projective integration allied with the patch scheme applied to forced diffusion in a medium with microscale heterogeneity in both space and time. This is more-or-less the first example of Eckhardt & Verfürth (2022) [§6.2].

Suppose the spatial microscale lattice is at points x_i , with constant spacing dx . With dependent variables $u_i(t)$, simulate the microscale lattice forced diffusion system

$$\frac{\partial u_i}{\partial t} = \frac{1}{dx^2} \delta[a_{i-1/2}(t) \delta u_i] + f_i(t), \quad (3.4)$$

in terms of the centred difference operator δ . The system has a microscale heterogeneity via the coefficients $a_{i+1/2}$ which has given periodicity ϵ in space, and periodicity ϵ^2 in time. Figure 3.12 shows an example patch simulation.

The approximate homogenised PDE is $U_t = A_0 U_{xx} + F$ with $U = 0$ at $x = 0, 1$. Its slowest mode is then $U = \sin(\pi x) e^{-A_0 \pi^2 t}$. When $A_0 = 3.3524$ as in Eckhardt then the rate of evolution is about 33 which is relatively fast on the simulation time-scale of $T = 1$. Let’s slow down the dynamics by reducing diffusivities by a factor of 30, so effectively $A_0 \approx 0.1$ and $A_0 \pi^2 \approx 1$.

Also, in the microscale fluctuations change the time variation to cosine, not

its square (because I cannot see the point of squaring it!).

The highest wavenumber mode on the macro-grid of patches, spacing H , is the zig-zag mode on $\dot{U}_i = A_0(U_{I+1} - 2U_I + U_{I-1})/H^2 + F_I$ which evolves like $U_I = (-1)^I e^{-\alpha t}$ for the fastest ‘slow rate’ of $\alpha = 4A_0^2/H^2$. When $H = 0.2$ and $A_0 \approx 0.1$ this rate is $\alpha \approx 10$.

Here use period $\epsilon = 1/100$ (so that computation completes in seconds, and because we have slowed the dynamics by 30). The patch scheme computes only on a fraction of the spatial domain. Projective integration computes only on a fraction of the time domain determined by the ‘burst length’.

3.9.1 Simulate heterogeneous diffusion systems

First establish the microscale heterogeneity has micro-period `mPeriod` on the spatial lattice, and coefficients inspired by Eckhardt2210.04536 §6.2. Set the phase of the heterogeneity so that each patch centre is a point of symmetry of the diffusivity. Then the heterogeneity is repeated to fill each patch. If an odd number of odd-periods in a patch, then the centre patch is a grid point of the field u , otherwise the centre patch is at a half-grid point.

```

98 clear all
99 %global OurCf2eps, OurCf2eps=true %option to save plots
100 mPeriod = 6
101 y = linspace(0,1,mPeriod+1)';
102 a = ( 3+cos(2*pi*y(1:mPeriod)) )/30
103 A0 = 1/mean(1./a) % roughly the effective diffusivity
```

The microscale diffusivity has an additional additive component of $+\frac{1}{30} \cos(2\pi t/\epsilon^2)$ which is coded into time derivative routine via global `microTimePeriod`.

Set the periodicity, via integer $1/\epsilon$, and other parameters.

```

116 nPeriodsPatch = 2 % any integer
117 rEpsilon = 100
118 dx = 1/(mPeriod*rEpsilon+1)
119 nSubP = nPeriodsPatch*mPeriod+2
120 tol=1e-9;
```

Set the time periodicity (global).

```

126 global microTimePeriod
127 microTimePeriod = 1/rEpsilon^2
```

Establish the global data struct `patches` for the microscale heterogeneous lattice diffusion system (3.4) solved on macroscale domain $[0, 1]$, with `nPatch` patches, and say fourth-order interpolation to provide the edge-values of the inter-patch coupling conditions. Distribute the patches either equispaced or chebyshev. Setting `patches.EdgyInt` true means the edge-values come from interpolating the opposite next-to-edge values of the patches (not the mid-patch values).

```

144 nPatch = 7
145 ordCC = 4
```

```

146 Dom = 'chebyshev'
147 global patches
148 configPatches1(@heteroDiffF,[0 1],Dom,nPatch ...
149     ,ordCC,dx,nSubP,'EdgyInt',true,'hetCoeffs',a);
150 DX = mean(diff(squeeze(patches.x(1,1,1,:))))

```

Set the forcing coefficients as the odd-periodic extensions, accounting for roundoff error in f2.

```

158 if 0 % given forcing
159     patches.f1=2*( patches.x-patches.x.^2 );
160     patches.f2=2*0.5+0*patches.x;
161 else% simple sine forcing
162     patches.f1=sin(pi*patches.x);
163     patches.f2=pi/2*sin(pi*patches.x);
164 end%if

```

Simulate Set the initial conditions of a simulation to be zero. Mark that edge of patches are not to be used in the projective extrapolation by setting initial values to NaN.

```

175 u0 = 0*patches.x;
176 u0([1 end],:) = nan;

```

Set the desired macro- and microscale time-steps over the time domain. The macroscale step is in proportion to the effective mean diffusion time on the macroscale, here $1/(A_0\pi^2) \approx 1$ so for macro-scale error less than 1% need $\Delta t < 0.24$, so use 0.1 say.

The burst time depends upon the sub-patch effective diffusion rate β where here rate $\beta \approx \pi^2 A_0/h^2 \approx 2000$ for patch width $h \approx 0.02$: use the formula from the Manual, with some extra factor, and rounded to the nearest multiple of the time micro-periodicity.

```

193 ts = linspace(0,1,21)
194 h=(nSubP-1)*dx;
195 beta = pi^2*A0/h^2 % slowest rate of fast modes
196 burstT = 2.5*log(beta*diff(ts(1:2)))/beta
197 burstT = max(10,round(burstT/microTimePeriod))*microTimePeriod +1e-12
198 %addpath('.../ProjInt')

```

Time the projective integration simulation.

```

204 tic
205 [us,tss,uss] = PIRK2(@heteroBurstF, ts, u0(:), burstT);
206 cputime=toc

```

Plot space-time surface of the simulation First, just a macroscale mesh plot—stereo pair.

```

216 xs=squeeze(patches.x);
217 Xs=mean(xs);
218 Us=squeeze(mean( reshape(us,length(ts),[],nPatch), 2,'omitnan')));

```

```

219 figure(1),clf
220 for k = 1:2, subplot(2,2,k)
221     mesh(ts,Xs(:),Us')
222     ylabel('space $x$'), xlabel('time $t$'), zlabel('$U(x,t)$')
223     colormap(0.8*hsv), axis tight, view(62-4*k,45)
224 end

```

Second, plot a surface detailing the microscale bursts—stereo pair. Do not bother with the patch-edge values. Optionally save to Figs folder.

```

232 xs([1 end],:) = nan;
233 for k = 1:2, subplot(2,2,2+k)
234     surf(tss,xs(:),uss', 'EdgeColor','none')
235     ylabel('space $x$'), xlabel('time $t$'), zlabel('$u(x,t)$')
236     colormap(0.7*hsv), axis tight, view(62-4*k,45)
237 end
238 ifOurCf2eps(mfilename)

```

3.9.2 heteroBurstF(): a burst of heterogeneous diffusion

This code integrates in time the derivatives computed by `heteroDiff` from within the patch coupling of `patchSys1`. Try `ode23`, although `ode45` may give smoother results. Sample every period of the microscale time fluctuations (or, at least, close to the period).

```

17 function [ts, ucts] = heteroBurstF(ti, ui, bT)
18     global microTimePeriod
19     [ts,ucts] = ode45( @patchSys1,ti+(0:microTimePeriod:bT),ui(:));
20 end

```


3.10 homoLanLif1D: computational homogenisation of a 1D heterogeneous Landau–Lifshitz by simulation on small patches

The Landau–Lifshitz equation describes the precessional motion of magnetization $\vec{M}$ in a solid (see *Landau–Lifshitz–Gilbert equation* in Wikipedia). In a medium with microscale heterogeneity $a(x)$, and with phenomenological damping parameter α , we explore the dynamics of $\vec{M}(x, t)$ governed by the nonlinear Landau–Lifshitz PDE (Leitenmaier & Runborg 2021, (1.1))⁵

$$\vec{M}_t = -\vec{M} \times \vec{H} - \alpha \vec{M} \times (\vec{M} \times \vec{H}), \quad \vec{H} := \vec{\nabla} \cdot (a \vec{\nabla} \vec{M}).$$

Note, for every x , $|\vec{M}(x, t)|$ is constant in time due to $\vec{M} \cdot \vec{M}_t = 0$ for every x, t . We normally set $|\vec{M}(x, 0)| = 1$.

Figure 3.13 shows an example simulation in time generated by the patch scheme applied to the above Landau–Lifshitz PDE on the spatial domain $[0, 1]$ with domain boundary conditions of 1-periodicity. The inter-patch coupling is realised by interpolation of the patch’s next-to-edge values to the patch opposite edges. Such coupling preserves symmetry in many systems (quartic interpolation appears to be the lowest order that generally gives good accuracy). With damping parameter $\alpha = 0.001$ then the largest few macroscale modes decay with rate roughly 0.1, and so are negligibly damped over a time of 0.1.

Suppose the spatial microscale lattice is at points x_i , with constant spacing dx . With dependent variables $\vec{M}_i(t)$, simulate the microscale lattice system

$$\vec{M}_{it} = -\vec{M}_i \times \vec{H}_i - \alpha \vec{M}_i \times (\vec{M}_i \times \vec{H}_i), \quad \vec{H}_i := \frac{1}{dx^2} \delta[a_{i-1/2} \delta \vec{M}_i],$$

in terms of the centred difference operator δ . The system has a microscale heterogeneity via the coefficients $a_{i+1/2}$ which we assume to have some given known periodicity (Leitenmaier & Runborg 2021, pp.6,27). Figure 3.13 shows a patch simulation of this system: observe the effects of the heterogeneity within each patch.

Parameters There are two closely related examples (Leitenmaier & Runborg 2021, pp.6,27), that we distinguish here with parameter `ex5p1`: set to either zero or one. The Landau–Lifshitz dissipation parameter α should be small. If the initial conditions are smooth, then `ode15s` has no problems for $\alpha = 0.001$.⁶

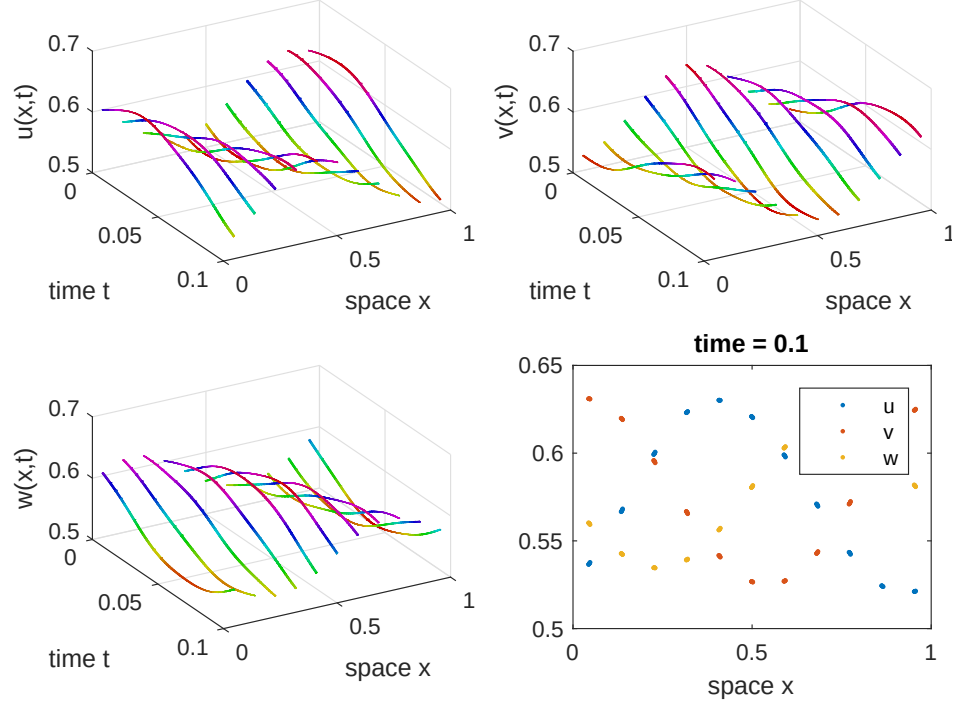
```
88 global alpha ex5p1
89 ex5p1 = 0; % set to 1 for L&O example of p.27
90 alpha = 0.001 % phenomenological damping parameter
```

The physical microscale periodicity of the heterogeneity is ϵ (ϵ is *not* the patch scale ratio):

⁵ Recall $a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$

⁶ But, add randomness to the initial conditions and the computation appears unstable with `ode15s` when $\alpha < 0.2$. However, `ode23` may be stable? for $\alpha = 0.01$ albeit expensively taking 10^7 time-steps per second (due to microscale oscillations of frequency up to 10^5 – 10^6).

Figure 3.13: magnetic field $\vec{M}(x, t) = (u, v, w)$ of the gap-tooth scheme applied to the heterogeneous Landau–Lifshitz PDE to show the emergent macroscale wave-like dynamics. This simulation uses eleven patches in space of size ratio 0.055. Compare the time $t = 0.1$ graph with Fig. 2.1 of [Leitenmaier & Runborg \(2021\)](#).



```
98 epsilon = 1/200/(1+ex5p1) %pp.6,27
```

3.10.1 Script code to simulate heterogeneous diffusion systems

This example script implements the following patch/gap-tooth scheme.

1. configPatches1
2. ode15s $\mapsto$ patchSys1 $\mapsto$ heteroLanLif1D
3. plot the simulation

First establish the microscale heterogeneity has micro-period `mPeriod` on the lattice with values of the column vector from [Leitenmaier & Runborg \(2021\)](#) [pp.6,27]. Later, the heterogeneity is repeated to fill each patch.

```
124 dx = 1/2000 %1/6000 %p.27
125 mPeriod = round(epsilon/dx)
126 a = 1 + 0.5*sin(2*pi*(0.5:mPeriod)'/mPeriod); %p.6
```

Establish the global data struct `patches` for the microscale heterogeneous lattice diffusion system (3.2) solved on 1-periodic domain, with maybe 24 patches, but 11 is enough, here each patch of size ratio to fit one period of the heterogeneity in each patch, and spectral inter-patch interpolation to provide the patch edge-values. Invoking `EdgyInt` means the edge-values come from interpolating the opposite next-to-edge values of the patches (not the mid-patch

values).

```

142 global patches
143 nPatch = 11 %24 %p.6, odd is slightly cleaner
144 nSubP = mPeriod+2
145 ratio = nPatch*epsilon
146 configPatches1(@heteroLanLif1D,[0 1],nan,nPatch ...
147     ,0,ratio,nSubP,'EdgyInt',true ...
148     ,'hetCoeffs',a);
149 assert(abs(dx-diff(patches.x(2:3)))<1e-10 ...
150     , 'microscale grid spacing error')

```

Simulate Set the initial conditions of a simulation to be that of [Leitenmaier & Runborg \(2021\)](#) [pp.6], except possibly perturbed by random microscale noise. Scale the initial conditions so that $|\vec{M}(x,0)| = 1$.

```

162 u0 = 0.5+exp(-0.1*cos(2*pi*(patches.x-0.32)));
163 v0 = 0.5+exp(-0.2*cos(2*pi*patches.x)) +0*randn(size(patches.x));
164 w0 = 0.5+exp(-0.1*cos(2*pi*(patches.x-0.75)));
165 M0 = [ u0 v0 w0 ] ./sqrt(u0.^2+v0.^2+w0.^2);
166 dM0dt = patchSys1(0,M0(:));

```

Integrate using standard integrators.

```

172 tic
173 [ts,Ms] = ode15s(@patchSys1, [0 0.1], M0(:));
174 cpuTime=toc
175 sizeMs=size(Ms)

```

Reshape results for processing. For simplicity, set edge values to `nans`. For the field values (which are rows in `Ms`) we need to reshape, permute, and reshape again.

```

184 xs = squeeze(patches.x);
185 Ms = reshape(Ms,length(ts),nSubP,3,nPatch);
186 Ms(:,[1 end],:,:) = nan; % nan patch edges
187 Ms = reshape( permute(Ms,[2 4 1 3]) ,[],length(ts),3);

```

Check on constancy of $|\vec{M}(x,t)|$ in time. The mean and standard deviation appears to show that, with `ode15s`, they are constant to errors typically 10^{-5} .

```

195 Mabs = sqrt( sum(Ms.^2,3) );
196 meanMabs = mean(Mabs(:),'omitnan')
197 stdevMabs = std(Mabs(:),'omitnan')

```

Plot space-time surface of the simulation Choose whether to save some plots, or not.

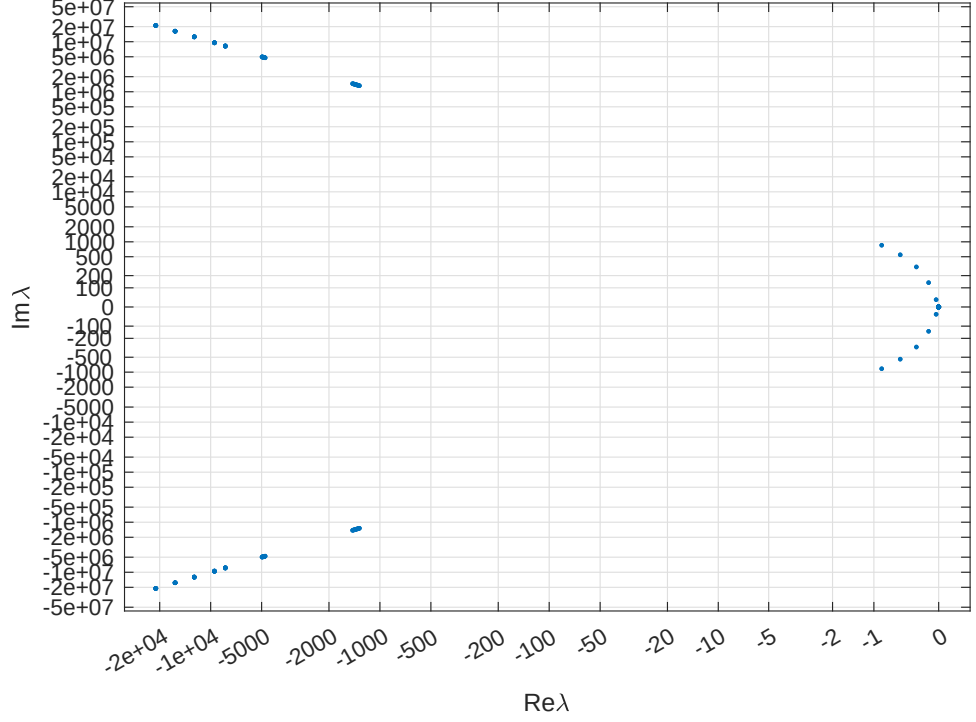
```

207 global OurCf2eps
208 OurCf2eps = false;

```

Subsampled surface over the macroscale duration of the simulation to show the propagation of the macroscale modes over the heterogeneous lattice.

Figure 3.14: spectrum of eigenvalues of the multiscale patch scheme applied to the Landau–Lifshitz PDE. The macroscale eigenvalues are clearly separated from those of the microscale sub-patch modes.



```

216 figure(1),clf
217 if length(ts)>50
218     [~,j]=min(abs(ts(:)-linspace(ts(1),ts(end),50)));
219     else j=1:length(ts); end
220 uvw='uvw';
221 for p=1:3
222     subplot(2,2,p)
223     mesh(ts(j),xs(:),Ms(:,j,p))
224     view(60,40), colormap(0.8*hsv)
225     xlabel('time $t$'), ylabel('space $x$')
226     zlabel(['$' uvw(p) '(x,t)$'])
227 end

```

Final time plot to compare with Fig. 2.1 of [Leitenmaier & Runborg \(2021\)](#).

```

234 subplot(2,2,4)
235 plot(xs(:),squeeze(Ms(:,end,:)),'.')
236 xlabel('space $x$'), legend(uvw(1),uvw(2),uvw(3))
237 title(['time = ' num2str(ts(end),4)])
238 ifOurCf2eps([mfilename 'uvw'])

```

3.10.2 Spectrum of the coded patch system

It appears the spectrum has the following properties as shown by [Figure 3.14](#), with $N = \text{nPatch}$ and $n = \text{nSubP} - 2$, and on base of $\vec{M} = \vec{1}/\sqrt{3}$.

- A (near) zero eigenvalue for each and every microscale lattice point (nN) due to $|\vec{M}(x, t)|$ being constant in time, for every x . Presumably near zero (roughly 10^{-2}) due to round-off error.
- $2N$ macroscale eigenvalues, including a pair of (near) zero eigenvalues of macroscale conservation, and others ranging from $27(-\alpha \pm i)$ to $(-6.4\alpha \pm 7.4i)(N - 1)^2$.
- $2(n - 1)N$ fast eigenvalues, more negative than about $-\alpha \cdot 10^6$ and higher frequency than about 10^6 . Presumably depends upon ϵ —the periodicity and patch size.

Form an equilibrium of $\vec{M}$ constant in space, then find the indices corresponding to patch interior points.

```

275 Me = 1+0.2*rand(1,3);
276 Me = Me./sqrt(sum(Me.^2,2))
277 Me = Me +0*patches.x;
278 Me([1 end],:,:) = nan;
279 i=find(~isnan(Me));
280 f0 = patchSys1(0,Me(:));
281 assert(abs( norm(f0(:)) )<1e-8,'not equilibrium')
```

Form the Jacobian by auto-differentiation or by numerical differentiation.

```

287 delta=1e-7;
288 nJac=length(i);
289 Jac=nan(nJac);
290 for j=1:nJac
291     M=Me; M(i(j))=M(i(j))+delta;
292     fj=patchSys1(0,M(:));
293     Jac(:,j)=(fj(i)-f0(i))/delta;
294 end
295 if exist('AutoDiff','class')
296     disp('**** computing Jacobian with AutoDiff')
297     Jac=AutoDiffJacobianAutoDiff(@(M) patchSys1(0,M),Me(:),i);
298     Jac=full(Jac(i,:));
299 else% compute Jac by finite-diff
300     Jac=nan(nJac);
301     for j=1:nJac
302         M=Me; M(i(j))=M(i(j))+delta;
303         fj=patchSys1(0,M(:));
304         Jac(:,j)=(fj(i)-f0(i))/delta;
305     end
306 end%if exist
```

Compute eigenvalues, sort, and count some groups according to ad hoc criteria.

```

313 eval = eig(Jac);
314 [~,k] = sort(abs(eval));
315 eval = eval(k);
316 nZero = sum(abs(eval)<1)
```

```

317 nCent = sum(abs(real(eval))<1e5*alpha)
318 nSlow = sum(abs(eval)<1e5)

Plot the spectrum of eigenvalues on quasi-log axes.

324 figure(2),clf
325 plot(real(eval),imag(eval),'.'');
326 xlabel('$\text{Re}\lambda$'), ylabel('$\text{Im}\lambda$')
327 quasiLogAxes(1,100);
328 ifOurCf2eps([mfilename 'Spec'])

```

3.10.3 heteroLanLif1D(): heterogeneous Landau–Lifshitz PDE

This function codes the lattice heterogeneous Landau–Lifshitz PDE (Leitenmaier & Runborg 2021, (1.1)) inside patches in 1D space. For 4D input array M storing the three components of $\vec{M}$ (via edge-value interpolation of `patchSys1`, Section 3.2), computes the time derivative at each point in the interior of a patch, output in M_t . The column vector of coefficients $c_i = 1 + \frac{1}{2} \sin(2\pi x_i/\epsilon)$ have previously been stored in struct `patches.cs`.

- With `ex5p1=0` computes the example EX1 (Leitenmaier & Runborg 2021, p.6).
- With `ex5p1=1` computes the first 'locally periodic' example (Leitenmaier & Runborg 2021, p.27).

```

30 function Mt = heteroLanLif1D(t,M,patches)
31     global alpha ex5p1
32     dx = diff(patches.x(2:3)); % space step

Compute the heterogeneous  $\vec{H} := \vec{\nabla} \cdot (a \vec{\nabla} \vec{M})$ 

38     a = patches.cs ...
39         +ex5p1*(0.1+0.25*sin(2*pi*(patches.x(2:end),:,:)-dx/2)+1.1));
40     H = diff(a.*diff(M))/dx^2;

```

At each microscale grid point, compute the cross-products $\vec{M} \times \vec{H}$ and $\vec{M} \times (\vec{M} \times \vec{H})$ to then give the time derivative $\vec{M}_t = -\vec{M} \times \vec{H} - \alpha \vec{M} \times (\vec{M} \times \vec{H})$ (Leitenmaier & Runborg 2021, (1.1)):

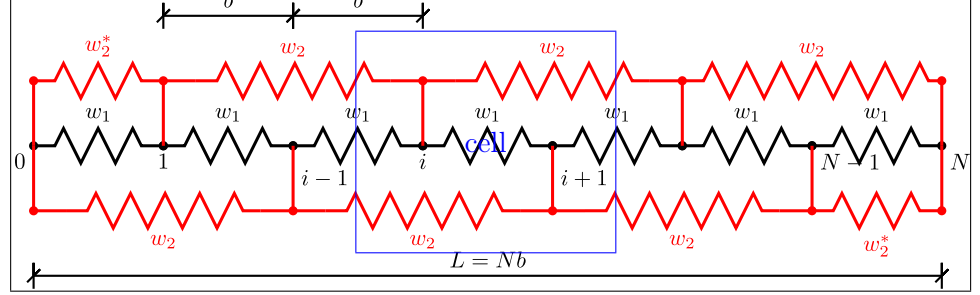
```

48     i = 2:size(M,1)-1; % interior points in a patch
49     MH=nan(size(H),'like',H); % preallocate for MxH
50     MH(:,3,:,:) = M(i,1,:,:).*H(:,2,:,:)-M(i,2,:,:).*H(:,1,:,:);
51     MH(:,2,:,:) = M(i,3,:,:).*H(:,1,:,:)-M(i,1,:,:).*H(:,3,:,:);
52     MH(:,1,:,:) = M(i,2,:,:).*H(:,3,:,:)-M(i,3,:,:).*H(:,2,:,:);
53     MMH=nan(size(H),'like',H); % preallocate for MxMxH
54     MMH(:,3,:,:) = M(i,1,:,:).*MH(:,2,:,:)-M(i,2,:,:).*MH(:,1,:,:);
55     MMH(:,2,:,:) = M(i,3,:,:).*MH(:,1,:,:)-M(i,1,:,:).*MH(:,3,:,:);
56     MMH(:,1,:,:) = M(i,2,:,:).*MH(:,3,:,:)-M(i,3,:,:).*MH(:,2,:,:);
57     Mt = nan(size(M),'like',M); % preallocate output array
58     Mt(i,:,:,:) = -MH-alpha*MMH;
59 end% function

```

Fin.

Figure 3.15: 1D arrangement of non-linear springs with connections to (a) next-to-neighbour node (Combescore 2022, Fig. 3(a)). The blue box is one micro-cell of one period, width $2b$, containing an odd and an even i .



3.11 Combescore2022: simulation and continuation of a 1D example nonlinear elasticity, via patches

Here we explore a nonlinear 1D elasticity problem with complicated microstructure. Executes a simulation. *But the main aim is to show how one may use the MatCont continuation toolbox (Govaerts et al. 2019) together with the Patch Scheme toolbox (Maclean et al. 2020) in order to explore parameter space by continuing branches of equilibria, etc.*

Figure 3.15 shows the microscale elasticity—adapted from Fig. 3(a) by Combescore (2022). Let the spatial microscale lattice be at rest at points x_i , with constant spacing b . With displacement variables $u_i(t)$, simulate the microscale lattice toy elasticity system with 2-periodicity: for $p = 1, 2$ (respectively black and red in Figure 3.15) and for every i ,

$$\begin{aligned} \epsilon_i^p &:= \frac{1}{pb}(u_{i+p/2} - u_{i-p/2}), & \sigma_i^p &:= w_p'(\epsilon_i^p), \\ \frac{\partial^2 u_i}{\partial t^2} &= \sum_{p=1}^2 \frac{1}{pb}(\sigma_{i+p/2}^p - \sigma_{i-p/2}^p), & w_p'(\epsilon) &:= \epsilon - M_p \epsilon^3 + \epsilon^5. \end{aligned} \quad (3.5)$$

The system has a microscale heterogeneity via the two different functions $w_p'(\epsilon)$ (Combescore 2022, §4):

- microscale ‘instability’ (structure) arises with $M_1 := 2$ and $M_2 := 1$ (Figures 3.16 and 3.19(b)); and
- large scale ‘instability’ (structure) arises with $M_1 := -1$ and $M_2 := 3$ (Figures 3.18 and 3.19(a)).

Microscale case Set $M_1 := 2$ and $M_2 := 1$. We fix the boundary conditions $u(0) = 0$ and parametrise solutions by $u(L)$. There are equilibria $u \approx u(L)x/L$, but under large compression (large negative $u(L)$) interesting structures develop. Figure 3.16 shows boundary layers with microscale variations develop for $u(L) < -13$. This figure plots a strain ϵ as the strain is nearly constant across the interior, so the boundary layers show up clearly. As $u(L)$ decreases further, Figure 3.16 shows the family of equilibria form complicated folds. Table 3.2 lists that MatCont also reports some branch

Figure 3.16: the case of microscale ‘instability’ appears as fluctuations close to both boundaries. As the system is physically compressed, the equilibrium curve has complicated folds, as shown here (and Figure 3.19(b)).

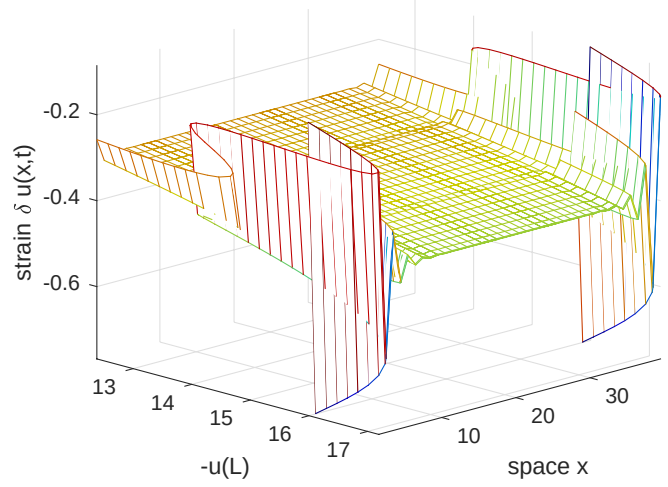


Table 3.2: Interesting equilibria for the cases of small scale instability: $M_1 := 2$, $M_2 := 1$ (Figures 3.16 and 3.19(b)). The rightmost column gives the $-u(L)$ parameter values for corresponding critical points in the three-patch code (Figure 3.17).

$-u(L)$	MatCont description	Patch
14.684	Branch point	14.599
14.702	Limit point	14.610
14.612	Neutral Saddle Equilibrium	-
14.063	Neutral Saddle Equilibrium	-
13.972	Limit point	13.817
13.988	Branch point	13.828
17.184	Branch point	17.197
-	Limit point	17.227
17.183	Neutral Saddle Equilibrium	17.211

points and neutral saddle equilibria in this same regime (see Figure 3.19(b)). I have not yet followed any of the branches.

The previous paragraph’s discussion is for a full domain simulation, albeit done through an imposed computational framework of physically abutting patches. Figure 3.17 shows the corresponding MatCont continuation for the patch scheme with $N = 3$ patches in the domain. Just three patches may well be reasonable as the structures in this problem are the two boundary layers, and a constant interior. Figure 3.17 shows the patch scheme reasonably resolves these. Table 3.2 also lists the special points, as reported by MatCont, in the equilibria of the patch scheme. The locations of these special points reasonably match those found by the full domain simulation.

Importantly, MatCont is about *ten times quicker to execute on the patches* than on the full domain code. This speed-up indicates that on larger scale problems the patch scheme could be very useful in continuation explorations.

Large scale case Set $M_1 := -1$ and $M_2 := 3$. We fix the boundary conditions $u(0) = 0$ and parametrise solutions by $u(L)$. There are equilibria $u \approx u(L)x/L$, but under large compression (large negative $u(L)$) interesting structures develop. Figure 3.18 shows an interior region of higher magnitude strain develops. Again, this figure plots a strain ϵ as the strain is nearly constant across the domain, so the interior structure shows up clearly. As $u(L)$

Figure 3.17: using just three patches, the case of microscale instability appears as fluctuations close to both boundaries. As the system is physically compressed, the equilibrium curve has complicated folds, as shown, and that approximately match Figure 3.16. But it is computed ten times quicker.

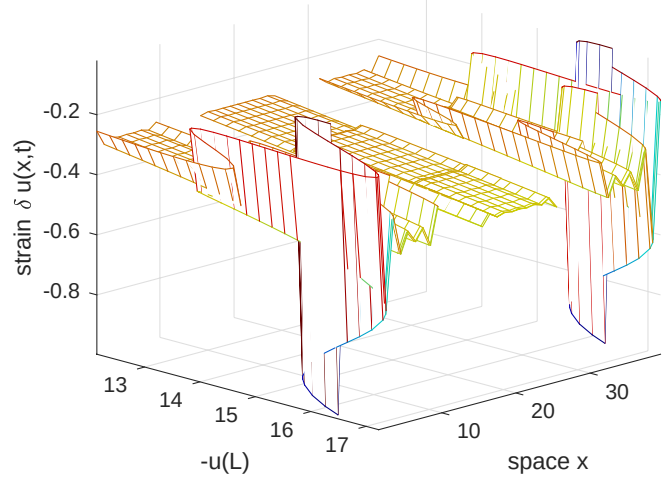
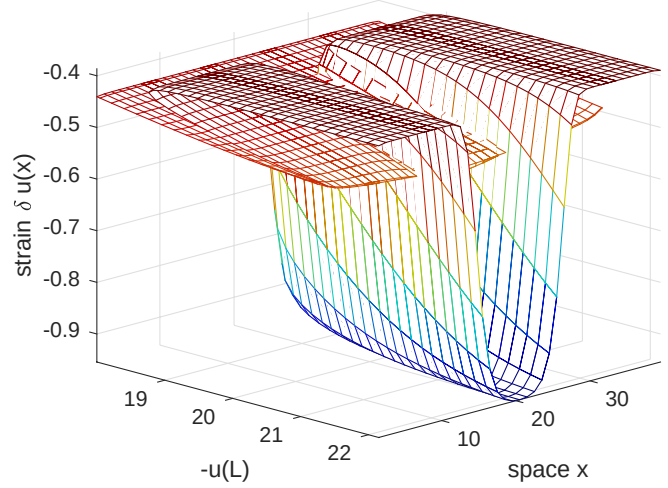


Figure 3.18: the case of large scale ‘instability’. Spatial structure appears in the middle of the domain. As the system is physically compressed, the equilibrium curve has complicated folds, as shown here and in Figure 3.19(a).



decreases further, Figure 3.18 shows the family of equilibria form complicated folds. Table 3.3 lists that MatCont also reports some branch points and neutral saddle equilibria in this regime (see Figure 3.19(a)). I have not yet followed any of the branches.

The patch scheme with $N = 3$ patches does not make reasonable predictions here. I suspect this failure is because the nontrivial interior structure here occupies too much of the domain to fit into one ‘small’ patch. Here the patch scheme may be useful if the physical domain is larger.

3.11.1 Configure heterogeneous toy elasticity systems

Set some physical parameters. Each cell is of width $dx := 2b$ as I choose to store u_i for odd i in $u((i+1)/2, 1, :)$ and for even i in $u(i/2, 2, :)$, that is,

Table 3.3: Interesting equilibria for the cases of large scale instability: $M_1 := -1$, $M_2 := 3$ (Figures 3.18 and 3.19(a)).

$-u(L)$	MatCont description
21.295	Limit point
18.783	Branch point
18.762	Neutral Saddle Equilibrium
18.761	Neutral Saddle Equilibrium
18.761	Limit point
18.934	Branch point
19.393	Branch point
19.928	Branch point
20.490	Branch point
21.055	Branch point
21.627	Branch point

the the physical displacements form the array

$$\mathbf{u} = \begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \\ u_5 & u_6 \\ \vdots & \vdots \end{bmatrix}.$$

Then corresponding velocities are adjoined as 3rd and 4th column.

```

229 clear all
230 global b M vis
231 b = 1 % separation of lattice points
232 N = 42 % # lattice steps in L
233 L = b*N % length of domain

```

The nonlinear coefficients of stress-strain are in array M, chosen by theCase.

```

240 theCase = 2
241 switch theCase
242 case 1, M = [0 0 0 0] % linear spring coefficients
243 case 2, M = [ 2 1 1 1] % micro scale instability??
244 case 3, M = [-1 3 1 1] % large scale instability??
245 end% switch
246 vis = 0.1 % does not appear to affect the equilibria
247 tEnd = 25

```

Patch parameters: here nSubP is the number of cells.

```

253 edgyInt = true
254 nSubP = 6, nPatch = 5 % gives full-domain on N=42, dx=2
255 %nSubP = 6, nPatch = 3 % patches for some crude comparison

```

Establish the global data struct `patches` for the microscale heterogeneous lattice elasticity system (3.5). Solved with `nPatch` patches, and interpolation (as high-order as possible) to provide the edge-values of the inter-patch coupling conditions.

```

268 global patches
269 configPatches1(@heteroNLE,[0 L], 'equispace', nPatch ...

```

```

270         ,0,2*b,nSubP,'EdgyInt',edgyInt);
271 xx = patches.x+[-1 1]*b/2; % staggered sub-cell positions

```

3.11.2 Simulate in time

Set the initial displacement and velocity of a simulation. Integrate some time using standard integrator.

```

284 u0 = [ sin(pi/L*xx) -0*0.14*cos(pi/L*xx) ];
285 tic
286 [ts,ust] = ode23(@patchSys1, tEnd*linspace(0,1,41), u0(:) ...
287               ,[],patches,0);
288 cpuIntegrateTime = toc

```

Plot space-time surface of the simulation To see the edge values of the patches, interpolate and then adjoin a row of `nans` between patches. Because of the odd/even storage we need to do a lot of permuting and reshaping. First, array of sub-cell coordinates in a column for each patch, separating patches also by an extra row of `nans`.

```

301 xs = reshape( permute( xx ,[2 1 3 4]), 2*nSubP,nPatch);
302 xs(end+1,:) = nan;

```

Interpolate patch edge values, at all times simultaneously by including time data into the 2nd dimension, and 2nd reshaping it into the 3rd dimension.

```

310 uvs = reshape( permute( reshape(ust ...
311               ,length(ts),nSubP,4,1,nPatch) ,[2 3 1 4 5]) ,nSubP,[],1,nPatch);
312 uvs = reshape( patchEdgeInt1(uvs) ,nSubP,4,[],nPatch);

```

Extract displacement field, merge the 1st two columns, permute the time variations to the 3rd, separate patches by NaNs, and merge spatial data into the 1st column.

```

320 us = reshape( permute( uvs(:,1:2,,:,:) ...
321               ,[2 1 4 3]) ,2*nSubP,nPatch,[]);
322 us(end+1,,:,:) = nan;
323 us = reshape(us,[],length(ts));

```

Plot space-time surface of displacements over the macroscale duration of the simulation.

```

330 figure(1), clf()
331 mesh(ts,xs(:,),us)
332 view(60,40), colormap(0.8*jet), axis tight
333 xlabel('time $t$'), ylabel('space $x$'), zlabel('$u(x,t)$')

```

Ditto for the velocity.

```

339 vs = reshape( permute( uvs(:,3:4,,:,:) ...
340               ,[2 1 4 3]) ,2*nSubP,nPatch,[]);
341 vs(end+1,,:,:) = nan;
342 vs = reshape(vs,[],length(ts));
343 figure(2), clf()

```

```

344     mesh(ts,xs(:),vs)
345     view(60,40), colormap(0.8*jet), axis tight
346     xlabel('time $t$'), ylabel('space $x$'), zlabel('$v(x,t)$')
347     drawnow

```

3.11.3 MatCont continuation

First, use `fsolve` to find an equilibrium at some starting compressive displacement—a compression that differs depending upon the case of nonlinearity.

```

359     muL0 = 12+6*(theCase==3)
360     u0 = [ -muL0*xx/L 0*xx ];
361     u0([1 end],:,:)=nan;
362     patches.i = find(~isnan(u0));
363     nVars=length(patches.i)
364     ueq=fsolve(@(v) dudtSys(0,v,muL0),u0(patches.i));

```

Start search for equilibria at other compression parameters. Starting from zero, need 1000+ to find both the large-scale and small-scale instability cases. But need less points when starting from parameter 12 or so.

```

373     disp('Searching for equilibria, may take 1000+ secs')
374     [uv0,vec0]=init_EP_EP(@matContSys,ueq,muL0,[1]);
375     opt=contset; % initialise MatCont options
376     opt=contset(opt,'Singularities',true); %to report branch points, p.24
377     opt=contset(opt,'MaxNumPoints',400); % restricts how far matcont goes
378     opt=contset(opt,'Backward',true); % strangely, needs to go backwards??
379     [uv,vec,s,h,f]=cont(@equilibrium, uv0, [], opt); %MatCont continuation

```

Post-process the report

```

386     disp('List of interesting critical points')
387     muLs=uv(nVars+1,:);
388     for j=1:numel(s)
389         disp([num2str(muLs(s(j).index),5) ' & ' s(j).msg ' \\\'])
390     end

```

Find a range of parameter and corresponding indices where all the critical points occur.

```

397     p1=muLs(end); pe=muLs(1);
398     if numel(s)>3, for j=2:numel(s)-1
399         p1=min(p1,muLs(s(j).index));
400         pe=max(pe,muLs(s(j).index));
401     end, end
402     pMid=(p1+pe)/2, pWid=abs(pe-p1)
403     iPars=find(abs(muLs(:)-pMid)<pWid);%include some either side

```

Choose an ‘evenly spaced’ subset of the range so we only plot up to sixty of the parameter values reported in the range.

```

411     nPars=numel(iPars)
412     dP=ceil((nPars-1)/60)

```

```

413  iP=1:dP:nPars;
414  muLP=muLs(iPars(iP));

Interpolate patch edge values, at all parameters simultaneously by including
parameter-wise data into the 2nd dimension, and 2nd reshaping it into the
3rd dimension.

422  uvs=nan(numel(iP),numel(u0));
423  uvs(:,patches.i)=uv(1:nVars,iPars(iP))';
424  uvs = reshape( permute( reshape(uvs ...
425      ,length(muLP),nSubP,4,1,nPatch) ,[2 3 1 4 5]) ,nSubP,[],1,nPatch);
426  uvs = reshape( patchEdgeInt1(uvs) ,nSubP,4,[],nPatch);

Extract displacement field, merge the 1st two columns, permute the parameter
variations to the 3rd, separate patches by NaNs, and merge spatial data into
the 1st column.

434  us = reshape( permute( uvs(:,1:2,:,:) ...
435      ,[2 1 4 3]) ,2*nSubP,nPatch,[]);
436  us(end+1,:,:) = nan;
437  us = reshape(us,[],length(muLP));

Plot space-time surface of displacements over the macroscale duration of the
simulation.

444  figure(4), clf()
445  mesh(muLP,xs(:),us)
446  view(60,40), colormap(0.8*jet), axis tight
447  xlabel('-u(L)'), ylabel('space $x$'), zlabel('$u(x)$')

Plot space-time surface of strain, differences in displacements, over the pa-
rameter variation.

454  figure(5), clf()
455  mesh(muLP,xs(1:end-1),diff(us))
456  view(45,20), colormap(0.8*jet), axis tight
457  xlabel('$-u(L)$'), ylabel('space $x$'), zlabel('strain $\Delta u(x)$')
458  ifOurCf2eps(['Comb22diffu' num2str(theCase)],[12 9])%optionally save

```

Labelled parameter plot Get the labelled 2D plots of [Figure 3.19](#) via MatCont's `cpl` function. In high-D problems it is unlikely that any one variable is a good thing to plot, so I show how to plot something else, here a strain. I use all the computed points so reform `uvs` (possibly better to have merged the critical points into the list of plotted parameters??).

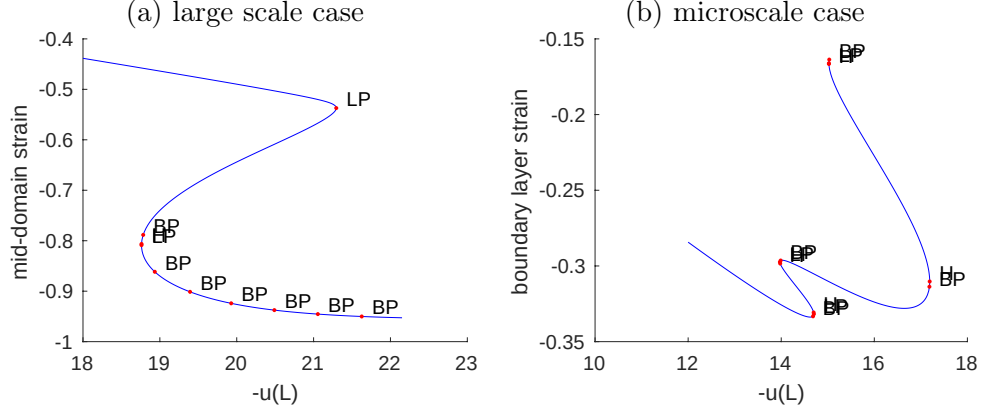
```

488  uvs = nan(numel(muLs),numel(u0));
489  uvs(:,patches.i) = uv(1:nVars,:);
490  uvs = reshape( uvs ,[],nSubP,4,nPatch);

```

As a function of the parameter, plot the strain in the middle of the domain (the middle of the middle patch), unless it is the microscale case when we plot a strain near the middle of the left boundary layer.

Figure 3.19: cross-sections through Figures 3.16 and 3.18: (a) large scale case, at the mid-point in space of Figure 3.18; (b) microscale case, in a boundary layer of Figure 3.16. These cross-sections are labelled with the various critical points.



```

499 if theCase==2, thePatch=1;
500 else thePatch=(nPatch+1)/2;
501 end%if
502 figure(7),clf
503 du = diff( uvs(:,nSubP/2,1:2,thePatch) ,1,3);
504 cpl([muLs;du'],[],s);
505 xlabel('$-u(L)$')
506 if thePatch==1, ylabel('boundary layer strain')
507 else ylabel('mid-domain strain')
508 end
509 ifOurCf2eps(['Comb22cpl' num2str(theCase)],[9 7])%optionally save

```

3.11.4 matContSys: basic function for MatCont analysis

This is the simple 'odefile' of the patch scheme wrapped around the microcode.

```

522 function out = matContSys%(t,coordinates,flag,y,z)
523 out{1} = [];%@init;
524 out{2} = @dudtSys;
525 out{3} = [];%@jacobian;
526 out{4} = [];%@jacobianp;
527 out{5} = [];%@hessians;
528 out{6} = [];%@hessiansp;
529 out{7} = [];
530 out{8} = [];
531 out{9} = [];
532 end% function matContSys

```

3.11.5 dudtSys(): wraps around the patch wrapper

This function adjoins `patches` to the argument list, places the variables within the patch structure, and then extracts their time derivatives to return. Used

by both MatCont and fsolve.

```
545 function ut = dudtSys(t,u,p)
546 global patches
```

The 4 here is the number of variables in each micro-cell, that is, notionally 'at' each x -grid point.

```
553 U=nan(1,4)+patches.x;
554 U(patches.i)=u(:);
555 Ut=patchSys1(t,U,patches,p);
556 ut=Ut(patches.i);
557 end
```

3.11.6 heteroNLE(): forced heterogeneous elasticity

This function codes the lattice heterogeneous example elasticity inside the patches. Computes the time derivative at each point in the interior of a patch, output in uvt.

```
573 function uvt = heteroNLE(t,uv,patches,muL)
574     if nargin<4, muL=0; end% default end displacement is zero
575     global b M vis
```

Separate state vector into displacement and velocity fields: u_{ijI} is the displacement at the j th point in the i th 2-cell in the I th patch; similarly for velocity v_{ijI} . That is, physically neighbouring points have different j , whereas physical next-to-neighbours have i different by one.

```
586     u=uv(:,1:2,:,:); v=uv(:,3:4,:,:); % separate u and v=du/dt
```

Provide boundary conditions, here fixed displacement and velocity in the left/right sub-cells of the leftmost/rightmost patches.

```
595 u(1,:,: ,1)=0;
596 v(1,:,: ,1)=0;
597 u(end,:,: ,end)=-muL;
598 v(end,:,: ,end)=0;
```

Compute the two different strain fields, and also a first derivative for some optional viscosity.

```
606     eps2 = diff(u)/(2*b);
607     eps1 = [u(:,2,:,:)-u(:,1,:,:), u([2:end 1],1,:,:)-u(:,2,:,:)]/b;
608     eps1(end,2,:,:)=nan; % as this value is fake
609     vx1 = [v(:,2,:,:)-v(:,1,:,:), v([2:end 1],1,:,:)-v(:,2,:,:)]/b;
610     vx1(end,2,:,:)=nan; % as this value is fake
```

Set corresponding nonlinear stresses

```
616     sig2 = eps2-M(2)*eps2.^3+M(4)*eps2.^5;
617     sig1 = eps1-M(1)*eps1.^3+M(3)*eps1.^5;
```

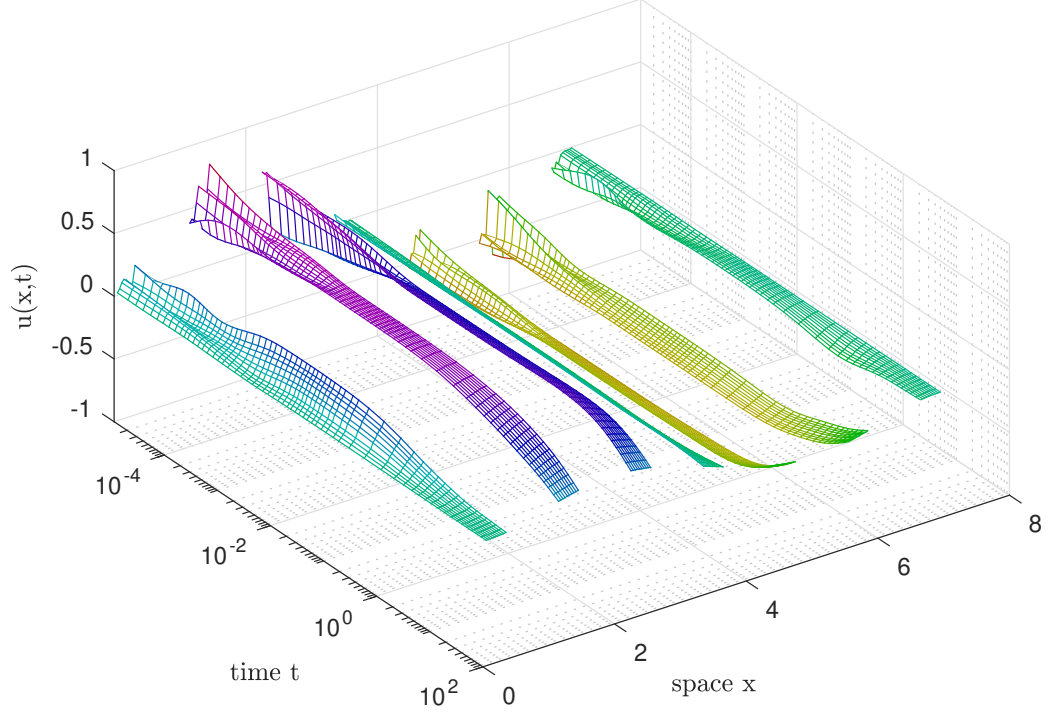
Preallocate output array, and fill in time derivatives of displacement and velocity, from velocity and gradient of stresses, respectively.

```

625     uvt = nan+uv;           % preallocate output array
626     i=2:size(uv,1)-1;
627     % rate of change of position
628     uvt(i,1:2,:,:)= v(i,:,:,:);
629     % rate of change of velocity +some artificial viscosity??
630     uvt(i,3:4,:,:)= diff(sig2) ...
631         +[ sig1(i,1,:,:)-sig1(i-1,2,:,:), diff(sig1(i,:,:,:),1,2)] ...
632         +vis*[ vx1(i,1,:,:)-vx1(i-1,2,:,:), diff(vx1(i,:,:,:),1,2) ];
633
634 end% function heteroNLE

```


Figure 3.20: hyper-diffusing field $u(x,t)$ in the patch scheme applied to microscale heterogeneous hyper-diffusion (Section 3.12). The log-time axis shows: $t < 10^{-2}$, rapid decay of sub-patch micro-structure; $10^{-2} < t < 1$, meso-time quasi-equilibrium; and $1 < t < 10^2$, slow decay of macroscale structures.



3.12 hyperDiffHetero: simulate a heterogeneous hyper-diffusion PDE in 1D on patches

Figure 3.20 shows an example simulation in time generated by the patch scheme applied to a heterogeneous version of the hyper-diffusion PDE. That such simulations makes valid predictions was established by Bunder et al. (2017) who proved that the scheme is accurate when the number of points in a patch is tied to a multiple of the periodicity of the pattern.

We aim to simulate the heterogeneous hyper-diffusion PDE

$$u_t = -D[c_1(x)Du] \quad \text{where operator } D := \partial_x(c_2(x)\partial_x), \quad (3.6)$$

for microscale periodic coefficients $c_l(x)$, and boundary conditions of $u = u_x = 0$ at $x = 0, L$. In this 1D space, the macroscale, homogenised, effective hyper-diffusion should be some unknown ‘average’ of these coefficients, but we use the patch scheme to provide a computational homogenisation. We discretise the PDE to a lattice of values $u_i(t)$, with lattice spacing dx , and governed by

$$\dot{u}_i = -D[c_{i1}Du_i] \quad \text{where operator } D := \delta(c_{i2}\delta)/dx^2$$

in terms of centred difference operator $\delta u_i := u_{i+1/2} - u_{i-1/2}$.

Set the desired microscale periodicity, and correspondingly choose random microscale diffusion coefficients (with some subscripts shifted by a half).

```

57 clear all
58 basename = mfilename
59 %global OurCf2eps, OurCf2eps=true %optional to save plots
60 nGap = 3 % controls size of gap between patches
61 nPtsPeriod = 5
62 dx = 0.5/nGap/nPtsPeriod

Create some random heterogeneous coefficients, log-uniform.

69 csVar = 1
70 cs = 0.2*exp( -csVar/2+csVar.*rand(nPtsPeriod,2) )

Establish global data struct patches for heterogeneous hyper-diffusion on a
finite domain with, on average, one patch per unit length. Use seven patches,
and use high-order interpolation with ordCC = 0.

80 nPatch = 7
81 nSubP = 2*nPtsPeriod+4 % or +2 for not-edgyInt
82 Len = nPatch;
83 ordCC = 0;
84 dom.type = 'equispace';
85 dom.bcOffset = 0.5 % for BC type
86 patches = configPatches1(@hyperDiffPDE,[0 Len],dom ...
87     ,nPatch,ordCC,dx,nSubP,'EdgyInt',true,'nEdge',2 ...
88     , 'hetCoeffs',cs);
89 xs=squeeze(patches.x);

```

Simulate in time Set an initial condition, and here integrate forward in time using a standard method for stiff systems—because of the simplicity of linear problems this method works quite efficiently here. Integrate the interface `patchSys1` ([Section 3.2](#)) to the microscale differential equations.

```

103 u0 = sin(2*pi/Len*patches.x).*rand(nSubP,1,1,nPatch);
104 tic
105 [ts,us] = ode15s(@patchSys1, [0 100], u0(:) ,[],patches);
106 simulateTime = toc
107 us = reshape(us,length(ts),numel(patches.x(:)),[]);

```

Plot the simulation in [Figure 3.20](#), using log-axis for time so we can see a little of both micro- and macro-dynamics.

```

116 figure(1),clf
117 xs([1:2 end-1:end],:) = nan;
118 t0=min(find(ts>1e-5));
119 mesh(ts(t0:3:end),xs(:),us(t0:3:end,:))', view(55,50)
120 colormap(0.7*hsv)
121 xlabel('time $t$'), ylabel('space $x$'), zlabel('$u(x,t)$')
122 ca=gca; ca.XScale='log'; ca.XLim=ts([t0 end]);
123 ifOurCf2eps([basename 'Uxt'])

```

Fin.

3.12.1 Heterogeneous hyper-diffusion PDE inside patches

As a microscale discretisation of hyper-diffusion PDE (3.6) $u_t = -D[c_1(x)Du]$, where heterogeneous operator $D = \partial_x(c_2(x)\partial_x)$.

```
138 function ut=hyperDiffPDE(t,u,patches)
139     dx=diff(patches.x(1:2)); % microscale spacing
```

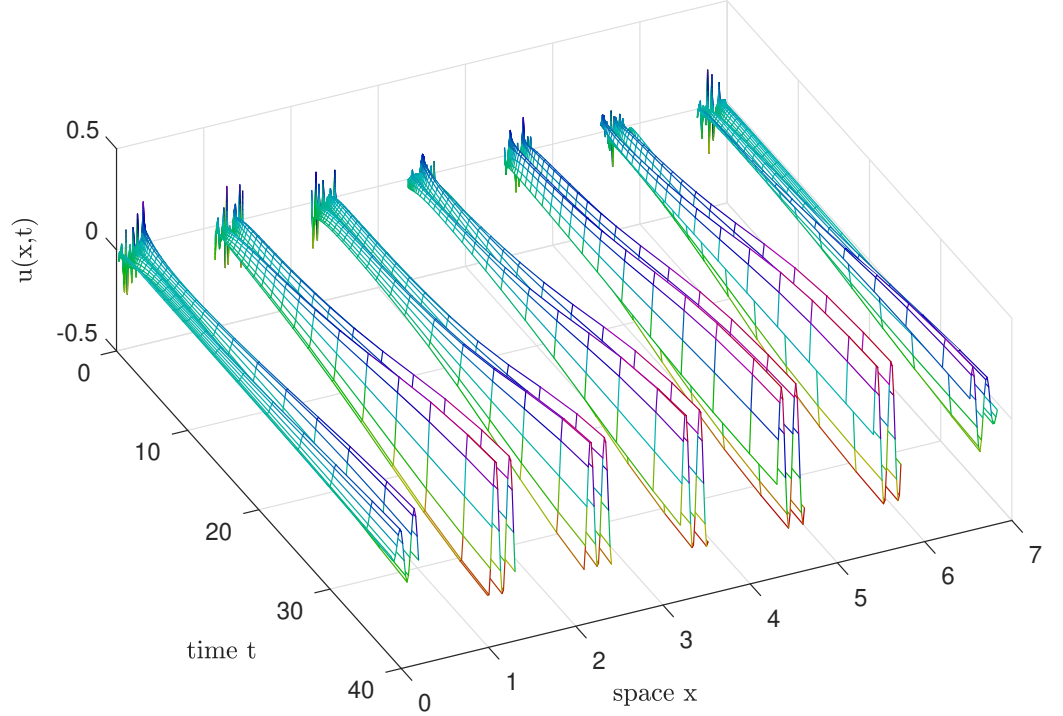
Code Dirichlet boundary conditions of zero function and derivative at left-end of left-patch, and right-end of right-patch. For slightly simpler coding, squeeze out the two singleton dimensions.

```
148     u = squeeze(u);
149     if ~patches.periodic % discretise BC u=u_x=0
150         u(1:2,1)=0;
151         u(end-1:end,end)=0;
152     end%if
```

Here code straightforward centred discretisation in space.

```
158     ut = nan+u; % preallocate output array
159     v = patches.cs(2:end,1).*diff(patches.cs(:,2).*diff(u))/dx^2;
160     ut(3:end-2,:) = -diff(patches.cs(2:end-1,2).*diff(v))/dx^2 ;
161 end
```

Figure 3.21: the pattern forming field $u(x, t)$ in the patch (gap-tooth) scheme applied to a microscale discretisation of the Swift–Hohenberg PDE (Section 3.13). Physically we see the rapid decay of much microstructure, but also the meso-time growth of sub-patch-scale patterns, wavenumber k_0 , that are modulated over the inter-patch distances and over long times.



3.13 SwiftHohenbergPattern: patterns of the Swift–Hohenberg PDE in 1D on patches

Figure 3.21 shows an example simulation in time generated by the patch scheme applied to the patterns arising from the Swift–Hohenberg PDE. That such simulations of patterns makes valid predictions was established by Bunder et al. (2017) who proved that the scheme is accurate when the number of points in a patch is just more than a multiple of the periodicity of the pattern.

Consider a lattice of values $u_i(t)$, with lattice spacing dx , and governed by a microscale centred discretisation of the Swift–Hohenberg PDE

$$\partial_t u = -(1 + \partial_x^2/k_0^2)^2 u + \text{Ra} u - u^3, \quad (3.7)$$

with boundary conditions of $u = u_x = 0$ at $x = 0, L$. For Ra just above critical, say $\text{Ra} = 0.1$, the system rapidly evolves to spatial quasi-periodic solutions with period ≈ 0.166 when wavenumber parameter $k_0 = 38$. On medium times these spatial oscillations grow to near equilibrium amplitude of $\sqrt{\text{Ra}}$, and over very long times the phases of the oscillations evolve in space to adapt to the boundaries.

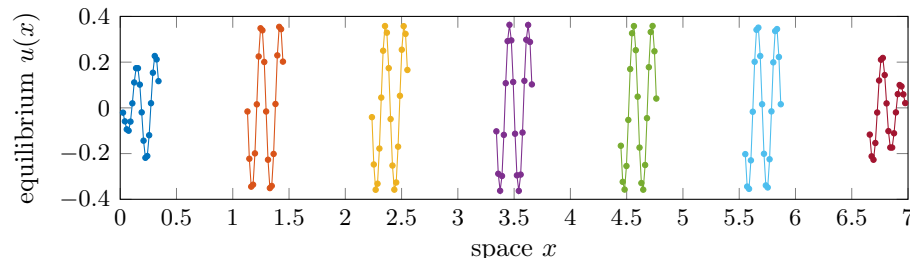
Set the desired microscale periodicity of the emergent pattern.

```

52 clear all, close all
53 %global OurCf2eps, OurCf2eps=true %optional to save plots

```

Figure 3.22: an equilibrium of the Swift–Hohenberg PDE on seven patches in 1D space. In the sub-patch patterns, there is a small phase shift in the patterns from patch to patch. And the amplitude of the pattern has to go to ‘zero’ at the boundaries.



```

54 Ra = 0.1 % Ra>0 leads to patterns
55 nGap = 3
56 %waveLength = 0.496688741721854 /nGap %for nPatch==5
57 waveLength = 0.497630331753555 /nGap %for nPatch==7
58 %waveLength = 0.5 /nGap %for periodic case
59 nPtsPeriod = 10
60 dx = waveLength/nPtsPeriod
61 k0 = 2*pi/waveLength

```

Establish global data struct `patches` for the Swift–Hohenberg PDE on some length domain. Use seven patches. Quartic (fourth-order) interpolation `ordCC = 4` provides values for the inter-patch coupling conditions.

```

72 nPatch = 7
73 nSubP = 2*nPtsPeriod+4
74 %nSubP = 2*nGap*nPtsPeriod+4 % full-domain
75 Len = nPatch;
76 ordCC = 4;
77 dom.type='equispace';
78 dom.bcOffset=0.5
79 patches = configPatches1(@SwiftHohenbergPDE,[0 Len],dom ...
80     ,nPatch,ordCC,dx,nSubP,'EdgyInt',true,'nEdge',2);
81 xs=squeeze(patches.x);

```

3.13.0.1 Find equilibrium with `fsolve`

Start the search from some guess.

```

103 fprintf('\n**** Find equilibrium with fsolve\n')
104 u = 0.4*sin(k0*patches.x);

```

But set the pairs of patch-edge values to `Nan` in order to use `patches.i` to index the interior sub-patch points as they are the variables.

```

112 u([1:2 end-1:end],:) = nan;
113 patches.i = find(~isnan(u));

```

Seek the equilibrium, and report the norm of the residual, via the generic patch system wrapper `theRes` (Section 3.15).

```

121 tic
122 [u(patches.i),res] = fsolve(@(v) theRes(v,patches,k0,Ra) ...
123     ,u(patches.i) ,optimoptions('fsolve','Display','off'));
124 solveTime = toc
125 normRes = norm(res)
126 assert(normRes<1e-6,'**** fsolve solution not accurate')

```

Plot the equilibrium see [Figure 3.22](#).

```

134 figure(1),clf
135 subplot(2,1,1)
136 plot(xs,squeeze(u),'-')
137 xlabel('space $x$'),ylabel('equilibrium $u(x)$')
138 ifOurCf2tex([mfilename 'Equilib'])%optionally save

```

3.13.0.2 Simulate in time

Set an initial condition, and here integrate forward in time using a standard method for stiff systems—because of the simplicity of linear problems this method works quite efficiently here. Integrate the interface `patchSys1` ([Section 3.2](#)) to the microscale differential equations.

```

155 fprintf('\n**** Simulate in time\n')
156 u0 = 0*patches.x+0.1*randn(nSubP,1,1,nPatch);
157 tic
158 [ts,us] = ode15s(@patchSys1, [0 40], u0(:) , [],patches,k0,Ra);
159 simulateTime = toc
160 us = reshape(us,length(ts),numel(patches.x(:)),[]);

```

Plot the simulation in [Figure 3.21](#).

```

167 figure(2),clf
168 xs([1:2 end-1:end],:) = nan;
169 mesh(ts(1:3:end),xs(:),us(1:3:end,:)) , view(65,60)
170 colormap(0.7*hsv)
171 xlabel('time $t$'), ylabel('space $x$'), zlabel('$u(x,t)$')
172 ifOurCf2eps([mfilename 'Uxt'])

```

Fin.

3.13.1 The Swift–Hohenberg PDE and BCs inside patches

As a microscale discretisation of Swift–Hohenberg PDE $u_t = -(1 + \partial_x^2/k_0^2)u + \text{Ra}u - u^3$, here code straightforward centred discretisation in space.

```

186 function ut=SwiftHohenbergPDE(t,u,patches,k0,Ra)
187     dx=diff(patches.x(1:2)); % microscale spacing
188     i=3:size(u,1)-2; % interior points in patches

```

Code Dirichlet boundary conditions of zero function and derivative, $u = u_x = 0$, at the left-end of the leftmost-patch, and the right-end of the rightmost-patch. For slightly simpler coding, squeeze out the two singleton dimensions.

```

198     u = squeeze(u);
199     u(1:2,1)=0;
200     u(end-1:end,end)=0;

    Here code straightforward centred discretisation in space.

206     ut=nan+u;           % preallocate output array
207     v = u(2:end-1,:)+diff(u,2)/dx^2/k0^2;
208     ut(i,:) = -( v(2:end-1,:)+diff(v,2)/dx^2/k0^2 ) ...
209         +Ra*u(i,:) -u(i,:).^3;
210 end

```

3.13.2 theRes(): wrapper function to zero for equilibria

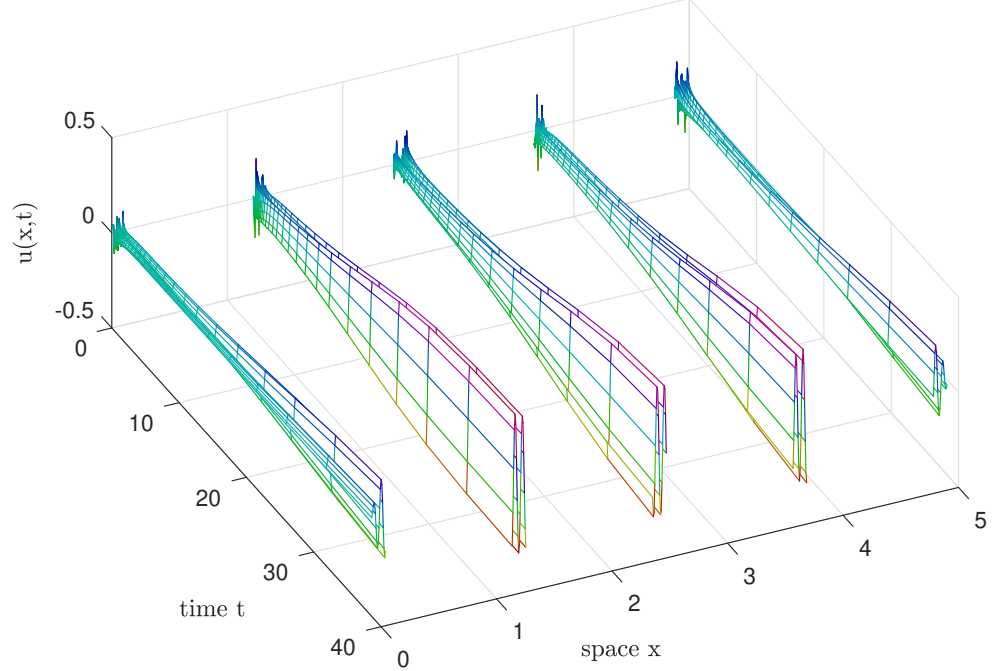
This functions converts a vector of values into the interior values of the patches, then evaluates the time derivative of the system at time zero, and returns the vector of patch-interior time derivatives.

```

225 function f=theRes(u,patches,k0,Ra)
226     v=nan(size(patches.x));
227     v(patches.i) = u;
228     f = patchSys1(0,v(:),patches,k0,Ra);
229     f = f(patches.i);
230 end%function theRes

```

Figure 3.23: the field $u(x,t)$ in the patch (gap-tooth) scheme applied to microscale heterogeneous Swift–Hohenberg PDE (Section 3.14). The heterogeneous coefficients are approximately uniform over $[0.9, 1.1]$. This heterogeneity has no noticeable affect on the simulation.



3.14 SwiftHohenbergHetero: patterns of a heterogeneous Swift–Hohenberg PDE in 1D on patches

Figure 3.23 shows an example simulation in time generated by the patch scheme applied to the patterns arising from a heterogeneous version of the Swift–Hohenberg PDE. That such simulations of patterns makes valid predictions was established by Bunder et al. (2017) who proved that the scheme is accurate when the number of points in a patch is tied to a multiple of the periodicity of the pattern.

Consider a lattice of values $u_i(t)$, with lattice spacing dx , arising from a microscale discretisation of the pattern forming, heterogeneous, Swift–Hohenberg PDE

$$\partial_t u = -D[c_1(x)Du] + \text{Ra} u - u^3, \quad D := 1 + \partial_x[c_2(x)\partial_x \cdot]/k_0^2, \quad (3.8)$$

where $c_\ell(x)$ have period $2\pi/k_0$. Coefficients c_ℓ are chosen iid random, nearly uniform, with mean near one. With mean one, the periodicity of c_ℓ approximately matches the periodicity of the resultant spatial pattern.

The current patch scheme coding preserves symmetry in the case of periodic patches (for every order of interpolation). For equispace and chebyshev options, the coupling currently fails symmetry.

Consider the spectrum in the symmetric cases of periodic patches (based upon only the cases $N = 5, 7$). There are $2N$ small eigenvalues, separated by a gap from the rest. In the homogeneous case, these occur as N pairs. With

small heterogeneity, they appear to split into $N - 1$ pairs, and two distinct. With stronger heterogeneity (say 0.5), they *often* appear to also split into two clusters, each of N eigenvalues, with one small-valued cluster, and one meso-valued cluster—curious. Further analysis with sparse approximation of the invariant spaces suggests the following:

- for homogeneous, the $2N$ modes are local oscillations in each patch, with two modes each corresponding to phase shifts of the possible oscillations;
- for heterogeneous
 - N eigenmodes appear to be one phase ‘locking’ to the heterogeneity; and
 - N eigenmodes appear to be other phase ‘locking’ to the heterogeneity. Unless it is something to do with the coupling, but then it only appears with heterogeneity.

Consider the spectrum with BCs of $u = u_{xx} = 0$ at ends. Non-symmetric so some eigenvalues are complex! For small or zero heterogeneity find $2N - 2$ eigenvalues are small. Effectively, two modes in each of $N - 2$ interior patches, and one mode each in the two end patches. With increasing heterogeneity (say above 0.3), the gap decreases as a couple (or some) of the small eigenvalues become larger in magnitude.

Consider the spectrum with BCs of $u = u_x = 0$ at ends. Non-symmetric so some eigenvalues are complex! For small or zero heterogeneity find $2N - 4$ eigenvalues are small. Effectively, two modes in each of $N - 2$ interior patches. With increasing heterogeneity (say above 0.4), half $(N - 2)$ of the small eigenvalues become larger in magnitude (presumably some phase ‘locking’ to the heterogeneity): effectively forms two clusters of modes.

Set the desired microscale periodicity of the patterns, here 0.062, and on the microscale lattice of spacing 0.0062, correspondingly choose random microscale material coefficients. The wavenumber of this microscale patterns is $k_0 \approx 101$.

```

103 clear all
104 %global OurCf2eps, OurCf2eps=true %optional to save plots
105 basename = ['r' num2str(floor(1e5*rem(now,1))) mfilename]
106 Ra = 0.1 % Ra>0 leads to patterns
107 nGap = 8 % controls size of gap between patches
108 waveLength = 0.496688741721854 /nGap %for nPatch==5
109 %waveLength = 0.497630331753555 /nGap %for nPatch==7
110 %waveLength = 0.5 /nGap %for periodic case
111 nPtsPeriod = 10
112 dx = waveLength/nPtsPeriod
113 k0 = 2*pi/waveLength

```

Create some random heterogeneous coefficients.

```

120 heteroVar = 0.99*[1 1] % must be <2
121 cl = 1./(1-heteroVar/2+heteroVar.*rand(nPtsPeriod,2));

```

```

122  cRange = quantile(c1,0:0.5:1)

Establish global data struct patches for heterogeneous Swift–Hohenberg PDE
with, on average, one patch per units length. Use seven patches to start
with. Quartic (fourth-order) interpolation ordCC = 4 provides values for
the inter-patch coupling conditions. Or use as high-order as possible with
ordCC = 0.

134  nPatch = 5
135  nSubP = 2*nPtsPeriod+4 % +2 for not-edgyInt
136  %nSubP = 2*nGap*nPtsPeriod+4 % approx full-domain
137  Len = nPatch;
138  ordCC = 0;
139  if 1, dom.type='periodic'
140  else % equispace currently ruins symmetry
141      dom.type='periodic';
142      dom.bcOffset=0.5
143  end
144  patches = configPatches1(@heteroSwiftHohenbergPDE,[0 Len],dom ...
145      ,nPatch,ordCC,dx,nSubP,'EdgyInt',true,'nEdge',2 ...
146      , 'hetCoeffs',c1);
147  xs=squeeze(patches.x);

```

3.14.0.1 Explore the Jacobian

Finds that with periodic patches, everything is symmetric. However, for equispace or chebyshev, the patch coupling is not symmetric—is this to be expected?

```

159  fprintf('\n**** Explore the Jacobian\n')
160  u0 = 0*patches.x;
161  u0([1:2 end-1:end],:) = nan;
162  patches.i = find(~isnan(u0));
163  nVars = numel(patches.i)
164  if exist('AutoDiff','class')
165      disp('**** computing Jacobian with AutoDiff')
166      Jac=AutoDiffJacobianAutoDiff( ...
167          @(u) SHres(u,patches,k0,0,0) ,u0(patches.i));
168      Jac=full(Jac);
169  else% compute Jac by finite-diff
170      Jac = nan(nVars);
171      for j=1:nVars
172          Jac(:,j)=SHres(double((1:nVars)==j),patches,k0,0,0);
173      end
174  end%if exist

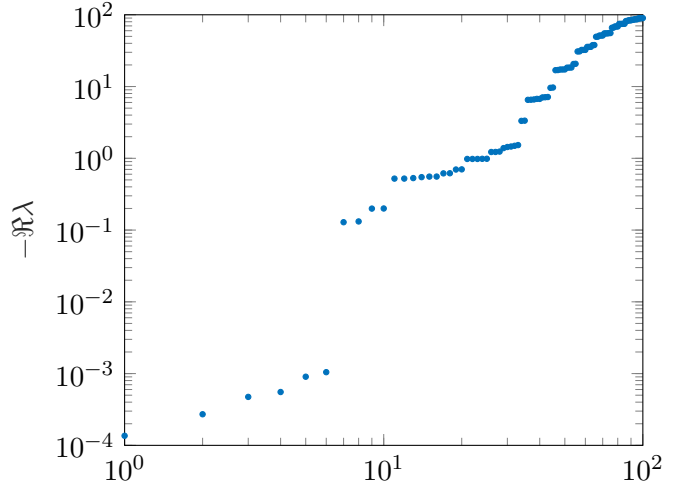
Check on the symmetry of the Jacobian

180  nonSymmetric = norm(Jac-Jac')
181  Jac(abs(Jac)<1e-12)=0;
182  antiJac = Jac-Jac';

```

Figure 3.24:

eigenvalues of the patch scheme on the heterogeneous Swift–Hohenberg PDE (linearised). With $N = 5$ patches and BCs of $u = u_x = 0$ at $x \in \{0, 5\}$, there are $2(N - 2) = 6$ small eigenvalues, $|\lambda| < 0.001$, corresponding to six slow modes in the interior.



```

183 antiJac(abs(antiJac)<1e-12)=0;
184 figure(6),clf
185 spy(Jac,','),hold on, spy(antiJac,'rx'),hold off
186 if nonSymmetric>5e-9, warning('failed symmetry'),
187 else Jac = (Jac+Jac')/2; %tweak to symmetry
188 end

```

Compute eigenvalues and eigenvectors.

```

194 figure(5),clf
195 [evec,mEval] = eig(-Jac , 'vector');
196 [~,j]=sort(real(mEval));
197 mEval=mEval(j); evec=evec(:,j);
198 loglog(real(mEval),',')
199 ylabel('$-\text{Re}\lambda$')
200 if OurCf2tex([basename 'Eval'])%optionally save

```

Explore sparse approximations of all the slowest together (lots of iterations required), or separately of the two clusters of the slowest (few iterations needed). First ascertain whether one or two clusters of small eigenvalues.

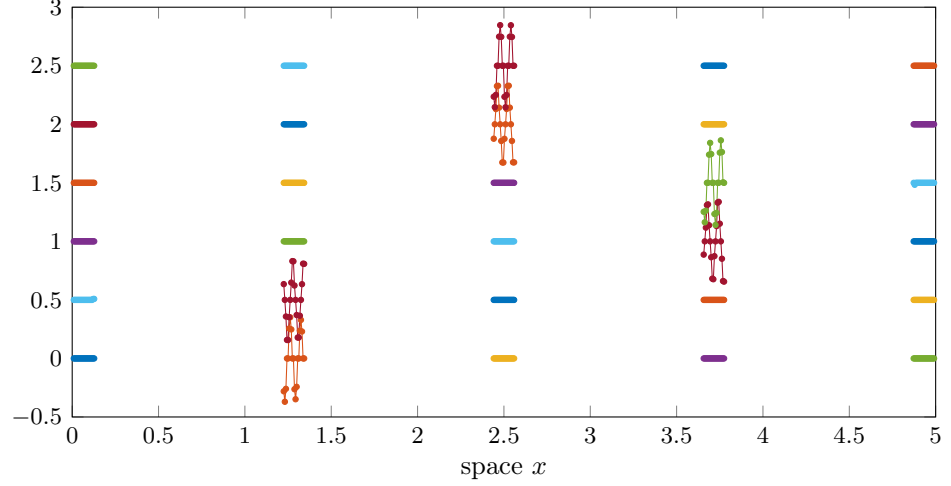
```

221 logGaps=diff(log10(real(mEval)));
222 [~,j]=sort(-logGaps);
223 %someLogGaps=[logGaps(j(1:5)) j(1:5)]
224 if logGaps(j(2))<0.4*logGaps(j(1)), nSlow=j(1)
225 else nSlow=min( sort(j(1:2)) , 3*nPatch)
226 end
227 log10Gap=logGaps(nSlow)
228 smallEvals=-mEval(1:nSlow(end)+2)

```

Second, make eigenvectors all real, sparsely approximate cluster modes via an algorithm developed from [Hu et al. \(2016\)](#), and plot. [Figure 3.25](#) shows that each pair of basis vectors are phase-shifted by 90° .

Figure 3.25: sparse approximations of the eigenvectors of the six slow modes of Figure 3.24. Plotted are sparse basis vectors for the invariant space spanned by the six slow eigenvectors: each basis vector shifted vertically to separate. Thus a fair approximation is that there are effectively two modes for each of the $N - 2 = 3$ interior patches.



```

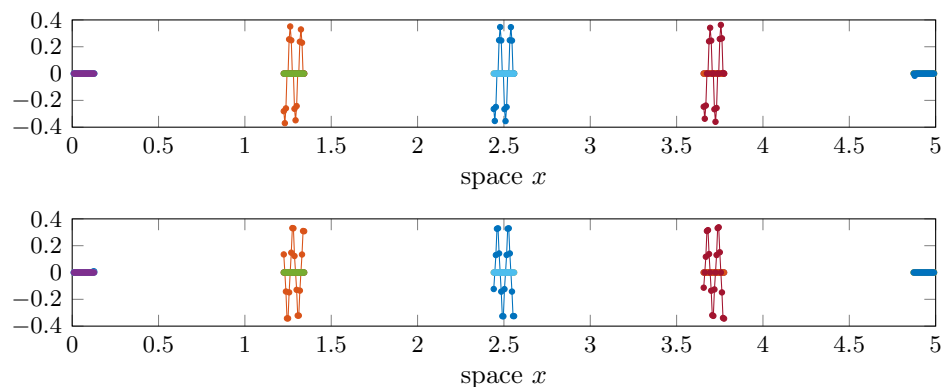
238 js=find(imag(mEval)>0);
239 evec(:,js)=imag(evec(:,js));
240 evec=real(evec);
241 if numel(nSlow)==1, S = spcart(evec(:,1:nSlow));
242 else S = spcart(evec(:,1:nSlow(1)));
243     S = [S spcart(evec(:,nSlow(1)+1:nSlow(2)))] ;
244 end;
245 figure(3),clf
246 vStep=ceil(max(abs(S(:)))*10+1)/10
247 for j=1:nSlow(end)
248     u0(patches.i)=S(:,j);
249     plot(xs,vStep*(j-1)+squeeze(u0),'.-'),hold on
250 end
251 hold off, xlabel('space $x$')
252 ifOurCf2tex([basename 'Evec'])%optionally save

Reorganise the eigenvectors to maybe clarify.

273 [i,j]=find(abs(S)>vStep/2);
274 j=find([1;diff(j)]);
275 [i,k]=sort(i(j));
276 figure(4)
277 for p=1:2
278     clf,subplot(2,1,1)
279     for j=p:2:numel(k)
280         u0(patches.i)=S(:,k(j));
281         plot(xs,squeeze(u0),'.-'),hold on
282     end% for j
283     hold off, xlabel('space $x$')

```

Figure 3.26: sparse basis approximations for the invariant subspace of the six slow modes of Figure 3.24. A replot of Figure 3.25 but with three of the basis vectors superimposed in each of the two panels.



```

284     ifOurCf2tex([basename 'Evec' num2str(p)])%optionally save
285 end%for p

```

3.14.0.2 Find an equilibrium with fsolve

Start the search from some guess.

```

308 fprintf('\n**** Find equilibrium with fsolve\n')
309 u = 0.4*sin(2*pi/waveLength*patches.x);

```

But set the pairs of patch-edge values to Nan in order to use `patches.i` to index the interior sub-patch points as they are the variables.

```

317 u([1:2 end-1:end],:) = nan;
318 patches.i = find(~isnan(u));

```

Seek the equilibrium, and report the norm of the residual, via the generic patch system wrapper `SHres` (Section 3.14.2).

```

326 tic
327 [u(patches.i),res] = fsolve(@(v) SHres(v,patches,k0,Ra,1) ...
328     ,u(patches.i) ,optimoptions('fsolve','Display','off'));
329 solveTime = toc
330 normRes = norm(res)
331 if normRes>1e-7, warning('residual large: bad equilibrium'),end

```

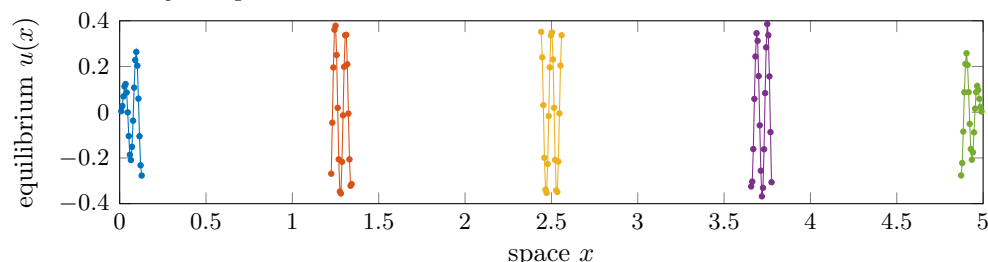
Plot the equilibrium see Figure 3.27.

```

339 figure(1),clf
340 subplot(2,1,1)
341 plot(xs,squeeze(u),'-')
342 xlabel('space $x$'),ylabel('equilibrium $u(x)$')
343 ifOurCf2tex([basename 'Equilib'])%optionally save

```

Figure 3.27: an equilibrium of the heterogeneous Swift–Hohenberg PDE determined by the patch scheme



3.14.0.3 Simulate in time

Set an initial condition, and here integrate forward in time using a standard method for stiff systems—because of the simplicity of linear problems this method works quite efficiently here. Integrate the interface `patchSys1` (Section 3.2) to the microscale differential equations.

```

368 fprintf('\n**** Simulate in time\n')
369 u0 = 0*sin(2*pi/waveLength*patches.x)+0.1*randn(nSubP,1,1,nPatch);
370 tic
371 [ts,us] = ode15s(@patchSys1, [0 40], u0(:) ,[],patches,k0,Ra,1);
372 simulateTime = toc
373 us = reshape(us,length(ts),numel(patches.x(:)),[]);

```

Plot the simulation in Figure 3.23.

```

380 figure(2),clf
381 xs([1:2 end-1:end],:) = nan;
382 mesh(ts(1:3:end),xs(:),us(1:3:end,:))', view(65,60)
383 colormap(0.7*hsv)
384 xlabel('time $t$'), ylabel('space $x$'), zlabel('$u(x,t)$')
385 ifOurCf2eps([basename 'Uxt'])

```

Fin.

3.14.1 Heterogeneous SwiftHohenberg PDE+BCs inside patches

As a microscale discretisation of Swift–Hohenberg PDE $u_t = -D[c_1(x)Du] + Ra u - u^3$, where heterogeneous operator $D = 1 + \partial_x(c_2(x)\partial_x)/k_0^2$.

```

399 function ut=heteroSwiftHohenbergPDE(t,u,patches,k0,Ra,cubic)
400     dx=diff(patches.x(1:2)); % microscale spacing
401     i=3:size(u,1)-2; % interior points in patches

```

Code a couple of different boundary conditions of zero function and derivative(s) at left-end of left-patch, and right-end of right-patch. For slightly simpler coding, squeeze out the two singleton dimensions.

```

410     u = squeeze(u);
411     if ~patches.periodic
412         switch 1

```

```

413     case 1 % these are u=u_x=0
414         u(1:2,1)=0;
415         u(end-1:end,end)=0;
416     case 2 % these are u=u_{xx}=0
417         u(1:2,1) = [-u(3,1); 0];
418         u(end-1:end,end) = [0; -u(end-2,end)];
419     end% case
420 end%if

```

Here code straightforward centred discretisation in space.

```

426     ut = nan(size(u),'like',u);          % preallocate output array
427     v = u(2:end-1,:)+diff(patches.cs(:    ,2).*diff(u))/dx^2/k0^2;
428     v = v.*patches.cs(2:end,1);
429     v = v(2:end-1,:)+diff(patches.cs(2:end-1,2).*diff(v))/dx^2/k0^2;
430     ut(i,:) = -v +Ra*u(i,:) -cubic*u(i,:).^3;
431 end

```

3.14.2 SHres(): a wrapper function

This functions converts a vector of values into the interior values of the patches, then evaluates the time derivative of the system at time zero, and returns the vector of patch-interior time derivatives.

```

446 function f=SHres(u,patches,k0,Ra,cubic)
447     v=nan(size(patches.x),'like',u);
448     v(patches.i) = u;
449     f = patchSys1(0,v(:),patches,k0,Ra,cubic);
450     f = f(patches.i);
451 end%function SHres

```

3.15 theRes(): wrapper function to zero for equilibria

This functions converts a vector of values into the interior values of the patches, then evaluates the time derivative of the system at time $t = 1$, and returns the vector of patch-interior time derivatives.

```

16 function f=theRes(u)
17     global patches
18     switch numel(size(patches.x))
19         case 4, pSys = @patchSys1;
20                 szv = size(patches.x);
21                 szii = 2:3;
22         case 5, pSys = @patchSys2;
23                 szv = size(patches.x+patches.y);
24                 szii = 3:4;
25         case 6, pSys = @patchSys3;
26                 szv = size(patches.x+patches.y+patches.z);
27                 szii = 4:5;
28         otherwise error('number of dimensions is somehow wrong')
29     end%switch
30     l = length(patches.nEdge);
31     szi = szv; szi(1:l) = szi(1:l)-2*patches.nEdge; % omit edges from count
32     nVars = prod(size(u))/prod(szi)/patches.nEnsem;
33     szv(szii) = [nVars patches.nEnsem]; % count nVars and ensem
34     assert(all(szv==round(szv)),['bad non-integer array size ' num2str(szv)])
35     v = nan(szv,'like',u);
36     v(patches.i) = u;
37     f = pSys(1,v(:),patches);
38     f = f(patches.i);
39 end%function theRes

```


3.16 `quasiLogAxes()`: transforms current axes of plot(s) to quasi-log

This function transforms the plots in the current axes. It rescales specified coordinates and labels the axes and its 2D or 3D plot(s). The original aim was to effectively show the complex spectrum of multiscale systems such as the patch scheme. The eigenvalues are over a wide range of magnitudes, but are signed. So we use a nonlinear `asinh` transformation of the axes, and then label the axes with reasonable ticks. This nonlinear rescaling is useful in other scenarios also.

```
22 function quasiLogAxes(xScale,yScale,zScale,cScale)
```

Input This function rescales the *current axes* (you may need to invoke `set()` to change to the axes you require). It rescales the existing plots (do not do further plots into the same axes).

- **xScale** (optional, default `inf`): if `inf`, then no transformation is done in the 'x'-coordinate. Otherwise, when **xScale** is not `inf`, transforms the plot *x*-coordinates with the `asinh()` function so that
 - for $|x| \lesssim x_{\text{scale}}$ the x-axis scaling is approximately linear, whereas
 - for $|x| \gtrsim x_{\text{scale}}$ the x-axis scaling is approximately signed-logarithmic.
- **yScale** (optional, default `inf`): corresponds to **xScale** for the second axis scaling.
- **zScale** (optional, default `inf`): corresponds to **xScale** for a third axis scaling if it exists.
- **cScale** (optional, default `inf`): corresponds to **xScale** but for a colormap, and colorbar scaling if one exists.

Output None, just the transformed plot.

Example If invoked with no arguments, then execute an example plot and its transformation.

```
61 if nargin==0
62     % generate some data
63     n=99; fast=(rand(n,1)<0.8);
64     z = -rand(n,1).*(1+1e3*fast)+1i*randn(n,1).*(5+1e2*fast);
65     % plot data and transform axes
66     plot(real(z),imag(z),'.'');
67     xlabel('real-part'), ylabel('imag-part')
68     title('un-transformed plot, pausing for 3 secs'), pause(3)
69     quasiLogAxes(1,10);
70     title('transformed plot')
71     return
72 end% example
```

Default values for scaling, `inf` denotes no transformation of that axis.

```
81  if nargin<4, cScale=inf; end
82  if nargin<3, zScale=inf; end
83  if nargin<2, yScale=inf; end
84  if nargin<1, xScale=inf; end
85  assert(class(xScale)=="double" ...
86         ,"May 2023 version of quasiLogAxes does not accept a handle")
```

4 Patches in 2D space

4.1 configPatches2(): configures spatial patches in 2D

Makes the struct `patches` for use by the patch/gap-tooth time derivative/step function `patchSys2()`. [Section 4.1.1](#) lists an example of its use.

```

19 function patches = configPatches2(fun,Xlim,Dom ...
20     ,nPatch,ordCC,dx,nSubP,varargin)
21 version = '2023-10-19';

```

Input If invoked with no input arguments, then executes an example of simulating a nonlinear diffusion PDE relevant to the lubrication flow of a thin layer of fluid—see [Section 4.1.1](#) for an example code.

- `fun` is the name of the user function, `fun(t,u,patches)` or `fun(t,u)` or `fun(t,u,patches,...)`, that computes time-derivatives (or time-steps) of quantities on the 2D micro-grid within all the 2D patches.
- `Xlim` array/vector giving the rectangular macro-space domain of the computation, namely $[Xlim(1), Xlim(2)] \times [Xlim(3), Xlim(4)]$. If `Xlim` has two elements, then the domain is the square domain of the same interval in both directions.

But, if `nPatch(n) = 1`, then in the spatial direction `n` the domain is the mean of `Xlim(2*n-1:2*n)` plus/minus half the patch-width in that direction.

- `Dom` sets the type of macroscale conditions for the patches, and reflects the type of microscale boundary conditions of the problem. If `Dom` is `NaN` or `[]`, then the field `u` is doubly macro-periodic in the 2D spatial domain, and resolved on equi-spaced patches. If `Dom` is a character string, then that specifies the `.type` of the following structure, with `.bcOffset` set to the default zero. Otherwise `Dom` is a structure with the following components.
 - `.type`, string, of either `'periodic'` (the default), `'equispace'`, `'chebyshev'`, `'usergiven'`. For all cases except `'periodic'`, users *must* code into `fun` the micro-grid boundary conditions that apply at the left/right/bottom/top edges of the leftmost/rightmost/bottommost/topmost patches, respectively.
 - `.bcOffset`, optional one, two or four element vector/array, in the cases of `'equispace'` or `'chebyshev'` the patches are placed so the left/right/top/bottom macroscale boundaries are aligned to the left/right/top/bottom edges of the corresponding extreme patches, but offset by `.bcOffset` of the sub-patch micro-grid spacing. For example, use `bcOffset=0` when the micro-code applies Dirichlet boundary values on the extreme edge micro-grid points, whereas use `bcOffset=0.5` when the microcode applies Neumann boundary conditions halfway between the extreme edge micro-grid points. Similarly for the top and bottom edges.

If `.bcOffset` is a scalar, then apply the same offset to all boundaries. If two elements, then apply the first offset to both x -

boundaries, and the second offset to both y -boundaries. If four elements, then apply the first two offsets to the respective x -boundaries, and the last two offsets to the respective y -boundaries.

- `.X`, optional vector/array with `nPatch(1)` elements, in the case 'usergiven' it specifies the x -locations of the centres of the patches—the user is responsible the locations makes sense.
- `.Y`, optional vector/array with `nPatch(2)` elements, in the case 'usergiven' it specifies the y -locations of the centres of the patches—the user is responsible the locations makes sense.
- `nPatch` determines the number of spatial patches: if scalar, then use the same number of patches in both directions, otherwise `nPatch(1:2)` gives the number of patches (an integer ≥ 1) in each direction.
- `ordCC` is the 'order' of interpolation for inter-patch coupling across empty space of the macroscale patch values to the edge-values of the patches: currently must be 0, 2, 4, ...; where 0 gives spectral interpolation.
- `dx` (real—scalar or two element) is usually the sub-patch micro-grid spacing in x and y . If scalar, then use the same `dx` in both directions, otherwise `dx(1:2)` gives the spacing in each of the two directions.

However, if `Dom` is `NaN` (as for pre-2023), then `dx` actually is `ratio` (scalar or two element), namely the ratio of (depending upon `EdgyInt`) either the half-width or full-width of a patch to the equi-spacing of the patch mid-points—adjusted a little when `nEdge > 1`. So either `ratio = 1/2` means the patches abut and `ratio = 1` is overlapping patches as in holistic discretisation, or `ratio = 1` means the patches abut. Small `ratio` should greatly reduce computational time.

- `nSubP` is the number of equi-spaced microscale lattice points in each patch: if scalar, then use the same number in both directions, otherwise `nSubP(1:2)` gives the number in each direction. If not using `EdgyInt`, then `nSubP./nEdge` must be odd integer(s) so that there is/are centre-patch lattice lines. So for the defaults of `nEdge = 1` and not `EdgyInt`, then `nSubP` must be odd.
- '`nEdge`', *optional* (integer—scalar or two element), default=1, the width of edge values set by interpolation at the edge regions of each patch. If two elements, then respectively the width in x, y -directions. The default is one (suitable for microscale lattices with only nearest neighbour interactions).
- `EdgyInt`, true/false, *optional*, default=false. If true, then interpolate to left/right/top/bottom edge-values from right/left/bottom/top next-to-edge values. If false or omitted, then interpolate from centre cross-patch lines.
- `nEnsem`, *optional-experimental*, default one, but if more, then an ensemble over this number of realisations.

- **hetCoeffs**, *optional*, default empty. Supply a 2D or 3D array of microscale heterogeneous coefficients to be used by the given microscale **fun** in each patch. Say the given array **cs** is of size $m_x \times m_y \times n_c$, where n_c is the number of different sets of coefficients. For example, in heterogeneous diffusion, $n_c = 2$ for the diffusivities in the *two* different spatial directions (or $n_c = 3$ for the diffusivity tensor). The coefficients are to be the same for each and every patch; however, macroscale variations are catered for by the n_c coefficients being n_c parameters in some macroscale formula.
 - If **nEnsem** = 1, then the array of coefficients is just tiled across the patch size to fill up each patch, starting from the (1,1)-point in each patch. Best accuracy usually obtained when the periodicity of the coefficients is a factor of **nSubP-2*nEdge** for **EdgyInt**, or a factor of **(nSubP-nEdge)/2** for not **EdgyInt**.
 - If **nEnsem** > 1 (value immaterial), then reset **nEnsem** := $m_x \cdot m_y$ and construct an ensemble of all $m_x \cdot m_y$ phase-shifts of the coefficients. In this scenario, the inter-patch coupling couples different members in the ensemble. When **EdgyInt** is true, and when the coefficients are diffusivities/elasticities in x and y directions, respectively, then this coupling cunningly preserves symmetry.
- **'parallel'**, true/false, *optional*, default=false. If false, then all patch computations are on the user's main CPU—although a user may well separately invoke, say, a GPU to accelerate sub-patch computations.

If true, and it requires that you have MATLAB's Parallel Computing Toolbox, then it will distribute the patches over multiple CPUs/cores. In MATLAB, only one array dimension can be split in the distribution, so it chooses the one space dimension x, y corresponding to the highest **nPatch** (if a tie, then chooses the rightmost of x, y). A user may correspondingly distribute arrays with property **patches.codist**, or simply use formulas invoking the preset distributed arrays **patches.x**, and **patches.y**. If a user has not yet established a parallel pool, then a 'local' pool is started.

Output The struct **patches** is created and set with the following components. If no output variable is provided for **patches**, then make the struct available as a global variable.¹

```

223 if nargout==0, global patches, end
224 patches.version = version;
```

- **.fun** is the name of the user's function **fun(t,u,patches)** or **fun(t,u)** or **fun(t,u,patches,...)**, that computes the time derivatives (or steps) on the patchy lattice.
- **.ordCC** is the specified order of inter-patch coupling.

¹When using **spmd** parallel computing, it is generally best to avoid global variables, and so instead prefer using an explicit output variable.

- `.periodic`: either true, for interpolation on the macro-periodic domain; or false, for general interpolation by divided differences over non-periodic domain or unevenly distributed patches.
- `.stag` is true for interpolation using only odd neighbouring patches as for staggered grids, and false for the usual case of all neighbour coupling—not yet implemented.
- `.Cwtsr` and `.Cwtsl`, only for macro-periodic conditions, are the `ordCC` × 2-array of weights for the inter-patch interpolation onto the right/top and left/bottom edges (respectively) with patch:macroscale ratio as specified or as derived from `dx`.
- `.x` (6D) is `nSubP(1) × 1 × 1 × 1 × nPatch(1) × 1` array of the regular spatial locations x_{iI} of the microscale grid points in every patch.
- `.y` (6D) is `1 × nSubP(2) × 1 × 1 × 1 × nPatch(2)` array of the regular spatial locations y_{jJ} of the microscale grid points in every patch.
- `.ratio` `1 × 2`, only for macro-periodic conditions, are the size ratios of every patch.
- `.nEdge` `1 × 2`, is the width of edge values set by interpolation at the edge regions of each patch, in the x, y -directions respectively.
- `.le`, `.ri`, `.bo`, `.to` determine inter-patch coupling of members in an ensemble. Each a column vector of length `nEnsem`.
- `.cs` either
 - [] 0D, or
 - if `nEnsem` = 1, `(nSubP(1) − 1) × (nSubP(2) − 1) × nc` 3D array of microscale heterogeneous coefficients, or
 - if `nEnsem` > 1, `(nSubP(1) − 1) × (nSubP(2) − 1) × nc × mxmy` 4D array of $m_x m_y$ ensemble of phase-shifts of the microscale heterogeneous coefficients.
- `.parallel`, logical: true if patches are distributed over multiple CPUs/cores for the Parallel Computing Toolbox, otherwise false (the default is to activate the *local* pool).
- `.codist`, *optional*, describes the particular parallel distribution of arrays over the active parallel pool.

4.1.1 If no arguments, then execute an example

```

308 if nargin==0
309 disp('With no arguments, simulate example of nonlinear diffusion')
```

The code here shows one way to get started: a user's script may have the following three steps (“ $\mapsto$ ” denotes function recursion).

1. `configPatches2`
2. `ode23` integrator $\mapsto$ `patchSys2` $\mapsto$ user's PDE

3. process results

Establish global patch data struct to interface with a function coding a nonlinear ‘diffusion’ PDE: to be solved on 6×4 -periodic domain, with 9×7 patches, spectral interpolation (0) couples the patches, with 5×5 points forming the micro-grid in each patch, and a sub-patch micro-grid spacing of 0.12 (relatively large for visualisation). [Roberts et al. \(2014\)](#) established that this scheme is consistent with the PDE (as the patch spacing decreases).

```

332 global patches
333 patches = configPatches2(@nonDiffPDE,[-3 3 -2 2] ...
334     , 'periodic', [9 7], 0, 0.12, 5 , 'EdgyInt', false);

```

Set an initial condition of a perturbed-Gaussian using auto-replication of the spatial grid.

```

341 u0 = exp(-patches.x.^2-patches.y.^2);
342 u0 = u0.*(0.9+0.1*rand(size(u0)));

```

Initiate a plot of the simulation using only the microscale values interior to the patches: optionally set x and y -edges to `nan` to leave the gaps between patches.

```

350 figure(1), clf, colormap(0.8*hsv)
351 x = squeeze(patches.x); y = squeeze(patches.y);
352 if 1, x([1 end],:) = nan; y([1 end],:) = nan; end

```

Start by showing the initial conditions of [Figure 4.1](#) while the simulation computes.

```

359 u = reshape(permute(squeeze(u0) ...
360     , [1 3 2 4]), [numel(x) numel(y)]);
361 hsurf = mesh(x(:),y(:),u');
362 axis([-3 3 -3 3 -0.03 1]), view(60,40)
363 legend('time = 0.00', 'Location', 'north')
364 xlabel('space $x$'), ylabel('space $y$'), zlabel('$u(x,y)$')
365 colormap(hsv)
366 ifOurCf2eps([mfilename 'ic'])

```

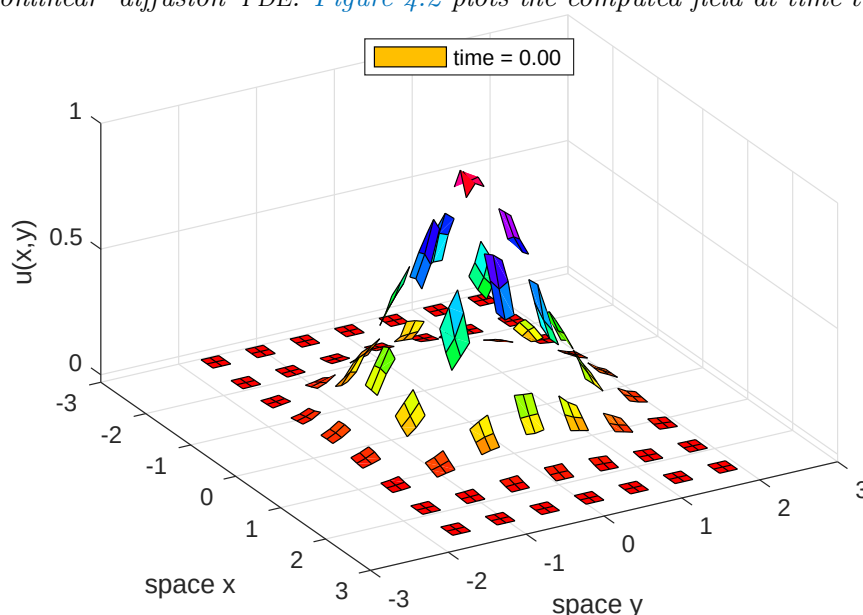
Integrate in time to $t = 4$ using standard functions. In MATLAB `ode15s` would be natural as the patch scheme is naturally stiff, but `ode23` is quicker ([Maclean et al. 2020](#), Fig. 4). Ask for output at non-uniform times because the diffusion slows.

```

383 disp('Wait to simulate nonlinear diffusion h_t=(h^3)_xx+(h^3)_yy')
384 drawnow
385 if ~exist('OCTAVE_VERSION','builtin')
386     [ts,us] = ode23(@patchSys2,linspace(0,2).^2,u0(:));
387 else % octave version is quite slow for me
388     lsode_options('absolute tolerance',1e-4);
389     lsode_options('relative tolerance',1e-4);
390     [ts,us] = odeOcts(@patchSys2,[0 1],u0(:));
391 end

```


Figure 4.1: initial field $u(x,y,t)$ at time $t = 0$ of the patch scheme applied to a nonlinear ‘diffusion’ PDE: Figure 4.2 plots the computed field at time $t = 3$.



Animate the computed simulation to end with Figure 4.2. Use `patchEdgeInt2` to interpolate patch-edge values.

```

399 for i = 1:length(ts)
400     u = patchEdgeInt2(us(i,:));
401     u = reshape(permute(squeeze(u) ...
402         ,[1 3 2 4]), [numel(x) numel(y)]);
403     set(hsurf,'ZData', u');
404     legend(['time = ' num2str(ts(i),'%.2f')])
405     pause(0.1)
406 end
407 ifOurCf2eps([mfilename 't3'])

```

Upon finishing execution of the example, exit this function.

```

422 return
423 end%if no arguments

```

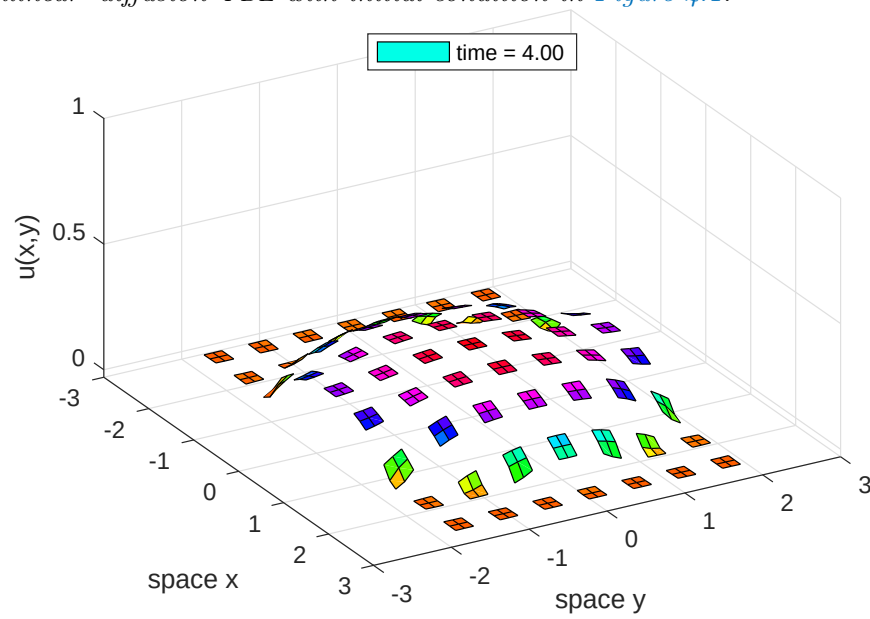
Example of nonlinear diffusion PDE inside patches As a microscale discretisation of $u_t = \nabla^2(u^3)$, code $\dot{u}_{ijkl} = \frac{1}{\delta x^2}(u_{i+1,j,k,l}^3 - 2u_{i,j,k,l}^3 + u_{i-1,j,k,l}^3) + \frac{1}{\delta y^2}(u_{i,j+1,k,l}^3 - 2u_{i,j,k,l}^3 + u_{i,j-1,k,l}^3)$.

```

13 function ut = nonDiffPDE(t,u,patches)
14     if nargin<3, global patches, end
15     u = squeeze(u); % reduce to 4D
16     dx = diff(patches.x(1:2)); % microgrid spacing
17     dy = diff(patches.y(1:2));
18     i = 2:size(u,1)-1; j = 2:size(u,2)-1; % interior patch points
19     ut = nan+u; % preallocate output array

```

Figure 4.2: field $u(x,y,t)$ at time $t = 3$ of the patch scheme applied to a nonlinear ‘diffusion’ PDE with initial condition in [Figure 4.1](#).



```

20     ut(i,j,,:,) = diff(u(:,j,,:,).^3,2,1)/dx^2 ...
21                   +diff(u(i,:,,:,).^3,2,2)/dy^2;
22 end

```

4.2 patchSys2(): interface 2D space to time integrators

To simulate in time with 2D spatial patches we often need to interface a users time derivative function with time integration routines such as `ode23` or `PIRK2`. This function provides an interface. Communicate patch-design variables ([Section 4.1](#)) either via the global struct `patches` or via an optional third argument. `patches` is required for the parallel computing of `spmd`, or if parameters are to be passed though to the user microscale function.

```

23 function dudt = patchSys2(t,u,patches,varargin)
24 if nargin<3, global patches, end

```

Input

- `u` is a vector/array of length $\text{prod}(\text{nSubP}) \cdot \text{nVars} \cdot \text{nEnsem} \cdot \text{prod}(\text{nPatch})$ where there are $\text{nVars} \cdot \text{nEnsem}$ field values at each of the points in the $\text{nSubP}(1) \times \text{nSubP}(2) \times \text{nPatch}(1) \times \text{nPatch}(2)$ grid.
- `t` is the current time to be passed to the user's time derivative function.
- `patches` a struct set by `configPatches2()` with the following information used here.
 - `.fun` is the name of the user's function `fun(t,u,patches,...)` that computes the time derivatives on the patchy lattice. The array `u` has size $\text{nSubP}(1) \times \text{nSubP}(2) \times \text{nVars} \times \text{nEnsem} \times \text{nPatch}(1) \times \text{nPatch}(2)$. Time derivatives must be computed into the same sized array, although herein the patch edge-values are overwritten by zeros.
 - `.x` is $\text{nSubP}(1) \times 1 \times 1 \times \text{lnPatch}(1) \times 1$ array of the spatial locations x_i of the microscale (i,j) -grid points in every patch. Currently it *must* be an equi-spaced lattice on both macro- and micro-scales.
 - `.y` is similarly $1 \times \text{nSubP}(2) \times 1 \times 1 \times 1 \times \text{nPatch}(2)$ array of the spatial locations y_j of the microscale (i,j) -grid points in every patch. Currently it *must* be an equi-spaced lattice on both macro- and micro-scales.
- `varargin`, optional, is arbitrary list of parameters to be passed onto the users time-derivative function as specified in `configPatches2`.

Output

- `dudt` is a vector/array of of time derivatives, but with patch edge-values set to zero. It is of total length $\text{prod}(\text{nSubP}) \cdot \text{nVars} \cdot \text{nEnsem} \cdot \text{prod}(\text{nPatch})$ and the same dimensions as `u`.

4.3 patchEdgeInt2(): sets 2D patch face values from 2D macroscale interpolation

Couples 2D patches across 2D space by computing their edge values via macroscale interpolation. Assumes patch edge values are determined by macroscale interpolation of the patch centre-plane values (Roberts et al. 2014, Bunder et al. 2021), or patch next-to-edge values which appears better (Bunder et al. 2020). This function is primarily used by patchSys3() but is also useful for user graphics.²

Communicate patch-design variables via a second argument (optional, except required for parallel computing of spmd), or otherwise via the global struct patches.

```

27 function u = patchEdgeInt2(u,patches)
28 if nargin<2, global patches, end

```

Input

- **u** is a vector/array of length $\text{prod}(\text{nSubP}) \cdot \text{nVars} \cdot \text{nEnsem} \cdot \text{prod}(\text{nPatch})$ where there are $\text{nVars} \cdot \text{nEnsem}$ field values at each of the points in the $\text{nSubP1} \cdot \text{nSubP2} \cdot \text{nPatch1} \cdot \text{nPatch2}$ multiscale spatial grid on the $\text{nPatch1} \cdot \text{nPatch2}$ array of patches.
- **patches** a struct set by configPatches2() which includes the following information.
 - **.x** is $\text{nSubP1} \times 1 \times 1 \times 1 \times \text{nPatch1} \times 1$ array of the spatial locations x_{iI} of the microscale grid points in every patch. Currently it *must* be an equi-spaced lattice on the microscale index i , but may be variable spaced in macroscale index I .
 - **.y** is similarly $1 \times \text{nSubP2} \times 1 \times 1 \times 1 \times \text{nPatch2}$ array of the spatial locations y_{jJ} of the microscale grid points in every patch. Currently it *must* be an equi-spaced lattice on the microscale index j , but may be variable spaced in macroscale index J .
 - **.ordCC** is order of interpolation, currently only $\{0, 2, 4, \dots\}$
 - **.periodic**, logical 2-array, indicates whether macroscale is periodic domain, or alternatively that the macroscale has left, right, top, bottom, front and back boundaries so interpolation is via divided differences.
 - **.stag** in $\{0, 1\}$ is one for staggered grid (alternating) interpolation. Currently must be zero.
 - **.Cwtsr** and **.Cwtsl** are the coupling coefficients for finite width interpolation in each of the x, y -directions—when invoking a periodic direction.
 - **.EdgyInt**, true/false, for determining patch-edge values by interpolation: true, from opposite-edge next-to-edge values (often

² Script patchEdgeInt2test.m verifies most of this code.

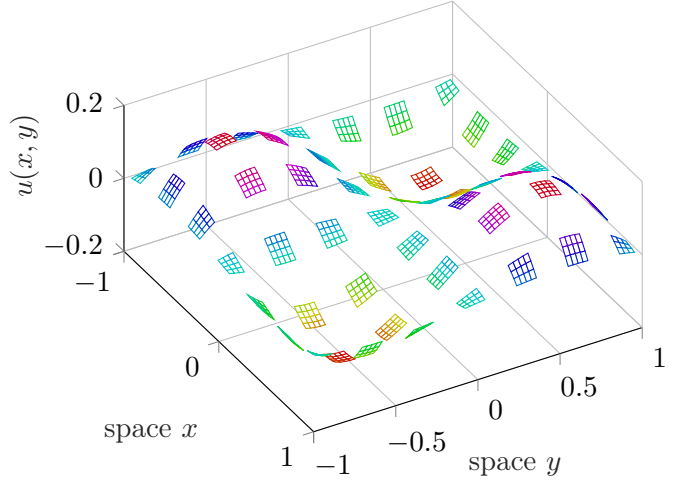
preserves symmetry); false, from centre cross-patch values (near original scheme).

- `.nEdge`, three elements, the width of edge values set by interpolation at the x, y -edge regions, respectively, of each patch (default is one all x, y -edges).
- `.nEnsem` the number of realisations in the ensemble.
- `.parallel` whether serial or parallel.

Output

- `u` is 6D array, `nSubP1 · nSubP2 · nVars · nEnsem · nPatch1 · nPatch2`, of the fields with edge values set by interpolation.

Figure 4.3: Equilibrium of the macroscale diffusion problem of Freese with Dirichlet zero-value boundary conditions (Section 4.4). The patch size is not small so we can see the patches.



4.4 monoscaleDiffEquil2: equilibrium of a 2D monoscale heterogeneous diffusion via small patches

Here we find the steady state $u(x, y)$, see Figure 4.3, to the heterogeneous PDE (inspired by Freese et al.³ §5.2)

$$u_t = A(x, y) \vec{\nabla} \vec{\nabla} u - f,$$

on domain $[-1, 1]^2$ with Dirichlet BCs, for coefficient pseudo-diffusion matrix

$$A := \begin{bmatrix} 2 & a \\ a & 2 \end{bmatrix} \quad \text{with } a := \text{sign}(xy) \text{ or } a := \sin(\pi x) \sin(\pi y),$$

and for forcing $f(x, y)$ such that the exact equilibrium is $u = x(1 - e^{1-|x|})y(1 - e^{1-|y|})$. But for simplicity, let's obtain $u = x(1 - x^2)y(1 - y^2)$ for which we code f later—as determined by this Reduce algebra code.

```
on gcd; factor sin;
u:=x*(1-x^2)*y*(1-y^2);
a:=sin(pi*x)*sin(pi*y);
f:=2*df(u,x,x)+2*a*df(u,x,y)+2*df(u,y,y);
```

Clear, and initiate globals.

```
58 clear all
59 global patches
60 %global OurCf2eps, OurCf2eps=true %option to save plot
```

Patch configuration Initially use 7×7 patches in the square $(-1, 1)^2$. For continuous forcing we may have small patches of any reasonable microgrid spacing—here the microgrid error dominates.

³ <http://arxiv.org/abs/2211.13731>

```

71 nPatch = 7
72 nSubP = 5
73 dx = 0.03

```

Specify some order of interpolation.

```

79 configPatches2(@monoscaleDiffForce2,[-1 1 -1 1], 'equispace' ...
80     ,nPatch ,4 ,dx ,nSubP , 'EdgyInt',true );

```

Compute the time-constant coefficient and time-constant forcing, and store them in struct `patches` for access by the microcode of [Section 4.4.1](#).

```

88 x=patches.x; y=patches.y;
89 patches.A = sin(pi*x).*sin(pi*y);
90 patches.fu = ...
91     +2*patches.A.*(9*x.^2.*y.^2-3*x.^2-3*y.^2+1) ...
92     +12*x.*y.*(x.^2+y.^2-2);

```

By construction, the PDE has analytic solution

```

98 uAnal = x.*(1-x.^2).*y.*(1-y.^2);

```

Solve for steady state Set initial guess of zero, with NaN to indicate patch-edge values. Index `i` are the indices of patch-interior points, and the number of unknowns is then its length.

```

111 u0 = zeros(nSubP,nSubP,1,1,nPatch,nPatch);
112 u0([1 end],:,:) = nan; u0(:,[1 end],:) = nan;
113 i = find(~isnan(u0));
114 nVariables = numel(i)

```

Could use `fsolve` for simplicity and robustness, via the generic patch wrapper `theRes()`. However, for such linear problems it is 100 times faster to use `AutoDiff` to get Jacobian, and then solve directly.

```

123 U0=AutoDiff(0*u0); % set an AD array of zeros
124 tic
125 F0 = patchSys2(1,U0); % evaluate patch system and its derivatives
126 Jac = getderivs(F0); % the Jacobian is the derivatives
127 uSoln = -Jac(i,i)\getvalue(F0(i)); % solve linear system
128 ADsolveTime = toc

```

Store the solution vector into the patches, and interpolate, but have not bothered to set boundary values so they stay NaN from the interpolation.

```

136 normSoln = norm(uSoln)
137 normError = norm(uSoln-uAnal(i))
138 u0(i) = uSoln;
139 f0 = patchSys2(1,u0);
140 normResidual = norm(f0(i),Inf)
141 assert(normResidual<1e-10)
142 u0 = patchEdgeInt2(u0);

```

Draw solution profile Separate patches with NaNs, then reshape arrays to suit 2D space surface plots.

```

153 figure(1), clf, colormap(0.8*hsv)
154 x(end+1,:,:) = nan; u0(end+1,:,:) = nan;
155 y(:,end+1,:) = nan; u0(:,end+1,:) = nan;
156 u = reshape(permute(squeeze(u0),[1 3 2 4]), [numel(x) numel(y)]);

    Draw the patch solution surface, with boundary-values omitted as already NaN
    by not bothering to set them.

163 mesh(x(:),y(:),u'); view(60,55)
164 xlabel('space $x$'), ylabel('space $y$'), zlabel('$u(x,y)$')
165 ifOurCf2tex(mfilename)%optionally save

```

4.4.1 monoscaleDiffForce2(): microscale discretisation inside patches of forced diffusion PDE

This function codes the lattice heterogeneous diffusion of the PDE inside the patches. For 6D input arrays u , x , and y , computes the time derivative at each point in the interior of a patch, output in ut .

```

182 function ut = monoscaleDiffForce2(t,u,patches)
183     dx = diff(patches.x(2:3)); % x space step
184     dy = diff(patches.y(2:3)); % y space step
185     i = 2:size(u,1)-1; % x interior points in a patch
186     j = 2:size(u,2)-1; % y interior points in a patch
187     ut = nan(size(u),'like',u); % preallocate output array

```

Set Dirichlet boundary value of zero around the square domain.

```

194     u( 1 ,:,:,: , 1 ,:)=0; % left edge of left patches
195     u(end,:,: ,end,:)=0; % right edge of right patches
196     u(:, 1 ,:,:,: , 1 )=0; % bottom edge of bottom patches
197     u(:,end,:,: ,end)=0; % top edge of top patches

```

Or code some function variation around the boundary, such as a function of y on the left boundary, and a (constant) function of x at the top boundary.

```

205     if 0
206         u(1,:,: ,1,:)=(1+patches.y)/2; % left edge of left patches
207         u(:,end,:,: ,end)=1; % top edge of top patches
208     end%if

```

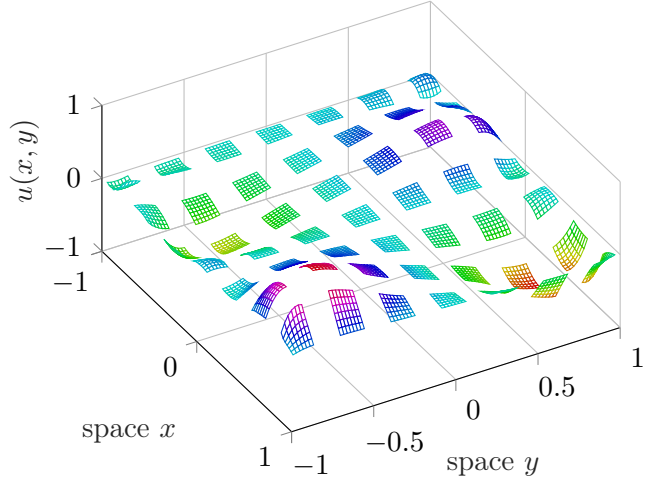
Compute the time derivatives via stored forcing and coefficients. Easier to code by conflating the last four dimensions into the one $,:$.

```

216     ut(i,j,:) ...
217     = 2*diff(u(:,j,:),2,1)/dx^2 + 2*diff(u(i,:,:),2,2)/dy^2 ...
218     + 2*patches.A(i,j,:).*( u(i+1,j+1,:) -u(i-1,j+1,:) ...
219     -u(i+1,j-1,:) +u(i-1,j-1,:) )/(4*dx*dy) ...
220     -patches.fu(i,j,:);
221 end%function monoscaleDiffForce2

```


Figure 4.4: Equilibrium of the multiscale diffusion problem of Freese with Dirichlet zero-value boundary conditions (Section 4.5). The patch size is not small so we can see the patches and the sub-patch grid. The solution $u(x, y)$ is boringly smooth.



4.5 twoscaleDiffEquil2: equilibrium of a 2D twoscale heterogeneous diffusion via small patches

Here we find the steady state $u(x, y)$ to the heterogeneous PDE (inspired by Freese et al.⁴ §5.3.1)

$$u_t = A(x, y) \vec{\nabla} \vec{\nabla} u - f,$$

on domain $[-1, 1]^2$ with Dirichlet BCs, for coefficient ‘diffusion’ matrix, varying with period 2ϵ on the microscale $\epsilon = 2^{-7}$, of

$$A := \begin{bmatrix} 2 & a \\ a & 2 \end{bmatrix} \quad \text{with } a := \sin(\pi x/\epsilon) \sin(\pi y/\epsilon),$$

and for forcing $f := (x + \cos 3\pi x)y^3$.

Clear, and initiate globals.

```

41 clear all
42 global patches
43 %global OurCf2eps, OurCf2eps=true %option to save plot

```

First establish the microscale heterogeneity has micro-period `mPeriod` on the spatial lattice. Set the phase of the heterogeneity so that each patch centre is a point of symmetry of the diffusivity. Then `configPatches2` replicates the heterogeneity to fill each patch.

```

56 mPeriod = 6
57 z = (0.5:mPeriod)'/mPeriod;
58 A = sin(2*pi*z).*sin(2*pi*z');

```

Set the periodicity, via ϵ , and other microscale parameters.

```

65 nPeriodsPatch = 1 % any integer
66 epsilon = 2^(-6) % 4 or 5 to see the patches
67 dx = (2*epsilon)/mPeriod
68 nSubP = nPeriodsPatch*mPeriod+2 % for edgy int

```

⁴ <http://arxiv.org/abs/2211.13731>

Patch configuration Say use 7×7 patches in $(-1,1)^2$, fourth order interpolation, and either ‘equispace’ or ‘chebyshev’:

```

79 nPatch = 7
80 configPatches2(@twoscaleDiffForce2,[-1 1], 'equispace' ...
81     ,nPatch ,4 ,dx ,nSubP , 'EdgyInt',true , 'hetCoeffs',A );

```

Compute the time-constant forcing, and store in struct `patches` for access by the microcode of [Section 4.6.1](#).

```

89 x = patches.x; y = patches.y;
90 patches.fu = 100*(x+cos(3*pi*x)).*y.^3;

```

Solve for steady state Set initial guess of zero, with NaN to indicate patch-edge values. Index `i` are the indices of patch-interior points, and the number of unknowns is then its length.

```

104 u0 = zeros(nSubP,nSubP,1,1,nPatch,nPatch);
105 u0([1 end],:,:) = nan; u0(:,[1 end],:) = nan;
106 i = find(~isnan(u0));
107 nVariables = numel(i)

```

Could use `fsolve` for simplicity and robustness, via the generic patch system wrapper `theRes` ([Section 3.15](#)). However, for such linear problems it is 100 times faster to use `AutoDiff` to get Jacobian, and then solve directly.

```

117 U0=AutoDiff(0*u0); % set an AD array of zeros
118 tic
119 F0 = patchSys2(1,U0); % evaluate patch system and its derivatives
120 Jac = getderivs(F0); % the Jacobian is the derivatives
121 uSoln = -Jac(i,i)\getvalue(F0(i)); % solve linear system
122 ADsolveTime = toc

```

Store the solution vector into the `patches`, and interpolate, but have not bothered to set boundary values so they stay NaN from the interpolation.

```

130 normSoln = norm(uSoln)
131 u0(i) = uSoln;
132 f0 = patchSys2(1,u0);
133 normResidual = norm(f0(i),Inf)
134 assert(normResidual<1e-10)
135 u0 = patchEdgeInt2(u0);

```

Draw solution profile Separate patches with NaNs, then reshape arrays to suit 2D space surface plots.

```

146 figure(1), clf, colormap(0.8*hsv)
147 x(end+1,:,:) = nan; u0(end+1,:,:) = nan;
148 y(:,end+1,:) = nan; u0(:,end+1,:) = nan;
149 u = reshape(permute(squeeze(u0),[1 3 2 4]), [numel(x) numel(y)]);

```

Draw the patch solution surface, with boundary-values omitted as already NaN by not bothering to set them.

```

156 mesh(x(:),y(:),u'); view(60,55)
157 xlabel('space $x$'), ylabel('space $y$'), zlabel('$u(x,y)$')
158 ifOurCf2tex(mfilename)%optionally save

```

4.5.1 twoscaleDiffForce2(): microscale discretisation inside patches of forced diffusion PDE

This function codes the lattice heterogeneous diffusion of the PDE inside the patches. For 6D input arrays u , x , and y , computes the time derivative at each point in the interior of a patch, output in ut .

```

177 function ut = twoscaleDiffForce2(t,u,patches)
178     dx = diff(patches.x(2:3)); % x space step
179     dy = diff(patches.y(2:3)); % y space step
180     i = 2:size(u,1)-1; % x interior points in a patch
181     j = 2:size(u,2)-1; % y interior points in a patch
182     ut = nan(size(u),'like',u); % preallocate output array

```

Set Dirichlet boundary value of zero around the square domain.

```

189     u( 1 ,:,:,: , 1 ,:)=0; % left edge of left patches
190     u(end,:,: ,end,:)=0; % right edge of right patches
191     u(:, 1 ,:,:,: , 1 )=0; % bottom edge of bottom patches
192     u(:,end,:,: ,end)=0; % top edge of top patches

```

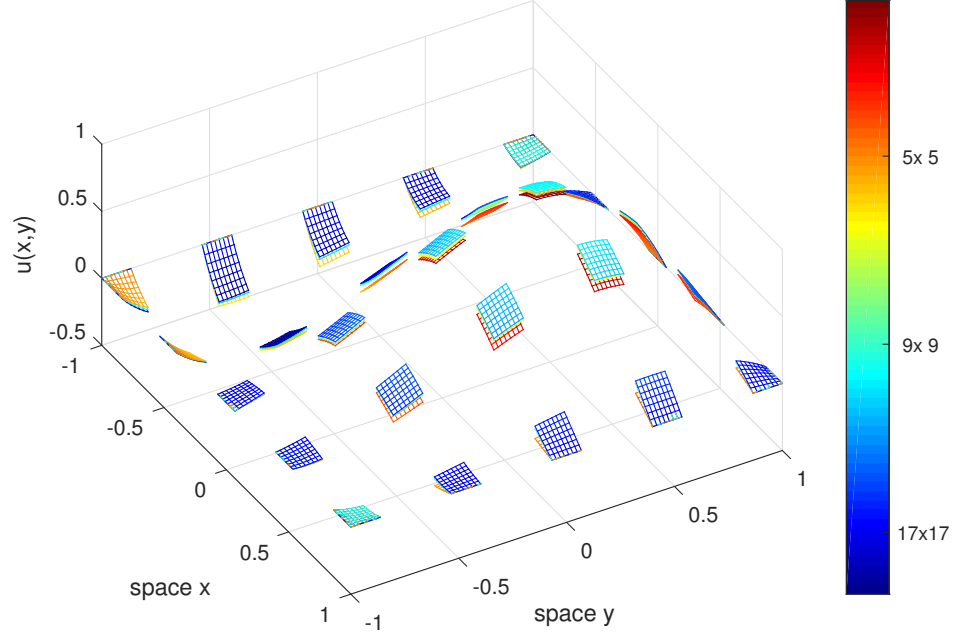
Compute the time derivatives via stored forcing and coefficients. Easier to code by conflating the last four dimensions into the one $,:$.

```

200     ut(i,j,:) ...
201     = 2*diff(u(:,j,:),2,1)/dx^2 +2*diff(u(i,:,:),2,2)/dy^2 ...
202     +2*patches.cs(i,j).*( u(i+1,j+1,:) -u(i-1,j+1,:) ...
203     -u(i+1,j-1,:) +u(i-1,j-1,:) )/(4*dx*dy) ...
204     -patches.fu(i,j,:);
205 end%function twoscaleDiffForce2

```

Figure 4.5: For various numbers of patches as indicated on the colorbar, plot the equilibrium of the multiscale diffusion problem of Freese with Dirichlet zero-value boundary conditions (Section 4.6). We only compare solutions only in these 25 common patches.



4.6 twoscaleDiffEquil2Errs: errors in equilibria of a 2D twoscale heterogeneous diffusion via small patches

Here we find the steady state $u(x, y)$ to the heterogeneous PDE (inspired by Freese et al.⁵ §5.3.1)

$$u_t = A(x, y) \vec{\nabla} \vec{\nabla} u + f,$$

on domain $[-1, 1]^2$ with Dirichlet BCs, for coefficient ‘diffusion’ matrix, varying with some microscale period ϵ (here $\epsilon \approx 0.24, 0.12, 0.06, 0.03$), of

$$A := \begin{bmatrix} 2 & a \\ a & 2 \end{bmatrix} \quad \text{with } a := \sin(\pi x/\epsilon) \sin(\pi y/\epsilon),$$

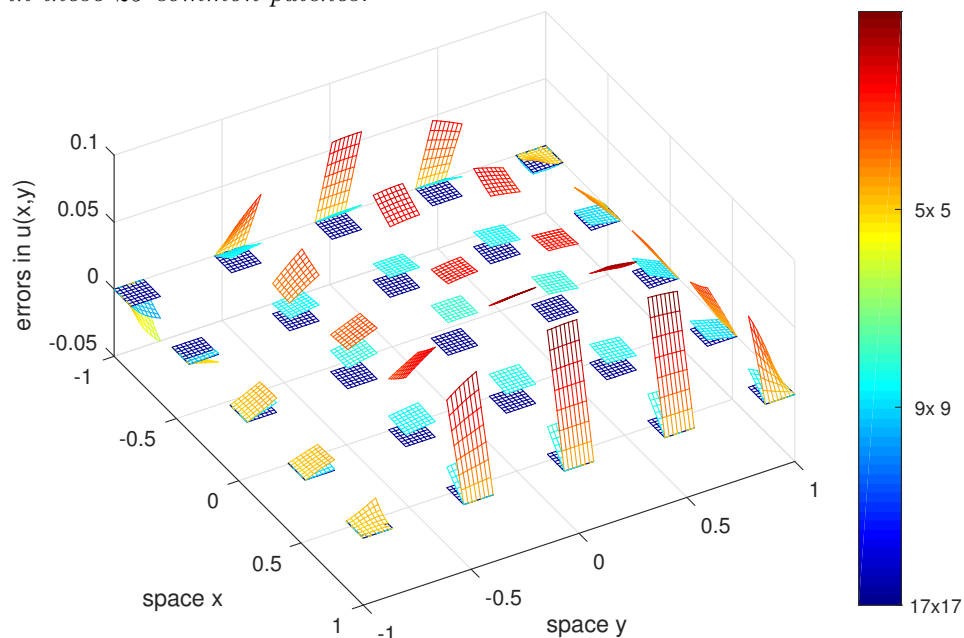
and for forcing $f := 10(x + y + \cos \pi x)$ (for which the solution has magnitude up to one).⁶

Here we explore the errors for increasing number N of patches (in both directions). Find mean-abs errors to be the following (for different orders of

⁵ <http://arxiv.org/abs/2211.13731>

⁶ Freese et al. had forcing $f := (x + \cos 3\pi x)y^3$, but here we want smoother forcing so we get meaningful results in a minute or two computation.⁷ For the same reason we do not invoke their smaller $\epsilon \approx 0.01$.

Figure 4.6: For various numbers of patches as indicated on the colorbar, plot the equilibrium of the multiscale diffusion problem of Freese with Dirichlet zero-value boundary conditions (Section 4.6). We only compare solutions only in these 25 common patches.



interpolation and patch distribution):

	N	5	9	17	33
equispace, 2nd-order	6E-2	3E-2	1E-2	3E-3	
equispace, 4th-order	3E-2	8E-3	7E-4	7E-5	
chebyshev, 4th-order	1E-2	2E-2	6E-3	2E-3	
usergiven, 4th-order	1E-2	2E-2	4E-3	n/a	
equispace, 6th-order	3E-2	1E-3	1E-4	2E-5	

Script start Clear, and initiate global patches. Choose the type of patch distribution to be either ‘equispace’, ‘chebyshev’, or ‘usergiven’. Also set order of interpolation (fourth-order is good start).

```

81 clear all
82 global patches
83 %global OurCf2eps, OurCf2eps=true %option to save plot
84 switch 1
85     case 1, Dom.type = 'equispace'
86     case 2, Dom.type = 'chebyshev'
87     case 3, Dom.type = 'usergiven'
88 end% switch
89 ordInt = 6

```

First configure the patch system Establish the microscale heterogeneity has micro-period `mPeriod` on the spatial lattice. Then `configPatches2` replicates the heterogeneity as needed to fill each patch.

```

100 mPeriod = 6
101 z = (0.5:mPeriod)'/mPeriod;
102 A = sin(2*pi*z).*sin(2*pi*z');

```

To use a hierarchy of patches with `nPatch` of 5, 9, 17, ..., we need up to N patches plus one dx to fit into the domain interval. Cater for up to some full-domain simulation—can compute $\log_2 N_{\max} = 5$ ($\epsilon = 0.06$) within minutes:

```

112 log2Nmax = 6 % >2 up to 6 OKish
113 nPatchMax=2^log2Nmax+1

```

Set the periodicity ϵ , and other microscale parameters.

```

120 nPeriodsPatch = 1 % any integer
121 nSubP = nPeriodsPatch*mPeriod+2 % for edgy int
122 epsilon = 2/(nPatchMax*nPeriodsPatch+1/mPeriod)
123 dx = epsilon/mPeriod

```

For various numbers of patches Choose five patches to be the coarsest number of patches. Define variables to store common results for the solutions from differing patches. Assign `Ps` to be the indices of the common patches: for equispace set to the five common patches, but for ‘chebyshev’ the only common ones are the three centre and boundary-adjacent patches.

```

136 us=[]; xs=[]; ys=[]; nPs=[];
137 for log2N=log2Nmax:-1:2
138     if log2N==log2Nmax
139         Ps=linspace(1,nPatchMax ...
140             ,5-2*all(Dom.type=='chebyshev'))
141     else Ps=(Ps+1)/2
142     end

```

Set the number of patches in $(-1, 1)$:

```

148 nPatch = 2^log2N+1

```

In the case of ‘usergiven’, we set the standard Chebyshev distribution of the patch-centres, which involves overlapping of patches near the boundaries! (instead of the coded chebyshev which has boundary layers of abutting patches, and non-overlapping Chebyshev between the boundary layers).

```

159 if all(Dom.type=='usergiven')
160     halfWidth = dx*(nSubP-1)/2;
161     X1 = -1+halfWidth; X2 = 1-halfWidth;
162     Dom.X = (X1+X2)/2-(X2-X1)/2*cos(linspace(0,pi,nPatch));
163     Dom.Y = Dom.X;
164 end

```

Configure the patches:

```

170 configPatches2(@twoscaleDiffForce2,[-1 1],Dom,nPatch ...
171     ,ordInt ,dx ,nSubP ,'EdgyInt',true ,'hetCoeffs',A );

```

Compute the time-constant forcing, and store in struct `patches` for access by the microcode of [Section 4.6.1](#).

```

179     if 1
180         patches.fu = 10*(patches.x+cos(pi*patches.x)+patches.y);
181     else patches.fu = 8+0*patches.x+0*patches.y;
182     end

```

Solve for steady state Set initial guess of either zero or a subsample of the previous, next-finer, solution. NaN indicates patch-edge values. Index `i` are the indices of patch-interior points, and the number of unknowns is then its length, circa 150,000.

```

193     if log2N==log2Nmax
194         u0 = zeros(nSubP,nSubP,1,1,nPatch,nPatch);
195     else u0 = u0(:,:,:,1:2:end,1:2:end);
196     end
197     u0([1 end],:,:) = nan; u0(:,[1 end],:) = nan;
198     i = find(~isnan(u0));
199     nVariables = numel(i)

```

Could solve via iterative solver `bicgstab`, using the generic patch system wrapper `theRes` ([Section 3.15](#)). However, for such linear problems it is 100 times faster to use `AutoDiff` to get Jacobian, and then solve directly.

```

209     U0=AutoDiff(0*u0); % set an AD array of zeros
210     tic
211     F0 = patchSys2(1,U0); % evaluate patch system and its derivatives
212     Jac = getderivs(F0); % the Jacobian is the derivatives
213     uSoln = -Jac(i,i)\getvalue(F0(i)); % solve linear system
214     ADsolveTime = toc

```

Store the solution into the patches, and give magnitudes—Inf norm is `max(abs())`.

```

221     normSoln = norm(uSoln,Inf)
222     u0(i) = uSoln;
223     f0 = patchSys2(1,u0);
224     normResidual = norm(f0(i),Inf)
225     assert(normResidual<1e-8,'bad solution')
226     u0 = patchEdgeInt2(u0);
227     u0( 1 ,:,:,:, 1 ,:)=0; % left edge of left patches
228     u0(end,:,:,:,end,:)=0; % right edge of right patches
229     u0(:, 1 ,:,:,:, 1 )=0; % bottom edge of bottom patches
230     u0(:,end,:,:,:,end)=0; % top edge of top patches

```

Concatenate the solution on common patches into stores.

```

236     us=cat(5,us,squeeze(u0(:,:,:,Ps,Ps)));
237     xs=cat(3,xs,squeeze(patches.x(:,:,:,Ps,:)));
238     ys=cat(3,ys,squeeze(patches.y(:,:,:,Ps)));
239     nPs = [nPs;nPatch];

```

End loop. Check micro-grids are aligned, then compute errors compared to the full-domain solution (or the highest resolution solution for the case of 'usergiven').

```

248 end%for log2N
249 assert(max(abs(reshape(diff(xs,1,3),[],1)))<1e-12,'x-coord failure')
250 assert(max(abs(reshape(diff(ys,1,3),[],1)))<1e-12,'y-coord failure')
251 errs = us-us(:,:,,1);
252 format short e
253 meanAbsErrs = mean(abs(reshape(errs,[],size(us,5))))
254 format short
255 ratioErrs = meanAbsErrs(2:end)./meanAbsErrs(1:end-1)

```

Plot solution in common patches First reshape arrays to suit 2D space surface plots, inserting nans to separate patches.

```

267 x = xs(:,:,1); y = ys(:,:,1); u=us;
268 x(end+1,:)=nan; y(end+1,:)=nan;
269 u(end+1,:)=nan; u(:,end+1,:)=nan;
270 u = reshape(permute(u,[1 3 2 4 5]),numel(x),numel(y),[]);

```

Plot the patch solution surfaces, with colour offset between surfaces (best if u -field has a range of one): blues are the full-domain solution, reds the coarsest patches.

```

278 figure(1), clf, colormap(jet)
279 for p=1:size(u,3)
280     mesh(x(:),y(:),u(:,:,p)',p+u(:,:,p)');
281     hold on;
282 end, hold off
283 view(60,55)
284 colorbar('Ticks',1:size(u,3) ...
285     , 'TickLabels',num2str(nPs)+"x"+num2str(nPs));
286 xlabel('space $x$'), ylabel('space $y$'), zlabel('$u(x,y)$')
287 ifOurCf2eps([mfilename 'us'])%optionally save

```

Plot error surfaces Plot the error surfaces, with colour offset between surfaces (best if u -field has a range of one): dark blue is the full-domain zero error, reds the coarsest patches.

```

299 err=u(:,:,1)-u;
300 maxAbsErr=max(abs(err(:)));
301 figure(2), clf, colormap(jet)
302 for p=1:size(u,3)
303     mesh(x(:),y(:),err(:,:,p)',p+err(:,:,p)'/maxAbsErr);
304     hold on;
305 end, hold off
306 view(60,55)
307 colorbar('Ticks',1:size(u,3) ...
308     , 'TickLabels',num2str(nPs)+"x"+num2str(nPs));
309 xlabel('space $x$'), ylabel('space $y$')

```



```

310 zlabel('errors in $u(x,y)$')
311 ifOurCf2eps(mfilename)%optionally save

```

4.6.1 twoscaleDiffForce2(): microscale discretisation inside patches of forced diffusion PDE

This function codes the lattice heterogeneous diffusion of the PDE inside the patches. For 6D input arrays u , x , and y , computes the time derivative at each point in the interior of a patch, output in ut .

```

330 function ut = twoscaleDiffForce2(t,u,patches)
331     dx = diff(patches.x(2:3)); % x space step
332     dy = diff(patches.y(2:3)); % y space step
333     i = 2:size(u,1)-1; % x interior points in a patch
334     j = 2:size(u,2)-1; % y interior points in a patch
335     ut = nan(size(u),'like',u); % preallocate output array

```

Set Dirichlet boundary value of zero around the square domain.

```

342     u( 1 ,:,:,: , 1 ,:)=0; % left edge of left patches
343     u(end,:,: ,end,:)=0; % right edge of right patches
344     u(:, 1 ,:,:,: , 1 )=0; % bottom edge of bottom patches
345     u(:,end,:,: ,end)=0; % top edge of top patches

```

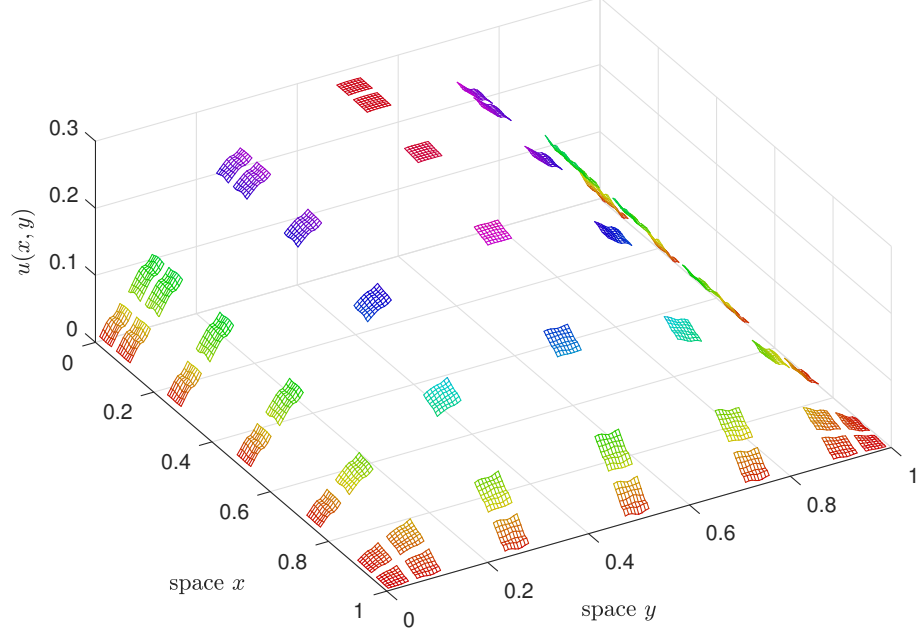
Compute the time derivatives via stored forcing and coefficients. Easier to code by conflating the last four dimensions into the one $,:$.

```

353     ut(i,j,:) ...
354     = 2*diff(u(:,j,:),2,1)/dx^2 +2*diff(u(i,:,:),2,2)/dy^2 ...
355     +2*patches.cs(i,j).*( u(i+1,j+1,:) -u(i-1,j+1,:) ...
356     -u(i+1,j-1,:) +u(i-1,j-1,:) )/(4*dx*dy) ...
357     +patches.fu(i,j,:);
358 end%function twoscaleDiffForce2

```

Figure 4.7: Equilibrium of the macroscale diffusion problem of Abdulle with boundary conditions of Dirichlet zero-value except for $x = 0$ which is Neumann (Section 4.7). Here the patches have a Chebyshev-like spatial distribution. The patch size is chosen large enough to see within.



4.7 abdulleDiffEquil2: equilibrium of a 2D multiscale heterogeneous diffusion via small patches

Here we find the steady state $u(x, y)$ to the heterogeneous PDE (inspired by Abdulle et al. 2020, §5.1)

$$u_t = \vec{\nabla} \cdot [a(x, y) \vec{\nabla} u] + 10,$$

on square domain $[0, 1]^2$ with zero-Dirichlet BCs, for coefficient ‘diffusion’ matrix, varying with period ϵ of (their (45))

$$a := \frac{2 + 1.8 \sin 2\pi x/\epsilon}{2 + 1.8 \cos 2\pi y/\epsilon} + \frac{2 + \sin 2\pi y/\epsilon}{2 + 1.8 \cos 2\pi x/\epsilon}.$$

Figure 4.7 shows solutions have some nice microscale wiggles reflecting the heterogeneity.

Clear, and initiate globals.

```

38 clear all
39 global patches
40 %global OurCf2eps, OurCf2eps=true %option to save plot

```

First establish the microscale heterogeneity has micro-period `mPeriod` on the spatial micro-grid lattice. Then `configPatches2` replicates the heterogeneity to fill each patch. (These diffusion coefficients should really recognise the half-grid-point shifts, but let’s not bother.)

```

53 mPeriod = 6
54 x = (0.5:mPeriod)'/mPeriod; y=x';
55 a = (2+1.8*sin(2*pi*x))./(2+1.8*sin(2*pi*y)) ...
56     +(2+    sin(2*pi*y))./(2+1.8*sin(2*pi*x));
57 diffusivityRange = [min(a(:)) max(a(:))]

```

Set the periodicity ϵ , here big enough so we can see the patches, and other microscale parameters.

```

64 epsilon = 0.04
65 dx = epsilon/mPeriod
66 nPeriodsPatch = 1 % any integer
67 nSubP = nPeriodsPatch*mPeriod+2 % when edgy int

```

Patch configuration Choose either Dirichlet (default) or Neumann on the left boundary in coordination with micro-code in [Section 4.7.1](#)

```

78 Dom.bcOffset = zeros(2);
79 if 1, Dom.bcOffset(1)=0.5; end% left Neumann

```

Say use 7×7 patches in $(0,1)^2$, fourth order interpolation, and either 'equispace' or 'chebyshev':

```

86 nPatch = 7
87 Dom.type='chebyshev';
88 configPatches2(@abdulleDiffForce2,[0 1],Dom ...
89     ,nPatch ,4 ,dx ,nSubP , 'EdgyInt',true , 'hetCoeffs',a );

```

Solve for steady state Set initial guess of zero, with NaN to indicate patch-edge values. Index i are the indices of patch-interior points, and the number of unknowns is then its length.

```

102 u0 = zeros(nSubP,nSubP,1,1,nPatch,nPatch);
103 u0([1 end],:,:) = nan; u0(:,[1 end],:) = nan;
104 i = find(~isnan(u0));
105 nVariables = numel(i)

```

Could use `fsolve` for simplicity and robustness, via the generic patch wrapper `theRes()`. However, for such linear problems it is many times faster to use `AutoDiff` to get Jacobian, and then solve directly.

```

114 U0=AutoDiff(0*u0); % set an AD array of zeros
115 tic
116 F0 = patchSys2(1,U0); % evaluate patch system and its derivatives
117 Jac = getderivs(F0); % the Jacobian is the derivatives
118 uSoln = -Jac(i,i)\getvalue(F0(i)); % solve linear system
119 ADsolveTime = toc
120 normSoln = norm(uSoln)

```

Store the solution vector into the patches, and interpolate, but have not bothered to set boundary values so they stay NaN from the interpolation.

```

128 normSoln = norm(uSoln)
129 u0(i) = uSoln;
130 f0 = patchSys2(1,u0);
131 normResidual = norm(f0(i),Inf)
132 assert(normResidual<1e-10)
133 u0 = patchEdgeInt2(u0);

```

Draw solution profile Separate patches with NaNs, then reshape arrays to suit 2D space surface plots.

```

144 figure(1), clf, colormap(0.8*hsv)
145 patches.x(end+1,:,:) = nan; u0(end+1,:,:) = nan;
146 patches.y(:,end+1,:) = nan; u0(:,end+1,:) = nan;
147 u = reshape(permute(squeeze(u0),[1 3 2 4]), ...
148             , [numel(patches.x) numel(patches.y)]);

```

Draw the patch solution surface, with boundary-values omitted as already NaN by not bothering to set them.

```

155 mesh(patches.x(:),patches.y(:),u'); view(60,55)
156 xlabel('space $x$'), ylabel('space $y$'), zlabel('$u(x,y)$')
157 ifOurCf2eps(mfilename) %optionally save plot

```

4.7.1 abdulleDiffForce2(): microscale discretisation inside patches of forced diffusion PDE

This function codes the lattice heterogeneous diffusion of the PDE inside the patches. For 6D input arrays u , x , and y , computes the time derivative at each point in the interior of a patch, output in ut .

```

176 function ut = abdulleDiffForce2(t,u,patches)
177     dx = diff(patches.x(2:3)); % x space step
178     dy = diff(patches.y(2:3)); % y space step
179     i = 2:size(u,1)-1; % x interior points in a patch
180     j = 2:size(u,2)-1; % y interior points in a patch
181     ut = nan(size(u),'like',u); % preallocate output array

```

Set Dirichlet boundary value of zero around the square domain, but also cater for zero Neumann condition on the left boundary.

```

189     u( 1 ,:,:,:, 1 ,:)=0; % left edge of left patches
190     u(end,:,:,end,:)=0; % right edge of right patches
191     u(:, 1 ,:,:,:, 1 )=0; % bottom edge of bottom patches
192     u(:,end,:,:,end)=0; % top edge of top patches
193     if 1, u(1,:,:,1,:)=u(2,:,:,1,:); end% left Neumann

```

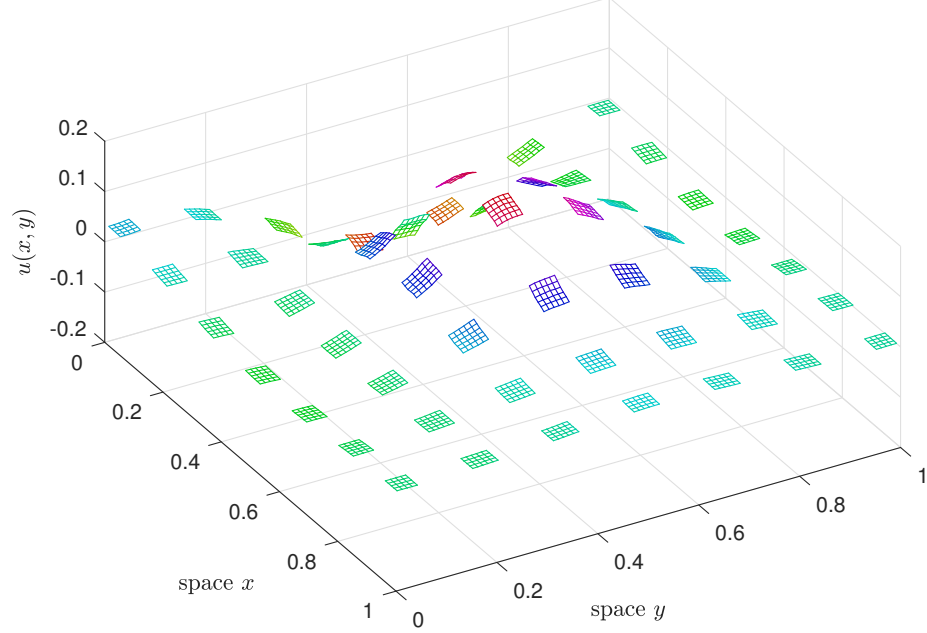
Compute the time derivatives via stored forcing and coefficients. Easier to code by conflating the last four dimensions into the one $,:$.

```

201     ut(i,j,:) = diff(patches.cs(:,j).*diff(u(:,j,:)))/dx^2 ...
202                 + diff(patches.cs(i,:).*diff(u(i,:,:),1,2),1,2)/dy^2 ...
203                 + 10;
204 end%function abdulleDiffForce2

```

Figure 4.8: Equilibrium of the macroscale diffusion problem of Bonizzoni et al. with Neumann boundary conditions of zero (Section 4.8). Here the patches have a equispaced spatial distribution. The microscale periodicity, and hence the patch size, is chosen large enough to see within.



4.8 randAdvectDiffEquil2: equilibrium of a 2D random heterogeneous advection-diffusion via small patches

Here we find the steady state $u(x, y)$ of the heterogeneous PDE (inspired by Bonizzoni et al.⁸ §6.2)

$$u_t = \mu_1 \nabla^2 u - (\cos \mu_2, \sin \mu_2) \cdot \vec{\nabla} u - u + f,$$

on domain $[0, 1]^2$ with Neumann boundary conditions, for microscale random pseudo-diffusion and pseudo-advection coefficients, $\mu_1(x, y) \in [0.01, 0.1]$ ⁹ and $\mu_2(x, y) \in [0, 2\pi)$, and for forcing

$$f(x, y) := \exp \left[-\frac{(x - \mu_3)^2 + (y - \mu_4)^2}{\mu_5^2} \right],$$

smoothly varying in space for fixed $\mu_3, \mu_4 \in [0.25, 0.75]$ and $\mu_5 \in [0.1, 0.25]$. The above system is dominantly diffusive for lengths scales $\ell < 0.01 = \min \mu_1$. Due to the randomness, we get different solutions each execution of this code. Figure 4.8 plots one example. A physical interpretation of the solution field is confounded because the problem is pseudo-advection-diffusion due to the varying coefficients being outside the $\vec{\nabla}$ operator.

Clear, and initiate globals.

```

50 clear all
51 global patches
52 %global OurCf2eps, OurCf2eps=true %option to save plot
```

⁸ <http://arxiv.org/abs/2211.15221>

⁹ More interesting microscale structure arises here for μ_1 a factor of three smaller.

First establish the microscale heterogeneity has micro-period `mPeriod` on the spatial lattice. Then `configPatches2` replicates the heterogeneity to fill each patch.

```

63 mPeriod = 4
64 mu1 = 0.01*10.^rand(mPeriod)
65 mu2 = 2*pi*rand(mPeriod)
66 cs = cat(3,mu1,cos(mu2),sin(mu2));
67 meanDiffAdvec=squeeze(mean(mean(cs)))

```

Set the periodicity ϵ , here big enough so we can see the patches, and other microscale parameters.

```

74 epsilon = 0.04
75 dx = epsilon/mPeriod
76 nPeriodsPatch = 1 % any integer
77 nSubP = nPeriodsPatch*mPeriod+2 % for edgy int

```

Patch configuration Say use 7×7 patches in $(0,1)^2$, fourth order interpolation, either ‘equispace’ or ‘chebyshev’, and the offset for Neumann boundary conditions:

```

89 nPatch = 7
90 Dom.type= 'equispace';
91 Dom.bcOffset = 0.5;
92 configPatches2(@randAdvecDiffForce2,[0 1],Dom ...
93     ,nPatch ,4 ,dx ,nSubP , 'EdgyInt',true , 'hetCoeffs',cs );

```

Compute the time-constant forcing, and store in struct `patches` for access by the microcode of [Section 4.8.1](#).

```

101 mu = [ 0.25+0.5*rand(1,2) 0.1+0.15*rand ]
102 patches.fu = exp(-((patches.x-mu(1)).^2 ...
103     +(patches.y-mu(2)).^2)/mu(3)^2);

```

Solve for steady state Set initial guess of zero, with NaN to indicate patch-edge values. Index `i` are the indices of patch-interior points, store in global patches for access by `theRes`, and the number of unknowns is then its number of elements.

```

118 u0 = zeros(nSubP,nSubP,1,1,nPatch,nPatch);
119 u0([1 end],:,:) = nan; u0(:,[1 end],:) = nan;
120 i = find(~isnan(u0));
121 nVariables = numel(i)

```

Could use `fsolve` for simplicity and robustness, via the generic patch wrapper `theRes()`. However, for such linear problems it is many times faster to use `AutoDiff` to get Jacobian, and then solve directly.

```

130 U0=AutoDiff(0*u0); % set an AD array of zeros
131 tic
132 F0 = patchSys2(1,U0); % evaluate patch system and its derivatives
133 Jac = getderivs(F0); % the Jacobian is the derivatives

```

```

134 uSoln = -Jac(i,i)\getvalue(F0(i)); % solve linear system
135 ADsolveTime = toc

```

Store the solution vector into the patches, and interpolate, but have not bothered to set boundary values so they stay NaN from the interpolation.

```

143 normSoln = norm(uSoln)
144 u0(i) = uSoln;
145 f0 = patchSys2(1,u0);
146 normResidual = norm(f0(i),Inf)
147 assert(normResidual<1e-10)
148 u0 = patchEdgeInt2(u0);

```

Draw solution profile Separate patches with NaNs, then reshape arrays to suit 2D space surface plots.

```

159 figure(1), clf, colormap(0.8*hsv)
160 patches.x(end+1,:,:) = nan; u0(end+1,:,:) = nan;
161 patches.y(:,end+1,:) = nan; u0(:,end+1,:) = nan;
162 u = reshape(permute(squeeze(u0),[1 3 2 4]), ...
163     , [numel(patches.x) numel(patches.y)]);

```

Draw the patch solution surface, with boundary-values omitted as already NaN by not bothering to set them.

```

170 mesh(patches.x(:),patches.y(:),u'); view(60,55)
171 xlabel('space $x$'), ylabel('space $y$'), zlabel('$u(x,y)$')
172 ifOurCf2eps(mfilename) %optionally save plot

```

4.8.1 randAdvecDiffForce2(): microscale discretisation inside patches of forced diffusion PDE

This function codes the lattice heterogeneous diffusion of the PDE inside the patches. For 6D input arrays u , x , and y , computes the time derivative at each point in the interior of a patch, output in ut .

```

191 function ut = randAdvecDiffForce2(t,u,patches)
192     dx = diff(patches.x(2:3)); % x space step
193     dy = diff(patches.y(2:3)); % y space step
194     i = 2:size(u,1)-1; % x interior points in a patch
195     j = 2:size(u,2)-1; % y interior points in a patch
196     ut = nan(size(u),'like',u); % preallocate output array

```

Set Neumann boundary condition of zero derivative around the square domain: that is, the edge value equals the next-to-edge value.

```

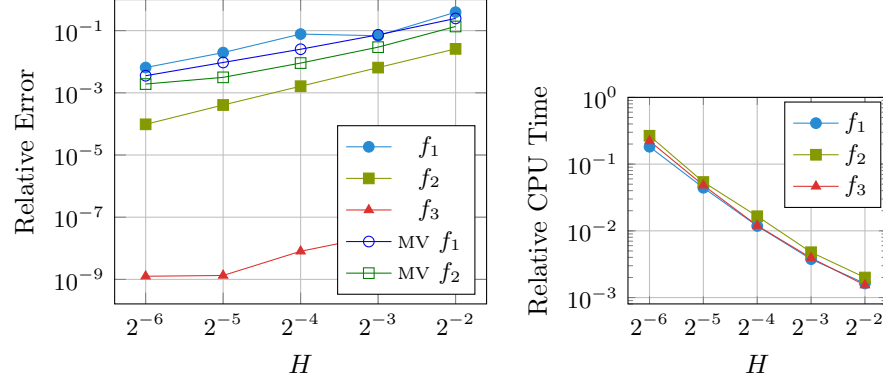
204     u( 1 ,:,:,:, 1 ,:)=u( 2 ,:,:,:, 1 ,:); % left edge of left patches
205     u(end,:,:,end,:)=u(end-1,:,:,end,:); % right edge of right patches
206     u(:, 1 ,:,:,:, 1 )=u(:, 2 ,:,:,:, 1 ); % bottom edge of bottom patches
207     u(:,end,:,:,end)=u(:,end-1,:,:,end); % top edge of top patches

```

Compute the time derivatives via stored forcing and coefficients. Easier to code by conflating the last four dimensions into the one $,:$.

```
215     ut(i,j,:) ...
216     = patches.cs(i,j,1).*(diff(u(:,j,:),2,1)/dx^2 ...
217       +diff(u(i,(:,)),2,2)/dy^2)...
218     -patches.cs(i,j,2).*(u(i+1,j,:)-u(i-1,j,:))/(2*dx) ...
219     -patches.cs(i,j,3).*(u(i,j+1,:)-u(i,j-1,:))/(2*dy) ...
220     -u(i,j,:) +patches.fu(i,j,:);
221 end%function randAdvecDiffForce2
```


Figure 4.9: results for the computational homogenisation of a forced, non-autonomous, 2D wave (Section 4.9). (left) relative RMS error of the patch scheme, each patch of width $1/128$, as a function of patch spacing H . The unfilled symbols are those of the energy norm from Maier & Verfürth (2021) (their Figure 5.1). (right) the relative compute time decreases very quickly in H as there are fewer patches spaced further apart.



4.9 homoWaveEdgy2: computational homogenisation of a forced, non-autonomous, 2D wave via simulation on small patches

This section addresses 2D waves in a media with microscale heterogeneity. It extends to waves the code for 2D diffusion. The results here favourably compare to the examples of Maier & Verfürth (2021).

Figure 4.9 summarises the results here. The left (larger) graph shows the error in the patch scheme decreasing with decreasing patch spacing H (increasing number of patches). Forcing f_1 and f_2 are as specified by §5.1 of Maier & Verfürth (2021), whereas f_3 here is f in their §5.2. For the case of forcing f_1 which is discontinuous in space (at $x = 0.4$), the errors are similar to that of Maier & Verfürth (2021)—compare the filled with unfilled circles. For the case of forcing f_2 which is continuous in the spatial domain, except for a second derivative discontinuity in its odd-periodic extension, the errors of the patch scheme are an order of magnitude better than that of Maier & Verfürth (2021)—compare the filled with unfilled squares. For the case of forcing f_3 which is smooth in the domain and in its odd-periodic extension, the patch scheme errors, roughly 10^{-8} , are at the tolerance of the time integration. Two caveats in a comparison with Maier & Verfürth (2021) are the slightly different norms used, and that they also address errors in the time integration, whereas here we use a standard adaptive integrator in order to focus purely on the spatial errors of the patch scheme.

Now let's code the simulation of the forced, non-autonomous, 2D wave. Maier & Verfürth (2021) have Dirichlet BCs of zero around the unit square, so replicate here by the odd periodic extension to the spatial domain $[-1, 1]^2$. In their §5.1, their microscale mesh step is $1/512 = 2^{-9}$. Coding that here results in a compute time of roughly 90 minutes, so here I provide a much coarser case that computes in only a few minutes: change as you please.

```

69 clear all
70 dx = 1/128 % 1/512=2-9 is the original, but takes 90 mins

The heterogeneity is of period four on the microscale lattice, so code a minimal
patch size that covers one period.

78 epsilon = 4*dx
79 nPeriodsPatch = 1
80 mPeriod = round(epsilon/dx)
81 nSubP = mPeriod*nPeriodsPatch+2

Choose which of three forcing functions to use

87 fn=2

Maier & Verfürth (2021) use varying number of macroscale grid steps from 4
to 64 on [0, 1] so here on [-1, 1] we use double the number patches in each
direction. Loop over the number of patches used, starting with the full domain
simulation, and then progressively coarsening the macroscale grid of patches.

100 nPatch = 2/epsilon/nPeriodsPatch
101 for iPat=0:9
102 if iPat>0, nPatch=nPatch/2, end
103 if nPatch<8, break, end

```

Set the periodic heterogeneous coefficient, isotropic:

$$a_\epsilon(t, x) = [3 + \sin(2\pi x/\epsilon) + \sin(2\pi t)] \cdot [3 + \sin(2\pi y/\epsilon) + \sin(2\pi t)],$$

which being in product form with two time-dependencies we store as the two spatially varying factors—although to preserve odd symmetry we phase shift the heterogeneity from sines to cosines. It is a user’s choice whether to code such spatial dependencies here with `cHetr` or within the time derivative function itself. In this case, I choose to code microscale heterogeneous coefficients here via `cHetr`, and the macroscale variation of f_i in the time derivative function.

Here the period of the heterogeneity is only four microscale lattice points in each direction (which is pretty inaccurate on the microscale, but immaterial as we and Maier & Verfürth (2021) only compare to the coded system on the microscale lattice, not to the PDE). With the following careful choices we ensure all the hierarchy of patch schemes both maintain odd symmetry, and also compute on grid points that are common with the full domain.

```

131 ratio = (nSubP-2)*dx/(2/nPatch)
132 Xleft=(1-ratio)/nPatch;
133 xmid=Xleft+dx*(0:mPeriod-1)'; % half-points
134 xi = Xleft+dx*(-0.5:mPeriod-1)'; % grid-points
135 % two components for ax, the x-dirn interactions
136 cHetr(:, :, 1) = (3+cos(2*pi*xmid/epsilon))+0*xmid';
137 cHetr(:, :, 2) = 0*xmid+(3+cos(2*pi*xi'/epsilon));
138 % two components for ay, the y-dirn interactions
139 cHetr(:, :, 3) = (3+cos(2*pi*xi/epsilon))+0*xmid';
140 cHetr(:, :, 4) = 0*xi+(3+cos(2*pi*xmid'/epsilon));

```

Configure patches using spectral interpolation. Quadratic interpolation did not seem significantly different for the case of discontinuous forcing f_1 .

```
149 configPatches2(@heteroWave2,[-1 1 -1 1],nan,nPatch ...
150               ,0,ratio,nSubP ,'EdgyInt',true , 'hetCoeffs',cHetr );
```

A check on the spatial geometry.

```
156 global patches
157 dxPat=diff(patches.x(1:2));
158 assert(abs(dx-dxPat)<1e-9,"dx mismatch")
```

Simulate Set the particular forcing function to use, and the zero initial conditions of a simulation.

```
168 patches.eff=fn;
169 clear uv0
170 uv0(:,:,1,1,:,:) = 0*patches.x+0*patches.y;
171 uv0(:,:,2,1,:,:) = 0*patches.x+0*patches.y;
```

Integrate using standard integrators. [Maier & Verfürth \(2021\)](#) use a scheme with fixed time-step of $\tau = 2^{-7} = 1/128$. Here `ode23` uses variable steps of about 0.0003, and takes 7 s for `nPatch=2*4` (whereas `ode15s` takes 149 s—even for the dissipating case), and takes 287 s for `nPatch=2*32` and roughly 4000 s for full domain `nPatch=2*128`.

```
183 disp('Now simulate over time, via simple ode23')
184 tic
185 [ts,us] = ode23(@patchSys2, linspace(0,1,11), uv0(:));
186 if iPat==0, odeTime0=toc
187 else relOdeTime(iPat)=toc/odeTime0
188 end
```

Compute error compared to full domain simulation Get spatial coordinates of patch-interior points, and reshape to column vectors.

```
198 i = 2:nSubP-1;
199 x = squeeze(patches.x(i,:,:, :, :));
200 y = squeeze(patches.y(:,i,:,:, :));
201 x=x(:); y=y(:);
```

At the final time of $t = 1$, get the row vector of data, form into the 6D array via the interpolation to the edges, and reshape patch-interior points to 2D spatial array.

```
209 uv = squeeze( patchEdgeInt2(us(end,:)));
210 u = squeeze( uv(i,i,1,:,:) );
211 u = reshape(permute(u,[1 3 2 4]),[numel(x) numel(y)]);
```

If this is the full domain simulation, then store as the reference solution.

```
218 if iPat==0
219     x0=x; y0=y; u0=u;
```

```

220     rms0=sqrt(mean(u0(:).^2))
221     else

    Else compute the error compared to the full domain solution. First find
    the indices of the full domain that match the spatial locations of the patch
    scheme.

229     [i,k] = find(abs(x0-x')<1e-9);
230     assert(length(i)==length(x),'find error in index i')
231     [j,k] = find(abs(y0-y')<1e-9);
232     assert(length(j)==length(y),'find error in index j')

    The RMS error over the surface is

238     errs=u-u0(i,j);
239     relrmserr(iPat)=sqrt(mean(errs(:).^2))/rms0
240     H(iPat)=2/nPatch
241 end%if iPat

    End the loop over the various number of patches, and return. Further, here
    not executed, code in the file animates the solution over time, and computes
    spectrum of the system.

251 end%for iPat
252 figure(1), clf
253 loglog(H,relrmserr,'o:'), grid on
254 xlabel('$H$'), ylabel('relative error')
255 %return

```

4.9.1 heteroWave2(): heterogeneous Waves

This function codes the lattice heterogeneous waves inside the patches. The forced wave PDE is

$$u_t = v, \quad v_t = \vec{\nabla}(a\vec{\nabla} \cdot u) + f$$

for scalars $a(t, x, y)$ and $f(t, x, y)$ where a has microscale variations. For 6D input arrays u , x , and y (via edge-value interpolation of `patchSys2`, [Section 4.2](#)), computes the time derivative at each point in the interior of a patch, output in ut . The four 2D arrays of heterogeneous interaction coefficients, c_{ijk} , have previously been stored in `patches.cs` (3D).

Supply patch information as a third argument (required by parallel computation), or otherwise by a global variable.

```

26 function ut = heteroWave2(t,u,patches)
27     if nargin<3, global patches, end

```

Microscale space-steps, and interior point indices.

```

33     dx = diff(patches.x(2:3)); % x micro-scale step
34     dy = diff(patches.y(2:3)); % y micro-scale step
35     i = 2:size(u,1)-1; % x interior points in a patch
36     j = 2:size(u,2)-1; % y interior points in a patch
37     assert(max(abs(u(:)))<9999,"u-field exploding")

```

Form coefficients here—odd periodic extension. To avoid slight errors in periodicity (in full domain simulation), first adjust any coordinates crossing $x = \pm 1$ or $y = \pm 1$.

```

47 x=patches.x; y=patches.y;
48 l=find(abs(x)>1); x(l)=x(l)-sign(x(l))*2;
49 l=find(abs(y)>1); y(l)=y(l)-sign(y(l))*2;

```

Then set at this time three possible forcing functions, although only use one depending upon `patches.eff`. Forcing f_1 and f_2 are as specified by §5.1 of [Maier & Verfürth \(2021\)](#), whereas f_3 here is f in their §5.2.

```

59 f1 = ( (abs(x)>0.4)*(20*t+230*t^2) ...
60       +(abs(x)<0.4)*(100*t+2300*t^2) ).*sign(x).*sign(y);
61 f2 = 20*t*x.*(1-abs(x)).*y.*(1-abs(y)) ...
62       +230*t^2*(sign(y).*x.*(1-abs(x))+sign(x).*y.*(1-abs(y)));
63 f3 = (5*t+50*t^2)*sin(pi*x).*sin(pi*y);

```

Also set the heterogeneous interactions at this time.

```

69 ax = (patches.cs(:, :, 1)+sin(2*pi*t)) ...
70       *(patches.cs(:, :, 2)+sin(2*pi*t));
71 ay = (patches.cs(:, :, 3)+sin(2*pi*t)) ...
72       *(patches.cs(:, :, 4)+sin(2*pi*t));

```

Reserve storage (using `nan+u` appears quickest), and then assign time derivatives for interior patch values due to the heterogeneous interaction and forcing.

```

81 ut = nan(size(u), 'like', u); % preallocate output array
82 ut(i,j,1,:) = u(i,j,2,:);
83 ut(i,j,2,:) ...
84 = diff(ax(:,j)).*diff(u(:,j,1,:),1,1)/dx^2 ...
85   +diff(ay(i,:)).*diff(u(i,:,1,:),1,2),1,2)/dy^2 ...
86   +(patches.eff==1)*f1(i,j, :, :) ...
87   +(patches.eff==2)*f2(i,j, :, :) ...
88   +(patches.eff==3)*f3(i,j, :, :) ...
89   + 1e-4*(diff(u(:,j,2,:),2,1)/dx^2+diff(u(i,:,2,:),2,2)/dy^2);
90 end% function

```

In the last line above, the slight damping of 10^{-4} causes microscale modes to decay at rate e^{-28t} , with frequencies 2000–5000, whereas macroscale modes decay with rates roughly 0.0005–0.05 with frequencies 10–100. This slight damping term may correspond to the weak damping of the backward Euler scheme adopted by [Maier & Verfürth \(2021\)](#) for time integration.

4.10 SwiftHohenberg2dPattern: patterns of the Swift–Hohenberg PDE in 2D on patches

Figures 4.10 to 4.15 show an example simulation in time generated by the patch scheme applied to the patterns arising from the 2D Swift–Hohenberg PDE.

Consider a lattice of values $u_i(t)$, with lattice spacing dx , and governed by a microscale centred discretisation of the Swift–Hohenberg PDE

$$\partial_t u = -(1 + \nabla^2/k_0^2)u + \text{Ra} u - u^3, \quad (4.1)$$

with various boundary conditions at $x, y = 0, L$. For Ra just above critical, say $\text{Ra} = 0.1$, the system rapidly evolves to spatial quasi-periodic solutions with period ≈ 0.24 when wavenumber parameter $k_0 = 26$. These spatial oscillations are here resolved on a micro-grid of spacing 0.042. On medium times these spatial oscillations grow to near equilibrium amplitude of $\sqrt{\text{Ra}}$, and over very long times the phases of the oscillations evolve in space to adapt to the boundaries.

Set the desired microscale periodicity, and correspondingly choose random microscale diffusion coefficients (with subscripts shifted by a half).

```

42 clear all
43 cMap=jet(64); cMap=0.8*cMap(7:end-7,:); % set colormap
44 basename = ['r' num2str(floor(1e5*rem(now,1))) mfilename]
45 %global OurCf2eps, OurCf2eps=true %optional to save plots
46 Ra = 0.2 % Ra>0 leads to patterns
47 nGapFac = 2
48 waveLength = 0.5/nGapFac
49 nPtsPeriod = 6
50 dx = waveLength/nPtsPeriod
51 k0 = 2.1*pi/waveLength

```

The above factor 2.1 is close to $3/\sqrt{2} = 2.1213$ for which $(\pm 1, \pm 2)$ modes have same linear growth-rate as $(\pm 2, 0)$ modes.

Establish global data struct `patches` for the Swift–Hohenberg PDE on some square domain. For simplicity, use five patches in each direction. Quartic (fourth-order) interpolation `ordCC = 4` provides values for the inter-patch coupling conditions. Set `bcOffset` for different boundary conditions around the square domain.

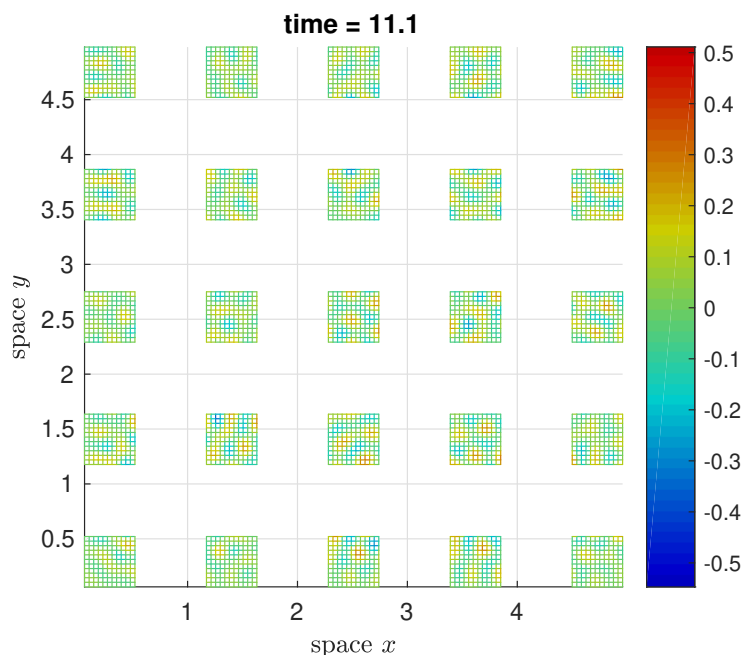
```

67 nPatch = 5
68 nSubP = 2*nPtsPeriod+4
69 Len = nPatch;
70 ordCC = 4;
71 dom.type='equispace';
72 dom.bcOffset=[0.5 0.5;1.0 1.5]
73 patches = configPatches2(@SwiftHohenbergPDE,[0 Len],dom ...
74     ,nPatch,ordCC,dx,nSubP,'EdgyInt',true,'nEdge',2);
75 xs=squeeze(patches.x);
76 ys=squeeze(patches.y);

```

Figure 4.10:

pattern field $u(x, y, t)$ in the patch scheme applied to a microscale discretisation of the 2D Swift–Hohenberg PDE. At this early time much of the random sub-patch microstructure has decayed leaving some random marginal modes starting to grow.



4.10.0.1 Simulate in time

Set an initial condition, and here integrate forward in time using a standard method for stiff systems. Integrate the interface `patchSys2` (Section 4.2) to the microscale differential equations (despite the extreme stiffness, `ode23` is ten times quicker than `ode15s`). Because pattern evolution is eventually phase-diffusion, here sample the pattern at quadratically varying times.

```

93 fprintf('\n**** Simulate in time\n')
94 u0 = 0.3*( -1+2*rand(size(patches.x+patches.y)) );
95 Ts=400*linspace(0,1,97).^2;
96 tic
97 [ts,us] = ode23(@patchSys2, Ts, u0(:),[],patches,k0,Ra);
98 simulateTime = toc
99 us = reshape(us',nSubP,nSubP,nPatch,nPatch,[]);

```

Plot the simulation such as that shown in Figures 4.10 to 4.15 First, reshape the data, omitting edge values.

```

135 xs([1:2 end-1:end],:) = nan;
136 ys([1:2 end-1:end],:) = nan;
137 us = reshape( permute(us,[1 3 2 4 5]) ...
138             ,nSubP*nPatch,nSubP*nPatch,[]);
139 uRange=[min(us(:)) max(us(:))];

```

Second, plot six examples of the evolving pattern, equi-spaced in time-index.

```

146 plots = round( 1+linspace(0,1,7)*(numel(ts)-1) )
147 for p=2:numel(plots)
148     figure(p),clf
149     mesh(xs(:),ys(:),us(:,:,plots(p)))

```

Figure 4.11:
 pattern field
 $u(x, y, t)$ in the
 patch scheme
 applied to a
 microscale dis-
 cretisation of
 the 2D Swift–
 Hohenberg PDE.
 By now the
 local sub-patch
 patterns have
 reached a quasi-
 equilibrium
 amplitude.

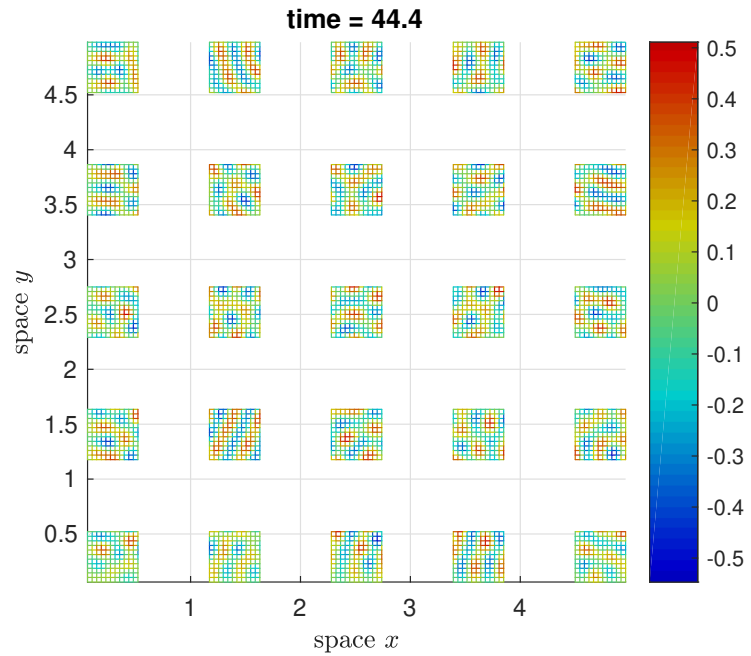
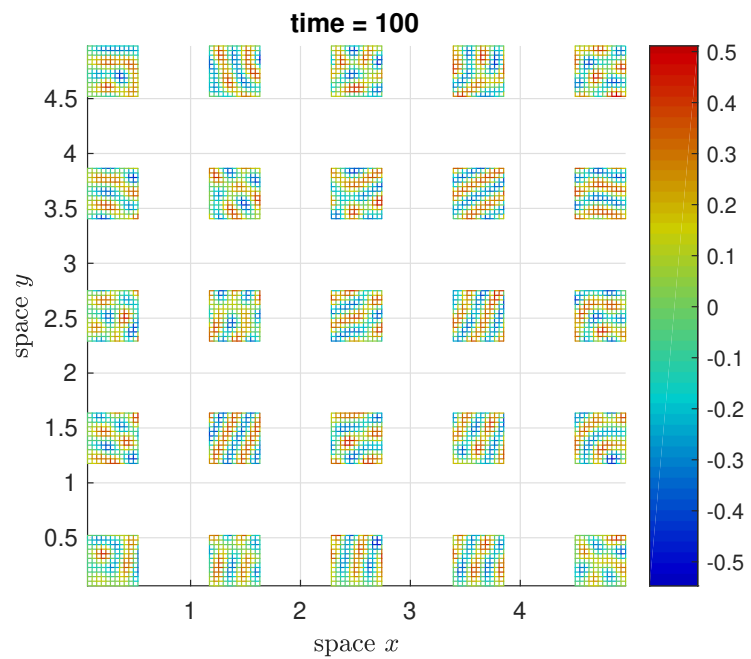
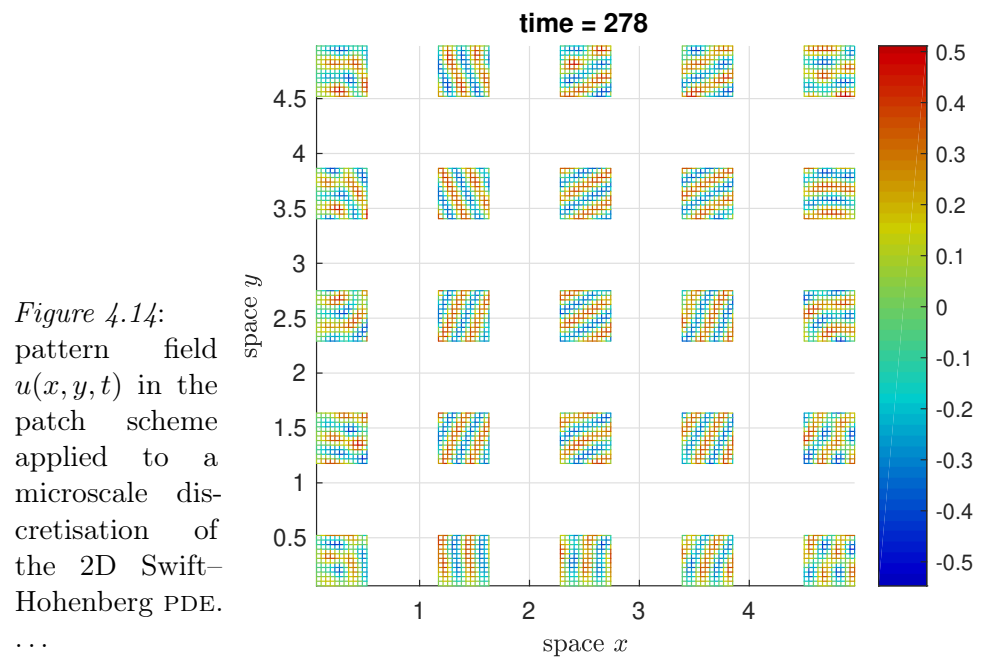
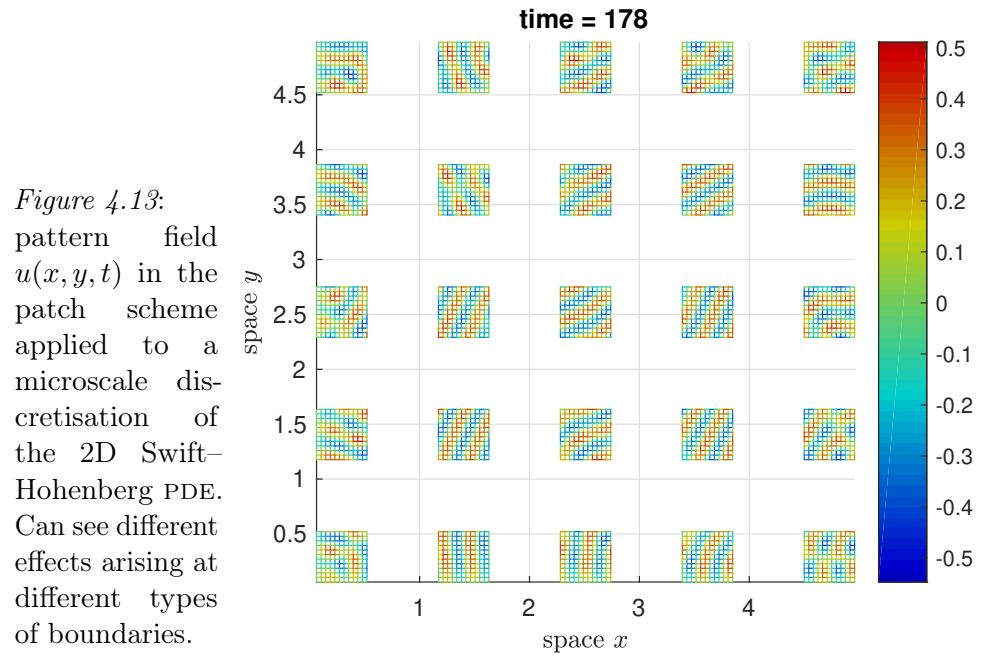
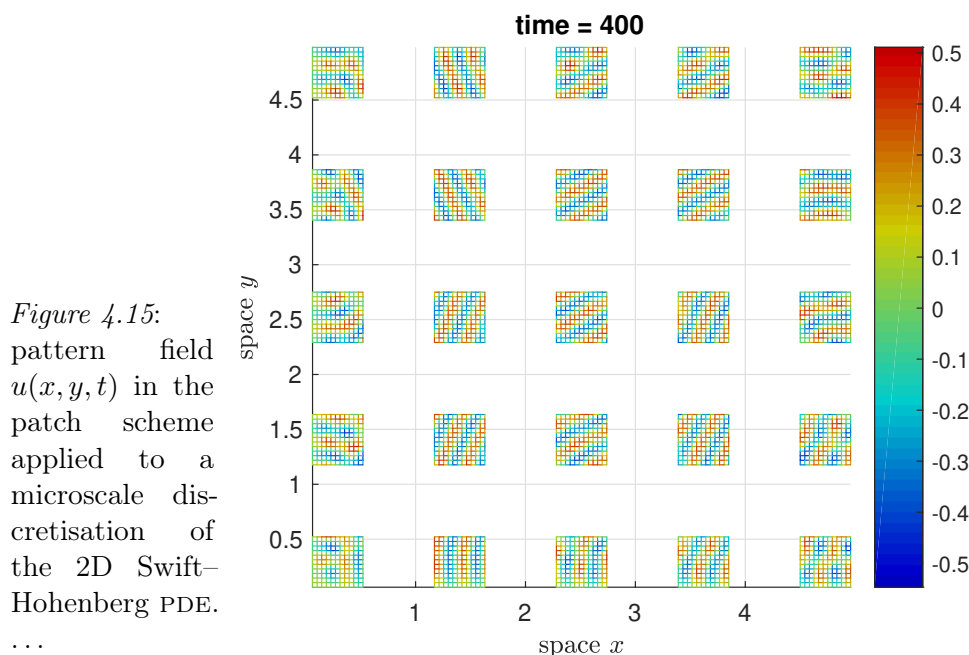


Figure 4.12:
 pattern field
 $u(x, y, t)$ in the
 patch scheme
 applied to a
 microscale dis-
 cretisation of
 the 2D Swift–
 Hohenberg PDE.
 Patterns within
 the patches are
 evolving to the
 preferred rolls,
 but with weak
 coupling to
 other patches.







```

150     axis equal, view(0,90)
151     caxis(uRange), colormap(cMap), colorbar
152     xlabel('space $x$'), ylabel('space $y$'), zlabel('$u(x,y,t)$')
153     title(['time = ' num2str(ts(plots(p)),3)])
154     ifOurCf2eps([basename num2str(p)], [12 11])
155 end%for p

```

Third, plot animation in time: starts after a key press.

```

161 %%
162 figure(1), clf
163 cf=mesh(xs(:),ys(:),us(:,:,1)');
164 axis equal, view(0,90)
165 caxis(uRange), colormap(cMap), colorbar
166 xlabel('space $x$'), ylabel('space $y$'), zlabel('$u(x,y,t)$')
167 title(['time = ' num2str(ts(1),3)])
168 ca=gca;
169 disp('Press any key to start animation'), pause
170 for p=2:numel(ts)
171     cf.ZData=us(:,:,p)';
172     cf.CData=us(:,:,p)';
173     ca.Title.String=['time = ' num2str(ts(p),3)];
174     pause(0.1)
175 end

```

Fin.

4.10.1 The Swift–Hohenberg PDE and BCs inside patches

As a microscale discretisation of Swift–Hohenberg PDE $u_t = -(1 + \nabla^2/k_0^2)^2 u + \text{Ra} u - u^3$, here code straightforward centred discretisation in space.

```

189 function ut=SwiftHohenbergPDE(t,u,patches,k0,Ra)
190     dx=diff(patches.x(1:2)); % microscale spacing
191     dy=diff(patches.y(1:2)); % microscale spacing
192     i=3:size(u,1)-2; % interior points in patches
193     j=3:size(u,2)-2; % interior points in patches

```

Code various boundary conditions. For slightly simpler coding, squeeze out the two singleton dimensions.

```

200     u = squeeze(u);
201     u(1:2,:,1,:)=0; % u=u_x=0 at x=0
202     u(:,1:2,:,1)=0; % u=u_y=0 at y=0
203     u(end-1,:,end,:)=0; % u=0 at x=L
204     u(end,:,end,:)= -u(end-2,:,end,:); % u_x=0 at x=L
205     u(:,end-1,:,end)= -u(:,end-2,:,end); % u_y=0 at y=L
206     u(:,end,:,end)= -u(:,end-3,:,end); % u_yyy=0 at y=L

```

Here code straightforward centred discretisation in space.

```

212     ut=nan+u; % preallocate output array
213     v = u(2:end-1,2:end-1,,:,:) ...
214         +( diff(u(:,2:end-1,,:,:),2,1)/dx^2 ...
215           +diff(u(2:end-1,,:,:),2,2)/dy^2 )/k0^2;
216     ut(i,j,,:,:) = -( v(2:end-1,2:end-1,,:,:) ...
217         +( diff(v(:,2:end-1,,:,:),2,1)/dx^2 ...
218           +diff(v(2:end-1,,:,:),2,2)/dy^2 )/k0^2 ) ...
219         +Ra*u(i,j,,:,:) -u(i,j,,:).^3;
220 end

```

5 Patches in 3D space

5.1 configPatches3(): configures spatial patches in 3D

Makes the struct `patches` for use by the patch/gap-tooth time derivative/step function `patchSys3()`, and possibly other patch functions. [Sections 5.1.1](#) and [5.4](#) list examples of its use.

```

20 function patches = configPatches3(fun,Xlim,Dom ...
21     ,nPatch,ordCC,dx,nSubP,varargin)
22 version = "2023-10-18";

```

Input If invoked with no input arguments, then executes an example of simulating a heterogeneous wave PDE—see [Section 5.1.1](#) for an example code.

- `fun` is the name of the user function, `fun(t,u,patches)` or `fun(t,u)` or `fun(t,u,patches,...)`, that computes time-derivatives (or time-steps) of quantities on the 3D micro-grid within all the 3D patches.
- `Xlim` array/vector giving the rectangular-cuboid macro-space domain of the computation: namely $[Xlim(1),Xlim(2)] \times [Xlim(3),Xlim(4)] \times [Xlim(5),Xlim(6)]$. If `Xlim` has two elements, then the domain is the cubic domain of the same interval in all three directions.

But, if `nPatch(n) = 1`, then in the spatial direction `n` the domain is the mean of `Xlim(2*n-1:2*n)` plus/minus half the patch-width in that direction.

- `Dom` sets the type of macroscale conditions for the patches, and reflects the type of microscale boundary conditions of the problem. If `Dom` is `NaN` or `[]`, then the field `u` is triply macro-periodic in the 3D spatial domain, and resolved on equi-spaced patches. If `Dom` is a character array, then that specifies the `.type` of the following structure, with `.bcOffset` set to the default zero. Otherwise `Dom` is a structure with the following components.
 - `.type`, char-array, of either `'periodic'` (the default, equivalently `'periodic '`), `'equispace'`, `'chebyshev'`, `'usergiven'`. For all cases except `'periodic'`, users *must* code into `fun` the micro-grid boundary conditions that apply at the left/right/bottom/top/back/front faces of the leftmost/rightmost/bottommost/topmost/backmost/frontmost patches, respectively.
 - `.bcOffset`, optional one, three or six element vector/array, in the cases of `'equispace'` or `'chebyshev'` the patches are placed so the left/right macroscale boundaries are aligned to the left/right faces of the corresponding extreme patches, but offset by `bcOffset` of the sub-patch micro-grid spacing. For example, use `bcOffset=0` when the micro-code applies Dirichlet boundary values on the extreme face micro-grid points, whereas use `bcOffset=0.5` when the microcode applies Neumann boundary conditions halfway between the extreme face micro-grid points. Similarly for the top, bottom, back, and front faces.

If `.bcOffset` is a scalar, then apply the same offset to all boundaries. If three elements, then apply the first offset to both x -boundaries, the second offset to both y -boundaries, and the third offset to both z -boundaries. If six elements, then apply the first two offsets to the respective x -boundaries, the middle two offsets to the respective y -boundaries, and the last two offsets to the respective z -boundaries.

- `.X`, optional vector/array with `nPatch(1)` elements, in the case 'usergiven' it specifies the x -locations of the centres of the patches—the user is responsible the locations makes sense.
- `.Y`, optional vector/array with `nPatch(2)` elements, in the case 'usergiven' it specifies the y -locations of the centres of the patches—the user is responsible the locations makes sense.
- `.Z`, optional vector/array with `nPatch(3)` elements, in the case 'usergiven' it specifies the z -locations of the centres of the patches—the user is responsible the locations makes sense.
- `nPatch` determines the number of spatial patches: if scalar, then use the same number of patches in all three directions, otherwise `nPatch(1:3)` gives the number (≥ 1) of patches in each direction.
- `ordCC` is the 'order' of interpolation for inter-patch coupling across empty space of the macroscale patch values to the face-values of the patches: currently must be 0, 2, 4, ...; where 0 gives spectral interpolation.
- `dx` (real—scalar or three elements) is usually the sub-patch micro-grid spacing in x , y and z . If scalar, then use the same `dx` in all three directions, otherwise `dx(1:3)` gives the spacing in each of the three directions.

However, if `Dom` is `NaN` (as for pre-2023), then `dx` actually is `ratio` (scalar or three elements), namely the ratio of (depending upon `EdgyInt`) either the half-width or full-width of a patch to the equi-spacing of the patch mid-points—adjusted a little when `nEdge` > 1. So either if not `EdgyInt` then `ratio` = $\frac{1}{2}$ means the patches abut and `ratio` = 1 is overlapping patches as in holistic discretisation, or if `EdgyInt` then `ratio` = 1 means the patches abut. Small `ratio` should greatly reduce computational time.

- `nSubP` is the number of equi-spaced microscale lattice points in each patch: if scalar, then use the same number in all three directions, otherwise `nSubP(1:3)` gives the number in each direction. If not using `EdgyInt`, then `nSubP./nEdge` must be odd integer(s) so that there is/are centre-patch lattice planes. So for the defaults of `nEdge` = 1 and not `EdgyInt`, then `nSubP` must be odd.
- '`nEdge`', *optional* (integer—scalar or three element), default=1, the width of face values set by interpolation at the face regions of each patch. If three elements, then respectively the width in x, y, z -directions.

The default is one (suitable for microscale lattices with only nearest neighbour interactions).

- **'EdgyInt'**, true/false, *optional*, default=false. If true, then interpolate to left/right/top/bottom/front/back face-values from right/left/bottom/top/back/front next-to-face values. If false or omitted, then interpolate from centre-patch planes.
- **'nEnsem'**, *optional-experimental*, default one, but if more, then an ensemble over this number of realisations.
- **'hetCoeffs'**, *optional*, default empty. Supply a 3D or 4D array of microscale heterogeneous coefficients to be used by the given microscale **fun** in each patch. Say the given array **cs** is of size $m_x \times m_y \times m_z \times n_c$, where n_c is the number of different arrays of coefficients. For example, in heterogeneous diffusion, $n_c = 3$ for the diffusivities in the *three* different spatial directions (or $n_c = 6$ for the diffusivity tensor). The coefficients are to be the same for each and every patch. However, macroscale variations are catered for by the n_c coefficients being n_c parameters in some macroscale formula.
 - If **nEnsem** = 1, then the array of coefficients is just tiled across the patch size to fill up each patch, starting from the (1, 1, 1)-point in each patch. Best accuracy usually obtained when the periodicity of the coefficients is a factor of **nSubP-2*nEdge** for **EdgyInt**, or a factor of **(nSubP-nEdge)/2** for not **EdgyInt**.
 - If **nEnsem** > 1 (value immaterial), then reset **nEnsem** := $m_x \cdot m_y \cdot m_z$ and construct an ensemble of all $m_x \cdot m_y \cdot m_z$ phase-shifts of the coefficients. In this scenario, the inter-patch coupling couples different members in the ensemble. When **EdgyInt** is true, and when the coefficients are diffusivities/elasticities in x, y, z -directions, respectively, then this coupling cunningly preserves symmetry.
- **'parallel'**, true/false, *optional*, default=false. If false, then all patch computations are on the user's main CPU—although a user may well separately invoke, say, a GPU to accelerate sub-patch computations.

If true, and it requires that you have MATLAB's Parallel Computing Toolbox, then **configPatches3** will distribute the patches over multiple CPUs/cores. In MATLAB, only one array dimension can be split in the distribution, so **configPatches3** chooses the one space dimension x, y, z corresponding to the highest **nPatch** (if a tie, then chooses the rightmost of x, y, z). A user may correspondingly distribute arrays with property **patches.codist**, or simply use formulas invoking the preset distributed arrays **patches.x**, **patches.y**, and **patches.z**. If a user has not yet established a parallel pool, then a 'local' pool is started.

Output The struct **patches** is created and set with the following components. If no output variable is provided for **patches**, then make the struct

available as a global variable.¹

```
233 if nargout==0, global patches, end
234 patches.version = version;
```

- `.fun` is the name of the user's function `fun(t,u,patches)` or `fun(t,u)` or `fun(t,u,patches,...)` that computes the time derivatives (or steps) on the patchy lattice.
- `.ordCC` is the specified order of inter-patch coupling.
- `.periodic`: either true, for interpolation on the macro-periodic domain; or false, for general interpolation by divided differences over non-periodic domain or unevenly distributed patches.
- `.stag` is true for interpolation using only odd neighbouring patches as for staggered grids, and false for the usual case of all neighbour coupling—not yet implemented.
- `.Cwtsr` and `.Cwtsl` are the `ordCC` $\times$ 3-array of weights for the inter-patch interpolation onto the right/top/front and left/bottom/back faces (respectively) with patch:macroscale ratio as specified or as derived from `dx`.
- `.x` (8D) is $\text{nSubP}(1) \times 1 \times 1 \times 1 \times 1 \times \text{nPatch}(1) \times 1 \times 1$ array of the regular spatial locations x_{iI} of the microscale grid points in every patch.
- `.y` (8D) is $1 \times \text{nSubP}(2) \times 1 \times 1 \times 1 \times 1 \times \text{nPatch}(2) \times 1$ array of the regular spatial locations y_{jJ} of the microscale grid points in every patch.
- `.z` (8D) is $1 \times 1 \times \text{nSubP}(3) \times 1 \times 1 \times 1 \times 1 \times \text{nPatch}(3)$ array of the regular spatial locations z_{kK} of the microscale grid points in every patch.
- `.ratio` 1×3 , for macro-periodic conditions, are the size ratios of every patch.
- `.nEdge` 1×3 , is the width of face values set by interpolation at the face regions of each patch, in the x, y, z -directions respectively.
- `.le, .ri, .bo, .to, .ba, .fr` determine inter-patch coupling of members in an ensemble. Each a column vector of length `nEnsem`.
- `.cs` either
 - [] 0D, or
 - if `nEnsem` = 1, $(\text{nSubP}(1) - 1) \times (\text{nSubP}(2) - 1) \times (\text{nSubP}(3) - 1) \times n_c$ 4D array of microscale heterogeneous coefficients, or
 - if `nEnsem` > 1, $(\text{nSubP}(1) - 1) \times (\text{nSubP}(2) - 1) \times (\text{nSubP}(3) - 1) \times n_c \times m_x m_y m_z$ 5D array of $m_x m_y m_z$ ensemble of phase-shifts of the microscale heterogeneous coefficients.

¹ When using `spmd` parallel computing, it is generally best to avoid global variables, and so instead prefer using an explicit output variable.

- `.parallel`, logical: true if patches are distributed over multiple CPUs/cores for the Parallel Computing Toolbox, otherwise false (the default is to activate the *local* pool).
- `.codist`, *optional*, describes the particular parallel distribution of arrays over the active parallel pool.

5.1.1 If no arguments, then execute an example

```

323 if nargin==0
324 disp('With no arguments, simulate example of heterogeneous wave')

```

The code here shows one way to get started: a user's script may have the following three steps (“ $\mapsto$ ” denotes function recursion).

1. `configPatches3`
2. `ode23` integrator $\mapsto$ `patchSys3` $\mapsto$ user's PDE
3. process results

Set random heterogeneous coefficients of period two in each of the three directions. Crudely normalise by the harmonic mean so the macro-wave time scale is roughly one.

```

342 mPeriod = [2 2 2];
343 cHetr = exp(0.9*randn([mPeriod 3]));
344 cHetr = cHetr*mean(1./cHetr(:))

```

Establish global patch data struct to interface with a function coding a nonlinear ‘diffusion’ PDE: to be solved on $[-\pi, \pi]^3$ -periodic domain, with 5^3 patches, spectral interpolation (0) couples the patches, each patch with micro-grid spacing 0.22 (relatively large for visualisation), and with 4^3 points forming each patch.

```

356 global patches
357 patches = configPatches3(@heteroWave3,[-pi pi] ...
358     , 'periodic' , 5, 0, 0.22, mPeriod+2 , 'EdgyInt', true ...
359     , 'hetCoeffs', cHetr);

```

Set a wave initial state using auto-replication of the spatial grid, and as [Figure 5.1](#) shows. This wave propagates diagonally across space. Concatenate the two u, v -fields to be the two components of the fourth dimension.

```

369 u0 = 0.5+0.5*sin(patches.x+patches.y+patches.z);
370 v0 = -0.5*cos(patches.x+patches.y+patches.z)*sqrt(3);
371 uv0 = cat(4,u0,v0);

```

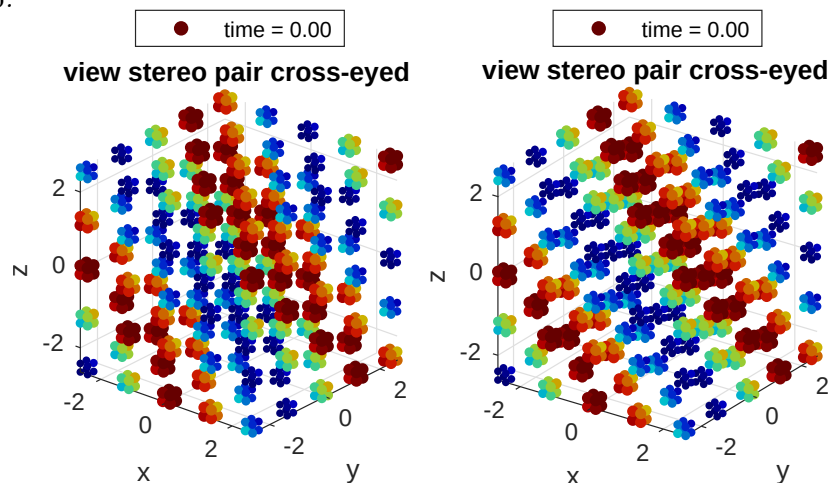
Integrate in time to $t = 6$ using standard functions. In Matlab `ode15s` would be natural as the patch scheme is naturally stiff, but `ode23` is much quicker ([Maclean et al. 2020](#), Fig. 4).

```

388 disp('Simulate heterogeneous wave u_tt=div[C*grad(u)]')
389 if ~exist('OCTAVE_VERSION','builtin')
390     [ts,us] = ode23(@patchSys3,linspace(0,6),uv0(:));
391 else %disp('octave version is very slow for me')

```

Figure 5.1: initial field $u(x, y, z, t)$ at time $t = 0$ of the patch scheme applied to a heterogeneous wave PDE: Figure 5.2 plots the computed field at time $t = 6$.



```

392     lsode_options('absolute tolerance',1e-4);
393     lsode_options('relative tolerance',1e-4);
394     [ts,us] = ode0cts(@patchSys3,[0 1 2],uv0(:));
395 end

```

Animate the computed simulation to end with Figure 5.2. Use `patchEdgeInt3` to obtain patch-face values in order to most easily reconstruct the array data structure.

Replicate x , y , and z arrays to get individual spatial coordinates of every data point. Then, optionally, set faces to `nan` so the plot just shows patch-interior data.

```

409 figure(1), clf, colormap(0.8*jet)
410 xs = patches.x+0*patches.y+0*patches.z;
411 ys = patches.y+0*patches.x+0*patches.z;
412 zs = patches.z+0*patches.y+0*patches.x;
413 if 1, xs([1 end],:,:)=nan;
414     xs(:,[1 end],:,:)=nan;
415     xs(:,:[1 end],:)=nan;
416 end;%option
417 j=find(~isnan(xs));

```

In the scatter plot, these functions `pix()` and `col()` map the u -data values to the size of the dots and to the colour of the dots, respectively.

```

425 pix = @(u) 15*abs(u)+7;
426 col = @(u) sign(u).*abs(u);

```

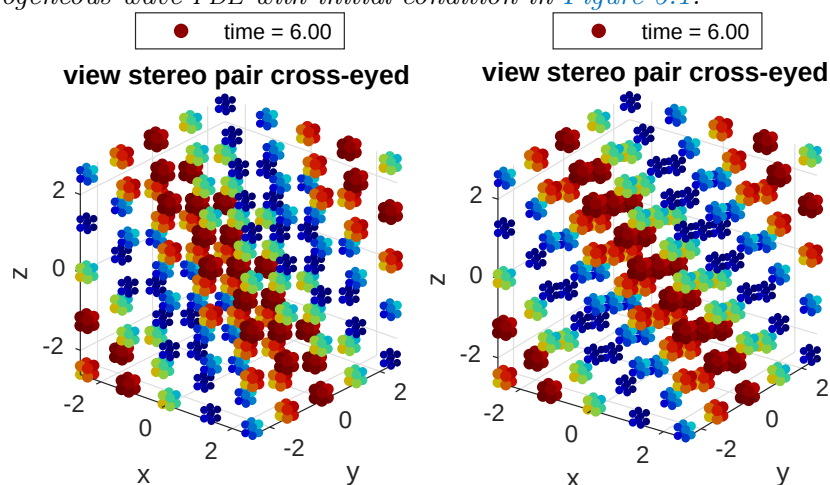
Loop to plot at each and every time step.

```

432 for i = 1:length(ts)
433     uv = patchEdgeInt3(us(i,:));
434     u = uv(:,:,:,1,:);

```

Figure 5.2: field $u(x, y, z, t)$ at time $t = 6$ of the patch scheme applied to the heterogeneous wave PDE with initial condition in [Figure 5.1](#).



```

435 for p=1:2
436     subplot(1,2,p)
437     if (i==1) | exist('OCTAVE_VERSION','builtin')
438         scat(p) = scatter3(xs(j),ys(j),zs(j),'filled');
439         axis equal, caxis(col([0 1])), view(45-5*p,25)
440         xlabel('$x$'), ylabel('$y$'), zlabel('$z$')
441         title('view stereo pair cross-eyed')
442     end % in matlab just update values
443     set(scat(p),'CData',col(u(j)) ...
444         , 'SizeData',pix((8+xs(j)-ys(j)+zs(j))/6+0*u(j)));
445     legend(['time = ' num2str(ts(i),'%4.2f')'],'Location','north')
446 end

```

Optionally save the initial condition to graphic file for [Figure 4.1](#), and optionally save the last plot.

```

454 if i==1,
455     ifOurCf2eps([mfilename 'ic'])
456     disp('Type space character to animate simulation')
457     pause
458 else pause(0.05)
459 end
460 end% i-loop over all times
461 ifOurCf2eps([mfilename 'fin'])

```

Upon finishing execution of the example, exit this function.

```

476 return
477 end%if no arguments

```

5.1.2 heteroWave3(): heterogeneous Waves

This function codes the lattice heterogeneous waves inside the patches. The wave PDE is

$$u_t = v, \quad v_t = \vec{\nabla}(C\vec{\nabla} \cdot u)$$

for diagonal matrix C which has microscale variations. For 8D input arrays u , x , y , and z (via edge-value interpolation of `patchSys3`, [Section 5.2](#)), computes the time derivative at each point in the interior of a patch, output in `ut`. The three 3D array of heterogeneous coefficients, c_{ijk}^x , c_{ijk}^y and c_{ijk}^z , have previously been stored in `patches.cs` (4D).

Supply patch information as a third argument (required by parallel computation), or otherwise by a global variable.

```
26 function ut = heteroWave3(t,u,patches)
27     if nargin<3, global patches, end
```

Microscale space-steps, and interior point indices.

```
33     dx = diff(patches.x(2:3)); % x micro-scale step
34     dy = diff(patches.y(2:3)); % y micro-scale step
35     dz = diff(patches.z(2:3)); % z micro-scale step
36     i = 2:size(u,1)-1; % x interior points in a patch
37     j = 2:size(u,2)-1; % y interior points in a patch
38     k = 2:size(u,3)-1; % z interior points in a patch
```

Reserve storage and then assign interior patch values to the heterogeneous diffusion time derivatives. Using `nan+u` appears quicker than `nan(size(u),patches.codist)`

```
46     ut = nan(size(u),'like',u); % preallocate output array
47     ut(i,j,k,1,:) = u(i,j,k,2,:);
48     ut(i,j,k,2,:) ...
49     =diff(patches.cs(:,j,k,1,:).*diff(u(:,j,k,1,:),1),1)/dx^2 ...
50     +diff(patches.cs(i,:,k,2,:).*diff(u(i,:,k,1,:),1,2),1,2)/dy^2 ...
51     +diff(patches.cs(i,j,:,3,:).*diff(u(i,j,:,1,:),1,3),1,3)/dz^2;
52 end% function
```

5.2 patchSys3(): interface 3D space to time integrators

To simulate in time with 3D spatial patches we often need to interface a users time derivative function with time integration routines such as `ode23` or `PIRK2`. This function provides an interface. Communicate patch-design variables ([Section 5.1](#)) either via the global struct `patches` or via an optional third argument. `patches` is required for the parallel computing of `spmd`, or if parameters are to be passed though to the user microscale function.

```

23 function dudt = patchSys3(t,u,patches,varargin)
24 if nargin<3, global patches, end

```

Input

- `u` is a vector/array of length $\text{prod}(\text{nSubP}) \cdot \text{nVars} \cdot \text{nEnsem} \cdot \text{prod}(\text{nPatch})$ where there are $\text{nVars} \cdot \text{nEnsem}$ field values at each of the points in the $\text{nSubP}(1) \times \text{nSubP}(2) \times \text{nSubP}(3) \times \text{nPatch}(1) \times \text{nPatch}(2) \times \text{nPatch}(3)$ spatial grid.
- `t` is the current time to be passed to the user's time derivative function.
- `patches` a struct set by `configPatches3()` with the following information used here.
 - `.fun` is the name of the user's function `fun(t,u,patches,...)` that computes the time derivatives on the patchy lattice. The array `u` has size $\text{nSubP}(1) \times \text{nSubP}(2) \times \text{nSubP}(3) \times \text{nVars} \times \text{nEnsem} \times \text{nPatch}(1) \times \text{nPatch}(2) \times \text{nPatch}(3)$. Time derivatives must be computed into the same sized array, although herein the patch edge-values are overwritten by zeros.
 - `.x` is $\text{nSubP}(1) \times 1 \times 1 \times 1 \times \text{nPatch}(1) \times 1 \times 1$ array of the spatial locations x_i of the microscale (i, j, k) -grid points in every patch. Currently it *must* be an equi-spaced lattice on both macro- and microscales.
 - `.y` is similarly $1 \times \text{nSubP}(2) \times 1 \times 1 \times 1 \times \text{nPatch}(2) \times 1$ array of the spatial locations y_j of the microscale (i, j, k) -grid points in every patch. Currently it *must* be an equi-spaced lattice on both macro- and microscales.
 - `.z` is similarly $1 \times 1 \times \text{nSubP}(3) \times 1 \times 1 \times 1 \times \text{nPatch}(3)$ array of the spatial locations z_k of the microscale (i, j, k) -grid points in every patch. Currently it *must* be an equi-spaced lattice on both macro- and microscales.
- `varargin`, optional, is arbitrary list of parameters to be passed onto the users time-derivative function as specified in `configPatches3`.

Output

- `dudt` is a vector/array of time derivatives, but with patch edge-values set to zero. It is of total length `prod(nSubP) · nVars · nEnsem · prod(nPatch)` and the same dimensions as `u`.

5.3 patchEdgeInt3(): sets 3D patch face values from 3D macroscale interpolation

Couples 3D patches across 3D space by computing their face values via macroscale interpolation. Assumes patch face values are determined by macroscale interpolation of the patch centre-plane values (Roberts et al. 2014, Bunder et al. 2021), or patch next-to-face values which appears better (Bunder et al. 2020). This function is primarily used by patchSys3() but is also useful for user graphics.²

Communicate patch-design variables via a second argument (optional, except required for parallel computing of spmd), or otherwise via the global struct patches.

```

27 function u = patchEdgeInt3(u,patches)
28 if nargin<2, global patches, end

```

Input

- **u** is a vector/array of length $\text{prod}(\text{nSubP}) \cdot \text{nVars} \cdot \text{nEnsem} \cdot \text{prod}(\text{nPatch})$ where there are $\text{nVars} \cdot \text{nEnsem}$ field values at each of the points in the $\text{nSubP1} \cdot \text{nSubP2} \cdot \text{nSubP3} \cdot \text{nPatch1} \cdot \text{nPatch2} \cdot \text{nPatch3}$ multiscale spatial grid on the $\text{nPatch1} \cdot \text{nPatch2} \cdot \text{nPatch3}$ array of patches.
- **patches** a struct set by configPatches3() which includes the following information.
 - **.x** is $\text{nSubP1} \times 1 \times 1 \times 1 \times 1 \times \text{nPatch1} \times 1 \times 1$ array of the spatial locations x_{iI} of the microscale grid points in every patch. Currently it *must* be an equi-spaced lattice on the microscale index i , but may be variable spaced in macroscale index I .
 - **.y** is similarly $1 \times \text{nSubP2} \times 1 \times 1 \times 1 \times 1 \times \text{nPatch2} \times 1$ array of the spatial locations y_{jJ} of the microscale grid points in every patch. Currently it *must* be an equi-spaced lattice on the microscale index j , but may be variable spaced in macroscale index J .
 - **.z** is similarly $1 \times 1 \times \text{nSubP3} \times 1 \times 1 \times 1 \times 1 \times \text{nPatch3}$ array of the spatial locations z_{kK} of the microscale grid points in every patch. Currently it *must* be an equi-spaced lattice on the microscale index k , but may be variable spaced in macroscale index K .
 - **.ordCC** is order of interpolation, currently only $\{0, 2, 4, \dots\}$
 - **.periodic** indicates whether macroscale is periodic domain, or alternatively that the macroscale has left, right, top, bottom, front and back boundaries so interpolation is via divided differences.
 - **.stag** in $\{0, 1\}$ is one for staggered grid (alternating) interpolation. Currently must be zero.

² Script patchEdgeInt3test.m verifies most of this code.

- `.Cwtsr` and `.Cwtsl` are the coupling coefficients for finite width interpolation in each of the x, y, z -directions—when invoking a periodic domain.
- `.EdgyInt`, true/false, for determining patch-edge values by interpolation: true, from opposite-edge next-to-edge values (often preserves symmetry); false, from centre cross-patch values (near original scheme).
- `.nEdge`, three elements, the width of edge values set by interpolation at the x, y, z -face regions, respectively, of each patch (default is one all x, y, z -faces).
- `.nEnsem` the number of realisations in the ensemble.
- `.parallel` whether serial or parallel.

Output

- `u` is 8D array, $\text{nSubP1} \cdot \text{nSubP2} \cdot \text{nSubP3} \cdot \text{nVars} \cdot \text{nEnsem} \cdot \text{nPatch1} \cdot \text{nPatch2} \cdot \text{nPatch3}$, of the fields with face values set by interpolation.

5.4 homoDiffEdgy3: computational homogenisation of a 3D diffusion via simulation on small patches

Simulate heterogeneous diffusion in 3D space on 3D patches as an example application. Then compute macroscale eigenvalues of the patch scheme applied to this heterogeneous diffusion to validate and to compare various orders of inter-patch interpolation.

This code extends to 3D the 2D code discussed in ???. First set random heterogeneous diffusivities of random (small) period in each of the three directions. Crudely normalise by the harmonic mean so the decay time scale is roughly one.

```
29 mPeriod = randi([3 4],1,3)
30 cHetr = exp(0.3*randn([mPeriod 3]));
31 cHetr = cHetr*mean(1./cHetr(:))
```

Configure the patch scheme with some arbitrary choices of domain, patches, size ratios. Use spectral interpolation as we test other orders subsequently. In 3D we appear to get only real eigenvalues by using edgy interpolation. What happens for non-edgy interpolation is unknown.

```
42 nSubP=mPeriod+2;
43 nPatch=[5 5 5];
44 configPatches3(@heteroDiff3, [-pi pi], nan, nPatch ...
45     ,0, 0.3, nSubP, 'EdgyInt',true ...
46     , 'hetCoeffs',cHetr );
```

5.4.1 Simulate heterogeneous diffusion

Set initial conditions of a simulation as shown in [Figure 5.3](#).

```
56 global patches
57 u0 = exp(-patches.x.^2/4-patches.y.^2/2-patches.z.^2);
58 u0 = u0.*(1+0.3*rand(size(u0)));
```

Integrate using standard integrators, unevenly spaced in time to better display transients.

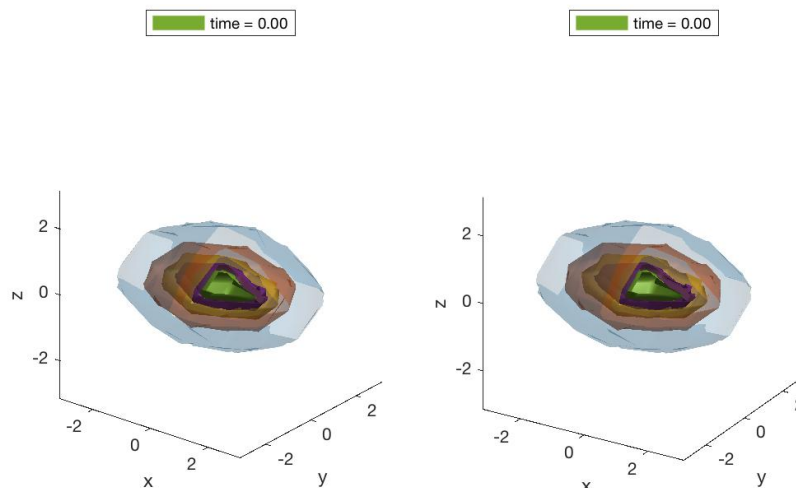
```
76 if ~exist('OCTAVE_VERSION','builtin')
77     [ts,us] = ode23(@patchSys3, 0.3*linspace(0,1,50).^2, u0(:));
78 else % octave version
79     [ts,us] = odeOcts(@patchSys3, 0.3*linspace(0,1).^2, u0(:));
80 end
```

Plot the solution as an animation over time.

```
88 figure(1), clf
89 rgb=get(gca,'defaultAxesColorOrder');
90 colormap(0.8*hsv)
```

Get spatial coordinates of patch interiors.

Figure 5.3: initial field $u(x,y,z,0)$ of the patch scheme applied to a heterogeneous diffusion PDE. Plotted are the isosurfaces at field values $u = 0.1, 0.3, \dots, 0.9$, with the front quadrant omitted so you can see inside. [Figure 5.4](#) plots the isosurfaces of the computed field at time $t = 0.3$.



```

96 x = reshape( patches.x([2:end-1],:,:,:) , [], 1);
97 y = reshape( patches.y(:, [2:end-1],:,:) , [], 1);
98 z = reshape( patches.z(:, :, [2:end-1],:) , [], 1);

```

For every time step draw the surface and pause for a short display.

```

105 for i = 1:length(ts)

```

Get the row vector of data, form into a 6D array, then omit patch faces, and reshape to suit the isosurface function. We do not use interpolation to get face values as the interpolation omits the corner edges and so breaks up the isosurfaces.

```

115     u = reshape( us(i,:) , [nSubP nPatch]);
116     u = u([2:end-1], [2:end-1], [2:end-1], :,:,:) ;
117     u = reshape( permute(u, [1 4 2 5 3 6]) ...
118               , [numel(x) numel(y) numel(z)]);

```

Optionally cut-out the front corner so we can see inside.

```

124     u( (x>0) & (y'<0) & (shiftdim(z,-2)>0) ) = nan;

```

The `isosurface` function requires us to transpose x and y .

```

131 v = permute(u, [2 1 3]);

```

Draw cross-eyed stereo view of some isosurfaces.

```

137     clf;
138     for p=1:2
139         subplot(1,2,p)
140         for iso=5:-1:1
141             isov=(iso-0.5)/5;
142             hsurf(iso) = patch(isosurface(x,y,z,v,isov));
143             isonormals(x,y,z,v,hsurf(iso))
144             set(hsurf(iso) , 'FaceColor',rgb(iso,:) ...
145                 , 'EdgeColor','none' ...
146                 , 'FaceAlpha',iso/5);
147             hold on
148         end
149         axis equal, view(45-7*p,25)
150         axis(pi*[-1 1 -1 1 -1 1])
151         xlabel('$x$'), ylabel('$y$'), zlabel('$z$')
152         legend(['time = ' num2str(ts(i),'%4.2f')'], 'Location','north')
153         camlight, lighting gouraud
154         hold off
155     end% each p
156     if i==1 % pause for the viewer
157         makeJpeg=false;
158         if makeJpeg, print(['Figs/' mfilename 't0'],'-djpeg'), end
159         disp('Press any key to start animation of isosurfaces')
160         pause
161     else pause(0.05)
162     end

    Finish the animation loop, and optionally output the isosurfaces of the final
    field, Figure 5.4.

178 end%for over time
179 if makeJpeg, print(['Figs/' mfilename 'tFin'],'-djpeg'), end

```

5.4.2 Compute Jacobian and its spectrum

Let's explore the Jacobian dynamics for a range of orders of interpolation, all for the same random patch design and heterogeneity. Except here use a small ratio as we do not plot and then the scale separation is clearest.

```

195 ratio = 0.025*(1+rand(1,3))
196 nSubP=randi([3 5],1,3)
197 nPatch=[3 3 3]
198 nEnsem = prod(mPeriod) % or just set one

```

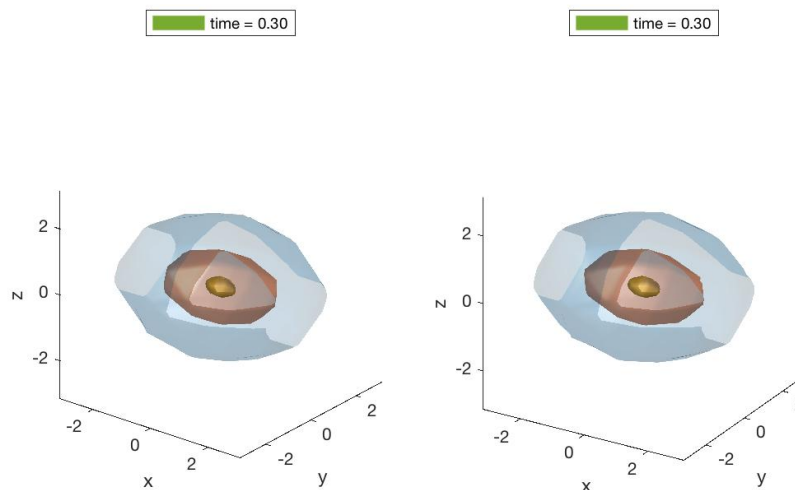
Find which elements of the 8D array are interior micro-grid points and hence correspond to dynamical variables.

```

205 u0 = zeros([nSubP,1,nEnsem,nPatch]);
206 u0([1 end],:,:,:) = nan;
207 u0(:,[1 end],:,:) = nan;
208 u0(:,:,[1 end],:) = nan;

```

Figure 5.4: final field $u(x, y, z, 0.3)$ of the patch scheme applied to a heterogeneous diffusion PDE. Plotted are the isosurfaces at field values $u = 0.1, 0.3, \dots, 0.9$, with the front quadrant omitted so you can see inside.



```

209     i = find(~isnan(u0));
210     nJac = length(i)
211     assert(nJac<9000 ...
212           , 'Jacobian is too big to quickly generate and analyse')

```

Store this many eigenvalues in array across different orders of interpolation.

```

219     nLeadEvals=prod(nPatch)+max(nPatch);
220     leadingEvals=[];

```

Evaluate eigenvalues for spectral as the base case for polynomial interpolation of order 2, 4, ...

```

228     maxords=6;
229     for ord=0:2:maxords
230         ord=ord

```

Configure with same heterogeneity.

```

236         configPatches3(@heteroDiff3, [-pi pi], nan, nPatch ...
237             , ord, ratio, nSubP, 'EdgyInt', true, 'nEnsem', nEnsem ...
238             , 'hetCoeffs', cHetr);

```

Construct the Jacobian of the scheme as the matrix of the linear transformation, obtained by transforming the standard unit vectors or via auto-differentiation.

```

246         tic

```

```

247     if exist('AutoDiff','class')
248         disp('**** using AutoDiff Jac')
249         Jac=AutoDiffJacobianAutoDiff(@(u) patchSys3(0,u),u0,i);
250         Jac=full(Jac(i,:));
251     else
252         Jac = nan(nJac);
253         for j = 1:nJac
254             u = u0(:)+(i(j)==(1:numel(u0)))';
255             tmp = patchSys3(0,u);
256             Jac(:,j) = tmp(i);
257         end
258     end%if exist
259     JacTime=toc

```

Test for symmetry, with error if we know it should be symmetric.

```

266     notSymmetric=norm(Jac-Jac')
267     % if notSymmetric>1e-7, spy(abs(Jac-Jac')>1e-7), end%??
268     assert(notSymmetric<1e-7,'failed symmetry')

```

Find all the eigenvalues (as `eigs` is unreliable), and put eigenvalues in a vector.

```

275     [evecs,evals] = eig((Jac+Jac')/2,'vector');
276     biggestImag=max(abs(imag(evals)));
277     if biggestImag>0, biggestImag=biggestImag, end

```

Sort eigenvalues on their real-part with most positive first, and most negative last. Store the leading eigenvalues in `egs`, and write out when computed all orders. The number of zero eigenvalues, `nZeroEv`, gives the number of decoupled systems in this patch configuration.

```

287     [~,k] = sort(-real(evals));
288     evals=evals(k); evecs=evecs(:,k);
289     if ord==0, nZeroEv=sum(abs(evals(:))<1e-5), end
290     % if ord==0, evec0=evecs(:,1:nZeroEv*nLeadEvals);
291     % else % find evec closest to that of each leading spectral
292     %     [~,k]=max(abs(evecs'*evec0));
293     %     evals=evals(k); % re-sort in corresponding order
294     % end
295     leadingEvals=[leadingEvals evals(nZeroEv*(1:nLeadEvals))];
296 end
297 disp('    spectral    quadratic    quartic    sixth-order ...')
298 leadingEvals=leadingEvals

```

5.4.3 heteroDiff3(): heterogeneous diffusion

This function codes the lattice heterogeneous diffusion inside the patches. For 8D input array `u` (via edge-value interpolation of `patchEdgeInt3`, such as by `patchSys3`, [Section 5.2](#)), computes the time derivative (3.1) at each point in the interior of a patch, output in `ut`. The three 3D array of diffusivities, c_{ijk}^x , c_{ijk}^y and c_{ijk}^z , have previously been stored in `patches.cs` (4+D).

Supply patch information as a third argument (required by parallel computation), or otherwise by a global variable.

```

23 function ut = heteroDiff3(t,u,patches)
24     if nargin<3, global patches, end

```

Microscale space-steps. Q: is using i,j,k slower than 2:end-1??

```

31     dx = diff(patches.x(2:3)); % x micro-scale step
32     dy = diff(patches.y(2:3)); % y micro-scale step
33     dz = diff(patches.z(2:3)); % z micro-scale step
34     i = 2:size(u,1)-1; % x interior points in a patch
35     j = 2:size(u,2)-1; % y interior points in a patch
36     k = 2:size(u,3)-1; % z interior points in a patch

```

Reserve storage and then assign interior patch values to the heterogeneous diffusion time derivatives. Using `nan+u` appears quicker than `nan(size(u),patches.codist)`

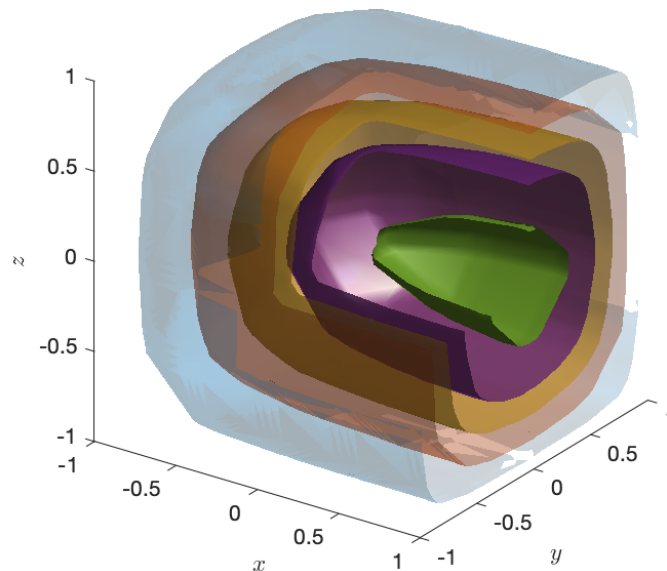
```

44     ut = nan(size(u),'like',u); % reserve storage
45     ut(i,j,k,:,:,:,:,:) ...
46     = diff(patches.cs(:,j,k,1,:)).*diff(u(:,j,k,:,:,:,:,:),1,1)/dx^2 ...
47     +diff(patches.cs(i,:,k,2,:)).*diff(u(i,:,k,:,:,:,:,:),1,2,1,2)/dy^2 ...
48     +diff(patches.cs(i,j,:,3,:)).*diff(u(i,j,:,:,:,:,:,:),1,3,1,3)/dz^2;
49 end% function

```

Fin.

Figure 5.5: macroscale of the random heterogeneous diffusion in 3D with boundary conditions of zero on all faces except for the Neumann condition on $x = 1$ (Section 5.5). The small patches are equispaced in space.



5.5 homoDiffBdryEquil3: equilibrium via computational homogenisation of a 3D heterogeneous diffusion on small patches

Find the equilibrium of a forced heterogeneous diffusion in 3D space on 3D patches as an example application. Boundary conditions are Neumann on the right face of the cube, and Dirichlet on the other faces. Figure 5.5 shows five isosurfaces of the 3D solution field.

Clear variables, and establish globals.

```

33 clear all
34 global patches
35 %global OurCf2eps, OurCf2eps=true %option to save plots

```

Set random heterogeneous diffusivities of random (small) period in each of the three directions. Crudely normalise by the harmonic mean so the decay time scale is roughly one.

```

46 mPeriod = randi([2 3],1,3)
47 cDiff = exp(0.3*randn([mPeriod 3]));
48 cDiff = cDiff*mean(1./cDiff(:))

```

Configure the patch scheme with some arbitrary choices of cubic domain, patches, and micro-grid spacing 0.02 (patches cover less than 1% of space). Use high order interpolation as few patches in each direction. Configure for Dirichlet boundaries except for Neumann on the right x -face.

```

60 nSubP = mPeriod+2;
61 nPatch = 7;
62 Dom.type = 'equispace';
63 Dom.bcOffset = zeros(2,3); Dom.bcOffset(2) = 0.5;
64 configPatches3(@microDiffBdry3, [-1 1], Dom ...
65     , nPatch, 0, 0.02, nSubP, 'EdgyInt',true ...
66     , 'hetCoeffs', cDiff );

```

Set forcing, and store in global patches for access by the microcode

```

74 patches.fu = 10*exp(-patches.x.^2-patches.y.^2-patches.z.^2);
75 patches.fu = patches.fu.*(1+rand(size(patches.fu)));

```

Solve for steady state Set initial guess of zero, with NaN to indicate patch-edge values. Index *i* are the indices of patch-interior points, store in global patches for access by `theRes`, and the number of unknowns is then its number of elements.

```

88 u0 = zeros([nSubP,1,1,nPatch,nPatch,nPatch]);
89 u0([1 end],:,:,:) = nan;
90 u0(:,[1 end],:,:) = nan;
91 u0(:,:,1 end),:) = nan;
92 i = find(~isnan(u0));
93 nVariables = numel(i)

```

Could use `fsolve` for simplicity and robustness, via the generic patch system wrapper `theRes` ([Section 3.15](#)). However, for such linear problems it is 100 times faster to use `AutoDiff` to get Jacobian, and then solve directly.

```

103 U0=AutoDiff(0*u0); % set an AD array of zeros
104 tic
105 F0 = patchSys3(1,U0); % evaluate patch system and its derivatives
106 Jac = getderivs(F0); % the Jacobian is the derivatives
107 uSoln = -Jac(i,i)\getvalue(F0(i)); % solve linear system
108 AdsolveTime = toc

```

Store the solution into the patches, and give magnitudes.

```

114 normSoln = norm(uSoln)
115 u0(i) = uSoln;
116 f0 = patchSys3(1,u0);
117 normResidual = norm(f0(i),Inf)
118 assert(normResidual<1e-10)
119 u0 = patchEdgeInt3(u0);

```

Plot isosurfaces of the solution

```

128 figure(1), clf
129 rgb=get(gca,'defaultAxesColorOrder');

```

Reshape spatial coordinates of patches.

```

135 x = patches.x(:); y = patches.y(:); z = patches.z(:);

```


Draw isosurfaces. Get the solution with interpolated faces, form into a 6D array, and reshape and transpose x and y to suit the isosurface function.

```

143 u = reshape( permute(squeeze(u0),[2 5 1 4 3 6]) ...
144             , [numel(y) numel(x) numel(z)]);
145 maxu=max(u(:)), minu=min(u(:))

```

Optionally cut-out the front corner so we can see inside.

```

151 u( (x'>0) & (y<0) & (shiftdim(z,-2)>0) ) = nan;

```

Draw some isosurfaces.

```

157 clf;
158 for iso=5:-1:1
159     isov=(iso-0.5)/5*(maxu-minu)+minu;
160     hsurf(iso) = patch(isosurface(x,y,z,u,isov));
161     isonormals(x,y,z,u,hsurf(iso))
162     set(hsurf(iso) , 'FaceColor',rgb(iso,:) ...
163         , 'EdgeColor','none' , 'FaceAlpha',iso/5);
164     hold on
165 end
166 hold off
167 axis equal, axis([-1 1 -1 1 -1 1]), view(35,25)
168 xlabel('$x$'), ylabel('$y$'), zlabel('$z$')
169 camlight, lighting gouraud
170 if OurCf2eps(mfilename) %optionally save plot
171 if exist('OurCf2eps') && OurCf2eps, print('-dpng',['Figs/' mfilename]), end

```

5.5.1 microDiffBdry3(): 3D forced heterogeneous diffusion with boundaries

This function codes the lattice forced heterogeneous diffusion inside the 3D patches. For 8D input array u (via edge-value interpolation of `patchEdgeInt3`, such as by `patchSys3`, [Section 5.2](#)), computes the time derivative at each point in the interior of a patch, output in ut . The three 3D array of diffusivities, c_{ijk}^x , c_{ijk}^y and c_{ijk}^z , have previously been stored in `patches.cs` (4D).

Supply patch information as a third argument (required by parallel computation), or otherwise by a global variable.

```

197 function ut = microDiffBdry3(t,u,patches)
198     if nargin<3, global patches, end

```

Microscale space-steps.

```

204 dx = diff(patches.x(2:3)); % x micro-scale step
205 dy = diff(patches.y(2:3)); % y micro-scale step
206 dz = diff(patches.z(2:3)); % z micro-scale step
207 i = 2:size(u,1)-1; % x interior points in a patch
208 j = 2:size(u,2)-1; % y interior points in a patch
209 k = 2:size(u,3)-1; % z interior points in a patch

```

Code microscale boundary conditions of say Neumann on right, and Dirichlet on left, top, bottom, front, and back (viewed along the z -axis).

```

217     u( 1 ,:,:,:,:, 1 ,:,:)=0; %left face of leftmost patch
218     u(end,:,:,:,:end,:,:)=u(end-1,:,:,:,:end,:,:); %right face of rightmost
219     u(:, 1 ,:,:,:,:, 1 ,:)=0; %bottom face of bottommost
220     u(:,end,:,:,:,:end,:)=0; %top face of topmost
221     u(:,:, 1 ,:,:,:,:, 1 )=0; %front face of frontmost
222     u(:,:,end,:,:,:,:end)=0; %back face of backmost

```

Reserve storage and then assign interior patch values to the heterogeneous diffusion time derivatives. Using `nan+u` appears quicker than `nan(size(u),patches.codist)`

```

230     ut = nan(size(u),'like',u); % reserve storage
231     ut(i,j,k,:) ...
232     = diff(patches.cs(:,j,k,1).*diff(u(:,j,k,:),1,1),1,1)/dx^2 ...
233       +diff(patches.cs(i,:,k,2).*diff(u(i,:,k,:),1,2),1,2)/dy^2 ...
234       +diff(patches.cs(i,j,:,3).*diff(u(i,j,:,1),1,3),1,3)/dz^2 ...
235       +patches.fu(i,j,k);
236 end% function

```

5.6 heteroDispersiveWave3: heterogeneous Dispersive Waves from 4th order PDE

This uses small spatial patches to simulate heterogeneous dispersive waves in 3D. The wave equation for $u(x, y, z, t)$ is the fourth-order in space PDE

$$u_{tt} = -\nabla^2(C\nabla^2 u)$$

for microscale variations in scalar $C(x, y, z)$.

Initialise some Matlab aspects.

```

21 clear all
22 cMap=jet(64); cMap=0.8*cMap(7:end-7,:); % set colormap
23 basename = [num2str(floor(1e5*rem(now,1))) mfilename]
24 %global OurCf2eps, OurCf2eps=true %optional to save plots

```

Set random heterogeneous coefficients of period two in each of the three directions. Crudely normalise by the harmonic mean so the macro-wave time scale is roughly one.

```

34 mPeriod = [2 2 2];
35 cHetr = exp(0.9*randn(mPeriod));
36 cHetr = cHetr*mean(1./cHetr(:))

```

Establish global patch data struct to interface with a function coding a fourth-order heterogeneous wave PDE: to be solved on $[-\pi, \pi]^3$ -periodic domain, with 5^3 patches, spectral interpolation (0) couples the patches, each patch with micro-grid spacing 0.22 (relatively large for visualisation), and with 6^3 points forming each patch. (Six because two edge layers on each of two faces, and two interior points for the PDE.)

```

50 global patches
51 patches = configPatches3(@heteroDispWave3,[-pi pi] ...
52     , 'periodic', 5, 0, 0.22, mPeriod+4, 'EdgyInt', true ...
53     , 'hetCoeffs', cHetr, 'nEdge', 2);

```

Set a wave initial state using auto-replication of the spatial grid, and as [Figure 5.6](#) shows. This wave propagates diagonally across space. Concatenate the two u, v -fields to be the two components of the fourth dimension.

```

64 u0 = 0.5+0.5*sin(patches.x+patches.y+patches.z);
65 v0 = -0.5*cos(patches.x+patches.y+patches.z)*3;
66 uv0 = cat(4,u0,v0);

```

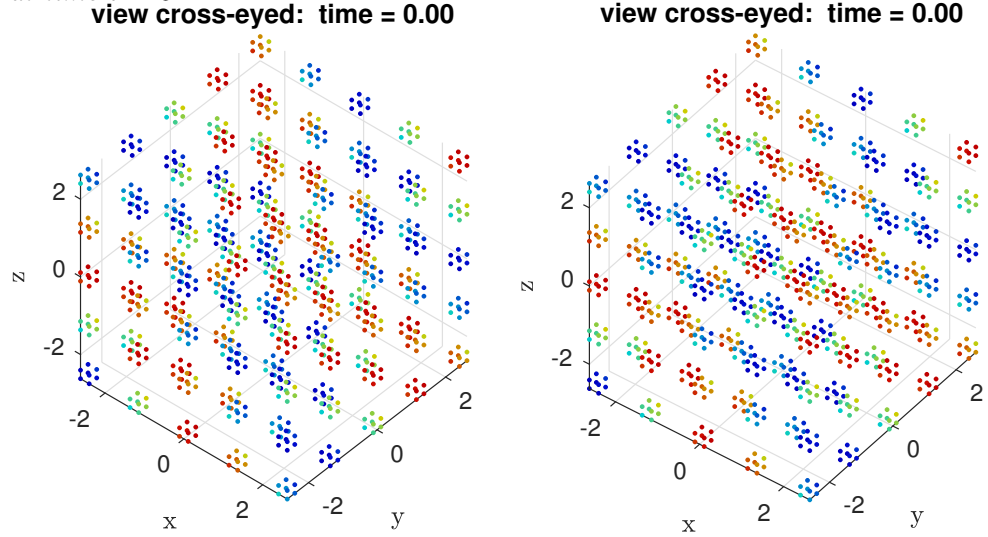
Integrate in time to $t = 6$ using standard functions. In Matlab `ode15s` would be natural as the patch scheme is naturally stiff, but `ode23` is much quicker ([Maclean et al. 2020](#), Fig. 4).

```

83 disp('Simulate heterogeneous wave u_tt=delsq[C*delsq(u)]')
84 tic
85 [ts,us] = ode23(@patchSys3,linspace(0,6),uv0(:));
86 simulateTime=toc

```

Figure 5.6: initial field $u(x, y, z, t)$ at time $t = 0$ of the patch scheme applied to a heterogeneous dispersive wave PDE: Figure 5.7 plots the computed field at time $t = 6$.



Animate the computed simulation to end with Figure 5.7. Use `patchEdgeInt3` to obtain patch-face values in order to most easily reconstruct the array data structure.

Replicate x , y , and z arrays to get individual spatial coordinates of every data point. Then, optionally, set faces to `nan` so the plot just shows patch-interior data.

```

100 %%
101 figure(1), clf, colormap(cMap)
102 xs = patches.x+0*patches.y+0*patches.z;
103 ys = patches.y+0*patches.x+0*patches.z;
104 zs = patches.z+0*patches.y+0*patches.x;
105 if 1, xs([1:2 end-1:end],:,:)=nan;
106     xs(:,[1:2 end-1:end],:,:)=nan;
107     xs(:,:,[1:2 end-1:end],:)=nan;
108 end;%option
109 j=find(~isnan(xs));

```

In the scatter plot, `col()` maps the u -data values to the colour of the dots.

```

116 col = @(u) sign(u).*abs(u);

```

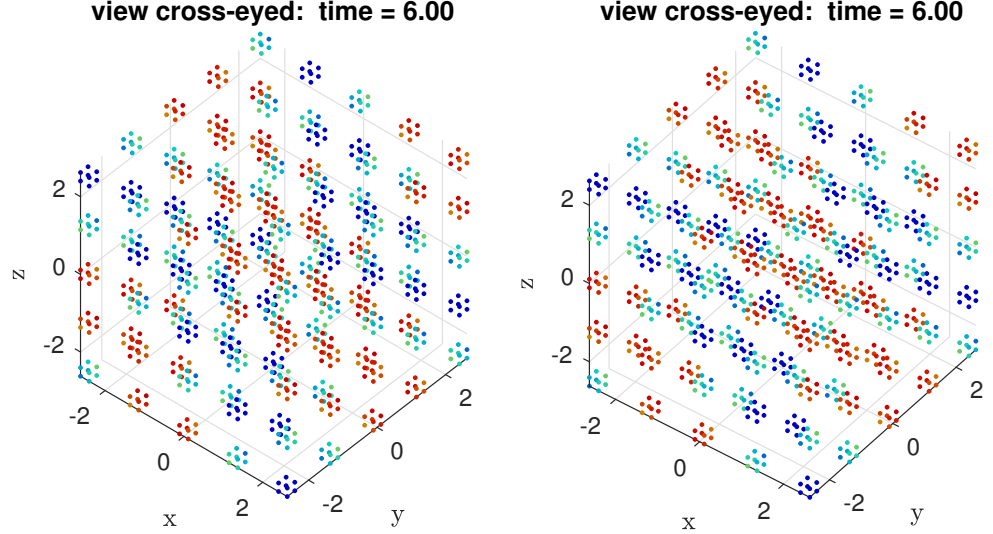
Loop to plot at each and every time step.

```

122 for i = 1:length(ts)
123     uv = patchEdgeInt3(us(i,:));
124     u = uv(:,:,:,1,:);
125     for p=1:2
126         subplot(1,2,p)
127         if (i==1)
128             scat(p) = scatter3(xs(j),ys(j),zs(j),'.');

```

Figure 5.7: field $u(x, y, z, t)$ at time $t = 6$ of the patch scheme applied to the heterogeneous dispersive wave PDE with initial condition in Figure 5.6.



```

129     axis equal, caxis(col([0 1])), view(45-4*p,42)
130     xlabel('$x$'), ylabel('$y$'), zlabel('$z$')
131     end
132     title(['view cross-eyed:  time = ' num2str(ts(i),'%4.2f')])
133     set( scat(p),'CData',col(u(j)) );
134     end

```

Optionally save the initial condition to graphic file for Figure 5.6, and optionally save the last plot.

```

142     if i==1,
143         ifOurCf2eps([basename 'ic'])
144         disp('Type space character to animate simulation')
145         pause
146     else pause(0.1)
147     end
148 end% i-loop over all times
149 ifOurCf2eps([basename 'fin'])

```

5.6.1 heteroDispWave3(): PDE function of 4th-order heterogeneous dispersive waves

This function codes the lattice heterogeneous waves inside the patches. The wave PDE for $u(x, y, z, t)$ and ‘velocity’ $v(x, y, z, t)$ is

$$u_t = v, \quad v_t = -\nabla^2(C\nabla^2 u)$$

for microscale variations in scalar $C(x, y, z)$. For 8D input arrays u , x , y , and z (via edge-value interpolation of `patchSys3`, Section 5.2), computes the time derivative at each point in the interior of a patch, output in `ut`. The 3D array of heterogeneous coefficients, C_{ijk} , c_{ijk}^y and c_{ijk}^z , have been stored in `patches.cs` (3D).

Supply patch information as a third argument (required by parallel computation), or otherwise by a global variable.

```
187 function ut = heteroDispWave3(t,u,patches)
188     if nargin<3, global patches, end
```

Micro-grid space steps.

```
194 dx = diff(patches.x(2:3));
195 dy = diff(patches.y(2:3));
196 dz = diff(patches.z(2:3));
```

First, compute $C\nabla^2 u$ into say u , using indices for all but extreme micro-grid points. We use a single colon to represent the last four array dimensions because the result arrays are already dimensioned.

```
205 I = 2:size(u,1)-1; J = 2:size(u,2)-1; K = 2:size(u,3)-1;
206 u(I,J,K,1,:) = patches.cs(I,J,K,1,:).*( diff(u(:,J,K,1,:),2,1)/dx^2 ...
207     +diff(u(I,:,K,1,:),2,2)/dy^2 +diff(u(I,J,:,1,:),2,3)/dz^2 );
```

Reserve storage, set lowercase indices to non-edge interior, and then assign interior patch values to the heterogeneous diffusion time derivatives.

```
215 ut = nan+u; % preallocate output array
216 i = I(2:end-1); j = J(2:end-1); k = K(2:end-1);
217 ut(i,j,k,1,:) = u(i,j,k,2,:); % du/dt=v
218 % dv/dt=delta^2 of above C*delta^2
219 ut(i,j,k,2,:) = -( diff(u(I,j,k,1,:),2,1)/dx^2 ...
220     +diff(u(i,J,k,1,:),2,2)/dy^2 +diff(u(i,j,K,1,:),2,3)/dz^2 );
221 end% function
```

6 Matlab parallel computation of the patch scheme

For large-scale simulations, we here assume you have a compute cluster with many independent computer processors linked by a high-speed network. The functions we provide in our toolbox aim to distribute computations in parallel across the cluster. *MATLAB's Parallel Computing Toolbox* empowers a reasonably straightforward way to implement this parallelisation.¹ The reason is that the patch scheme ([Chapter 3](#)) has a clear domain decomposition of assigning relatively few patches to each processor.

The examples listed herein are all *Proof of Principle*: as coded they are all small enough that non-parallel execution is here much quicker than the parallel execution. One needs significantly larger and/or more detailed problems than these examples before parallel execution is effective.

As in all parallel cluster computing, interprocessor communication time all too often dominates. It is important to reduce communication as much as possible compared to computation. Consequently, parallel computing is only effective when there is a very large amount of microscale computation done on each processor per communication—all of the examples listed herein are quite small and so the parallel computation of these is much slower than serial computation. We guesstimate that the microscale code may need, per time-step, of the order of many millions of operations per processor in order for the parallelisation to be useful.

To help minimise communication in time-dependent problems we have drafted a special integrator `RK2mesoPatch`, [Section 6.4](#), that communicates between patches only on a meso-time ([Bunder et al. 2016](#)).

¹ This parallelisation is not written for, nor tested for, Octave.

6.1 chanDispSpmdd: simulation of a 1D shear dispersion via simulation on small patches across a channel

Simulate 1D shear dispersion along long thin channel, dispersion that is emergent from micro-scale dynamics in 2D space. Use 1D patches as a Proof of Principle example of parallel computing with `spmd`. In this shear dispersion, although the micro-scale diffusivities are one-ish, the shear causes an effective longitudinal ‘diffusivity’ of the order of Pe^2 —which is typically much larger than the micro-scale diffusivity (Taylor 1953, e.g.).

The spatial domain is the channel (large) L -periodic in x and $|y| < 1$. Seek to predict a concentration field $c(x, y, t)$ satisfying the linear advection-diffusion PDE

$$\frac{\partial c}{\partial t} = -Pe u(y) \frac{\partial c}{\partial x} + \frac{\partial}{\partial x} \left[\kappa_x(y) \frac{\partial c}{\partial x} \right] + \frac{\partial}{\partial y} \left[\kappa_y(y) \frac{\partial c}{\partial y} \right]. \quad (6.1)$$

where Pe denotes a Peclet number, parabolic advection velocity $u(y) = \frac{3}{2}(1 - y^2)$ with noise, and parabolic diffusivity $\kappa_x(y) = \kappa_y(y) = (1 - y^2)$ with noise. The noise is to be multiplicative and log-normal to ensure advection and diffusion are all positive, and to be periodic in x .

For a microscale computation we discretise in space with x -spacing δx , and n_y points over $|y| < 1$ with spacing $\delta y := 2/n_y$ at $y_j := -1 + (j - \frac{1}{2})\delta y$, $j = 1 : n_y$. Our microscale discretisation of PDE (6.1) is then

$$\begin{aligned} \frac{\partial c_{ij}}{\partial t} &= -Pe u(y_j) \frac{c_{i+1,j} - c_{i-1,j}}{2\delta x} + \frac{d_{i,j+1/2} - d_{i,j-1/2}}{\delta y} + \frac{D_{i+1/2,j} - D_{i-1/2,j}}{\delta x}, \\ d_{ij} &:= \kappa_y(y_j) \frac{c_{i,j+1/2} - c_{i,j-1/2}}{\delta y}, \quad D_{ij} := \kappa_x(y_j) \frac{c_{i+1/2,j} - c_{i-1/2,j}}{\delta x}. \end{aligned} \quad (6.2)$$

These are coded in Section 6.1.4 for the computation.

Choose one of four cases:

- **theCase=1** is corresponding code without parallelisation (in this toy problem it is much the quickest because there is no expensive interprocessor communication);
- **theCase=2** illustrates that `RK2mesoPatch` invokes `spmd` computation if parallel has been configured.
- **theCase=3** shows how users explicitly invoke `spmd`-blocks around the time integration.
- **theCase=4** invokes projective integration for long-time simulation via short bursts of the micro-computation, bursts done within `spmd`-blocks for parallel computing.

First, clear all to remove any existing globals, old composites, etc—although a parallel pool persists. Then choose the case.

```
75 clear all
76 theCase = 1
```


The micro-scale PDE is evaluated at positions y_j across the channel, $|y| < 1$. The even indexed points are the collocation points for the PDE, whereas the odd indexed points are the half-grid points for specification of y -diffusivities.

```

86 ny = 7
87 y = linspace(-1,1,2*ny+1);
88 yj = y(2:2:end);

```

Set micro-scale advection (array 1) and diffusivity (array 2) with (roughly) parabolic shape (Watt & Roberts 1995, MacKenzie & Roberts 2003, e.g.). Here modify the parabola by a heterogeneous log-normal factor with specified period along the channel: modify the strength of the heterogeneity by the coefficient of `randn` from zero to perhaps one: coefficient 0.3 appears a good moderate value. Remember that `configPatches1` reshapes `cHetr` to 2D.

```

101 mPeriod = 4
102 cHetr = shiftdim([3/2 1],-1).*(1-y.^2) ...
103      .*exp(0.3*randn([mPeriod 2*ny+1 2]));

```

Configure the patch scheme with some arbitrary choices of domain, patches, size ratios. Choose some random order of interpolation to see the alternatives. Set `patches` information to be global so the info can be used for Cases 1–2 without being explicitly passed as arguments. Choose the parallel option if not Case 1, which invokes `spmd`-block internally, so that field variables become *distributed* across cpus.

```

118 if theCase<=2, global patches, end
119 nPatch=15
120 nSubP=2+mPeriod
121 ratio=0.2+0.2*(theCase<4)
122 Len=nPatch/ratio
123 ordCC=2*randi([0 3])
124 disp('**** Setting configPatches1')
125 patches = configPatches1(@chanDispMicro, [0 Len], nan ...
126      , nPatch, ordCC, ratio, nSubP, 'EdgyInt',true ...
127      , 'hetCoeffs',cHetr , 'parallel',(theCase>1) );

```

When using parallel then additional parameters to `patches` should be set within a `spmd` block (because `patches` is a co-distributed structure).

```

135 Peclet = 10
136 if theCase==1, patches.Pe = Peclet;
137 else      spmd, patches.Pe = Peclet; end
138 end

```

6.1.1 Simulate heterogeneous advection-diffusion

Set initial conditions of a simulation as shown in Figure 6.1.

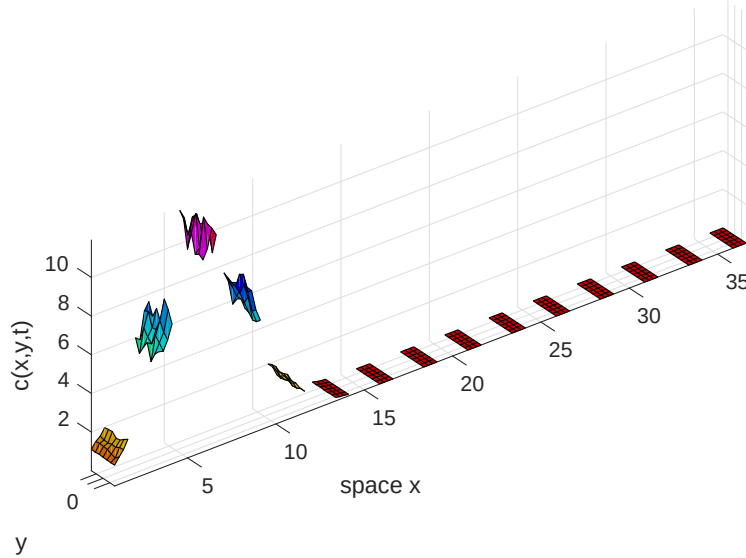
```

149 disp('**** Set initial condition and test dc0dt =')
150 if theCase==1

```

Without parallel processing, invoke the usual operations.

Figure 6.1: initial field $u(x, y, 0)$ of the patch scheme applied to a heterogeneous advection-diffusion PDE. Figure 6.2 plots the roughly smooth field values at time $t = 4$. In this example the patches are relatively large, ratio 0.4, for visibility.



```

156     c0 = 10*exp(-(ratio*patches.x-2.5).^2/2) +0*yj;
157     c0 = c0.*(1+0.2*rand(size(c0)));
158     dc0dt = patchSys1(0,c0);

```

With parallel, we must use an `spmd`-block for computations: there is no difference in cases 2–4 here. Also, we must sometimes use `patches.codist` to explicitly code how to distribute new arrays over the cpus. Now `patchSys1` does not invoke `spmd` so higher level code must, as here. Even if `patches` is global, inside `spmd`-block we *must* pass it explicitly as a parameter to `patchSys1`.

```

171     else, spmd
172         c0 = 10*exp(-(ratio*patches.x-2.5).^2/2) +0*yj;
173         c0 = c0.*(1+0.2*rand(size(c0),patches.codist));
174         dc0dt = patchSys1(0,c0,patches)
175     end%spmd
176 end%if theCase

```

Integrate in time, either via the automatic `ode23` or via `RK2mesoPatch` which reduces communication between patches. By default, `RK2mesoPatch` does ten micro-steps for each specified meso-step in `ts`. For stability: with noise up to 0.3, need micro-steps less than 0.005; with noise 1, need micro-steps less than 0.0015.

```

198     warning('Integrating system in time, wait patiently')
199     ts=4*linspace(0,1);

```

Go to the selected case.

```

205     switch theCase

```

1. For non-parallel, we could use `RK2mesoPatch` as indicated below, but instead choose to use standard `ode23` as here `patchSys1` accesses patch information via global `patches`. For post-processing, reshape each and every row of the computed solution to the correct array size—namely that of the initial condition.

```

217 case 1
218 % [cs,uerrs] = RK2mesoPatch(ts,c0);
219 [ts,cs] = ode23(@patchSys1,ts,c0(:));
220 cs=reshape(cs,[length(ts) size(c0)]);

```

2. In the second case, `RK2mesoPatch` detects a parallel patch code has been requested, but has only one cpu worker, so it auto-initiates an `spmd`-block for the integration. Both this and the next case return *composite* results, so just keep one version of the results.

```

232 case 2
233 cs = RK2mesoPatch(ts,c0);
234 cs = cs{1};

```

3. In this third case, a user could merge this explicit `spmd`-block with the previous one that sets the initial conditions.

```

243 case 3,spmd
244 cs = RK2mesoPatch(ts,c0,[],patches);
245 end%spmd
246 cs = cs{1};

```

4. In this fourth case, use Projective Integration (PI) over long times (`PIRK4` also works). Currently the PI is done serially, with parallel `spmd`-blocks only invoked inside function `aBurst()` ([Section 6.3.3](#)) to compute each burst of the micro-scale simulation. For a Peclet number of ten, the macro-scale time-step needs to be less than about 0.5 (which here is very little projection)—presumably the mean advection in a macro-step needs to be less than about the patch spacing. The function `microBurst()` here interfaces to `aBurst()` ([Section 6.1.3](#)) in order to provide shaped initial states, and to provide the patch information.

```

264 case 4
265 microBurst = @(tb0,xb0,bT) ...
266 aBurst(tb0,reshape(xb0,size(c0)),patches);
267 ts = 0:0.7:5
268 cs = PIRK2(microBurst,ts,gather(c0(:)));
269 cs = reshape(cs,[length(ts) size(c0)]);

```

End the four cases.

```

276 end%switch theCase

```

6.1.2 Plot the solution

Optionally set to save some plots to file.

```

287 if 0, global OurCf2eps, OurCf2eps=true, end

```

Animate the computed solution field over time

```
293 figure(1), clf, colormap(0.8*hsv)
```

First get the x -coordinates and omit the patch-edge values from the plot (because they are not here interpolated).

```
301 if theCase==1, x = patches.x;
302 else, spmd
303     x = gather( patches.x );
304     end%spmd
305     x = x{1};
306 end
307 x([1 end],:,:) = nan;
```

For every time step draw the concentration values as a set of surfaces on 2D patches, with a short pause to display animation.

```
315 nTimes = length(ts)
316 for l = 1:nTimes
```

At each time, squeeze sub-patch data into a 3D array, permute to get all the x -variation in the first two dimensions, and reshape into x -variation for each and every (y).

```
325     c = reshape( permute( squeeze( ...
326         cs(1,:,:,,:) ) , [1 3 2] ) , numel(x),ny);
```

Draw surface of each patch, to show both micro-scale and macro-scale variation in space.

```
333     if l==1
334         hp = surf(x(:),yj,c');
335         axis([0 Len -1 1 0 max(c(:))])
336         axis equal
337         xlabel('space $x$'), ylabel('$y$'); zlabel('$c(x,y,t)$')
338         ifOurCf2eps([mfilename 't0'])
339         legend(['time = ' num2str(ts(l),'%4.2f')] ...
340             , 'Location', 'north')
341         disp('**** pausing, press blank to animate')
342         pause
343     else
344         hp.ZData = c';
345         legend(['time = ' num2str(ts(l),'%4.2f')])
346         pause(0.1)
347     end
```

Finish the animation loop, and optionally save the final plot to file, [Figure 6.2](#).

```
363 end%for over time
364 ifOurCf2eps([mfilename 'tFin'])
```

Macro-scale view Plot a macro-scale mesh of the predictions: at each of a selection of times, for every patch, plot the patch-mean value at the mean- x .

Figure 6.2: final field $c(x, y, 4)$ of the patch scheme applied to a heterogeneous advection-diffusion PDE (6.1) with heterogeneous factor log-normal, here distributed $\exp[\mathcal{N}(0, 1)]$.

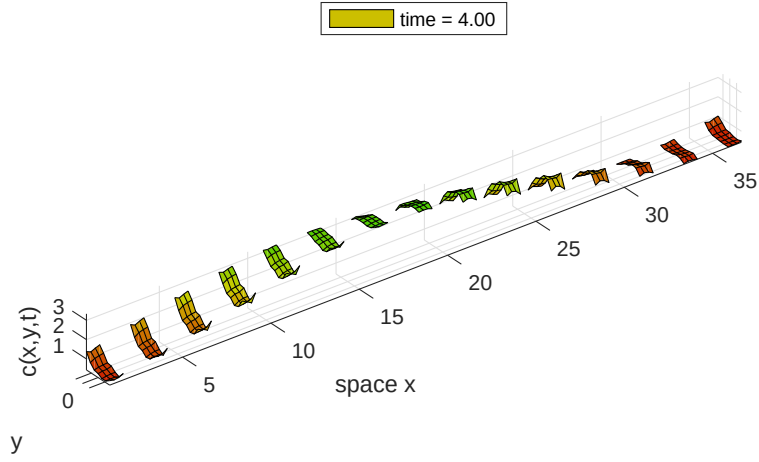
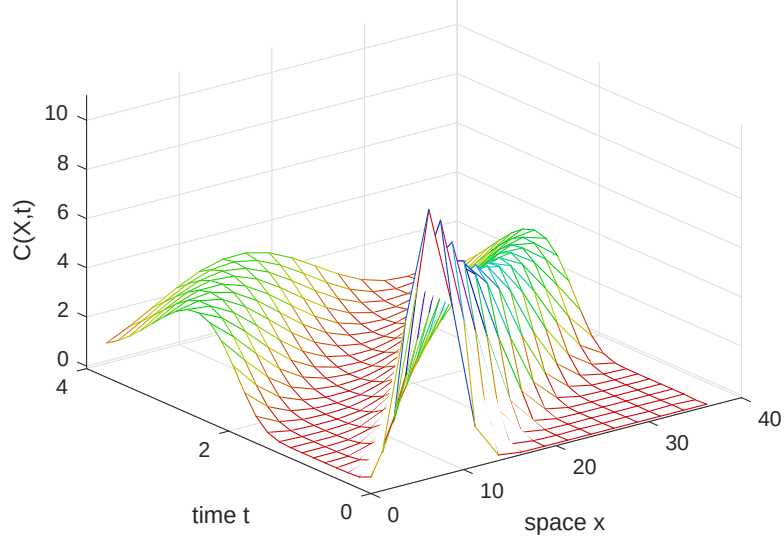


Figure 6.3: macro-scale view of heterogeneous advection-diffusion PDE along a (periodic) channel obtained via the patch scheme.



```

374 figure(2), clf, colormap(0.8*hsv)
375 X = squeeze(mean(x(2:end-1,:,:,:)));
376 C = squeeze(mean(mean(cs(:,2:end-1,:,:,:),2),3));
377 j = 1:ceil(nTimes/30):nTimes;
378 mesh(X,ts(j),C(j,:));
379 xlabel('space $x$'),ylabel('time $t$'),zlabel('$C(X,t)$')
380 zlim([-0.1 11])
381 ifOurCf2eps([mfilename 'Macro'])

```

6.1.3 microBurst function for Projective Integration

Projective Integration stability appears to require bursts longer than 0.2. Each burst is done in parallel processing. Here use RK2mesoPatch to take

take meso-steps, each with default ten micro-steps so the micro-scale step is 0.0033. With macro-step 0.5, these parameters usually give stable projective integration.

```

404 function [tbs,xbs] = aBurst(tb0,xb0,patches)
405     normx=max(abs(xb0(:)));
406     disp(['* aBurst t=' num2str(tb0) ' |x|=' num2str(normx)])
407     assert(normx<20,'solution exploding')
408     tbs = tb0+(0:0.033:0.2);
409     spmd
410         xb0 = codistributed(xb0,patches.codist);
411         xbs = RK2mesoPatch(tbs,xb0,[],patches);
412     end%spmd
413     xbs=reshape(xbs{1},length(tbs),[]);
414 end%function

```

Fin.

6.1.4 chanDispMicro(): heterogeneous 2D advection-diffusion in a long thin channel

This function codes the lattice heterogeneous diffusion inside the patches. For 4D input arrays of concentration c and spatial lattice x (via edge-value interpolation of `patchSys1`, [Section 3.2](#)), computes the time derivative (6.2) at each point in the interior of a patch, output in `ct`. The heterogeneous advections and diffusivities, $u_i(y_j)$ and $\kappa_i(y_{j+1/2})$, have previously been merged and stored in the one array `patches.cs` (2D).

```

22 function ct = chanDispMicro(t,c,p)
23     [nx,ny,~,~]=size(c); % micro-grid points in patches
24     ix = 2:nx-1;         % x interior points in a patch
25     dx = diff(p.x(2:3)); % x space step
26     dy = 2/ny;           % y space step
27     ct = nan+c;          % preallocate output array
28     pcs = reshape(p.cs,nx-1,[],2);

```

Compute the cross-channel flux using ‘ghost’ nodes at channel boundaries, so that the flux is zero at $y = \pm 1$ either because the boundary values are replicated so the differences are zero, or because the diffusivities in `cs` are zero at the channel boundaries.

```

38     ydif = pcs(ix,1:2:end,2) ...
39         .*(c(ix,[1:end end],:,:)-c(ix,[1 1:end],:,:))/dy;

```

Now evaluate advection-diffusion time derivative (6.2). Could use upwind advection and no longitudinal diffusion, or, as here, centred advection and diffusion.

```

48     ct(ix,:,:,) = (ydif(:,2:end,:)-ydif(:,1:end-1,:))/dy ...
49     + diff(pcs(:,2:2:end,2).*diff(c))/dx^2 ...
50     - p.Pe*pcs(ix,2:2:end,1).*(c(ix+1,:,:,)-c(ix-1,:,:,))/(2*dx);
51 end% function

```

6.2 rotFilmSpmd: simulation of a 2D shallow water flow on a rotating heterogeneous substrate

As an example application, consider the flow of a shallow layer of fluid on a solid flat rotating substrate, such as in spin coating (Wilson et al. 2000, Oron et al. 1997, §II.K, e.g.) or large-scale shallow water waves (Dellar & Salmon 2005, Hereman 2009, e.g.). Let $\vec{x} = (x, y)$ parametrise location on the rotating substrate, and let the fluid layer have thickness $h(\vec{x}, t)$ and move with depth-averaged horizontal velocity $\vec{v}(\vec{x}, t) = (u, v)$. We take as given (with its simplified physics) that the (non-dimensional) governing set of PDEs is the nonlinear system (Bunder & Roberts 2018, eq. (1), e.g.)

$$\frac{\partial h}{\partial t} = -\nabla \cdot (h\vec{v}), \quad (6.3a)$$

$$\frac{\partial \vec{v}}{\partial t} = \begin{bmatrix} -b & f \\ -f & -b \end{bmatrix} \vec{v} - (\vec{v} \cdot \nabla) \vec{v} - g\nabla h + \vec{\nabla} \cdot (\nu \vec{\nabla} \vec{v}), \quad (6.3b)$$

where $b(\vec{x})$ represents heterogeneous ‘bed’ drag, f is the Coriolis coefficient, g is the acceleration due to gravity, $\nu(\vec{x})$ is a heterogeneous ‘kinematic viscosity’, and we neglect surface tension.

The aim is to simulate the macroscale dynamics which (for constant b) is approximately that of the nonlinear diffusion $\partial h / \partial t \approx \frac{gb}{b^2 + f^2} \vec{\nabla} \cdot (h \vec{\nabla} h)$ (Bunder & Roberts 2018, eq. (2)). But there is no known algebraic closure for the macroscale in the case of heterogeneous $b(\vec{x})$ and $\nu(\vec{x})$, nonetheless the patch scheme automatically predicts a sensible macroscale for such heterogeneous dynamics (Figure 6.5).

For the microscale computation, Section 6.2.4 discretises the PDEs (6.3) in space with x, y -spacing $\delta x, \delta y$.

Choose one of four cases:

- **theCase=1** is corresponding code without parallelisation (in this toy problem it is much the quickest because there is no expensive communication);
- **theCase=2** illustrates that `RK2mesoPatch` invokes `spmd` computation if parallel has been configured.
- **theCase=3** shows how users explicitly invoke `spmd`-blocks around the time integration.
- **theCase=4** invokes projective integration for long-time simulation via short bursts of the micro-computation, bursts done within `spmd`-blocks for parallel computing.

First, clear all to remove any existing globals, old composites, etc—although a parallel pool persists. Then choose the case.

```
71 clear all
72 theCase = 1
```

Set micro-scale bed drag (array 1) and diffusivity (arrays 2–3) to be a heterogeneous log-normal factor with specified period: modify the strength

of the heterogeneity by the coefficient of `randn` from zero to perhaps one: coefficient 0.3 appears a good moderate value.

```
82 mPeriod = 5
83 bnu = shiftdim([1 0.5 0.5],-1) ...
84     .*exp(0.3*randn([mPeriod mPeriod 3]));
```

Configure the patch scheme with these choices of domain, patches, size ratios—here each patch is square in space. In Cases 1–2, set `patches` information to be global so the info can be used without being explicitly passed as arguments.

```
96 if theCase<=2, global patches, end
```

In Case 4, double the size of the domain and use more separated patches accordingly, to maintain the spatial microscale grid spacing to be 0.055. Here use fourth order edge-based coupling between patches. Choose the parallel option if not Case 1, which invokes `spmd`-block internally, so that field variables become *distributed* across cpus.

```
108 nSubP = 2+mPeriod
109 nPatch = 9
110 ratio = 0.2+0.2*(theCase<4)
111 Len = 2*pi*(1+(theCase==4))
112 disp('**** Setting configPatches2')
113 patches = configPatches2(@rotFilmMicro, [0 Len], nan ...
114     , nPatch, 4, ratio, nSubP, 'EdgyInt',true ...
115     , 'hetCoeffs',bnu , 'parallel',(theCase>1) );
```

When using parallel, any additional parameters to `patches`, such as physical parameters for the microcode, must be set within a `spmd` block (because `patches` is a co-distributed structure). Here set frequency of substrate rotation, and strength of gravity.

```
125 f = 5, g = 1
126 if theCase==1, patches.f = f; patches.g = g;
127 else spmd, patches.f = f; patches.g = g; end
128 end
```

6.2.1 Simulate heterogeneous advection-diffusion

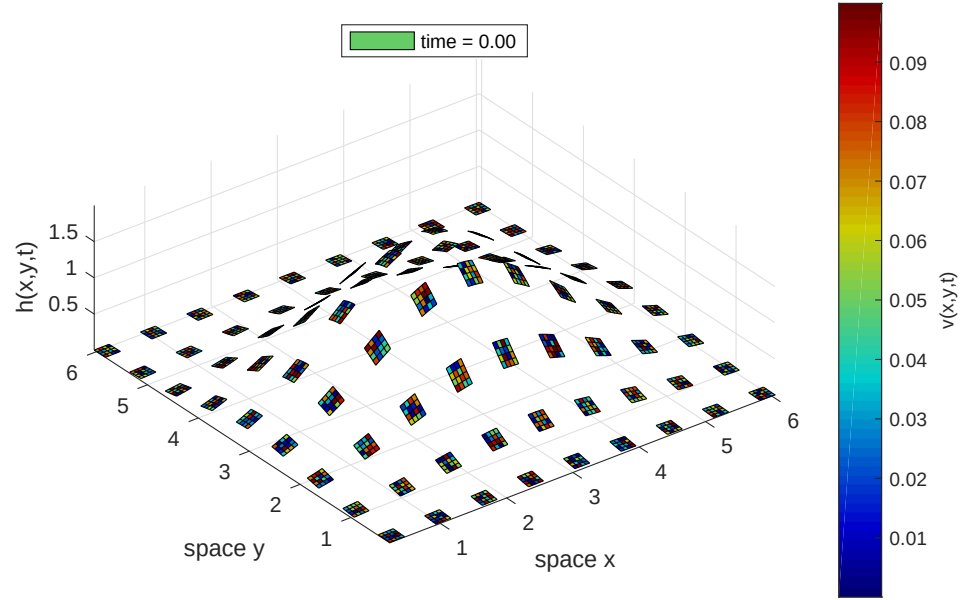
Set initial conditions of a simulation as shown in [Figure 6.4](#). Here the initial condition is a (periodic) quasi-Gaussian in h and zero velocity $\vec{v}$, with additive random perturbations.

```
141 disp('**** Set initial condition and test dhuv0dt =')
142 if theCase==1
```

When not parallel processing, invoke the usual operations. Here add a random noise to the velocity field, but keep $h(x, y, 0)$ smooth as shown by [Figure 6.4](#). The `shiftdim(...,-1)` moves the given row-vector of coefficients into the third dimension to become coefficients of the fields (h, u, v) , respectively.

```
153     huv0 = shiftdim([0.5 0 0],-1) ...
154     .*exp(-cos(patches.x)/2-cos(patches.y));
```


Figure 6.4: initial field $h(x,y,0)$ of the patch scheme applied to the heterogeneous, shallow water, rotating substrate, PDE (6.3). The micro-scale sub-patch colour displays the initial y -direction velocity field $v(x,y,0)$. Figure 6.5 plots the roughly smooth field values at time $t = 6$. In this example the patches are relatively large, ratio 0.4, for visibility.



```

155     huv0 = huv0+0.1*shiftdim([0 1 1],-1).*rand(size(huv0));
156     dhuv0dt = patchSys2(0,huv0);

```

With parallel, we must use an `spmd`-block for computations: there is no difference in Cases 2–4 here. Also, we must sometimes explicitly tell functions how to distribute some initial condition arrays over the cpus. Now `patchSys2` does not invoke `spmd` so higher level code must, as here. Even if `patches` is global, inside an `spmd`-block we *must* pass `patches` explicitly as a parameter to `patchSys2`.

```

170 else, spmd
171     huv0 = shiftdim([0.5 0 0],-1) ...
172         .*exp(-cos(patches.x)/2-cos(patches.y));
173     huv0 = huv0+0.1*rand(size(huv0),patches.codist);
174     dhuv0dt = patchSys2(0,huv0,patches)
175     end%spmd
176 end%if theCase

```

Integrate in time, either via the automatic `ode23` or via `RK2mesoPatch` which reduces communication between patches. By default, `RK2mesoPatch` does ten micro-steps for each specified meso-step in `ts`. For stability: with noise up to 0.3, need micro-steps less than 0.0003; with noise 1, need micro-steps less than 0.0001.

```

201 warning('Integrating system in time, wait a minute')
202 ts=0:0.003:0.3;

```

Go to the selected case.

208 `switch theCase`

1. For non-parallel, we could use `RK2mesoPatch` as indicated below, but instead choose to use standard `ode23` as here `patchSys2` accesses patch information via global `patches`. For post-processing, reshape each and every row of the computed solution to the correct array size—namely that of the initial condition.

```
220 case 1
221 %     tic,[huvs,uerrs] = RK2mesoPatch(ts,huv0);toc
222     [ts,huvs] = ode23(@patchSys2,[0 4],huv0(:));
223     huvs=reshape(huvs,[length(ts) size(huv0)]);
```

2. In the second case, `RK2mesoPatch` detects a parallel patch code has been requested, but has only one cpu worker, so it auto-initiates an `spmd`-block for the integration. Both this and the next case return *composite* results, so just keep one version of the results.

```
235 case 2
236     huvs = RK2mesoPatch(ts,huv0);
237     huvs = huvs{1};
```

3. In this third case, a user could merge this explicit `spmd`-block with the previous one that sets the initial conditions.

```
246 case 3,spmd
247     huvs = RK2mesoPatch(ts,huv0,[],patches);
248     end%spmd
249     huvs = huvs{1};
```

4. In this fourth case, use Projective Integration (PI). Currently the PI is done serially, with parallel `spmd`-blocks only invoked inside function `aBurst()` (Section 6.2.3) to compute each burst of the micro-scale simulation. The macro-scale time-step needs to be less than about 0.1 (which here is not much projection). The function `microBurst()` interfaces to `aBurst()` (Section 6.2.3) in order to provide shaped initial states, and to provide the patch information.

```
264 case 4
265     microBurst = @(tb0,xb0,bT) ...
266         aBurst(tb0,reshape(xb0,size(huv0)),patches);
267     ts = 0:0.1:1
268     huvs = PIRK2(microBurst,ts,gather(huv0(:)));
269     huvs = reshape(huvs,[length(ts) size(huv0)]);
```

End the four cases.

276 `end%switch theCase`

6.2.2 Plot the solution

Optionally set to save some plots to file.

```
287 if 0, global OurCf2eps, OurCf2eps=true, end
```

Animate the computed solution field over time

```
293 figure(1), clf, colormap(0.8*jet)
```

First get the x -coordinates and omit the patch-edge values from the plot (because they are not here interpolated).

```
300 if theCase==1, x = patches.x;
301             y = patches.y;
302 else, spmd
303     x = gather( patches.x );
304     y = gather( patches.y );
305 end%spmd
306 x = x{1}; y = y{1};
307 end
308 x([1 end],:,:,:) = nan;
309 y(:,[1 end],:,:) = nan;
```

Draw the field values as a patchy surface evolving over 100–200 time steps.

```
316 nTimes = length(ts)
317 for l = 1:ceil(nTimes/200):nTimes
```

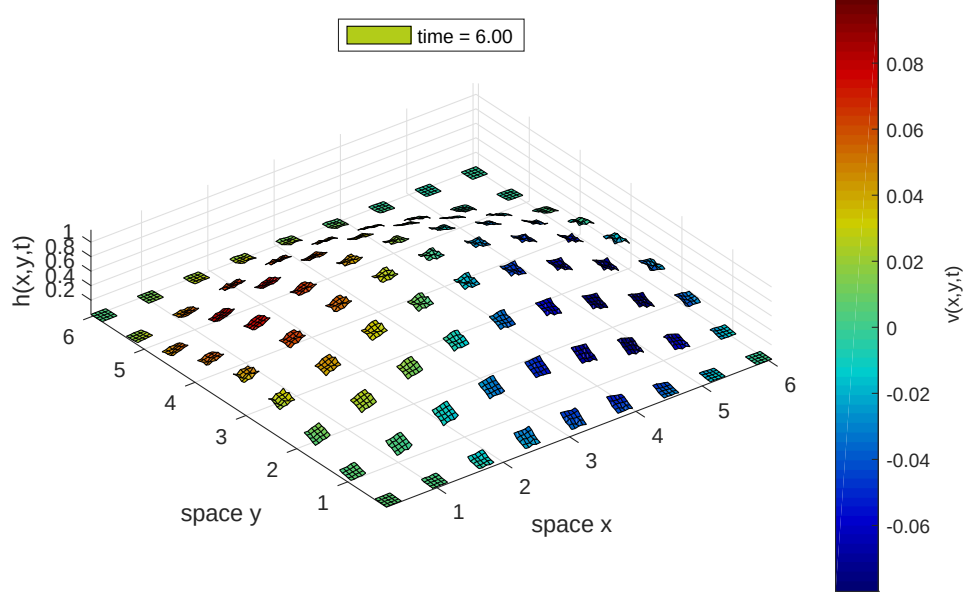
At each time, squeeze sub-patch data fields into three 4D arrays, permute to get all the x/y -variations in the first/last two dimensions, and then reshape to 2D.

```
325 h = reshape( permute( squeeze( ...
326     huvs(1,:,:,1,1,:,:) ) , [1 3 2 4]) , numel(x),numel(y));
327 u = reshape( permute( squeeze( ...
328     huvs(1,:,:,2,1,:,:) ) , [1 3 2 4]) , numel(x),numel(y));
329 v = reshape( permute( squeeze( ...
330     huvs(1,:,:,3,1,:,:) ) , [1 3 2 4]) , numel(x),numel(y));
```

Draw surface of each patch, to show both micro-scale and macro-scale variation in space. Colour the surface according to the velocity v in the y -direction.

```
338 if l==1
339     hp = surf(x(:),y(:),h',v');
340     axis([0 Len 0 Len 0 max(h(:))])
341     c = colorbar; c.Label.String = 'v(x,y,t)';
342     legend(['time = ' num2str(ts(l),'%4.2f')] ...
343         , 'Location', 'north')
344     axis equal
345     xlabel('space $x$'), ylabel('space $y$'), zlabel('$h(x,y,t)$')
346     ifOurCf2eps([mfilename 't0'])
347     disp('**** pausing, press blank to begin animation')
348     pause
349 else
350     hp.ZData = h'; hp.CData = v';
351     legend(['time = ' num2str(ts(l),'%4.2f')])
```

Figure 6.5: final field $h(x, y, 6)$, coloured by $v(x, y, 6)$, of the patch scheme applied to the heterogeneous, shallow water, rotating substrate, PDE (6.3) with heterogeneous factors log-normal, here distributed $\exp[\mathcal{N}(0, 1)]$.



```

352     pause(0.1)
353     end

```

Finish the animation loop, and optionally save the final plot to file, [Figure 6.5](#).

```

370 end%for over time
371 ifOurCf2eps([mfilename 'tFin'])

```

6.2.3 microBurst function for Projective Integration

Projective Integration stability appears to require bursts longer than 0.01. Each burst is done in parallel processing. Here use `RK2mesoPatch` to take take meso-steps, each with default ten micro-steps so the micro-scale step is 0.0003. With macro-step 0.1, these parameters usually give stable projective integration.

```

388 function [tbs,xbs] = aBurst(tb0,xb0,patches)
389     normx=max(abs(xb0(:)));
390     disp(['* aBurst t=' num2str(tb0) ' |x|=' num2str(normx)])
391     assert(normx<20,'solution exploding')
392     tbs = tb0+(0:0.003:0.015);
393     spmd
394         xb0 = codistributed(xb0,patches.codist);
395         xbs = RK2mesoPatch(tbs,xb0,[],patches);
396     end%spmd
397     xbs=reshape(xbs{1},length(tbs),[]);
398 end%function

```

Fin.

6.2.4 rotFilmMicro(): 2D shallow water flow on a rotating heterogeneous substrate

This function codes the heterogeneous shallow water flow (6.3) inside 2D patches. The PDEs are discretised on the multiscale lattice in terms of evolving variables h_{ijIJ} , u_{ijIJ} and v_{ijIJ} . For 6D input array `huv` (via edge-value interpolation of `patchEdgeInt2()`, Section 4.2), computes the time derivatives (6.3) at each point in the interior of a patch, output in `huvt`. The heterogeneous bed drag and diffusivities, b_{ij} and ν_{ij} , have previously been merged and stored in the array `patches.cs` ($2D \times 3$): herein `patches` is named `p`.

```

24 function huvt = rotFilmMicro(t,huv,p)
25     [nx,ny,~]=size(huv); % micro-grid points in patches
26     i = 2:nx-1;          % x interior points in a patch
27     j = 2:ny-1;          % y interior points in a patch
28     dx = diff(p.x(2:3)); % x space step
29     dy = diff(p.y(2:3)); % y space step
30     huvt = nan+huv;      % preallocate output array

```

Set indices of fields in the arrays. Need to store different diffusivity values for the x, y -directions as they are evaluated at different points in space.

```

38     h=1; u=2; v=3;
39     b=1; nux=2; nuy=3;

```

Use a staggered micro-grid so that $h(i,j) = h_{ij}$, $u(i,j) = u_{i+1/2,j}$, and $v(i,j) = v_{i,j+1/2}$. We need the following to interpolate some quantities to other points on the staggered micro-grid. But the first two statements fill-in two needed corner values because they are not (currently) interpolated by `patchEdgeInt2()`.

```

51 huv(1,ny,u,:,:,) = huv(2,ny,u,:,:,)+huv(1,ny-1,u,:,:,) ...
52                     -huv(2,ny-1,u,:,:,);
53 huv(nx,1,v,:,:,) = huv(nx,2,v,:,:,)+huv(nx-1,1,v,:,:,) ...
54                     -huv(nx-1,2,v,:,:,);
55 v4u = (huv(i,j-1,v,:,:,)+huv(i+1,j,v,:,:,) ...
56         +huv(i,j,v,:,:,)+huv(i+1,j-1,v,:,:,))/4;
57 u4v = (huv(i,j+1,u,:,:,)+huv(i-1,j,u,:,:,) ...
58         +huv(i,j,u,:,:,)+huv(i-1,j+1,u,:,:,))/4;
59 h2u = (huv(2:nx,:,h,:,:,)+huv(1:nx-1,:,h,:,:,))/2;
60 h2v = (huv(:,2:ny,h,:,:,)+huv(:,1:ny-1,h,:,:,))/2;

```

Evaluate conservation of mass PDE (6.3a) (needing averages of h at half-grid points):

```

67     huvt(i,j,h,:,:,) = ...
68         - (h2u(i,j ,:,:,) .* huv(i ,j,u,:,:,) ...
69           - h2u(i-1,j,:,:,) .* huv(i-1,j,u,:,:,) )/dx ...
70         - (h2v(i,j ,:,:,) .* huv(i,j ,v,:,:,) ...
71           - h2v(i,j-1,:,:,) .* huv(i,j-1,v,:,:,))/dy ;

```

Evaluate the x -direction momentum PDE (6.3b) (needing to interpolate component v to u -points):

```

79     huvt(i,j,u,:,:,:) = ...
80         - p.cs(i,j,b).*huv(i,j,u,:,:,:) + p.f.*v4u ...
81         - huv(i,j,u,:,::).*(huv(i+1,j,u,:,::)-huv(i-1,j,u,:,::))/(2*dx) ...
82         - v4u.*(huv(i,j+1,u,:,::)-huv(i,j-1,u,:,::))/(2*dy) ...
83         - p.g*(huv(i+1,j,h,:,::)-huv(i,j,h,:,::))/dx ...
84         + diff(p.cs(:,j,nux).*diff(huv(:,j,u,:,::),[],1),[],1)/dx^2 ...
85         + diff(p.cs(i,:,nuy).*diff(huv(i,:,u,:,::),[],2),[],2)/dy^2 ;

```

Evaluate the y -direction momentum PDE (6.3b) (needing to interpolate component u to v -points):

```

93     huvt(i,j,v,:,:,:) = ...
94         - p.cs(i,j,b).*huv(i,j,v,:,::) - p.f.*u4v ...
95         - u4v.*(huv(i+1,j,v,:,::)-huv(i-1,j,v,:,::))/(2*dx) ...
96         - huv(i,j,v,:,::).*(huv(i,j+1,v,:,::)-huv(i,j-1,v,:,::))/(2*dy) ...
97         - p.g*(huv(i,j+1,h,:,::)-huv(i,j,h,:,::))/dy ...
98         + diff(p.cs(:,j,nux).*diff(huv(:,j,v,:,::),[],1),[],1)/dx^2 ...
99         + diff(p.cs(i,:,nuy).*diff(huv(i,:,v,:,::),[],2),[],2)/dy^2 ;
100 end% function

```

6.3 homoDiff31spmd: computational homogenisation of a 1D dispersion via parallel simulation on small 3D patches of heterogeneous diffusion

Simulate effective dispersion along 1D space on 3D patches of heterogeneous diffusion as a Proof of Principle example of parallel computing with `spmd`. With only one patch in each of the y, z -directions, the solution simulated is strictly periodic in y, z with period `ratio`: there are only macro-scale variations in the x -direction. The discussion here only addresses issues with `spmd` parallel computing. For discussion on the 3D patch scheme with heterogeneous diffusion, see code and documentation for `homoDiffEdgy3` in [Section 5.4](#).

Choose one of four cases:

- `theCase=1` is corresponding code without parallelisation (in this toy problem it is much the quickest because there is no expensive communication);
- `theCase=2` for minimising coding by a user of `spmd`-blocks;
- `theCase=3` is for users happier to explicitly invoke `spmd`-blocks.
- `theCase=4` invokes projective integration for long-time simulation via short bursts of the micro-computation, bursts done within `spmd`-blocks for parallel computing.

First, clear all to remove any existing globals, old composites, etc—although a parallel pool persists. Then choose the case.

```
48 clear all
49 theCase = 1
```

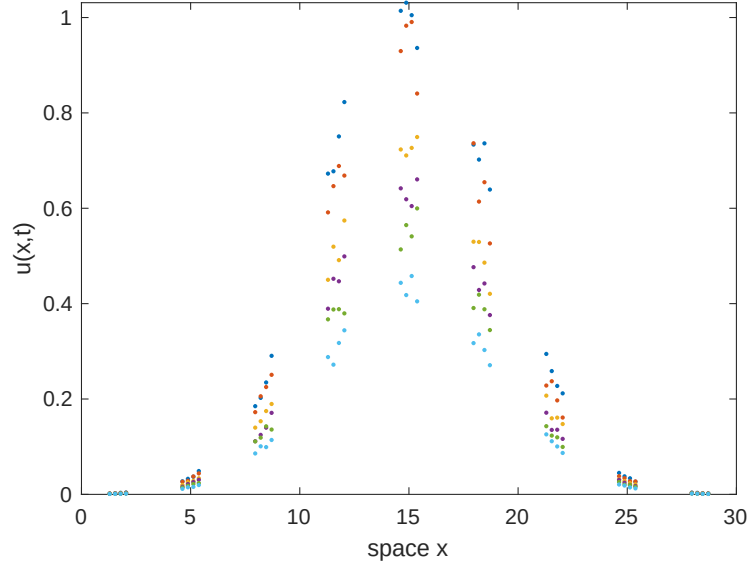
Set micro-scale heterogeneity with various spatial periods in the three directions.

```
57 mPeriod = [4 3 2] %1+randperm(3)
58 cHetr = exp(0.3*randn([mPeriod 3]));
59 cHetr = cHetr*mean(1./cHetr(:))
```

Configure the patch scheme with some arbitrary choices of domain, patches, size ratios—here each patch is a unit cube in space. Choose some random order of interpolation. Set `patches` information to be global so the info can be used for Case 1 without being explicitly passed as arguments. Choose the parallel option if not Case 1, which invokes `spmd`-block internally, so that field variables become *distributed* across cpus.

```
73 if any(theCase==[1 2]), global patches, end
74 nSubP=mPeriod+2
75 nPatch=[9 1 1]
76 ratio=0.3
77 Len=nPatch(1)/ratio
78 ordCC=2*randi([0 3])
79 disp('**** Setting configPatches3')
80 patches = configPatches3(@heteroDiff3,[0 Len 0 1 0 1], nan ...
```

Figure 6.6: initial field $u(x, y, z, 0)$ of the patch scheme applied to a heterogeneous diffusion PDE. The vertical spread indicates the extent of the structure in u in the cross-section variables y, z . [Figure 6.7](#) plots the nearly smooth field values at time $t = 0.4$.



```

81         , nPatch, ordCC, [ratio 1 1], nSubP, 'EdgyInt', true ...
82         , 'hetCoeffs', cHetr , 'parallel', (theCase>1) );

```

6.3.1 Simulate heterogeneous diffusion

Set initial conditions of a simulation as shown in [Figure 6.6](#).

```

92 disp('**** Set initial condition and testing du0dt =')
93 if theCase==1
    Without parallel processing, invoke the usual operations.
99     u0 = exp( -(patches.x-Len/2).^2/Len ...
100              -patches.y.^2/2-patches.z.^2 );
101     u0 = u0.*(1+0.2*rand(size(u0)));
102     du0dt = patchSys3(0,u0);

```

With parallel, must use an `spmd`-block for computations: there is no difference in cases 2–4 here. Also, we must sometimes explicitly code how to distribute some new arrays over the cpus. Now `patchSys3` does not invoke `spmd` so higher level code must, as here. Even if `patches` is global, inside `spmd`-block we must pass it explicitly as a parameter to `patchSys3`.

```

115 else, spmd
116     u0 = exp( -(patches.x-Len/2).^2/Len ...
117              -patches.y.^2/2-patches.z.^2/4 );
118     u0 = u0.*(1+0.2*rand(size(u0),patches.codist));
119     du0dt = patchSys3(0,u0,patches);
120     end%spmd
121 end%if theCase

```


Integrate in time. Use non-uniform time-steps for fun, and to show more of the initial rapid transients.

Alternatively, use `RK2mesoPatch` which reduces communication between patches, recalling that, by default, `RK2mesoPatch` does ten micro-steps for each specified step in `ts`. For unit cube patches, need micro-steps less than about 0.004 for stability.

```
144 warning('Integrating system in time, wait patiently')
145 ts=0.4*linspace(0,1,21).^2;
```

Go to the selected case.

```
151 switch theCase
```

1. For non-parallel, we could use `RK2mesoPatch` as indicated below, but instead choose to use standard `ode23` as here `patchSys3` accesses patch information via global `patches`. For post-processing, reshape each and every row of the computed solution to the correct array size—that of the initial condition.

```
163 case 1
164 % [us,uerrs] = RK2mesoPatch(ts,u0);
165 [ts,us] = ode23(@patchSys3,ts,u0(:));
166 us=reshape(us,[length(ts) size(u0)]);
```

2. In the second case, `RK2mesoPatch` detects a parallel patch code has been requested, but has only one cpu worker, so it auto-initiates an `spmd`-block for the integration. Both this and the next case return *composite* results, so just keep one version of the results.

```
178 case 2
179 us = RK2mesoPatch(ts,u0);
180 us = us{1};
```

3. In this third case, a user could merge this explicit `spmd`-block with the previous one that sets the initial conditions.

```
189 case 3,spmd
190 us = RK2mesoPatch(ts,u0,[],patches);
191 end%spmd
192 us = us{1};
```

4. In this fourth case, use Projective Integration (PI) over long times (PIRK4 also works). Currently the PI is done serially, with parallel `spmd`-blocks only invoked inside function `aBurst()` ([Section 6.3.3](#)) to compute each burst of the micro-scale simulation. A macro-scale time-step of about 3 seems good to resolve the decay of the macro-scale ‘homogenised’ diffusion.² The function `microBurst()` here interfaces to `aBurst()` ([Section 6.3.3](#)) in order to provide shaped initial states, and to provide the patch information.

² Curiously, `PIG()` appears to suffer unrecoverable instabilities with its variable step size!

```

210 case 4
211     microBurst = @(tb0,xb0,bT) ...
212         aBurst(tb0 ,reshape(xb0,size(u0)) ,patches);
213     ts = 0:3:51
214     us = PIRK2(microBurst,ts,gather(u0(:)));
215     us = reshape(us,[length(ts) size(u0)]);

End the four cases.

222 end%switch theCase

```

6.3.2 Plot the solution

Optionally save some plots to file.

```

233 if 0, global OurCf2eps, OurCf2eps=true, end

Animate the solution field over time. Since the spatial domain is long in  $x$ 
and thin in  $y, z$ , just plot field values as a function of  $x$ .

241 figure(1), clf
242 if theCase==1
243     x = reshape( patches.x(2:end-1,:,:), [],1);
244 else, spmd
245     x = reshape(gather( patches.x(2:end-1,:,:), []), [],1);
246     end%spmd
247     x = x{1};
248 end

For every time step draw the field values as dots and pause for a short display.

255 nTimes = length(ts)
256 for l = 1:length(ts)

At each time, squeeze interior point data into a 4D array, permute to get all
the  $x$ -variation in the first two dimensions, and reshape into  $x$ -variation for
each and every  $(y, z)$ .

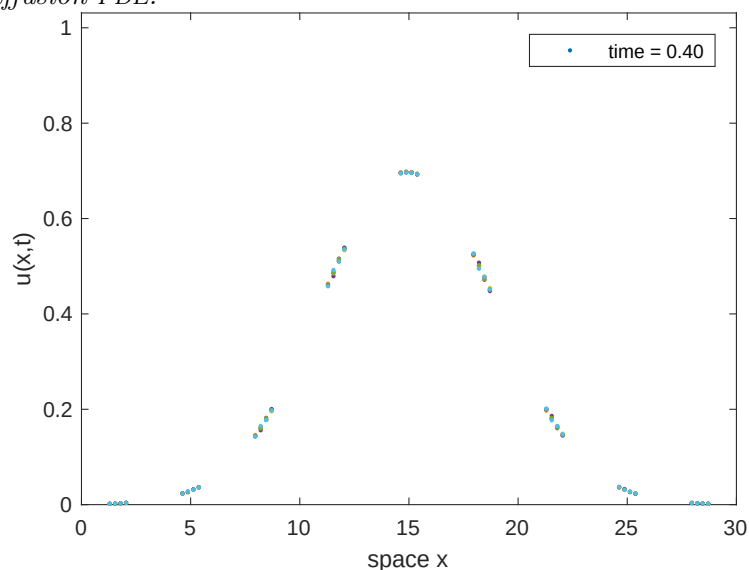
265     u = reshape( permute( squeeze( ...
266         us(l,2:end-1,2:end-1,2:end-1,:) ) , [1 4 2 3]) , numel(x), []);

Draw point data to show spread at each cross-section, as well as macro-scale
variation in the long space direction.

273     if l==1
274         hp = plot(x,u,'.');
275         axis([0 Len 0 max(u(:))])
276         xlabel('space $x$'), ylabel('$u(x,y,z,t)$')
277         ifOurCf2eps([mfilename 't0'])
278         legend(['time = ' num2str(ts(l),'%4.2f')])
279         disp('**** pausing, press blank to animate')
280         pause
281     else
282         for p=1:size(u,2), hp(p).YData=u(:,p); end

```

Figure 6.7: final field $u(x, y, z, 0.4)$ of the patch scheme applied to a heterogeneous diffusion PDE.



```

283     legend(['time = ' num2str(ts(1), '%4.2f')])
284     pause(0.1)
285 end

```

Finish the animation loop, and optionally output the final plot, [Figure 6.7](#).

```

298 end%for over time
299 ifOurCf2eps([mfilename 'tFin'])

```

6.3.3 microBurst function for Projective Integration

Projective Integration stability seems to need bursts longer than 0.2. Here take ten meso-steps, each with default ten micro-steps so the micro-scale step is 0.002. With macro-step 3, these parameters usually give stable projective integration (but not always).

```

315 function [tbs,xbs] = aBurst(tb0,xb0,patches)
316     normx=max(abs(xb0(:)));
317     disp(['aBurst t = ' num2str(tb0) ' |x| = ' num2str(normx)])
318     assert(normx<10,'solution exploding')
319     tbs = tb0+(0:0.02:0.2);
320     spmd
321         xb0 = codistributed(xb0,patches.codist);
322         xbs = RK2mesoPatch(tbs,xb0,[],patches);
323     end%spmd
324     xbs=reshape(xbs{1},length(tbs),[]);
325 end%function

```

Fin.

6.4 RK2mesoPatch()

This is a Runge–Kutta, 2nd order, integration of a given deterministic system of ODEs on patches. It invokes meso-time updates of the patch-edge values in order to reduce interpolation costs, and uses a linear variation in edge-values over the meso-time-step (Bunder et al. 2016, case $Q = 2$). This function is aimed primarily for large problems executed on a computer cluster to markedly reduce expensive communication between computers.

If using within projective integration, it appears quite tricky to get all the time-steps chosen appropriately. One has to choose times for: the micro-scale time-step, the meso-time interval between communications, the longer meso-time burst length, and the macro-scale integration time-step.

```

27 function [xs,errs] = RK2mesoPatch(ts,x0,nMicro,patches)
28 if nargin<4, global patches, end

```

Input

- `patches.fun()` is a function such as `dxdt=fun(t,x,patches)` that computes the right-hand side of the ODE $d\vec{x}/dt = \vec{f}(t, \vec{x})$ where $\vec{x}$ is a vector/array, t is a scalar, and the result $\vec{f}$ is a correspondingly sized vector/array.
- `x0` is an vector/array of initial values at the time `ts(1)`.
- `ts` is a vector of meso-scale times to compute the approximate solution, say in $\mathbb{R}^\ell$ for $\ell \geq 2$.
- `nMicro`, optional, default 10, is the number of micro-time-steps taken for each meso-scale time-step.
- `patches` struct set by `configPatchesn` and provided as either as parameter, or as a global variable.

Output

- `xs`, 5/7/9D (depending upon `nD`) array of length $\ell \times \dots$ of approximate solution vector/array at the specified times. But, if using parallel computing via `spmd`, then `xs` is a *composite* 5/7/9D array, so outside of an `spmd`-block access a single copy of the array via `xs{1}`. Similarly for `errs`.
- `errs`, column vector in $\mathbb{R}^\ell$ of local error estimate for the step from t_{k-1} to t_k .

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